



NI 43-101 TECHNICAL REPORT
LALOR & SNOW LAKE MANITOBA, CANADA
MARCH 29, 2021, EFFECTIVE AS OF JANUARY 1, 2021

Prepared under the supervision of:

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CAUTIONARY NOTE REGARDING FORWARD LOOKING INFORMATION

This Technical Report contains "forward-looking statements" and "forward-looking information" (collectively, "forward-looking information") within the meaning of applicable Canadian and United States securities legislation. All information contained in this Technical Report, other than statements of current and historical fact, is forward-looking information. Often, but not always, forward-looking information can be identified by the use of words such as "plans", "expects", "budget", "guidance", "scheduled", "estimates", "forecasts", "strategy", "target", "intends", "objective", "goal", "understands", "anticipates" and "believes" (and variations of these or similar words) and statements that certain actions, events or results "may", "could", "would", "should", "might" "occur" or "be achieved" or "will be taken" (and variations of these or similar expressions). All of the forward-looking information in this Technical Report is qualified by this cautionary note.

Forward-looking information includes, but is not limited to, our objectives, strategies, intentions and expectations regarding Hudbay's Lalor mine and Snow Lake operations, including the anticipated commissioning and ramp-up of the New Britannia mill, forecasted production and projected capital and operating costs based on the life of mine plans for Lalor and the Snow Lake operations, the forecasted cash costs of production, planned and potential optimization opportunities, including the Stall mill recovery improvement project and the potential to increase the future production rate at Lalor, the potential to convert mineral resources to mineral reserves and to identify additional mineral resources through exploration, the anticipated effect of external factors on revenue, such as commodity prices, estimation of mineral reserves and resources, mine life projections, reclamation costs, economic outlook, government regulation of mining operations, and expectations regarding labour and community relations. Forward-looking information is not, and cannot be, a guarantee of future results or events. Forward-looking information is based on, among other things, opinions, assumptions, estimates and analyses that, while considered reasonable by us at the date the forward-looking information is provided, inherently are subject to significant risks, uncertainties, contingencies and other factors that may cause actual results and events to be materially different from those expressed or implied by the forward-looking information.

The material factors or assumptions that we identified and were applied by us in drawing conclusions or making forecasts or projections set out in the forward-looking information include, but are not limited to:

- the schedule for the refurbishment, commissioning and ramp-up of the New Britannia mill;
- the success of mining, processing, exploration and development activities;
- the scheduled maintenance and availability of our Snow Lake processing facilities;
- the accuracy of geological, mining and metallurgical estimates;
- anticipated metals prices and the costs of capital projects and production;
- the supply and demand for metals we produce;
- the supply and availability of third party processing facilities for our concentrate;
- the supply and availability of all forms of energy and fuels at reasonable prices;
- the availability of transportation services at reasonable prices;
- no significant unanticipated operational or technical difficulties;
- the availability of personnel for our exploration, development and operational projects and ongoing employee relations;
- maintaining good relations with the communities surrounding Lalor mine and no labour disruptions;
- no significant unanticipated challenges with stakeholders at our various projects;
- no significant unanticipated events or changes relating to regulatory, environmental, health and safety matters;
- no contests over title to our properties, including as a result of rights or claimed rights of aboriginal peoples;

- no significant unanticipated litigation; and
- no significant and continuing adverse changes in general economic conditions or conditions in the financial markets (including commodity prices and foreign exchange rates).

The risks, uncertainties, contingencies and other factors that may cause actual results to differ materially from those expressed or implied by the forward-looking information may include, but are not limited to, risks generally associated with the mining industry, such as economic factors (including future commodity prices, currency fluctuations, energy prices and general cost escalation), risks related to the new Lalor and Snow Lake mine plan including the schedule for the refurbishment, commissioning and ramp-up of the New Britannia mill and the Stall mill recovery improvement project, risks related to the transition of our work force from the 777 mine to Snow Lake following the closure of the 777 mine, risks related to the negotiation of new collective bargaining agreement with the labour unions representing our employees in Manitoba, uncertainties related to the development and operation of our projects, dependence on key personnel and employee and union relations, risks related to political or social unrest or change, risks in respect of aboriginal and community relations, rights and title claims, operational risks and hazards, including unanticipated environmental, industrial and geological events and developments and the inability to insure against all risks, failure of plant, equipment, processes, transportation and other infrastructure to operate as anticipated, depletion of Hudbay's reserves, volatile financial markets that may affect our ability to obtain additional financing on acceptable terms, the failure to obtain required approvals or clearances from government authorities on a timely basis, uncertainties related to the geology, continuity, grade and estimates of mineral reserves and resources, and the potential for variations in grade and recovery rates, uncertain costs of reclamation activities, Hudbay's ability to comply with its pension and other post-retirement obligations, our ability to abide by the covenants in our debt instruments and other material contracts, tax refunds, hedging transactions, as well as the risks discussed under the heading "Risk Factors" in our most recent Annual Information Form dated March 29, 2021.

Should one or more risk, uncertainty, contingency or other factor materialize, or should any factor or assumption prove incorrect, actual results could vary materially from those expressed or implied in the forward-looking information. Accordingly, you should not place undue reliance on forward-looking information. We do not assume any obligation to update or revise any forward-looking information after the date of this Technical Report or to explain any material difference between subsequent actual events and any forward-looking information, except as required by applicable law.

Hudbay uses certain non-IFRS financial performance measures in its financial reports and in this 43-101 Technical Report. Cash cost and sustaining cash cost per pound of copper and ounce of gold produced are shown because the company believes they help investors and management assess the performance of its operations, including the margin generated by the operations and the company. Unit operating costs are shown because these measures are used by the company as a key performance indicator to assess the performance of its mining and processing operations. These measures do not have a meaning prescribed by IFRS and are therefore unlikely to be comparable to similar measures presented by other issuers. These measures should not be considered in isolation or as a substitute for measures prepared in accordance with IFRS and are not necessarily indicative of operating profit or cash flow from operations as determined under IFRS. Other companies may calculate these measures differently. For further details on these measures, including reconciliations of historical unit operating costs and cash costs per pound of copper produced to the most comparable IFRS measures, please refer to page 53 of Hudbay's management's discussion and analysis for the three and twelve months ended December 31, 2020 available on SEDAR at www.sedar.com.

This 43-101 Technical Report contains references to both United States dollars and Canadian dollars. All dollar amounts referenced, unless otherwise indicated, are expressed in constant Canadian dollars.

SIGNATURE PAGE

This Technical Report titled "NI 43-101 Technical Report, Lalor and Snow Lake Operations, Manitoba, Canada", dated March 29th 2021 and effective as of January 1st 2021, was prepared under the supervision and signed by the following author:

Dated this 29th day of March, 2021.

/s/ Olivier Tavchandjian

Signature of Qualified Person

Olivier Tavchandjian, P. Geo.
Vice President, Exploration and Geology
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1. SUMMARY

Hudbay is a Canadian integrated mining company with assets in North and South America. Hudbay operates multiple properties in the Province of Manitoba. Current operations near Snow Lake include the Lalor underground mine, the Stall ore concentrator, a tailings impoundment area and other ancillary facilities that support the operation including a paste plant at Lalor mine. The New Britannia mill which will produce a copper concentrate as well as gold-silver doré is currently being refurbished with a planned start of production in the second half of 2021.

This Technical Report has been prepared to support the public disclosure of Mineral Resources and Mineral Reserves at the Lalor Mine and nearby Snow Lake satellite deposits and to provide an updated mine plan that achieves up to 5,300 tpd of base metal, gold and copper-gold ore feed to the Stall and New Britannia concentrators. Refurbishing the New Britannia mill will significantly increase gold production from Lalor, extend the life of the Snow Lake operations through the processing of gold production from the WIM and 3 Zone deposits and enable new gold and copper-gold exploration opportunities in the Snow Lake region by having an operating processing facility with substantially higher gold, silver and copper recoveries. In addition, the recently discovered 1901 deposit, located between the former producing Chisel mine and Lalor mine and within trucking distance to surface will provide supplementary feed for several years in order to maximize the use of the processing capacity at both the Stall and New Britannia mills.

The Snow Lake satellite deposits that host additional mineral reserve estimates and are included in the consolidated mine plan of the Snow Lake operations include the 1901, WIM and 3 Zone deposits. Additionally, mineral resources have also been estimated for the Pen II, Watts and New Britannia properties. These satellite deposits are not currently part of the mine plan and may provide an opportunity in addition to the Lalor and 1901 inferred mineral resource estimates to increase Hudbay's mineral reserves in the area and further expand production and/or extend the operating lives of the Stall and New Britannia processing facilities.

1.1. QUALIFIED PERSON (QP)

This Technical Report has been prepared in accordance with National Instrument Form 43-101F1. The qualified person ("QP") who supervised the preparation of this Technical Report is Olivier Tavchandjian, P. Geo., Hudbay's Vice-President, Exploration and Geology.

1.2. PROPERTY LOCATION, OWNERSHIP AND DESCRIPTION

For all the properties described in this Technical Report, Hudbay owns a 100% interest and there are no royalties payable other than those potentially payable to the province. Surface rights are held under general permits and are sufficient for purposes of our development plans. These properties and their associated infrastructures are located within 60 km of the town of Snow Lake, approximately 200 km by road east of the city of Flin Flon where the nearest full service commercial airport is located.

All these properties are accessible by road from the town of Snow Lake. The Snow Lake area has a typical mid-continental climate, with short summers and long, cold winters. Climate generally has only a minor effect on local exploration and mining activities. The area is surrounded by water bodies. Topography shows gentle relief that rarely exceeds 10 m.

The town of Snow Lake is a full-service community with available housing, hospital, police, fire department, potable water system, restaurants and stores. To house non-local employees during their work rotations, the company provides a camp located in town which services Hudbay employees and contractors for the mine and mill operations. Other infrastructure in the area includes provincial roads, a 115 kV Manitoba Hydro power grid within four kilometres of Lalor and Manitoba Telecom land line and cellular phone service.

Hudbay operates two concentrators in the Snow Lake / Flin Flon area: the Stall mill located approximately 16 km from Lalor and the Flin Flon mill located 200 km to the west. While the Flin Flon mill provides an

opportunity to truck additional production from Lalor that the Stall mill cannot accommodate, the recent operating performance of the Stall mill through 2020 has resulted in minimum tonnage being trucked to Flin Flon from Snow Lake over the past year as the Stall mill has been operating close to the 3,900tpd produced by Lalor mine

Hudbay also operates a zinc metallurgical plant in Flin Flon with a capacity of 115,000 tonnes per annum of refined zinc. However, Hudbay intends to cease operations and put the zinc plant and the Flin Flon mill on care and maintenance at the end of Q2 2022, when Hudbay's 777 mine closes. After the zinc plant closure, the concentrates produced from the Stall and Flin Flon mills will be sold to market.

The New Britannia gold mill is located approximately 16 km east of the Lalor mine and it is currently being refurbished with a new copper flotation plant under construction. Hudbay plans to complete the refurbishment, commissioning and ramp-up of the New Britannia mill by the third quarter of 2021, adding a new copper flotation circuit with a nominal throughput capacity of 1,500tpd.

1.3. HISTORY

The Snow Lake area has a long exploration and mining history. Exploration in the Lalor-Chisel area has been occurring since the 1950s and the Chisel Basin area has hosted four past producing mines. This basin is also the host of the Lalor deposit. Lalor commenced initial ore production from the ventilation shaft in August 2012, only five years after its initial discovery hole and achieved commercial production from the main shaft in the third quarter of 2014. Since 2012, the mine has produced 8.8 million tonnes.

Gold was first discovered in 1914 approximately 20 km to the southeast of Snow Lake and in 1917, the Moose Horn-Ballast claims produced the first gold in Manitoba. First mine construction at the New Britannia site started in 1945 and in March 1949, the mine was opened as the Nor-Acme mine. Production continued until 1958. 4.9 Mt were mined at an average gold grade of 4.4 g/t and Nor-Acme mill recovered approximately 610,000 Oz of gold during this production period. TVX and High River formed a joint venture to reopen the mine and TVX became the operator. Full production from the main shaft was achieved in August 1996. Through various transactions, Kinross became the operator of the New Britannia mine-mill complex. Production ceased at the end of September 2004 and the mill was put on care and maintenance in 2005 due to a low gold price environment after producing 1.6 million ounces of gold. The New Britannia complex was subsequently acquired by QMX Gold Corporation.

Hudbay acquired a 100% interest in the New Britannia mill, and related mineral properties, from QMX Gold Corporation in 2015.

1.4. GEOLOGY AND MINERALIZATION

The properties of interest for this Technical Report are all located within the Trans-Hudson Orogen of the Flin Flon Greenstone Belt. The volcanic assemblages consist of mafic to felsic volcanic rocks with intercalated volcanogenic sedimentary rocks.

The volcanogenic massive sulphide (VMS) deposits located near the town of Snow Lake have been subdivided into two different groups: Cu-Zn-rich (Cu-Zn, Cu-Zn-Au) and Zn-Cu-rich (Zn-Pb-Cu-Ag) types. The Cu-Zn-rich deposits mainly occur in the Anderson sequence and the Zn-Cu-rich deposits occur in the Chisel sequence. The Watts River Deposit, located 52km east-southeast of the town of Snow Lake lies in the eastern portion of the Flin Flon-Snow Lake Greenstone belt and is a stratabound accumulation of sulphides that precipitated in a depositional environment similar to the base metal deposits of the Snow Lake mining camp.

Mineralization of the lode-gold vein-type deposits are hosted in the Amisk group mafic and felsic volcanic rocks which are structurally controlled and associated with shear zones, faults, fold hinges and axial planes that host simple to complex vein systems. The mineralization is associated with lithological contacts of contrasting properties in the sequence of interlayered volcanic and volcanoclastic rocks.

1.5. EXPLORATION

Since the last NI 43-101 Technical Report published in 2019, exploration drilling at Lalor has focused on adding and converting inferred mineral resource estimates with a strong emphasis on confirming the continuity of the gold mineralization. In parallel, surface drilling has been successful in discovering and defining mineral resources for both base metal and gold mineralization at the 1901 deposit and also confirming mineral resources at the Watts deposit.

Exploration drilling will continue in 2021 both at Lalor and at the 1901 deposit in order to confirm and expand the known mineral resource and mineral reserve estimates. Exploration drilling will also follow-up on some initial successes obtained between Lalor and the former producing Chisel mine while efforts continue to discover new base metal and gold deposits in the Snow Lake region.

1.6. DRILLING, SAMPLING, ASSAYING, QA/QC AND VALIDATION

At Lalor, over 3,300 drill holes totaling more than 580,000 m were included in the Lalor database to support the mineral resource and mineral reserve estimates while drilling supporting the 1901, Watts, Pen II and Wim mineral resource and mineral reserve estimates totals 51,000m, 25,000m, 2,000 m and 43,000 m respectively. For the New Britannia mineral resource estimates including the 3 Zone and Birch zones, over 730,000 m of drilling completed after 1995 were used. Drilling at all properties is a combination of NQ and BQ diamond drill holes, surveyed with either Reflex downhole tools or Gyro for deeper/longer holes.

At Lalor, 1901, Watts, Pen II and Wim, the core was photographed before samples were split and bagged for shipment before dispatch to the laboratories, QA/QC samples were inserted into the sample stream. Hudbay's practice involves insertion of the following every 100 samples, 5 blanks, 5 duplicates, 5 base metal standards and 2 gold standards.

Sample preparation has been conducted at three different laboratories over time. Since September 2016, nearly all samples are prepared and assayed at Bureau Veritas in Vancouver. All drill core samples analyzed since the last technical report (NI 43-101 published on March 28, 2019) have been sent for analysis at Bureau Veritas while the SGS laboratory in Vancouver was used as the umpire laboratory for quality control purposes.

Results from the QA/QC program for standards, blanks, duplicates and external checks show that the program has been working effectively for the Lalor, 1901, Watts, Pen II and Wim properties, meeting industry standards and the data used provides a representative and unbiased basis for resource modeling purposes.

Most of the drill cores and chips assays from the New Britannia mine were tested by the previous owners from 1995 to 2003 and were completed at the on-site mill laboratory using a fire assay/atomic absorption finish (FA/AA) method. Standard, blank and duplicate assay samples were added to each batch of 21 samples for drill core and to each batch of 24 samples for chip samples. The sampling and analytical procedures conformed to the industry standards at the time, and these were adequate to ensure a representative determination for the type of gold mineralization identified on the property. In 2019, 6 holes drilled by Hudbay at 3 Zone confirmed previous drilling results.

It is the QP's opinion that the data is adequate and acceptable for use in the estimation of mineral resources at all the properties covered by this Technical Report.

1.7. MINERAL PROCESSING AND METALLURGICAL TESTING

The Stall concentrator is an operating plant running at steady state and, as a result, several of the initial metallurgical test results and assumptions have been revised to reflect the operating experience and performance of the plant over the past six years of operation in processing the ore produced from the Lalor mine. The Stall concentrator is producing a copper concentrate grade of 18 to 20% copper at 83 to 85% recovery and a zinc concentrate grade of 51% zinc at 90 to 93% recovery. 55 to 62% of the gold and silver are recovered in the copper concentrate as co-products. Over the life of Lalor mine, copper, gold and silver grade will increase and the average zinc grade will decrease from Lalor. This trend will partially be offset in 2026, when the 1901 deposit enters production and feed Stall with zinc rich mineralization.

Extensive metallurgical testing was conducted in 2019 and 2020 to demonstrate the technical viability and economic benefits of some changes to the process flowsheet of the Stall mill to improve metal recoveries and/or concentrate grade from historical performance. The main changes to the flowsheet are planned to be implemented by the end of 2022 and include the addition of Jameson cells to increase copper rougher and cleaner capacity, the addition of a talc preflotation circuit, an increase in the zinc circuit cleaning capacity and froth washing and an increase in recovery of free gold through the addition of a Knelson gravity concentrator on the copper regrind cyclone underflow. The total cost to implement these upgrades is estimated to be \$19 million (C\$24 million) and is expected to increase Stall's copper recoveries to between 91% and 95%, gold recoveries to between 64% and 70%, and silver recoveries to between 65% and 74%.

In addition, further testwork is underway to include a lead recovery stage into the sequential flotation circuit. This testwork will consider the benefits of a copper-lead-zinc sequential flowsheet utilizing TNC 312 as the copper circuit collector and 3418A as the lead circuit collector. These further changes to the flowsheet are expected to improve metal recovery and the benefits of this addition have not yet been incorporated into the present life of mine plan, Hudbay anticipates a short payback on this additional limited investment at Stall.

In 2020 a metallurgical testwork program was conducted by Blue Coast Research to cover composites representing low grade, medium grade and high grade of the two zinc rich lenses of the 1901 deposit. A subsample of each of the composite samples was ground to a p80 of 100µm and submitted for mineralogical analysis. Mineralogical analysis, and flotation tests were completed on each of the six composites and confirmed that the metallurgical performance of the Stall concentrator for the Lalor base metal lenses was applicable to the 1901 deposit, including the potential benefit of a lead recovery stage in the flotation circuit.

Should the Watts or Pen II Zones be developed, it is expected that they could be milled at the Stall Concentrator and blended with ore from the Lalor Mine.

To establish the future performance of the New Britannia mill, two major metallurgical test work programs were undertaken between 2015 and 2020 by SGS Lakefield and Blue Coast Research. The first laboratory program was conducted at SGS in 2015-2016 to develop a flowsheet, design criteria, and metallurgical forecast information for the gold and copper-gold rich ore types. All sample composites were submitted for mineralogical analysis by QEMSCAN to identify minerals and their liberation as well as for Bond rod and Bond ball mill work tests, gravity concentration test, rougher and cleaner flotation tests, cyanidation tests, and rolling bottle leach tests. In addition, CIP modelling was completed to predict the gold extraction performance of a CIP or CIL circuit. Cyanide destruction tests using the SO₂/Air process were also carried out following standard SGS procedure of completing batch tests first to confirm applicability and to optimize retention times and reagent requirements. These results were confirmed and further refined in 2019-2020 at Blue Coast Research facilities. The testwork program was conducted on 13 composites and included additional mineralogy, flotation, cyanidation, and cyanide destruction tests.

Metallurgical testwork conducted in 2019 on WIM and 3 zone has confirmed that this mineralization is also amenable to successful beneficiation at the New Britannia mill. Four composites were created for each deposit and submitted for mineralogical, comminution and flotation as well as leach test work and gravity concentration in the case of 3 Zone. These tests have been used to confirm the copper, gold and silver recoveries applied in the life of mine plan for these two satellite deposits.

It is anticipated that any mineral resource from the 1901 gold zones and New Britannia property would also be processed at the New Britannia mill, by-passing the copper flotation circuit in the case of New Britannia zones and achieving similar gold recoveries as per historical performance of the mine.

1.8. MINERAL RESOURCE ESTIMATES

Lalor, 1901, Watts and Pen II

For the Lalor, 1901, Watts and Pen II properties, the construction of the 3D resource model and the estimation of mineral resources were performed by Hudbay personnel following Hudbay procedures in compliance with best industry standards and the CIM guidelines, using Leapfrog® version 4.2.3 and MineSight® version 15, two industry standard commercial geological and mining software.

The construction of the mineralized envelopes was solely based on the type of mineralization intersected. To this effect, assay intervals with high Au grade but no zinc or copper were excluded from base metal lenses and vice versa. Sample coding was checked to ensure proper tagging of the solids to actual drill hole locations. The mineral envelopes were used as hard boundaries in all cases for grade interpolation purposes.

Density values were generated from robust multi-regression formulas where no actual measurement was taken. Assay intervals were regularized by compositing drill hole data within each interpreted mineralized envelope. Although most of the drill holes were assayed in 1 m intervals, a composite length of 2.5 m was selected as more appropriate to conduct interpolation into the 5mx5mx5m block size selected by the mine engineering team as the optimum selective mining unit (SMU).

Exploratory data analysis (EDA) includes basic statistical evaluation of the composites for zinc (Zn), gold (Au), copper (Cu), silver (Ag) and density (SG). The EDA was conducted separately for each mineralized envelope and aimed to identify sub-domains that did not support the use of hard contacts for grade interpolation purposes but that justified being used for grade capping, block model validation, smoothing assessment and correction and resource classification.

The deciles¹ analysis (Parish) method was used to define high-grade outliers and to confirm the need for grade capping. This analysis was conducted on the 2.5 m composites for each sub-domain independently. Down-hole and directional variograms for Zn, Au, Cu, Ag, and SG were created for each individual mineral envelope. A linear combination of a nugget and two nested spherical models were adjusted in all cases. Once generated, a systematic visual check was conducted to ensure that the search ellipsoid would be correctly oriented with respect to the geometry of the mineral envelopes.

The grade estimation process was validated for each envelope to ensure appropriate honouring of the input data and subsequent unbiased resource reporting and use of the model for reserve estimation. The validation steps included systematic visual investigations, checks for global bias and over-smoothing. Where the model was found to be over-smoothed, an indirect log-normal correction was applied to the kriged estimates.

The regression slope values obtained from the kriging of zinc (in the case of base metals lenses) or gold (in the case of gold zones) grade estimates was used as the primary criteria for resource classification. The block by block coding assignment was then smoothed to remove isolated blocks of one category within another.

For the present estimates, Hudbay has implemented a stringent approach to establish the potential for economic extraction of its resource reporting for underground deposits. With this approach, the potential for economic extraction of the mineral resource estimates at Lalor, 1901, Watts, and Pen II are reported within the constraint of a 'stope optimization envelope'. This excludes small isolated individual blocks above the economic cut-off criteria from the resource estimate and includes some 'geological dilution' that would need to be included in the economic envelope to maintain minimum spatial continuity requirements to define mineable shapes.

¹ I.S. Parrish (1997) Mining Engineering Journal, Geologist's Gordon Knot: to cut or not to cut (pp. 45-49)

The parameters used as input to define the stope optimization envelope cover all the relevant technical and economic constraints including minimum stope and waste pillar dimensions and a NSR value calculation for each block based on anticipated metal recoveries, long-term metal price forecast and operating and capital costs based on the 2021 Lalor mine and Stall concentrator budgets. In all cases, two NSR values were calculated for each block to assess and compare the value of the blocks going to the Stall mill (no material difference between the two) or going to the new Britannia mill. The mineral resource estimates are reported to ensure that each potential stope would cover all its associated operating mining and milling costs.

The resulting mineral resource estimates for the Lalor, 1901, Watts and Pen II properties are presented in Table 1-1. All measured and indicated mineral resource estimates at Lalor and 1901 were either converted to mineral reserve estimates or deemed non-economic after applying the resource to reserve appropriate conversion factors.

**TABLE 1-1: SUMMARY OF MINERAL RESOURCE ESTIMATES FOR THE LALOR, 1901, WATTS AND PEN II DEPOSITS
(AS OF JANUARY 1ST, 2021)**

Lalor, 1901, Watts and Pen II Mineral Resource Estimates (Exclusive of Mineral Reserves) – January 1, 2021						
			Tonnes	Cu (%)	Zn (%)	Au (g/t) Ag (g/t)
Base Metal	Indicated	PEN II	470,000	0.49	8.89	0.3 6.8
		Subtotal	470,000	0.49	8.89	0.3 6.8
	Inferred	Lalor	590,000	0.31	3.48	2.8 55.0
		1901	310,000	0.85	6.44	2.0 24.6
		Watts	3,150,000	2.34	2.58	1.0 31.0
		PEN II	130,000	0.37	9.81	0.3 6.8
		Subtotal	4,180,000	1.88	3.22	1.3 33.2
Gold	Inferred	Lalor	5,610,000	1.17	0.35	4.6 26.0
		1901	480,000	0.72	0.55	6.7 37.3
		Subtotal	6,090,000	1.13	0.37	4.8 26.9

Notes:

1. Totals may not add up correctly due to rounding.
2. Mineral resources that are not mineral reserves do not have demonstrated economic viability.
3. Mineral resources in the above tables do not include mining dilution or recovery factors.
4. Base metal mineral resources are estimated based on the assumption that they would be processed at the Stall concentrator while gold mineral resources are estimated based on the assumption that they would be processed at the New Britannia concentrator, which is currently being refurbished.
5. Metal prices of US\$1.10 per pound zinc, US\$1,500 per ounce gold, US\$3.10 per pound copper, and US\$18.00 per ounce silver with an exchange rate of 1.30 C\$/US\$ were used to estimate mineral resources at Lalor.
6. 1901, Watts, and Pen II mineral resources were initially estimated using metal price assumptions that vary marginally over the assumptions used to estimate mineral resources at Lalor. In the QP's opinion, the combined impact of these small variations does not have any impact on the mineral resource estimates.
7. Lalor mineral resources are estimated using a minimum NSR cut-off of \$105 per tonne for waste filled mining areas and a minimum of \$116 per tonne for paste filled mining areas.
8. 1901 mineral resources are estimated using a minimum NSR cut-off of C\$140 per tonne.
9. Watts mineral resources are estimated using a minimum NSR cut-off of \$150 per tonne, assuming processing recoveries of 90% for copper, 80% for zinc, 70% for gold and 70% for silver.
10. Pen II mineral resources are estimated using a minimum NSR cut-off of \$75 per tonne and assume that the mineral resources would be amenable to processing at the Stall mill.

WIM and 3 Zone

Golder Associates ("Golder") prepared a resource model for the Wim property in 2015 and for the 3 Zone property in 2019 using Datamine® version 2.1.1547.0, an industry standard commercial geological and mining software. The procedures used to create and validate the resource block model are compliant with best industry standards and the CIM guidelines. A similar methodology was used in both cases.

A smooth and continuous 3D mineralized envelope was built using a copper equivalency cut-off grade in the case of WIM and a 2g/t gold grade in the case of 3 Zone to guide the construction of the mineralized

envelope while including material below cut-off where necessary to maintain the continuity of the envelope. Scatter plots were used to identify assay outliers and for grade capping. Missing density values were generated using a linear regression formula from either the Fe grade or from the Cu grade for the WIM deposit while a constant density value was used for 3 Zone where there is no significant correlation between density and gold content. A composite length of 1 m was selected to conduct interpolation into the blocks in both deposits. Compositing was weighted by density at WIM.

A block size of 15 m east-west (strike), 3 m north-south (thickness), and 5 m elevation was selected. Grade for Cu, Zn, Au, Ag and density (SG) was estimated using an inverse distance interpolant (IDW2) in the case of WIM and an ordinary kriging estimate in the case of 3 Zone in their respective unfolded space and interpolation was weighted by density in the case of WIM. A nearest neighbor (NN) interpolation was also performed to provide declustered sample grade for block model validation. The search ellipsoids were defined from pairwise variograms modeled by a linear combination of a nugget and nested spherical structures.

Resource classification was assigned based on the search parameters used for grade interpolation. The block by block coding assignment was then smoothed to remove isolated blocks of indicated within areas of mostly inferred category and vice versa.

For WIM and 3 Zone, there are no mineral resource estimates being reported exclusive of the mineral reserve estimates. All measured and indicated mineral resource estimates that were deemed non-economic after applying the resource to reserve appropriate conversion factors were simply depleted from the resource inventory.

New Britannia Mine

For the former New Britannia, mine and its satellite gold deposits, the historical resource estimate performed by Kinross and by Alexis Minerals followed a conventional and industry standard approach and have been independently validated in 2018 by WSP Engineering ("WSP"). A block model of 15 ft. x 15 ft. x 15 ft. was created for the main mine upper and lower zones while a 10 ft. x 10 ft. x 10 ft. block model was created for the Birch Zone. The cut-off grades for the resource have been estimated over a 6-ft. minimum true width with a variable cut-off by zone as summarized in 1-6. The variation in the cut-off grade is related to new mining versus remnant mining. The New Britannia mill will have been refurbished before the mineral resource estimates would potentially be mined.

Given that WSP had to rely on historical documentation for some of the technical information supporting the estimation of the mineral resource estimates, the tonnes and grades previously estimated by Kinross and Alexis Minerals as measured and indicated resources were downgraded to an inferred category. Mineral resources that are not mineral reserves do not have demonstrated economic viability

TABLE 1-2: GOLD RESOURCE ESTIMATE FOR THE NEW BRITANNIA MINE AND BIRCH DEPOSIT ⁽¹⁾⁽²⁾⁽³⁾

Zone	Location	Cut-off (Au g/t)	Tonnes	Grade
Lower Dick NW	1780 -	3.3	7,000	4.76
Main Mine 1780 +	1780 +	3.5	1,142,000	4.55
Lower Dick NW	780 -	3.3	213,000	4.55
Lower Dick EW	1780 -	2	823,000	4.59
Lower Ruttan	1780 -	2	568,000	4.28
Birch	Birch	3.3	569,000	4.42
Total Inferred			3,322,000	4.49

1. Totals may not add up correctly due to rounding.
2. Mineral resources that are not mineral reserves do not have demonstrated economic viability.
3. Mineral resources in the above tables do not include mining dilution or recovery factors.

1.9. MINERAL RESERVE ESTIMATES

No mineral reserve estimates are reported for the Pen II, Watts, 1901 gold zones and New Britannia Mine deposits.

The following steps were followed in developing the reserve estimates at Lalor, 1901, WIM and 3 Zone:

- Calculate two payable (NSR) values for each individual block in the resource model depending on whether processing would occur at the Stall concentrator or at the New Britannia concentrator, using long-term metal prices, concentrator recoveries, metal payability and downstream smelter treatment and refining costs assumptions.
- Design stopes in the Deswik Stope Optimizer, considering depleted mineral resources, existing workings, resource categories and mine and mill operations costs. Dilution and recovery are estimated and applied at this step. Stopes are designed for both the Stall concentrator option and the New Britannia concentrator option.
- Considering grades, value and location in the mine, assign stopes to either Stall or New Britannia concentrator.
- Establish stope economics using a secondary NSR calculation where, along with mine and mill operations costs, mine capital, waste development and offsite administration costs are applied to each stope.
- Assign whether stopes can be upgraded to mineral reserves based on resource classification.
- Design ore development required for mining the reserves. Deplete development from the stopes. Interrogate grades of designed development for inclusion in mineral reserves. Sequence and schedule development and stope production for input to a financial Life of Mine (LOM) study to support mineral reserve economics.

The above methodology takes into consideration the different ore types and the milling options for the mine's future production and considers the various ore types found at these deposits.

The mineral reserve estimates exclude the mined out mineral resources, non-recoverable pillars (rib, post and sill) within mined out areas, mineral resources that are sterilized or not recoverable due to previous mining and stopes based on inferred mineral resource estimates.

A reconciliation between tonnes and grade predicted from the resource model and actual mine production credited by the mills for the first 9 months of 2020 shows a close comparison on all parameters. The reported tonnes and grade from the resource model are globally within 6% of actual production and on the conservative side for zinc and copper, i.e. under-estimate the quantity of metal recovered by the mill by approximately 10% reflecting the past focus for grade control on base metal production at Lalor which will evolve to gold going forward.

The author considers that the mineral reserves as classified and reported comply with all disclosure in accordance with the CIM guidelines requirements and definitions. The author is not aware of any mining, metallurgical, infrastructure, permitting or other relevant factors that could materially affect the mineral reserve estimate.

The Lalor mine, 1901, WIM and 3 Zone mineral reserves as of January 1, 2021 are summarized in 1-7.

TABLE 1-3: SUMMARY OF LALOR, 1901, WIM AND 3 ZONE MINERAL RESERVES

Snow Lake Mineral Reserve Estimates – January 1, 2021						
			Tonnes	Cu (%)	Zn (%)	Au (g/t) Ag (g/t)
Base Metal	Proven	Lalor	6,860,000	0.50	5.79	2.6 29.4
		1901	890,000	0.40	6.61	2.0 28.1
		Subtotal	7,740,000	0.49	5.88	2.5 29.2
	Probable	Lalor	1,190,000	0.64	4.32	3.2 31.9
		1901	690,000	0.25	9.49	1.4 29.9
		Subtotal	1,880,000	0.50	6.23	2.6 31.1
Gold	Proven	Lalor	3,950,000	0.60	1.03	5.2 27.7
		Subtotal	3,950,000	0.60	1.03	5.2 27.7
	Probable	Lalor	3,630,000	1.16	0.53	5.7 27.9
		WIM	2,450,000	1.63	0.25	1.6 6.3
		3 Zone	660,000	-	-	4.2 -
		Subtotal	6,740,000	1.22	0.38	4.0 17.3
Base Metal and Gold	Proven and Probable	Lalor	15,620,000	0.69	3.25	4.0 28.8
		1901	1,580,000	0.34	7.87	1.7 28.9
		WIM	2,450,000	1.63	0.25	1.6 6.3
		3 Zone	660,000	-	-	4.2 -
		Total	20,310,000	0.75	3.14	3.5 25.2

Notes:

1. Totals may not add up correctly due to rounding.
2. Metal prices of US\$1.10 per pound zinc, US\$1,500 per ounce gold, US\$3.10 per pound copper, and US\$18.00 per ounce silver with an exchange rate of 1.30 C\$/US\$ were used to confirm the economic viability of the mineral reserve estimates.
3. Mineral reserves are estimated using a minimum NSR cut-off of \$105 per tonne for waste filled mining areas and a minimum of \$116 per tonne for paste filled mining areas.
4. For Lalor, individual stope gold grades were capped at 10 g/t, as a prudent estimate until reserves to mill reconciliations can establish that the high-grade gold can indeed be entirely recovered. This capping method resulted in the reduction of the global gold reserve grade by approximately 3%.
5. WIM mineral reserves are estimated assuming processing recoveries of 98% for copper, 88% for gold, and 70% for silver based on processing through New Britannia's flotation and tails leach circuits.
6. 3 Zone mineral reserves are estimated assuming processing recoveries of 85% for gold based on processing through New Britannia's leach circuit.

1.10. MINING METHODS AND PRODUCTION SCHEDULE

Geology, geotechnical information, orebody geometry, productivity and mining experience are the key drivers to select the best suited mining methods at Lalor mine and the Snow Lake satellite deposits including considerations for lateral development, production, backfilling and ore transportation.

At Lalor, the hanging wall and footwall rocks are in general of good quality allowing the use of mechanized drilling and blasting techniques. The mineralized lenses dip at an average of 30°, but locally varying from 10 to 55°. Mining methods currently in use include cut and fill and longhole open stope. Paste backfill is used to increase recovery and accelerate the mining cycle. Lower grade areas will be filled with rock from waste development as all rock produced during the development cycle is disposed as backfill in the mining cycle.

Ore is mucked using Load Haul and Dump (LHD) loaders which are operated remotely in inaccessible areas. Ore is loaded into underground haul trucks at the stoping level by LHD's or from ore passes and transported to the ore handling system at the production shaft for hoisting to surface. Approximately 600 to 1000 tpd of material is planned to be trucked to surface during the life of mine production schedule. Ore delivered to the production shaft is sized to less than 0.55 m by one of two rock breakers. Ore is hoisted from the mine by two 16 tonne capacity bottom dump skips in balance. Gold ore and base Metal ore is hoisted and stockpiled separately. On surface, ore is truck hauled to a primary crusher at the Chisel North mine site, crushed to less than 0.15 m, and will then be trucked to either the Stall, Flin Flon or New

Britannia concentrator for processing. An external primary crusher is being commissioned at Stall mill that will service all ore types and once commissioned, will deem the primary crusher at Chisel North mine redundant and it will be removed from property.

Mine development includes sub-horizontal tunneling and a ramp system to provide access between levels.

Main levels are developed parallel to and in the footwall of the ore zones. Where possible, main levels are located to provide access to multiple ore zones and are connected to haul ramps to allow mechanized equipment to travel from level to level. Main ventilation and ore pass raises are developed using a raisebore and/or Alimak climber. Drain, paste backfill and electrical cable holes are drilled from level to level using longhole or raisebore drills and are reamed to designed diameter.

Paste backfill is currently used at Lalor to increase recovery and accelerate the mining cycle and was a contributing factor to achieve a steady state of 4,500 tpd since the first quarter of 2019. Lalor mine production is approximately 20% by cut and fill mining and 80% by longhole bulk mining methods.

The Lalor mine production rate is expected to ramp-up to 5,300 tpd during the second half of 2022, following the closure of the 777 mine and the transition of personnel and equipment from the 777 mine to Lalor.

1901, WIM, and 3 Zone mining is based on development and stope designs used at 777 and Lalor mines, and the combined Life of Mine (LOM) production schedule is shown in Table 1-4. The Deswik software was used to assist with the LOM sequencing and scheduling to generate the production schedule. Historic mining rates and systems, equipment or crew's capabilities were applied and levelled. From this output, adjustments were made to further balance the capabilities to create the final plan.

TABLE 1-4: LOM PRODUCTION SCHEDULE

Year	Lalor					1901					WIM					3 Zone	
	Tonnes	Au (g/t)	Ag (g/t)	Cu (%)	Zn (%)	Tonnes	Au (g/t)	Ag (g/t)	Cu (%)	Zn (%)	Tonnes	Au (g/t)	Ag (g/t)	Cu (%)	Zn (%)	Tonnes	Au (g/t)
2021	1,627,500	3.54	26.43	0.70	4.24	-	-	-	-	-	-	-	-	-	-	-	-
2022	1,627,500	4.01	27.04	0.77	3.65	-	-	-	-	-	-	-	-	-	-	-	-
2023	1,855,000	4.09	26.07	0.74	3.02	-	-	-	-	-	-	-	-	-	-	-	-
2024	1,855,000	4.26	26.94	0.64	3.00	-	-	-	-	-	-	-	-	-	-	-	-
2025	1,855,000	4.59	27.19	0.91	2.49	-	-	-	-	-	-	-	-	-	-	-	-
2026	1,763,001	4.20	27.76	0.65	2.79	91,999	2.29	38.16	0.23	6.26	-	-	-	-	-	-	-
2027	1,475,329	4.13	31.71	0.83	3.13	334,187	2.04	42.67	0.27	6.16	-	-	-	-	-	-	-
2028	1,284,108	3.63	34.46	0.51	3.57	343,393	1.96	21.83	0.38	8.65	-	-	-	-	-	-	-
2029	1,273,696	3.40	32.42	0.48	3.84	353,804	1.43	25.41	0.32	9.09	-	-	-	-	-	-	-
2030	822,886	3.79	32.51	0.50	3.40	312,205	1.58	21.84	0.44	5.77	104,294	0.76	4.64	1.22	0.09	-	-
2031	183,027	5.24	33.68	0.58	2.06	145,381	1.14	31.52	0.29	12.53	413,553	1.24	6.01	1.62	0.18	-	-
2032	-	-	-	-	-	-	-	-	-	-	438,130	1.55	5.66	1.47	0.32	-	-
2033	-	-	-	-	-	-	-	-	-	-	438,020	1.74	6.51	1.72	0.42	-	-
2034	-	-	-	-	-	-	-	-	-	-	401,392	1.82	6.67	1.71	0.28	37,564	3.40
2035	-	-	-	-	-	-	-	-	-	-	316,455	1.82	6.92	1.71	0.17	218,824	4.17
2036	-	-	-	-	-	-	-	-	-	-	287,784	1.68	6.76	1.67	0.13	219,102	4.17
2037	-	-	-	-	-	-	-	-	-	-	48,783	1.87	6.86	1.70	0.13	186,565	4.46
Total	15,622,046	4.02	28.80	0.69	3.25	1,580,969	1.73	28.88	0.34	7.87	2,448,411	1.60	6.31	1.63	0.25	662,055	4.21

1.11. RECOVERY METHODS

The Stall concentrator complex is located approximately 16 km east of the Lalor Mine. Conventional crushing, grinding and flotation operations are used to process the ore. The nominal throughput rate of 3,500 tpd has been consistently exceeded in 2020 with production reaching 3,800t/d in a sustainable manner. The mill operates 24 hours per day, 365 days per year, with scheduled downtime for maintenance as required. The concentrator produces a copper concentrate with gold and silver credits and a zinc concentrate. Both concentrates are shipped by truck to Flin Flon. From there the copper concentrate is loaded onto rail cars and shipped to third party smelters. The zinc concentrate will continue to be trucked to Hudbay's zinc plant in Flin Flon which produces pure zinc metal until its planned closure in 2022. This refinery was refitted with the world's first two-stage pressure leach operation for recovering zinc from zinc sulphide concentrates in 1993. No roasters are used in this zinc extraction process and no sulfur dioxide (SO₂) gas is produced. After 2022, the zinc concentrate will be loaded onto rail cars and shipped to third party smelters. Tailings from the flotation circuit are utilized to produce a cemented paste backfill for use

underground. Tailings not required for paste backfill will continue to be deposited into the existing Anderson TIA.

The New Britannia mill is located approximately 16 km north of the Lalor mine. Conventional crushing, grinding, flotation, cyanide leach and recovery methods will be used to process the ore. The nominal throughput rate when the refurbishment program will be completed is 1,500 tpd starting in mid 2021. The plant will operate 24 hours per day for 362 days per year, with an availability of 92%. The concentrator will produce a copper concentrate with gold and silver credits and doré gold-silver bars. The copper concentrate will be shipped by truck to Flin Flon and will then be delivered to third party smelters. It is anticipated that, once New Britannia is operational, the copper concentrate produced at the Stall mill will be dewatered at New Britannia. The doré bars will be shipped offsite using conventional third-party transportation.

1.12. MARKET STUDIES AND CONTRACTS

Lalor produces the majority of its zinc concentrate and copper concentrate with gold and silver credits from its Stall mill. The refurbished New Britannia mill will process ore from the Lalor gold zone to produce a copper concentrate and gold-silver doré bar.

Currently, the Lalor zinc concentrate is delivered by truck to Hudbay's operations in Flin Flon and processed at the Flin Flon zinc plant into refined zinc metal and sold to customers in North America. The economic assumptions presume the same into 2022. During or beyond 2022, it is anticipated that Lalor's zinc concentrate will be sold to third party North American refineries.

The Lalor copper concentrate produced at the Stall mill is sold directly to a copper smelter in North America under a long-term agreement. The Lalor copper concentrate is delivered by rail. Lalor copper concentrate expected to be produced from the refurbished New Britannia mill has not been committed. It is anticipated that all sales contracts for the sale of copper concentrate produced from the refurbished New Britannia mill will be at standard industry terms.

The refurbished New Britannia mill will produce a gold doré commencing in 2022. It is forecast to contain approximately 69% silver, 26% gold and 5% other elements. The doré bar will be delivered and sold to refineries using conventional third-party transportation at standard industry terms.

Engineering, supply and construction contracts are initiated, managed and administrated by Hudbay's Manitoba Business Unit. Hudbay has a marketing division that is responsible for administering the company's marketing and sales of concentrates and metals. As well, Hudbay conducts ongoing research on metal prices and sales terms as part of normal business and long-range planning process to achieve market terms. Contract terms used in the Lalor financial evaluation are based on this research.

In 2020, Hudbay entered into a gold forward sale and prepay arrangement and received an upfront payment of \$115 million to fund the refurbishment of the New Britannia gold mill. As a result, Hudbay is obligated to deliver a total of 79,954 gold ounces in 2022 and 2023 or approximately 3,331 gold ounces per month over a 24-month period from January 2022 to December 2023. The upfront payment was based on gold forward curve prices averaging approximately \$1,682 per ounce.

1.13. ENVIRONMENTAL STUDIES, PERMITTING AND SOCIAL OR COMMUNITY IMPACT

The permits required for the current Lalor operation, including the Lalor mine, Stall concentrator and Anderson tailings facility have all been issued and remain valid.

At this time, there are no known environmental concerns which could adversely affect Hudbay's ability to operate the Lalor mine. Since the mine site is nearby existing facilities in the Snow Lake area, the Lalor mine was able to utilize infrastructure, services, and previously disturbed land associated with permitted, pre-existing and current mining operations in the Snow Lake area. The Lalor mine and associated projects are designed to minimize the potential impact on the surrounding environment by keeping the footprint of

the operations as small as possible and by using existing licensed facilities for the withdrawal of water and disposal of wastes.

Initial proposals for baseline work at Wim have been prepared by AECOM. Once complete these environmental studies will form the basis of the required approvals needed to advance this project should it be deemed viable.

3 Zone is part of the New Britannia site. Significant environmental studies of the area are available, and additional environmental assessments would be utilized to augment our understanding of property and any potential offsite impacts. Approvals to advance this project would be through Provincial regulators as part of an alteration of the existing Environment Act Licence for the property.

The 1901 deposit would leverage all existing surface and underground development near Lalor operations. Significant environmental baseline work has recently been conducted by AECOM and in conjunction with the significant number of past studies will be used to gain approvals for this development should it prove viable.

The re-commissioning of the New Britannia concentrator involves the placement of the Lalor gold ore tailings in the fully licensed Anderson tailings facility or the Lalor Paste fill plant via a pipeline connecting Stall concentrator to Lalor mine. All environmental permits are in place for the refurbishment of the New Britannia mill and associated infrastructure.

Initial environmental reviews and assessments will need to be completed at Watts. Permitting efforts will be completed prior to the project advancement. Hudbay plans to engage qualified consulting firms to carry out baseline studies when there is a better understanding on when the Watts potential mine production would best fit into the integrated business plan of the Snow Lake operations of Hudbay.

Hudbay engaged AECOM to carry out baseline studies for the Pen II property. Once completed, these studies will comply with the regulatory requirements and will be part of the permitting submission process should the Pen II Zone project be advanced.

Based on Hudbay's long-term (more than 50 years) mining experience in the Snow Lake region, and baseline studies to date, there is no known First Nation or Aboriginal hunting, fishing, trapping or other traditional use of the land in the zone of potential influence for the Lalor mine and associated facilities. Post closure, all water quality and earthen structures will be monitored and inspected in order to ensure the sites' conditions meet the applicable regulatory requirements.

1.14. CAPITAL AND OPERATING COSTS

The capital expenditures required to execute the mine plans at Lalor and 1901 amount to \$598M (including \$123M in growth projects) during the 2021-2025 period. Growth capital is represented by the New Britannia mill refurbishment and construction of a pipeline corridor to support the planned increase in gold production, as well as Stall recovery improvement projects. Also included in the five-year total mentioned above as sustaining capital is \$54M for the development of the 1901 deposit from the Chisel-Lalor ramp.

From 2026 onwards, there is an additional \$215M in sustaining capital required at Lalor and 1901 for capitalized mine development, replacement/acquisition of mining equipment and for other capitalized expenditures related to milling and environmental activities. Sustaining capital costs for the new mine plans aim at preserving mineralization with higher precious metals content for extraction following commissioning of the New Britannia mill. As a result, capitalized development costs are relatively higher during the first few years as priority is given to maximizing base metals output. Overall, this approach results in a superior economic value for the Snow Lake Operations.

The WIM mine development plan contemplates construction activities occurring in 2029, followed by commissioning in 2030 and ramp-up to maximum production rate by end of 2031. The capital expenditures required at WIM and at 3 Zone for refurbishing the existing mining infrastructures and for further developing the deposit totals \$167M over the combined life of the projects.

Operating cost estimates are based on the 2021 budget figures for Lalor and were developed by Hudbay with a bottom-up approach and using quotes from local suppliers, Manitoba operations experience, labor costs within the region, and actual 2020 costs. The operating costs include all the relevant mining, milling and refining activities as well as the attributable general and administrative costs to the Snow Lake complex production. Labour costs constitute the largest component of the mining and milling operating costs. Labour costs include both employee and contractor costs arising from mining, milling and maintenance activities. Until the closure of the zinc plant in 2022, the zinc concentrate produced from Stall will be refined into finished zinc to maximize the realizable value from zinc produced. An operating cost has been allocated to the Snow Lake operations for the refining of the zinc concentrates produced from Lalor mine. Post 2022, the zinc concentrate will be sold directly to market. General and administration costs are comprised of an allocation of shared services, administration, and other office costs to the Snow Lake operations based on relative usage versus the Flin Flon operations.

Table 1-5 illustrates that Snow Lake's significant by-product credits will reduce cash operating costs and sustaining cash costs on both a zinc and gold basis. During the first six years of operation with New Britannia (2022-2027), the Snow Lake operations will produce approximately 182,000 ounces of gold annually at a cash cost, net of by-product credits, of \$US411/oz. This positions Lalor to be one of the lowest cost gold mines in Canada.

TABLE 1-5: CASH AND SUSTAINING UNIT CASH COSTS

CASH COSTS	Cash Costs on a Zinc Basis			Cash Costs on a Gold Basis		
	Contained Zinc Mlbs	Cash Costs US\$/lb	Sustaining Cash Costs US\$/lb	Contained Gold '000s oz	Cash Costs US\$/oz	Sustaining Cash Costs US\$/oz
2021	134	(0.23)	0.48	115	(275)	550
2022	112	(0.77)	0.18	160	361	1,027
2023	102	(1.21)	(0.56)	190	434	784
2024	101	(1.16)	(0.54)	191	440	766
2025	78	(1.88)	(1.09)	208	393	690
2026	101	(0.88)	(0.23)	184	454	812
2027	126	(0.44)	(0.06)	162	382	680
2028	145	0.08	0.34	123	443	749
2029	158	0.26	0.58	111	488	953
2030	92	0.07	0.29	97	627	834
2031	44	0.16	0.55	42	647	1,056
2032	-	-	-	19	669	1,271
2033	-	-	-	22	518	571
2034	-	-	-	24	543	1,028
2035	-	-	-	41	642	1,074
2036	-	-	-	39	754	1,015
2037	-	-	-	25	1,140	1,170
LOM	1,193	(0.60)	(0.02)	1,753	421	812

Note:

1. Cash costs and sustaining cash costs are presented net of by-product credits. By-product credits are calculated using the following assumptions: zinc price of \$1.20/lb in 2021, \$1.15/lb in 2022, \$1.10/lb in 2023 and long-term; copper price of \$3.75/lb in 2021, \$3.30/lb in 2022, \$3.10/lb in 2023 and long-term; silver price of \$25/oz in 2021, \$23/oz in 2022, \$20/oz in 2023, \$19/oz in 2024 and \$18/oz for the long-term; C\$/US\$ exchange rate of 1.27 in 2021, 1.28 in 2022, 1.29 in 2023 and 1.30 for long-term.

1.15. INTERPRETATION AND CONCLUSION

Lalor Mine has been in production continuously since mid-2012 and has operated without major interruptions. As of the end of 2020, the mine has produced 8.8 million tonnes. After a continuous production ramp-up cycle, the mine has reached its steady state optimal ore production of around 4,500 tpd. More recently, during an unplanned operating shut down of 6 weeks at our 777 mine in Flin Flon, Hudbay temporarily re-allocated personnel and equipment from 777 to Lalor and achieved consistently during this period a production rate of 5,300t/d with a combination of hoisting and trucking. This performance has built sufficient confidence to include a ramp-up to 5,300t/d to the life of mine plan for Lalor starting in the second half of 2022 after the closure of the 777 mine.

The strategy for the 2021 mine plan at Lalor is to mine base metals of higher grade zinc in 2021 while at the same time advancing the development required to start mining the gold and copper gold zones. Once the New Britannia mill becomes fully operational in the second half of 2021, the mine plan will target a production of 1,500 tonnes per day of gold ore, supplemented by 3,800 tonnes per day of base metal ore. The total production rate of Lalor is maintained at 5,300t/d from the second half of 2022 until 2026, after which the 1901 ore will supplement the base metal production to the Stall mill to allow a reduction in production at Lalor while maintaining the total Snow Lake camp production of 5,300t/d until 2029 followed by lower production rates until depletion of the mineral reserve estimates. While the Stall mill closes in 2031 based on mineral reserve estimates only, the New Britannia mill remains in operation for an additional six years until 2038 supported by the WIM and 3 zone mining production.

All the relevant capital and operating expenditures required to execute the life of mine plan of the Snow Lake operations have been accounted for in the demonstration of economic viability of the mineral reserve estimates including growth and sustaining capital costs and operating costs for the mining, milling and refining activities as well as the attributable general and administrative costs.

The author considers that the mineral resources and reserves as classified and reported comply with all disclosure in accordance with requirements and CIM Definitions. The author is not aware of any mining, metallurgical, infrastructure, permitting or other relevant factors that could materially affect the mineral reserve estimate.

The recent discovery of new zinc and gold mineralization zones at the 1901 deposit between Lalor and the former Chisel mine, the addition of mineral resource estimates at the Watts deposits and with its significant landholding position in the Flin Flon and Snow Lake belts, Hudbay remains in a strong position to continue to increase its resources and reserves in the short and medium term.

1.16. RECOMMENDATIONS

Significant value has been unlocked by leveraging Hudbay's exploration, technical and operational expertise, its existing portfolio of processing infrastructures and several inexpensive acquisitions.

On the exploration front, the author recommends to pursue in-mine exploration to incrementally add both base metal and gold rich inferred resources where lenses are still open in their along strike or down dip directions. The in-mine drill program should also continue to convert enough inferred mineral resource estimates to an indicated category so that they can be converted to mineral reserve estimates. In addition, surface exploration should continue along the Chisel-1901-Lalor trend which has the potential to define additional inferred resources in the Snow Lake camp over the next one to two years.

The Watts deposit located within 200km of the Snow Lake processing facilities offer additional opportunities to maximise the use of the Stall and New Britannia mills and/or extend their operating life.

As the Lalor mine matures and additional satellite ore sources are identified to supplement the feed blend to the Stall and new Britannia mills, several metallurgical projects are being advanced to match the configuration of the mill to the requirement of the ore including addressing the increasing Pb:Cu ratio with a traditional sequential Cu-Pb-Zn flotation circuit at Stall, increasing gold recovery at Stall, confirming the amenability of the 1901 gold lenses to processing at the New Britannia mill and improving flotation recoveries at Stall through reduction of the feed size.

Historical production at the New Britannia mill saw throughput rates of up to 2,000tpd based on the availability of ore from the underground operation. With the increase in the mineral reserves of the Lalor gold zones alongside additional satellite deposits, an opportunity exists to increase the throughput rate in the future. The study to date has identified that the tailings pipeline, pre-leach thickener and various minor pumping systems would require upgrade to accommodate the increased tonnage rate. In addition to the processing equipment upgrades the study includes the upgrade of the pipeline corridor road to accommodate ore haulage from Stall to New Britannia to remove traffic from the public access road.

2. INTRODUCTION & TERMS OF REFERENCE

This Technical Report has been prepared for Hudbay to support the public disclosure of Mineral Resources and Mineral Reserves at the Lalor Mine and nearby Snow Lake satellite deposits and to provide an updated mine plan that contemplates 5,300 tpd of base metal, gold and copper-gold ore feed to the Stall and New Britannia concentrators.

Hudbay is a Canadian integrated mining company with assets in North and South America, principally focused on the discovery, production and marketing of base and precious metals. Hudbay's objective is to maximize shareholder value through efficient operations, organic growth and accretive acquisitions, while maintaining its financial strength.

Hudbay operates multiple properties in the Province of Manitoba. Operations near Flin Flon include the 777 Mine, an ore concentrator and a zinc plant with all required tailings disposal facilities. Current operations near Snow Lake include the Lalor underground mine as well as an ore concentrator, a tailings impoundment area and other ancillary facilities that support the operation. The property is located approximately 16 km by road west of the town of Snow Lake, Manitoba. Hudbay owns a 100% interest in the Lalor property through one Mineral Lease and eight Order in Council Leases to the south of the mine.

Refurbishing the New Britannia mill is expected to significantly increase gold production from Lalor and enable new gold and copper-gold exploration opportunities in the Snow Lake region by having an operating processing facility with substantially higher gold and copper recoveries. New Britannia was placed on care and maintenance in 2005 by its previous owner after producing 1.6 million ounces of gold. Refurbished, it will create additional value for Hudbay in the Flin Flon and Snow Lake regions as Hudbay pursues low-risk brownfield development opportunities.

The Wim deposit was acquired by Hudbay in the third quarter of 2018 for approximately C\$0.5 million. Wim is a copper-gold deposit that starts from surface and is located approximately 15 km by road from the New Britannia mill. Hudbay has developed a mine plan for the joint operation of the WIM and 3 Zone deposit and conducted metallurgical testing to confirm the technical and economic viability of these two small mines accessible by underground ramp to feed the New Britannia mill after the richest portions of the Lalor reserves and resources have been depleted.

The recently discovered 1901 deposit located approximately half way between the former producing Chisel mine and Lalor mine will be accessed by horizontal underground development from an existing ramp currently used for haulage of equipment and manpower between Chisel and Lalor. Prefeasibility level of engineering conducted on the measured and indicated mineral resource estimates of the 1901 deposit have confirmed the technical feasibility and economic viability of the project that will contribute to an increase in the total Snow Lake camp production from 4,500 to 5,300 tonnes per day between 2022 and 2027. The deposit hosts additional base metal and gold rich inferred resources that further enhance the production capacity and extend the life of the Snow Lake camp.

New Britannia is a former producing gold mine that produced approximately 600,000 oz between 1949 and 1958 and an additional 800,000 oz between 1995 and 2005. Significant mineral resources remain accessible at New Britannia as well as in the nearby Birch and No. 3 Zone with some investments in existing mining infrastructure. WSP was engaged in 2018 to audit and restate the historical resource estimates previously reported for these deposits. Hudbay continues to re-evaluate the technical and economic viability of the existing mineral resources and the potential to process this material at the New Britannia mill.

The Watts deposit is a medium copper-gold-zinc deposit accessible by ramp and located approximately 50 km Southeast of Snow Lake. Watts could constitute a supplemental source of feed for the Stall mill. Watts would be accessible by ramp and is 100% owned by Hudbay and free of any encumbrance. When the available capacity at the Stall mills for this deposit has been more firmly established, Hudbay will conduct in due course metallurgical testing, infill drilling and technical studies to confirm its technical and economic viability.

Pen II is a low tonnage and high-grade zinc deposit that starts from surface and is located approximately 6 km by road from the Stall mill. Pen II could constitute a supplemental source of feed for the Stall mill.

2.1. QUALIFIED PERSON (QP)

This Technical Report has been prepared in accordance with National Instrument Form NI 43-101F1. The QP who supervised the preparation of this Technical Report is Olivier Tavchandjian, P. Geo., Hudbay's Vice-President, Exploration and Geology.

Mr. Tavchandjian is not independent of Hudbay, and this is not an independent technical report. Nevertheless, Hudbay is a "producing issuer" as defined in NI 43-101. As such, this Technical Report is not required to be prepared by or under the supervision of an independent QP.

Mr. Tavchandjian has been directly involved on a regular basis with the exploration, geology, resource-modelling, mine planning as well as with the estimation of operating and capital costs for the Lalor mine, WIM, 3 Zone and 1901 deposits as well as with the Watts and Pen II deposits. Mr. Tavchandjian has visited Lalor mine on a regular basis since September 2017 as well as the core storage facilities and the relevant internal and external laboratories and has directly overseen the mineral resource and reserve estimation process and the recent exploration drill programs conducted by Hudbay at the 1901, 3 Zone and Watts properties.

Todd McCracken, P. Geo for WSP has prepared the mineral resource estimates for the New Britannia mine site. Mr. Tavchandjian has reviewed the work conducted by WSP and is satisfied that the mineral resource estimates for the New Britannia properties were estimated in compliance with NI 43-101.

2.2. SOURCES OF INFORMATION

Sources of information for geological and mineral resources include: core drilling and sampling data, underground development and mapping, and assay and geochemistry analysis.

Sources of information for the mineral reserves include: the mineral resource block model, actual production and cost data, budget projections, life of mine inventory based on stope geometry parameters and mine development sequence.

Sources of information for the metallurgy, processing and economic analysis include: the actual operating data for the current processing facilities, extensive metallurgical test work and engineering pre-feasibility studies done by AECOM in 2018 for the refurbishment of the New Britannia mill and its associated infrastructures which were 75% completed by the end of February 2021 and by Stantec for the mine plan of the 1901 deposit.

2.3. UNIT ABBREVIATIONS

Units of measurement in this Technical Report conform to the SI (metric) system unless otherwise noted. Table 2-1 above, lists the notable unit abbreviations utilized in this Technical Report.

TABLE 2-1: UNIT ABBREVIATIONS

Abbreviation	Unit	Abbreviation	Unit
\$C or C\$	Canadian dollars	m	Metre
\$US or US\$	US dollars	M	million
%	Percent	m ASL	metres above sea level
°C	degree Celsius	m ²	squared metre
µm	micrometre or micron	m ³	cubic metre
BTU	British thermal unit	m ³ /hr	cubic metre per hour
CFM or cfm	Cubic feet per minute	mL	metre level
dmt	Dry metric tonnes	mm	millimetre
ft	foot	ml	millilitres
g	gram	MVA	Mega volt amp
g/t	gram per metric tonne	MW	Megawatt
Ga	billion years	nT	nanotesla
gpm	gallon per minute	oz	Troy ounces
Ha	hectare	ppm	parts per million
HP or hp	Horsepower	psi	Pounds per square inch
hr	hour	t	metric tonne
kg	kilogram	tpd	metric tonnes per day
km	kilometre	US GPM	United States gallon per minute
km/hr	kilometre per hour	US\$ or \$US	United States dollar
kV	kilovolt	V	Volt
kW	kilowatt	wmt	Wet metric tonne
L/min	litres per minute	Zn Eq	Percent
lb	pound (unit of weight)		

2.4. ACRONYMS AND ABBREVIATIONS

Abbreviation of company names and other notable terms used in the report are as shown in Table 2-2.

TABLE 2-2: ACRONYMS AND ABBREVIATION

Abbreviation	Term	Abbreviation	Term	Abbreviation	Term
3D	Three-Dimensional	EEM	Environmental Effects Monitoring	NSS	Near solid sulphide
AAS	Atomic Absorption Spectrometry	EM	Electromagnetic	OIC	Order In Council
AEP	Advanced Exploration Project	ES	Emission spectrograph	OES	Optical Emission Spectrometry
Ag	Silver	Fe	Iron	OK	Ordinary Kriging
ANFO	Ammonium nitrate	FOB	Fine ore bin	OLS	Ordinary least square
As	Arsenic	GPSS	Global Positioning Satellite System	OREAS	Ore Research and Exploration
ASL	Above Sea Level	HCl	Hydrogen chloride	P. Eng.	Professional Engineer
Au	Gold	HNO ₃	Nitric acid	Pb	Lead
Au Eq	Gold Equivalency	Hudbay	Collectively all Hudbay Minerals Inc. subsidiaries and business groups	PR	Provincial Road
AV	average	ICP	Inductively Coupled Plasma	QAQC	Quality Assurance and Quality Control
BLTMF	Birch Lake Tailings Management Facility	LCT	Lock cycle test	QP	Qualified Person
BQ	BQ drill core size 36.4mm	IDW	Inverse Distance Squared Weighted	Reflex	Reflex E-Z Shot
BV	Bureau Veritas	LHD	Load Haul and Dump	ROM	Run of mine
CBV	Certified Best Value	LIMS	Information management system	R ²	Coefficient of determination
CIM	Canadian Institute of Mining, Metallurgy and Petroleum	LOD	Limit of detection	RTK	Real Time Kinematic
CIP/CIL	Carbon-in-pulp / Carbon-in-leach	LOM	Life of Mine	SG	Specific Gravity
CRF	Cemented waste rock backfill	MBU	Manitoba Business Unit	SS	Solid sulphide
CRM	Certified Reference Materials	MCCN	Mathias Colomb First Nation	TIA	Tailings Impoundment Area
Cu	Copper	MIBC	Methyl isobutyl carbinol	RE	Relative Error
CV	Coefficient of Variation	ML	Mineral Lease	TMI	Total Magnetic Intensity
DH	Drill hole	MS	Mass Spectrometry	UCS	Unconfined compressive strength
DE	Data entry	NoA	Notice of Alteration	URF	Unconsolidated waste rock backfill
DGPS	Differential Global Positioning System	NI	National Instrument	UTM	Universal Transverse Mercator
EAL	Environmental Act Licence	NN	Nearest Neighbour	VMS	Volcanogenic Massive Sulphide
EDA	Exploratory data analysis	NSR	Net smelter return	Zn	Zinc
EDM	Electronic distance measurement	NQ	NQ drill core size 47.6mm	Zn Eq	Zinc Equivalency

2.5. OTHER CONTRIBUTIONS

TABLE 2-3: CONTRIBUTIONS FOR THIS REPORT

Section	Description	Participants	Responsible
1	Summary	Olivier Tavchandjian	Olivier Tavchandjian
2	Introduction	Olivier Tavchandjian	Olivier Tavchandjian
3	Reliance on Other Experts	Olivier Tavchandjian	Olivier Tavchandjian
4	Property Description and Location	Kenda Kozar	Olivier Tavchandjian
5	Accessibility, Climate, Local Resources, Infrastructure and Physiography	Neil Richardson	Olivier Tavchandjian
6	History	Neil Richardson	Olivier Tavchandjian
7	Geological Setting and Mineralization	Keith Harrison, John Townend, Neil Richardson	Olivier Tavchandjian
8	Deposit Types	Keith Harrison, John Townend, Neil Richardson	Olivier Tavchandjian
9	Exploration	Keith Harrison, John Townend, Neil Richardson	Olivier Tavchandjian
10	Drilling	Keith Harrison, John Townend, Neil Richardson	Olivier Tavchandjian
11	Sample Preparation, Analyses and Security	Reg Yaworski, Janelle Grant, Sean Bawden	Olivier Tavchandjian
12	Data Verification	Reg Yaworski, Janelle Grant, Sean Bawden	Olivier Tavchandjian
13	Mineral Processing and Metallurgical Testing	Matthew Taylor	Olivier Tavchandjian
14	Mineral Resource Estimates	Janelle Grant, Sean Bawden, Marc-Andre Brulotte, Olivier Tavchandjian	Olivier Tavchandjian
15	Mineral Reserve Estimates	Brent Christensen, Mike Tsafaras	Olivier Tavchandjian
16	Mining Methods	Brent Christensen, Mike Tsafaras	Olivier Tavchandjian
17	Recovery Methods	Matthew Taylor	Olivier Tavchandjian
18	Project Infrastructure	Robert Carter	Olivier Tavchandjian
19	Market Studies and Contracts	Eugene Lee, Jon Douglas	Olivier Tavchandjian
20	Environmental Studies, Permitting and Social or Community Impact	Jay Cooper	Olivier Tavchandjian
21	Capital and Operating Costs	Brent Christensen, Robert Carter	Olivier Tavchandjian
22	Economic Analysis	n/a	Olivier Tavchandjian
23	Adjacent Properties	Neil Richardson	Olivier Tavchandjian
24	Other Relevant Data and Information	Olivier Tavchandjian	Olivier Tavchandjian
25	Interpretation and Conclusions	Olivier Tavchandjian	Olivier Tavchandjian
26	Recommendations	Olivier Tavchandjian	Olivier Tavchandjian

3. RELIANCE ON OTHER EXPERTS

Standard professional procedures were followed when preparing the contents of this Technical Report. Data used in this Technical Report has been verified where possible and the QP has no reason to believe that the data was not collected in a professional manner and no information has been withheld that would affect the conclusions made herein.

The information, conclusions, opinions, and estimates contained herein are based on:

- Information available to Hudbay at the time of preparation of this Technical Report, and
- Assumptions, conditions, and qualifications as set forth in this Technical Report.

4. PROPERTY LOCATION AND DESCRIPTION

4.1. LALOR MINE, 1901 AND PEN II DEPOSITS

The Lalor mine is located 16 km by road west of the town of Snow Lake at 54°52'24" north latitude, 100°09'01" west longitude and 303 mASL on National Topographic System (NTS) map 63K/16 (Figure 4 1).

Hudbay owns a 100% interest in the Lalor property through one Mineral Lease and eight Order in Council Leases to the south of the property totalling 946 hectares.

Hudbay holds the exclusive right to the minerals, other than quarry minerals, and the mineral access rights required for working the lands, mining and producing minerals from the Lalor mine. The surface tenure required for the mining operations have all been approved.

The 1901 deposit is located approximately 2.5km south-southeast of Lalor mine at 54°51.5'04.4" north latitude, 100°07'57.3" west longitude and 301 mASL on National Topographic System (NTS) map 63K/16. The deposit area is accessed by drill trail off the Lalor mine road to the south-southwest.

Hudbay holds the mineral rights to the claims that host the 1901 deposit and other contiguous claims in the area covering a total of 201 hectares.

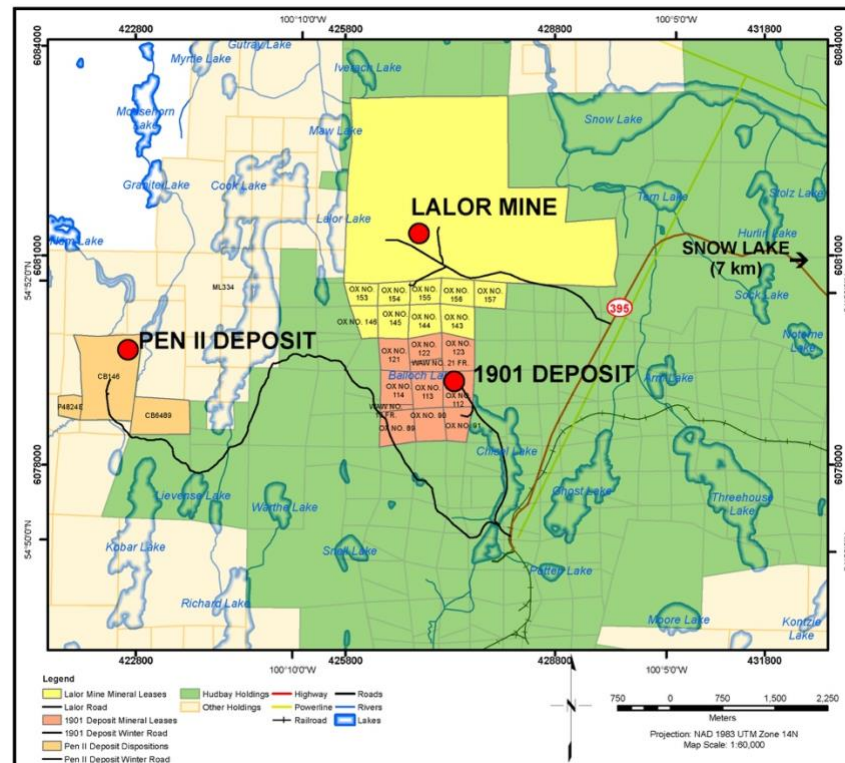
The Pen II deposit is located 12 km west of the town of Snow Lake at 54°51'28" north latitude and 100°12'20" west longitude and 296 mASL on NTS map 63K/16.

Hudbay holds the mineral rights to claim CB146 which hosts the Pen II deposit. Other contiguous claims are held by Hudbay Minerals in the area and they comprise of one fraction and one regular claim for a total of 196 Ha (Table 4-1 and Figure 4-1)Table 4-1:

TABLE 4-1: LIST OF MINERAL LEASES, OIC AND MINERAL CLAIMS FOR THE LALOR MINE, 1901 DEPOSIT AND PEN II DEPOSIT

Project	Name	Number	Holder	Expiry Date	Hectares
Lalor Mine	OX NO. 157	M5729	Hudbay Minerals Inc.	12/09/2022	17.4
Lalor Mine	OX NO. 153	M5778	Hudbay Minerals Inc.	4/08/2023	14.6
Lalor Mine	OX NO. 154	M5779	Hudbay Minerals Inc.	4/08/2023	15
Lalor Mine	OX NO. 155	M5780	Hudbay Minerals Inc.	4/08/2023	15.4
Lalor Mine	OX NO. 156	M5781	Hudbay Minerals Inc.	4/08/2023	18.3
Lalor Mine	OX NO. 143	M7278	Hudbay Minerals Inc.	9/06/2023	21.3
Lalor Mine	OX NO. 144	M7279	Hudbay Minerals Inc.	9/06/2023	20.5
Lalor Mine	OX NO. 145	M7280	Hudbay Minerals Inc.	9/06/2023	21.1
Lalor Mine	OX NO. 146	M7281	Hudbay Minerals Inc.	9/06/2023	14.9
Lalor Mine		ML334	Hudbay Minerals Inc.	3/29/2033	787.9
1901 Deposit	OX NO. 89	M7232	Hudbay Minerals Inc.	9/06/2023	24.1
1901 Deposit	OX NO. 90	M7233	Hudbay Minerals Inc.	9/06/2023	20.6
1901 Deposit	OX NO. 91	M7234	Hudbay Minerals Inc.	9/06/2023	17.3
1901 Deposit	OX NO. 112	M7253	Hudbay Minerals Inc.	9/06/2023	22.4
1901 Deposit	OX NO. 113	M7254	Hudbay Minerals Inc.	9/06/2023	23.2
1901 Deposit	OX NO. 114	M7255	Hudbay Minerals Inc.	9/06/2023	22
1901 Deposit	OX NO. 121	M7260	Hudbay Minerals Inc.	9/06/2023	19.6
1901 Deposit	OX NO. 122	M7261	Hudbay Minerals Inc.	9/06/2023	16.9
1901 Deposit	OX NO. 123	M7262	Hudbay Minerals Inc.	9/06/2023	23.4
1901 Deposit	WAW NO. 19 FR.	M7325	Hudbay Minerals Inc.	9/06/2023	3.9
1901 Deposit	WAW NO. 21 FR.	M7327	Hudbay Minerals Inc.	9/06/2023	7.6
Pen II Deposit		CB146	Hudbay Minerals Inc.	2/09/2025	126
Pen II Deposit		CB6489	Hudbay Minerals Inc.	7/16/2024	62
Pen II Deposit	PEN 4824 FR.	P4824E	Hudbay Minerals Inc.	6/13/2025	9

FIGURE 4-1: LOCATION OF THE LALOR, 1901 AND PEN II DEPOSITS MINERAL LEASES AND MINERAL CLAIMS



Land Use Permitting

General Permits are issued and administered by the Province of Manitoba Real Estate Services Division, Manitoba Department of Finance. Provided all terms and conditions of the General Permit are met, including payment of annual fees, the permit is automatically renewed for a 1-year term on an annual basis. Quarry Leases are issued and administered by the Province of Manitoba Mines Branch.

Two General Permits (GP59093 and GP63483) and one Quarry Lease are held by Hudbay and are required to carry on mining activities at Lalor. In detail, these are:

GP59093: This General Permit covers an area of 4.05Ha (4.0 km x 5m) and includes an all-weather access road, accommodates a 4 km 25 kV transmission line, a 4 km discharge line, a 4 km fresh water line, a 200m x 200m parking lot and an additional access road.

GP63483: The General Permit is related to the mine site.

QL-1928: QL-1928 was renewed November 26, 2017 for a 10-year term and provides the holder with the exclusive right to explore for, develop, and produce clay, gravel, rock or stone. QL-1928 will expire November 2027 and provided this lease is still required it can be renewed for another 10 year term.

General Permit and Quarry Lease status for the Lalor property is shown in Table 4-2

TABLE 4-2 GENERAL PERMITS AND QUARRY LEASE FOR LALOR

Permit/Quarry Number	Holder	Issue Dates	Hectares	Annual Fees (Excludes GST)
GP59093	Hudbay	31-Dec-07	4.05	\$415
GP63483	Hudbay	10-Jun-10	159.37	\$798
QL-1928	Hudbay	26-Nov-17	11.00	\$297
Total			174.42	\$1,510

4.2. NEW BRITANNIA MINE AND OTHER SNOW LAKE GOLD PROPERTIES

Since 2015, Hudbay owns 100% interest in the Snow Lake Gold Properties through two Mineral Leases and 57 Mineral Claims covering a total area of 1,232 hectares. The New Britannia Mine-Mill Complex is located on the north side of the Town of Snow Lake at 54°53'10" north latitude and 100°01'24" west longitude and 293 mASL on NTS map 63K/16 (Table 4-2)

The two mining leases have been legally surveyed and remain in good standing providing that the annual taxes and lease fees are paid. The lease fees are CAN\$ 12.00/Ha each year and are subject to a 1.38% NSR, based on production, payable to W. Bruce Dunlop.

The following summarizes the sequence of corporate transactions and mergers leading to the current ownership and consolidation of the property:

Prior to December 22, 2006, the mining leases and the Hudbay Option claims were held by 1126774 Ontario Limited (1126774 Ontario) which was held by Kinross and High River Gold Ltd. in a joint venture, with each party holding 50%.

On December 22, 2006, 1126774 Ontario was combined with Pegasus Mines Ltd. (Pegasus), with the merged entity retaining the name Pegasus.

Pegasus was wholly-owned by the New Britannia Mine Joint Venture (NBM-JV) which consisted of Piper Capital Inc. (Piper) owning 60%, and Garson Resources Ltd. (Garson Resources) owning 40%. Prior to the establishment of the NBM-JV, Garson Resources Ltd. held the Squall Lake claims.

On June 28, 2007, Piper and Garson Resources merged to form Garson Gold Corp.

On September 19, 2007, Pegasus changed its name to New Britannia Mines Ltd.

The terms of the agreement with Kinross have been described in the previous Micon Technical Report (Lewis, 2006).

At the time of the original transaction with Kinross, the claims were optioned to Kinross by Hudbay. Exercise of the option required a one-time cash payment of CAN\$ 400,000 and a royalty consisting of 1.5% of the gross proceeds from the sale or disposition of all metals mined and removed from the optioned claims. However, Hudbay has retained the rights to the base metals on the mineral claims and should a base metal deposit be discovered and exploited on the claims, then Garson would retain a 1.5% royalty.

On December 31, 2007, the CAN\$ 400,000 option payment was made and, subject to the 1.5% royalty, the Hudbay claims are now 100%-owned by NBM Ltd. Prior to conducting any production mining, the claims would have to be either incorporated into the existing mineral lease ML-61 or incorporated into a new mining lease. Micon indicated that due to the pre-existing royalty on ML-61, the claims should be incorporated into a new mining lease (Lewis, 2009). This would create a boundary which could be surveyed underground so that no conflicts between the two separate royalty holders would exist.

The Squall Lake group of claims north of Snow Lake comprises 14 contiguous, unpatented mining claims with an area of 899 Ha and is 100% owned by Hudbay, subject to various underlying royalties. The

property contains a net profits royalty (NPI) interest payable on commencement of commercial production of 4% to a private individual, an additional 6% NPI payable to American Barrick (now Barrick), and a 30% NPI payable to Barrick up to a maximum of \$550,000. The property also contains a production royalty of \$0.10 per ton payable to a private individual on products milled from 6 of the claims. No mineral resource estimates have been estimated in this Technical Report for the Squall Lake group of claims.

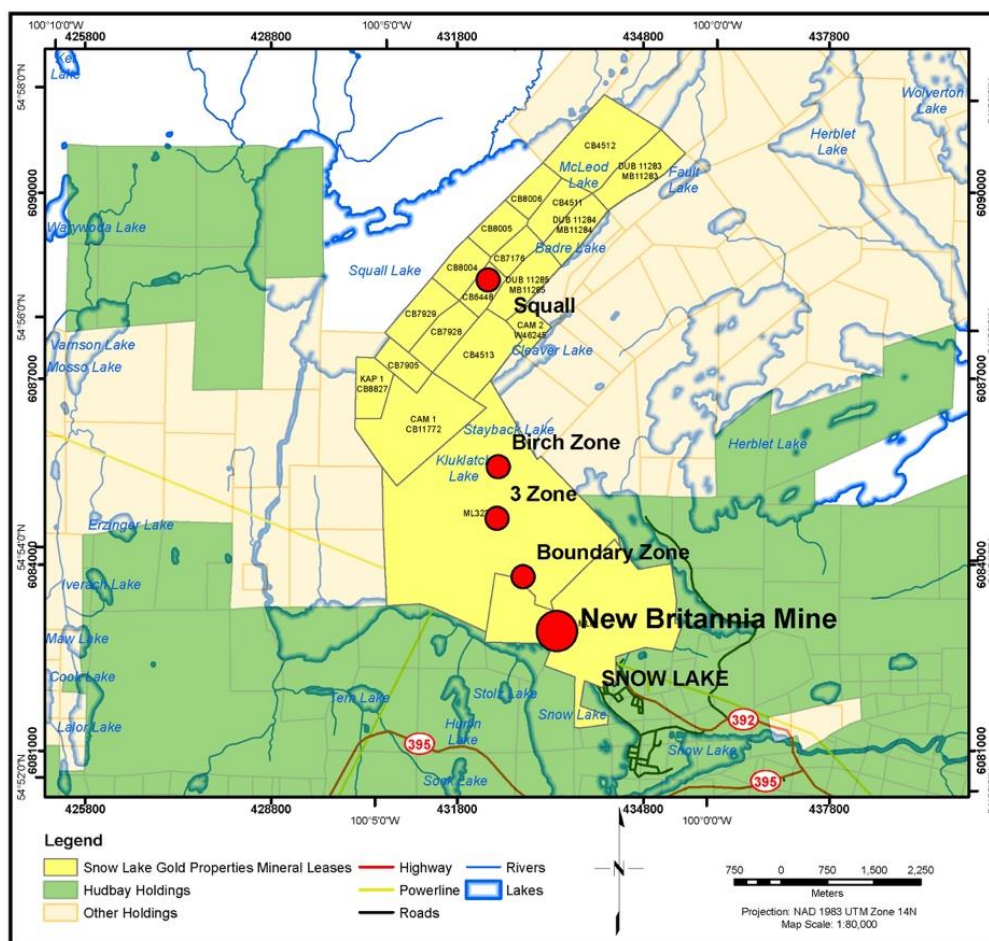
The mineral claims require annual assessment work in the amount of CAN\$ 25.00/Ha to keep them in good standing beyond their current expiry dates. In Manitoba, when the claim holder exceeds the minimum annual assessment expenditures for the mineral claims, surplus expenditures can be banked and utilized in future years to meet the annual assessment requirements.

Mineral lease ML-61 (NBM) and mineral lease ML-323 are located on the NTS map sheet 63K/16NE. The mineral claims currently held by Hudbay are located on NTS map sheets 63K/16NE, 63J/13NW, and 63K-16SE. The Squall Lake claims are located on NTS map sheet 63K/16NE Table 4-3 and Figure 4-2)

TABLE 4-3: LIST OF MINERAL LEASES AND MINERAL CLAIMS FOR THE SNOW LAKE GOLD AND SQUALL LAKE PROPERTIES

Project	Name	Number	Holder	Expiry	Hectares
Snow Lake Gold		ML323	Hudbay Minerals Inc.	3/28/2037	432
Snow Lake Gold		ML61	Hudbay Minerals Inc.	4/01/2034	830
Squall Area	CAM 1	CB11772	Hudbay Minerals Inc.	3/16/2022	195
Squall Area		CB4511	Hudbay Minerals Inc.	9/05/2026	43
Squall Area		CB4512	Hudbay Minerals Inc.	9/05/2026	152
Squall Area		CB4513	Hudbay Minerals Inc.	9/05/2026	119
Squall Area		CB6448	Hudbay Minerals Inc.	9/05/2026	33
Squall Area		CB7176	Hudbay Minerals Inc.	9/05/2026	33
Squall Area		CB7905	Hudbay Minerals Inc.	3/11/2022	45
Squall Area		CB7928	Hudbay Minerals Inc.	5/13/2022	42
Squall Area		CB7929	Hudbay Minerals Inc.	7/08/2026	43
Squall Area		CB8004	Hudbay Minerals Inc.	7/08/2026	43
Squall Area		CB8005	Hudbay Minerals Inc.	7/08/2026	43
Squall Area		CB8006	Hudbay Minerals Inc.	7/08/2026	43
Squall Area	KAP 1	CB8827	Hudbay Minerals Inc.	11/23/2024	48
Squall Area	DUB 11283	MB11283	Hudbay Minerals Inc.	10/13/2029	75
Squall Area	DUB 11284	MB11284	Hudbay Minerals Inc.	10/13/2029	53
Squall Area	DUB 11285	MB11285	Hudbay Minerals Inc.	10/13/2029	63
Squall Area	CAM 2	W46245	Hudbay Minerals Inc.	3/16/2026	17

FIGURE 4-2: LOCATION OF THE SNOW LAKE AND SQUALL LAKE MINERAL LEASES AND MINERAL CLAIMS



4.3.WATTS DEPOSIT

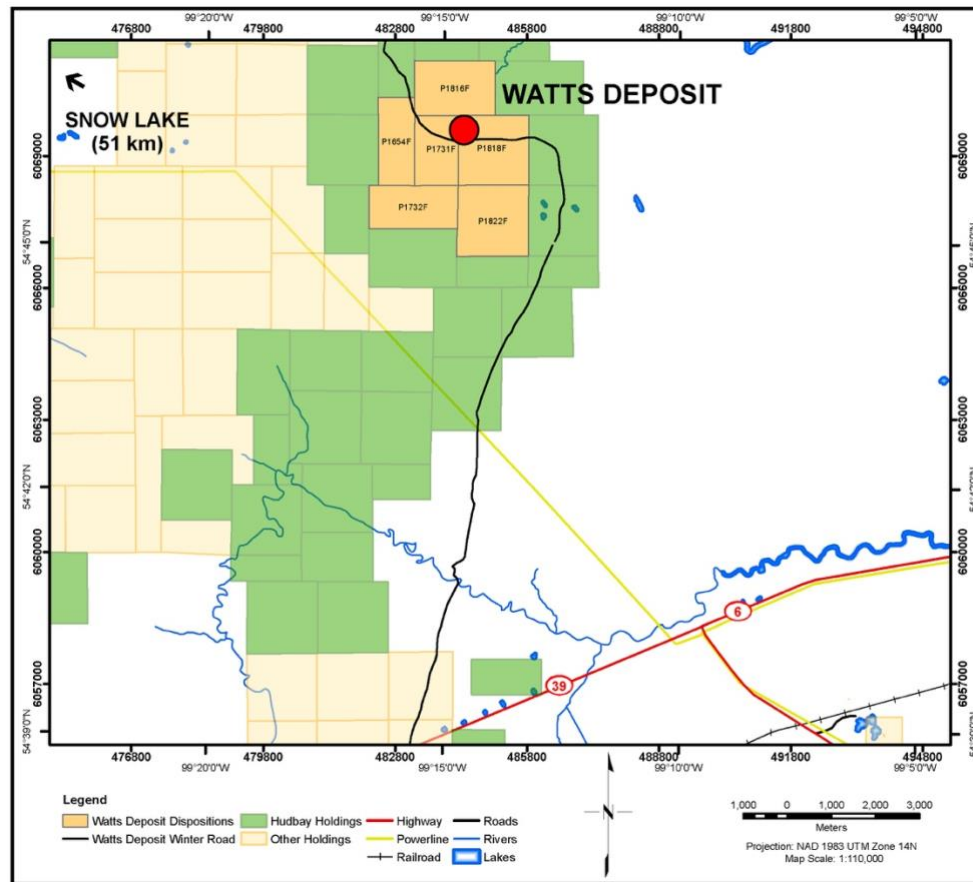
The Watts deposit is located approximately 52km east-southeast of the town site of Snow Lake, Manitoba and is centred on Latitude 54° 45' 51" North and Longitude 99° 15' 44" West and 260 mASL.

A group of six contiguous claims encompass the Watts River Property. The six claims cover an area of 1,282 hectares and are in good standing. Hudbay Minerals Inc. holds 100% of the mineral rights (Table 4-4).

TABLE 4-4: LIST OF MINERAL CLAIMS FOR THE WATTS DEPOSIT

Project	Name	Number	Holder	Expiry Date	Hectares
Watts Deposit	KUS 1732	P1732F	Hudbay Minerals Inc.	5/24/2024	197.7
Watts Deposit	KUS 1654	P1654F	Hudbay Minerals Inc.	4/19/2023	166
Watts Deposit	KUS 1818	P1818F	Hudbay Minerals Inc.	5/01/2022	255.2
Watts Deposit	KUS 1731	P1731F	Hudbay Minerals Inc.	5/24/2022	157.5
Watts Deposit	KUS 1816	P1816F	Hudbay Minerals Inc.	5/01/2027	227
Watts Deposit	KUS 1822	P1822F	Hudbay Minerals Inc.	5/01/2030	262.4

FIGURE 4-3: LOCATION OF THE WATTS DEPOSIT MINERAL CLAIMS



4.4. WIM DEPOSIT

The Wim deposit is located 16 km north of the Town of Snow Lake at 55°1'34" north latitude and 100°2'50" west longitude and 316 mASL on NTS map 63N/01 (Table 4-5).

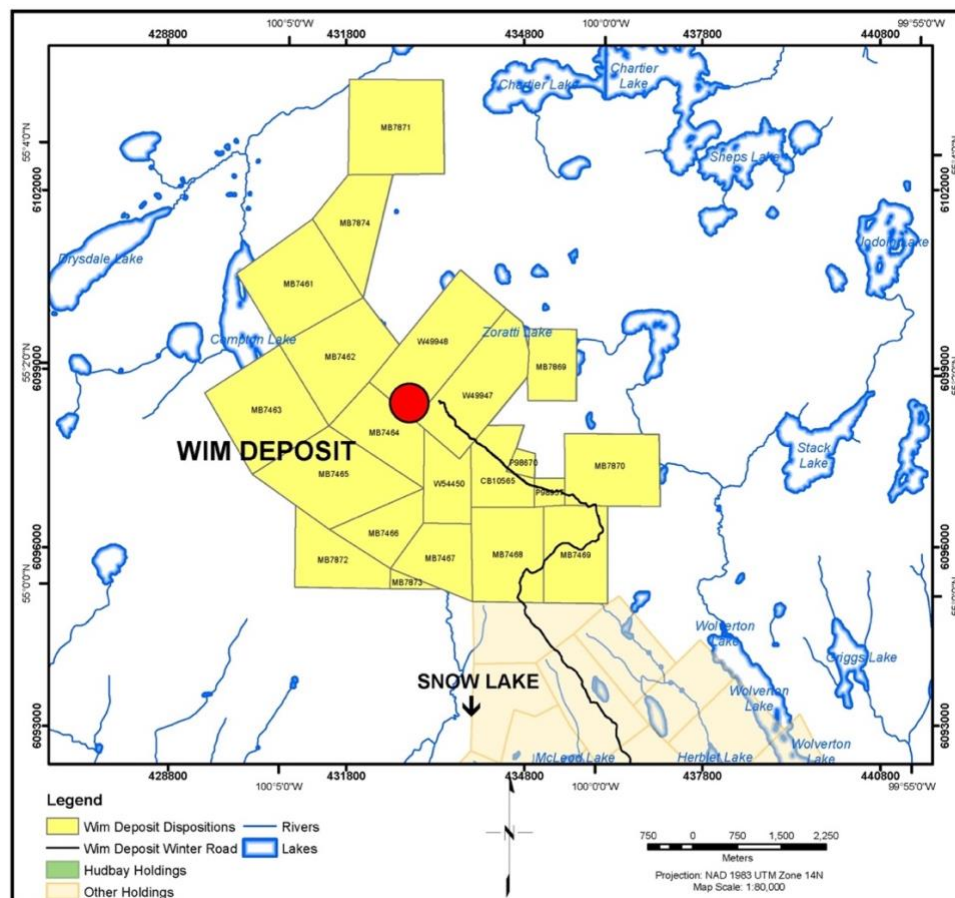
The Wim deposit is 100% owned by Hudbay. On October 31, 2018 it was announced that Alexandria had entered into a purchase agreement with Hudbay to sell the Hudvam and Wim properties. Previously, on December 22, 2014, Alexandria signed an Arrangement Agreement with Murgor Resources Inc. (Murgor) to acquire all Murgor's shares and assets (the acquisition was completed on March 9, 2015).

Hudbay holds the mineral rights to 21 claims for a total 3,149 Ha. The Wim deposit is hosted by claims W49947 and W49948 (Table 4-5).

TABLE 4-5: LIST OF MINERAL CLAIMS FOR THE WIM DEPOSIT

Project	Name	Number	Holder	Expiry	Hectares
Wim Deposit	ZYL 10565	CB10565	Hudbay Minerals Inc.	8/06/2029	81
Wim Deposit	WIM 7461	MB7461	Hudbay Minerals Inc.	2/05/2029	200
Wim Deposit	WIM 7462	MB7462	Hudbay Minerals Inc.	2/05/2029	172
Wim Deposit	WIM 7463	MB7463	Hudbay Minerals Inc.	2/05/2029	195
Wim Deposit	WIM 7464	MB7464	Hudbay Minerals Inc.	2/03/2029	173
Wim Deposit	WIM 7465	MB7465	Hudbay Minerals Inc.	2/04/2029	256
Wim Deposit	WIM 7466	MB7466	Hudbay Minerals Inc.	2/05/2029	100
Wim Deposit	WIM 7467	MB7467	Hudbay Minerals Inc.	2/05/2029	116
Wim Deposit	WIM 7468	MB7468	Hudbay Minerals Inc.	2/05/2029	192
Wim Deposit	WIM 7469	MB7469	Hudbay Minerals Inc.	2/05/2029	190
Wim Deposit	MUG 7869	MB7869	Hudbay Minerals Inc.	9/12/2029	97
Wim Deposit	MUG 7870	MB7870	Hudbay Minerals Inc.	9/12/2029	192
Wim Deposit	MUG 7871	MB7871	Hudbay Minerals Inc.	4/05/2029	256
Wim Deposit	MUG 7872	MB7872	Hudbay Minerals Inc.	4/05/2029	138
Wim Deposit	MUG 7873	MB7873	Hudbay Minerals Inc.	4/05/2029	16
Wim Deposit	MUG 7874	MB7874	Hudbay Minerals Inc.	4/05/2029	157
Wim Deposit	WIM 880	P98670	Hudbay Minerals Inc.	6/05/2029	21
Wim Deposit	NEW 9	P98957	Hudbay Minerals Inc.	7/02/2029	21
Wim Deposit	WIN 49947	W49947	Hudbay Minerals Inc.	5/21/2029	228
Wim Deposit	WIN 49948	W49948	Hudbay Minerals Inc.	5/21/2029	243
Wim Deposit	ZYL 54450	W54450	Hudbay Minerals Inc.	12/22/2029	105

FIGURE 4-4: LOCATION OF THE WIM DEPOSIT MINERAL CLAIMS



5. ACCESSIBILITY, CLIMATE, LOCAL RESOURCES, INFRASTRUCTURE AND PHYSIOGRAPHY

5.1. ACCESSIBILITY

The Lalor mine, its nearby satellite deposits and all the infrastructures associated with operations at Snow Lake are located approximately 200 km by road east of the city of Flin Flon and are within or close to the town of Snow Lake, Manitoba. Access to the mineral deposits is described individually below (Figure 5-1)

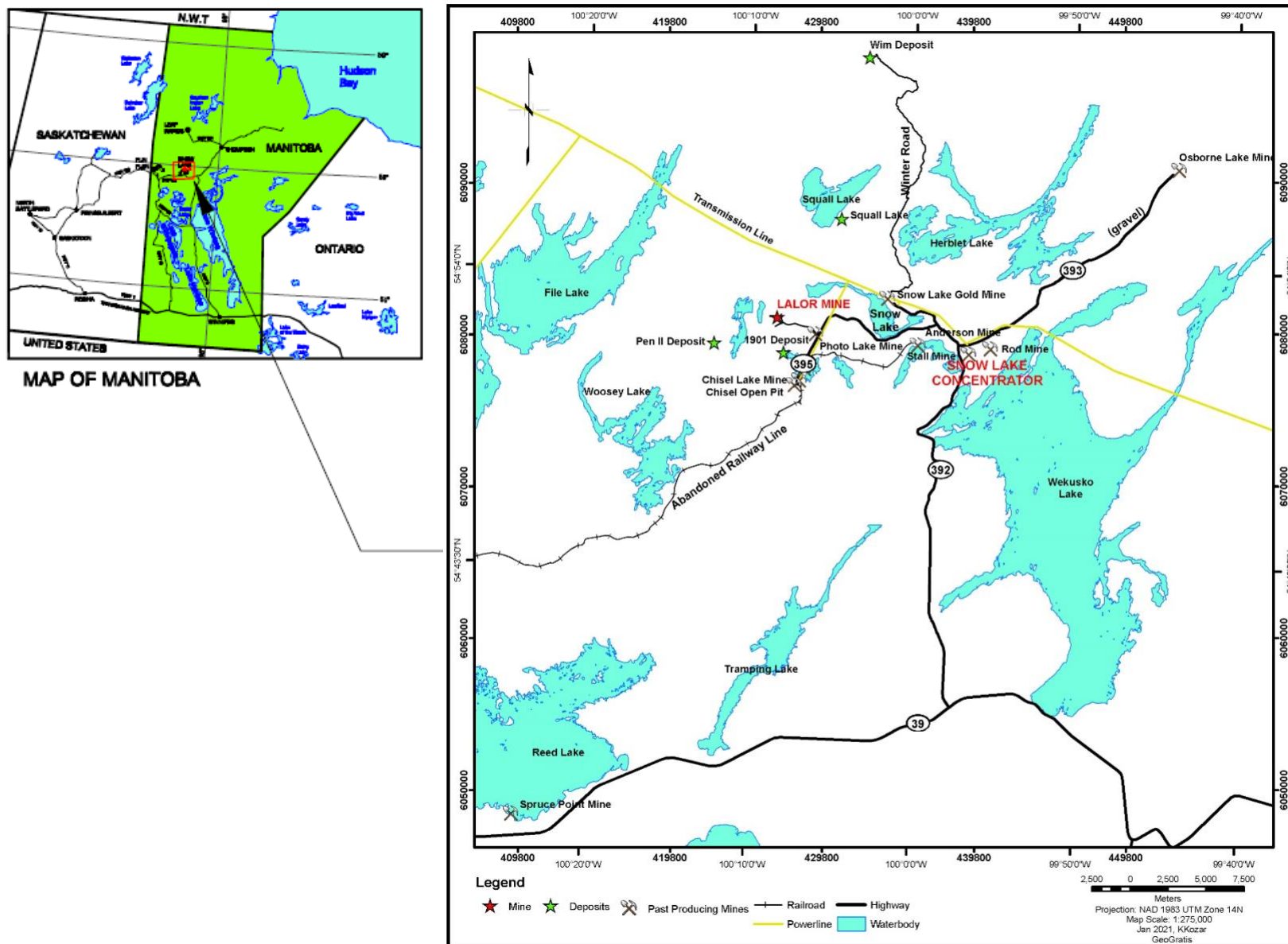
Access to Lalor mine is by travelling ~29.5km north along Provincial Road (PR) #392 off from PR #39 towards the Town of Snow Lake. Turn left on PR #395 for 9km till Lalor mine access road.

- Access to the 1901 deposit site is by traveling to the end of PR #395 and taking a winter road across Chisel Lake or a snowmobile / quad trail for ~5km from around the past producing Chisel open pit to the site.
- Access to the Pen II site is from Provincial Roads #392 and #395, located past the old Chisel mine site.
- The New Britannia Mine-Mill Complex is located on the north side of the Town of Snow Lake and accessed from paved Cedar Avenue.
- Access to the Wim site is from the Town of Snow Lake, travelling north from the New Britannia mine site and over a tailings dam road along the Manitoba Hydro Transmission line.
- Access to the Watts River deposit is located approximately 52 km east-southeast of the town site of Snow Lake, Manitoba and is accessed by helicopter or a 17km winter road northeast from Ponton, Manitoba, where Provincial Highway #6 meets Provincial Highway #39. Alternate access can be obtained by utilizing the Manitoba Hydro powerline route that provides Snow Lake with power and passes within a few kilometres of the Property.

5.2. CLIMATE

The Snow Lake area has a typical mid-continental climate, with short summers and long, cold winters. Climate generally has only a minor effect on local exploration and mining activities. The nearest Environment Canada weather station is located near Baker's Narrows at the Flin Flon airport, approximately 16 km southeast of Flin Flon and approximately 100 km west of Snow Lake. The average annual temperature at the Baker's Narrows weather station is 0.1°C. The average summer temperature is approximately 17°C, and the average winter temperature is -14°C. The lowest monthly average temperature occurs in January at -21.1°C and the highest monthly average temperature is in July at 18.3°C. Freeze-up of small bays and lakes occurs in mid-November, with breakup occurring in mid-May. There is an average of 115 frost-free days. On average, 45.7 cm of precipitation falls annually, of which, 35% is snow. Since 1960, monthly precipitations have ranged from zero to a high of 18.1 cm, with a maximum daily precipitation of 7.82 cm. Average monthly winds for the area range from 10km/hr to 13km/hr, with 40% of the winds originating from the northwest, northeast or north.

FIGURE 5-1: SNOW LAKE REGIONAL MAP



5.3. LOCAL RESOURCES

The nearest community is the town of Snow Lake, Manitoba. The community of 899 (Statistics Canada, 2016 census) has 498 private dwellings. There are two cottage subdivisions located on Wekusko Lake along PR #392, as well as residences at Herb Lake Landing, located approximately 40 km south of the town. There are also a small number of seasonal remote cottages located near lakes throughout the area.

Snow Lake community services include a health facility staffed with two doctors, an ambulance and a fire truck, a grocery store, two hotels/motels, three service stations.

The nearest larger centres (5,000+ residents) are Flin Flon (208 km), The Pas (219 km) and Thompson (260 km), all accessible by paved highway. There is a 1,100m by 20m un-serviced gravel municipal airstrip located approximately 16 km from Snow Lake along PR #393. A small craft charter service operates out of the community of Snow Lake, where small planes and helicopters can be chartered. Rental vehicles are available at the Flin Flon airport. The nearest full-service commercial airport is located at Baker's Narrows near Flin Flon, approximately 185 km from Snow Lake. The nearest international airport is located in Winnipeg, approximately 700 km from Snow Lake.

There is no rail in the immediate area of Snow Lake. The nearest rail access is at Wekusko siding, approximately 65 km southeast of Snow Lake. Wekusko is accessible by an all-weather road. A gravel rail bed (ties and rail removed) connects the Stall concentrator to Chisel Lake mine and continues to a rail line at Optic Lake siding, approximately 65 km west of Chisel Lake. Optic Lake is not road accessible.

To house non-local employees during their work rotations, the company provides a camp located in the Town of Snow Lake which services Hudbay employees and contractors for the mine and mill operations.

Outland Camps in Snow Lake currently consist of an upper camp section with 7 dorms and a lower camp section with 4 dorms and a Kitchen and dining hall. The camp has a total capacity of 436 rooms. There is a recreation trailer that has recently been turned into a Covid-19 testing facility and an exercise trailer currently closed due to Covid-19.

Utilities are provided by the town of Snow Lake, including potable water, sewage and electricity. Heating is electrical and propane. The camp includes parking spots for employees.

5.4. EXISTING INFRASTRUCTURE

Hudbay operates a zinc metallurgical plant in Flin Flon, Manitoba, approximately 200 km from Snow Lake with a capacity of 115,000 tpa of refined zinc. However, Hudbay intends to cease operations and put the zinc plant on care and maintenance at the end of the second quarter of 2022.

Hudbay also operates two ore concentrators: The Stall mill located approximately 16 km from Lalor and the Flin Flon mill is located in Flin Flon. The Stall mill is currently operating seven days per week, producing at or above 3,800 tpd. During the 2021-2022 period, although 100% of Lalor's production will be processed at the Stall or New Britannia mills, any additional production could be trucked to the Flin Flon mill until this facility shuts down at the same time as the zinc plant closes. The Stall mill has two circuits with design capacities of 900 tpd and 2,180 tpd. At the Flin Flon mill, the ore from Lalor has been in the past processed separately as it is free of the encumbrances that affect the other ore processed in Flin Flon from the 777 mine, i.e. ore subject to the Silver Wheaton precious metal streaming agreement.

Both concentrators have two flotation circuits producing a zinc concentrate and a copper concentrate. The zinc concentrate produced at Stall is hauled by truck to Hudbay metallurgical facilities in Flin Flon. The copper concentrates are sold to market. The tailings from the Stall mill are deposited in the Anderson Tailings Impoundment Area or used at the paste plant. The permitted Hudbay Anderson TIA, located approximately 13 km from Lalor, is used for tailings disposal.

The Lalor paste plant was commissioned in the third quarter of 2018, is located northeast of the existing head frame complex, and is capable of delivering 165 tph solids (tails) or 93 m³/hr of paste. The paste

plant is capable of varying the binder content in the paste to provide flexibility in the strength gained from the paste where higher and early strength may be required depending on mining method.

Tails are currently pumped from the Stall concentrator to the Anderson TIA and diverted, as needed, to the Anderson booster station. Two pipelines are installed between the Anderson booster pump station and the paste plant located at Lalor mine site, which are situated approximately 13 km apart.

General area infrastructure includes provincial roads, 115 kV power to within four kilometres of Lalor provided by Manitoba Hydro and telephone land line and cellular phone service provided by Manitoba Telecom.

The Lalor mine is located 3.5 km from the Hudbay Chisel North mine. Chisel North infrastructures include a mined out open pit used for waste rock disposal, fresh (process) water sources, pumps and waterlines, 4160V and 550V power, mine discharge water lines, a 2,500 gpm water treatment plant with retention areas, and mine buildings including offices and a change house. These facilities are used for geological core processing, surface mobile equipment shop, project offices, and the crushing of Lalor ore. The Chisel site is also the location of two electrical transformers 115 kV to 25 kV that feed Lalor mine.

The infrastructure on site at Lalor includes, the main office change house, headframe, hoistroom, down cast fans, exhaust fans, main pump station, potable water treatment plant, sewage treatment plant and several other smaller buildings for purposes such as health and safety and additional change house space. The New Britannia mill is located approximately 16 km north of the Lalor mine and it is currently under construction (refurbishment) with a planned start-up of operations in the spring of 2021. The mill infrastructure consists mainly of three buildings and houses the crusher and mill/gold circuits. There are two existing crusher buildings: primary and secondary, each approximately of 108m² and 136m² respectively. The primary building houses the jaw crusher, and the secondary building houses the cone crusher and screen deck, each building with associated conveyors. The existing mill building is approximately 1,939m² and contains the grinding and gold circuits, elution, EW and refining. The outside equipment includes thickener and leach tanks.

The 1901 deposit is located approximately 1,000m half-way off the ramp that connects the Chisel and Lalor mines. The deposit would be accessed by lateral development from the ramp and would use the existing surface infrastructures.

The 3 Zone deposit is located on the past producing New Britannia mine site in the town of Snow Lake and has existing infrastructure of a ventilation raise, decline and underground workings to approximately the 130mL. A total of 219,152 tonnes was mined from this area from 1995-1996. The New Britannia mill is within 2km on an all-weather road.

The WIM deposit is located approximately 16km north of the town of Snow Lake and is currently within trucking distance to either Stall or New Britannia mill. Currently no infrastructure exists on the claims but available in Snow Lake and can be brought into the property.

The Pen II deposit is located approximately 4km west southwest of Lalor mine and 12km west of the Town of Snow Lake. Currently no infrastructure exists on the claims but is available in Snow Lake or from Lalor mine site and brought into the property.

Access to the Watts property is via helicopter or a 17km winter road northeast from Ponton, Manitoba, where Provincial Highway #6 meets Provincial Highway #39. The winter trail utilizes the Manitoba Hydro powerline route that provides Snow Lake with power and passes within a few kilometres of the property.

In all cases, the surface rights are sufficient to accommodate mining activities. Given the fact that ore from all the deposits will be processed at either Stall or New Brit mills, the power and water requirements will be sufficient for the treatment of ore and the disposal of tailings and waste material. Given their relative proximity to Snow Lake, mining personal availability will not be an issue.

5.5. PHYSIOGRAPHY

The Snow Lake mining camp is located in the Boreal Shield Ecozone, the largest ecozone in Canada, extending as a broad inverted arch from northern Saskatchewan east to Newfoundland. This area is surrounded by water bodies including the Snow, File, Woosey, Anderson and Wekusko Lakes, all located in the Churchill River Upland Ecoregion in the Wekusko Ecodistrict.

The dominant soils are well to excessively drained dystic brunisols that have developed on shallow, sandy and stony veneers of water-worked glacial till overlying bedrock. Significant areas consist of peat-filled depressions with very poorly drained typic and terric fibrisolic and mesisolic organic soils overlying loamy to clayey glaciolacustrine sediments.

Topography shows gentle relief that rarely exceeds 10 m at approximately 300 mASL, with depressional lowlands.

6. HISTORY

6.1. LALOR & THE CHISEL BASIN AREA

Exploration in the Lalor-Chisel area began in the 1950s. The Chisel Basin area hosted four previously producing mines; Chisel Lake, Chisel Open Pit, Photo Lake and Chisel North. This basin is also the host of the Lalor deposit.

In early 2007, hole DUB168 was drilled almost vertically to test a 2003 surveyed Crone Geophysics deep penetrating pulse electromagnetic anomaly and intersected 45 m of mineralization starting at a downhole depth of 782 m. Drilling at Lalor has been continuous since the discovery of mineralization on the property and as of the date of this Technical Report, totals in excess of 500,000 drilled metres.

Lalor commenced initial ore production from the ventilation shaft in August 2012 and achieved commercial production from the main shaft in the third quarter of 2014. Table 6-1 summarizes the ore production by year at Lalor from 2012 to 2020.

TABLE 6-1: LALOR MINE PRODUCTION

	Tonnes	Zn%	Cu%	Au g/t	Ag g/t
2012	73,000	11.83	0.63	1.67	19.3
2013	401,000	9.44	0.84	1.21	19.4
2014	552,000	8.52	0.88	2.29	23.8
2015	934,000	8.18	0.71	2.53	21.4
2016	1,086,000	7.01	0.62	2.24	21.6
2017	1,293,000	7.73	0.68	1.93	23.2
2018	1,260,000	6.25	0.74	2.19	25.4
2019	1,537,000	6.35	0.75	2.16	25.5
2020	1,654,000	5.73	0.74	2.56	25.3
Total	8,790,000	7.02	0.73	2.22	23.8

In February 2019, Hudbay exploration discovered massive sulfide mineralization at the 1901 deposit. Initial inferred mineral resource estimates were declared for this deposit in August 2019 for two base metal lenses. Follow-up drilling in 2020 resulted in upgrading base metal inferred resource estimates to measured and indicated categories and adding new inferred mineral resources both for the known two base metal lenses and for two new gold rich lenses. In 2020, Hudbay also completed pre-feasibility study on the engineering and mineral processing of the 1901 measured and indicated mineral resources.

6.2. NEW BRITANNIA AND SNOW LAKE DEPOSITS

Gold was first discovered in 1914 approximately 20 km to the southeast of Snow Lake on the eastern shores of Wekusko Lake, also known as Herb Lake. In 1917, the Moose Horn-Ballast claims produced the first gold in Manitoba. First significant industrial gold production came from the Laguna mine which was in production intermittently until 1939 and produced 59k oz of gold.

Mine construction at the New Britannia site first started in 1945. In March 1949, the mine was opened as Nor-Acme mine. Production continued until 1958. 4.9Mt were mined at an average grade of 4.4g/t and the Nor-Acme mill recovered approximately 610,000 oz of gold during this production period.

Between 1958 and the early 1990s, various companies worked on the property with the intention to bring the mine back into production, however, these attempts were unsuccessful.

TVX and High River formed a joint venture in 1994 to reopen the mine. TVX became the operator of the mine in late 1994 and ore from the secondary mining areas (No. 3 Zone and Birch pit) located on mineral

lease ML-323 was used to bring the mine and mill into production prior to the completion of the rehabilitation and construction of the main shaft.

The New Britannia mill poured the first gold bullion in November 1995 which was obtained from the gold produced at No. 3 Zone. When production commenced in 1995, the name of the Nor-Acme Mine was changed to the New Britannia Mine. Production ceased at No. 3 Zone in May 1996 and production from the Birch pit was limited to the spring and summer of 1996. Full production from the main shaft was achieved in August 1996. Through various transactions, Kinross became the operator of the New Britannia mine-mill complex. Production mining ceased at the end of September 2004 and the mill was put on care and maintenance in 2005 due to a low gold price environment after producing 1.6 million oz of gold.

6.3. WIM DEPOSIT

Initial exploration on the property was comprised of airborne electromagnetic (EM) and radiometric surveys carried out by HBED in 1961, which identified several geophysical conductors. An extensive program of staking, line cutting, ground geophysical surveying and diamond drilling over high priority anomalies immediately followed.

VMS mineralization was discovered on the Wim deposit in December 1961 by diamond drilling of a ground EM anomaly delineated during the ground follow-up of an airborne EM survey. A total of one hundred and fifty-six (156) drillholes for 38,185 meters were completed from 1962 to 1991. In 2006 Murgor Resources Inc. optioned the property from Hudbay and completed a drilling program from 2007 to 2010 for sixty-four (64) diamond drill holes for a total of 19,332 metres. Most of this drilling was focused on defining the known mineralization and completing a NI 43-101 resource estimate. This historical mineral resource estimate included an indicated resource of 3.9 million tonnes @ 1.7% Cu, 0.3% Zn, 1.6g/t Au and 6.7g/t Ag and inferred mineral resource of 0.7 million tonnes @ 1%Cu, 0.4% Zn, 1.8g/t Au and 4.7g/t Ag.

In 2019, Hudbay completed seven (7) metallurgical holes for a total of 2,533 metres. Mineralization from these holes was sent off for metallurgical testwork. The WIM deposit hosts a copper-gold rich sulphide lens within amphibolite grade altered volcanic rocks.

6.4. PEN II

The first surface geophysics completed at Pen II was a Turam survey done in 1973. Early drilling in 1970's focused on this geophysical Turam survey, without significant results. Two airborne surveys were subsequently completed in 1984 and 1999. Hudbay completed Thirty-nine (39) holes between 1983 and 2007 for a total of 8,764 metres including the discovery hole which followed up an airborne and HLEM anomaly on the property. Callinan Mines Limited option the property in 1999 to 2000 and drilled three (3) holes for a total of 893.3 metres. Callinex later dropped the option. Between 2017 and 2019, Hudbay Exploration completed sixty-two (62) diamond drill holes to expand and define the resource limits for a total of 11,620 metres.

6.5. WATTS

The property was first staked in 1995 by HBED to cover anomalies located by a 1993 SPECTREM airborne EM survey. Diamond drilling of an HLEM anomaly in 1995 by HBED resulted in the discovery of the Watts River Zone. HBED drilled a total of 9,930m in 25 holes on the Watts River deposit between 1995 and 1997.

The project was optioned to Fort Knox – Berland Resources in 2000/2001. They completed seven (7) drill holes for a total of 2,476m. Only two holes were drilled on the main horizon, while most of the program focused on the newly discovered east zone located 400m in the hanging wall of the main zone. HBED implemented a new drilling program beginning in 2005 as a follow up to previous drill results and drilled a total of 9,284m in 20 holes between 2005 and 2007. In 2019 Hudbay completed six (6) holes for 2,673 metres testing the extension of the mineralized envelopes

7. GEOLOGICAL SETTING & MINERALIZATION

7.1. REGIONAL GEOLOGY

The properties of interest for this Technical Report are all within the Paleoproterozoic Flin Flon Greenstone Belt

Figure 7-1) and are overlain by a thin veneer of Pleistocene glacial/fluvial sediments. Located within the Trans-Hudson Orogen, the Flin Flon Greenstone Belt consists of a variety of distinct 1.92 to 1.87 Ga (billion years) tectonostratigraphic assemblages. These include juvenile arc, back-arc, ocean-floor, ocean-island and evolved volcanic arc assemblages that were amalgamated to form the Amisk accretionary collage prior to the emplacement of intermediate to granitoid plutons and subsequent deformation (Syme et al., 1998). The volcanic assemblages consist of mafic to felsic volcanic rocks with intercalated volcanogenic sedimentary rocks.

At the eastern end of the Flin Flon Greenstone Belt, the Snow Lake area includes a lithologically and structurally diverse sequence of deformed and metamorphosed volcanic, sedimentary and intrusive rocks. This sequence is dominated by 1.84–1.81 Ga fold-thrust-style tectonics and by a southwest-verging allochthon of volcanic rocks, the Snow Lake arc assemblage.

The Snow Lake arc assemblage that hosts the producing and past-producing mines in the Snow Lake area is a 20 km wide by 6 km thick section that records a temporal evolution in geodynamic setting from 'primitive arc' (Anderson sequence to the south) to 'mature arc' (Chisel sequence) to 'arc-rift' (Snow Creek sequence to the northeast, Bailes and Galley, 2007). The Anderson and Chisel sequences represent the geological units of interest (Figure 7-2).

The Anderson sequence is a bimodal primitive arc sequence dominated by basaltic andesite, which contains at least three rhyolite flow centres: Daly, Sneath and Anderson-Stall. The mafic and felsic volcanic rocks of the Anderson sequence are intruded by the Sneath Lake subvolcanic intrusive complex (Figure 7-2). The Anderson-Stall and Daly rhyolites occur directly above the Sneath Lake intrusion near its two ends.

The Watts River deposit is located in the eastern end of the Paleoproterozoic meta-volcanic rocks of the Flin Flon Domain. Due to a thick and pervasive cover of Quaternary glacial deposits and a thin veneer of Ordovician limestone the Paleoproterozoic rocks in the vicinity of the deposit are mainly unexposed. The meta-greywacke of the Central Kisseynew subdomain was extrapolated from their exposures on Grass river beneath the Watts River area as far south as North Moose lakes (Leclaire, et al., 1997). The Watts River deposit is likely hosted by the greywacke gneisses of the Kisseynew domain.

FIGURE 7-1: GEOLOGY OF MANITOBA

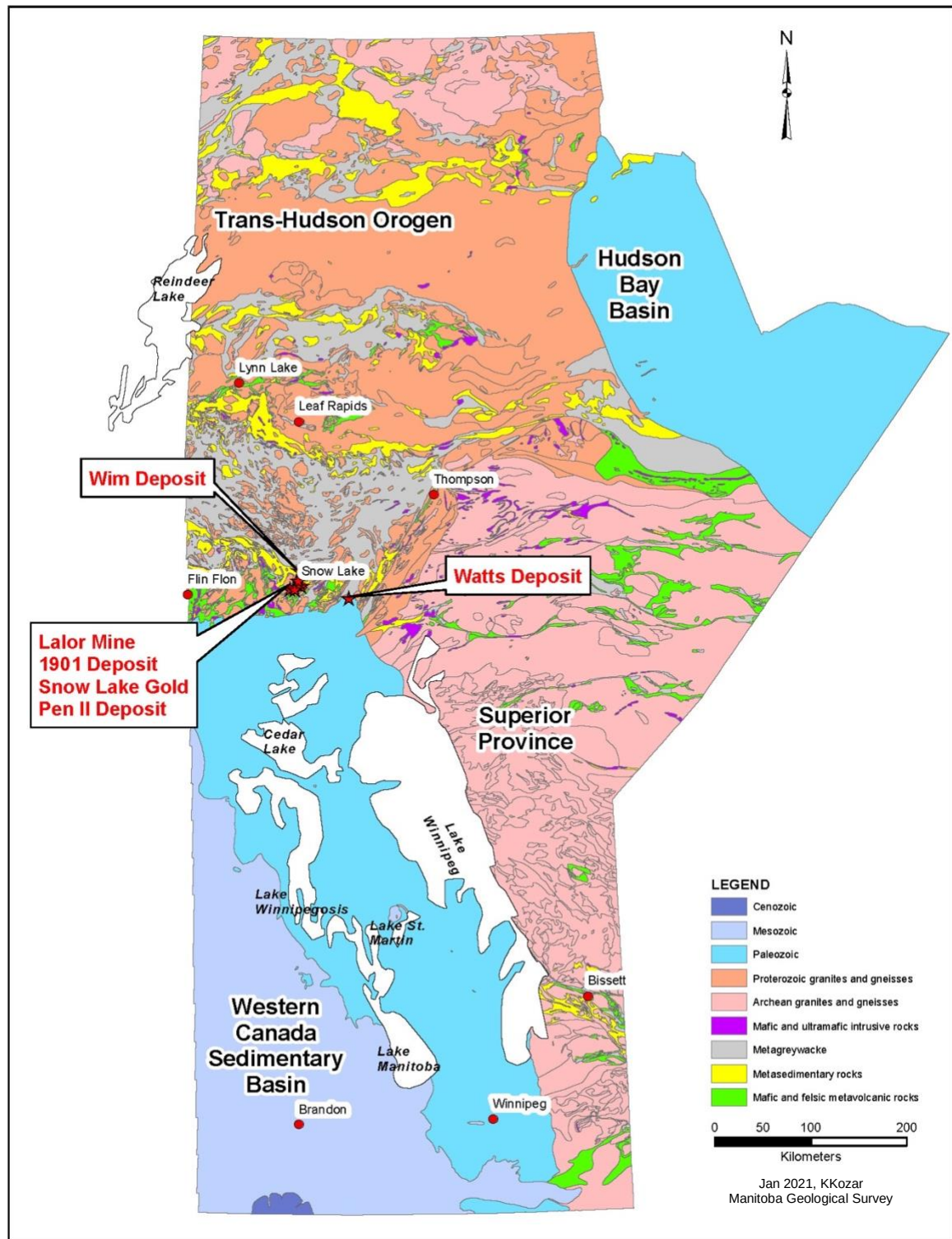
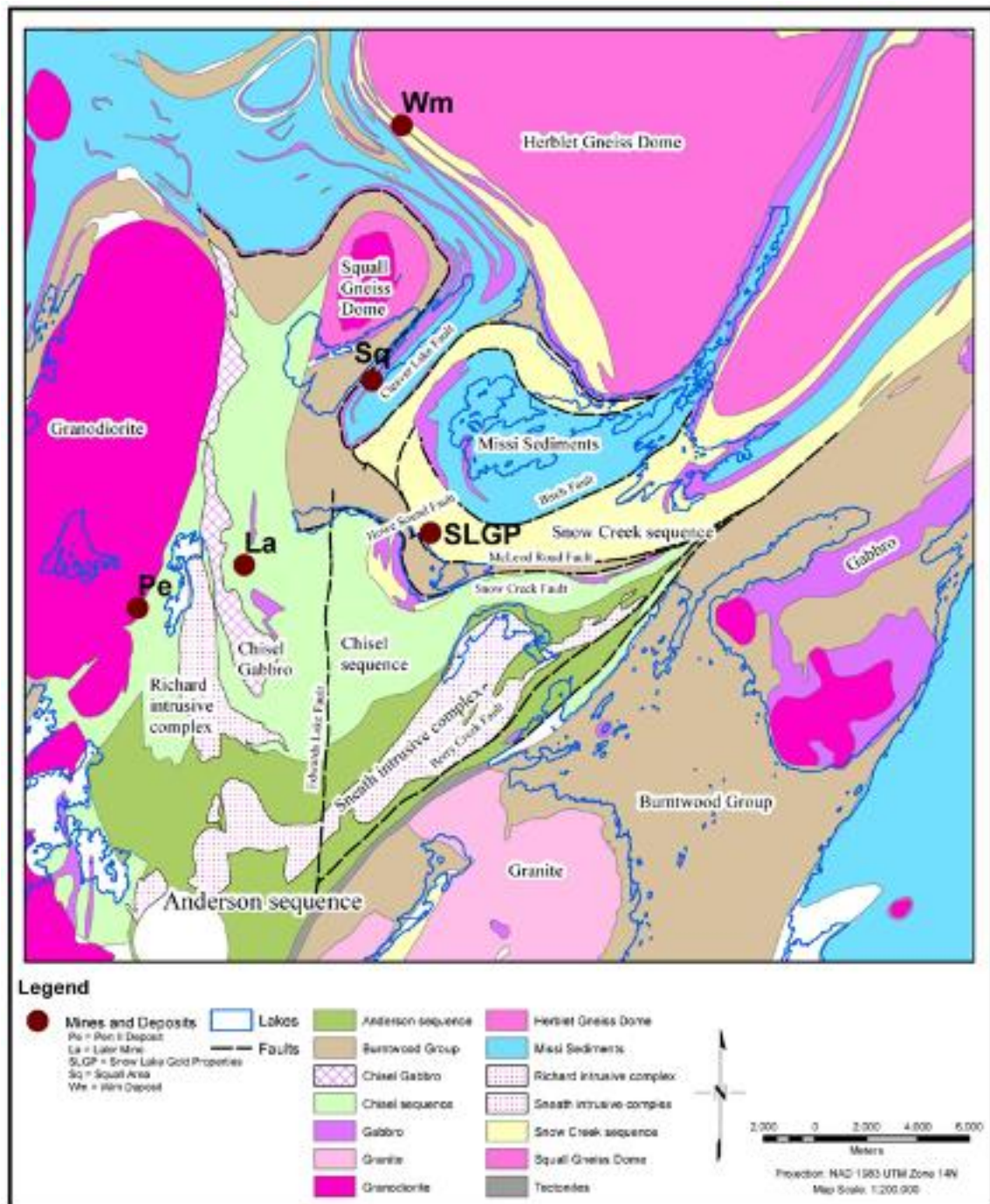


FIGURE 7-2: SNOW LAKE AREA (AFTER BAILES AND GALLEY, 2011)



VMS Deposits

The volcanogenic massive sulphide (VMS) deposits located near the town of Snow Lake have been subdivided into two different groups: Cu-Zn-rich (Cu-Zn, Cu-Zn-Au) and Zn-Cu-rich (Zn-Pb-Cu-Ag) types. The Cu-Zn-rich deposits mainly occur in the Anderson sequence and the Zn-Cu-rich deposits occur in the Chisel sequence.

Cu-Zn-rich deposits and occurrences in the Anderson sequence are confined almost entirely to within the Anderson-Stall and Daly rhyolite centres. All the developed deposits are within the larger Anderson-Stall rhyolite. The Sneath Lake synvolcanic tonalite in the Anderson Lake area shows a steady increase in both intensity of alteration of supracrustal rocks and of the Cu/Zn ratio (Cu/Cu+Zn) contained in the VMS deposits as the distance to the pluton decreases. The variation of alteration and metal Cu/Zn ratio is interpreted to be the result from higher temperature for pluton-proximal portions of the regional synvolcanic hydrothermal system (Bailes and Galley, 1996).

There are seven Zn-rich VMS deposits located within the mature arc Chisel sequence: Lalor, Chisel, Chisel North, 1901, Pen II, Photo and Ghost. The seven deposits generally overlie the Powderhouse dacite and local discrete rhyolite flow complexes. They have been subjected to polyphase deformation.

The Wim deposit, located north of Snow Lake, consists of one large copper-gold (zinc and silver) VMS lens occurring in an early Proterozoic island-arc assemblage of felsic and mafic volcanic rocks that stretches for an exposed length of 250 km east-west and 75 km north-south, on the western flank of the Herblet gneiss dome (Figure 7-2).

The Watts River Deposit is located 52km east-southeast of the town of Snow Lake. The deposit lies in the eastern portion of the Flin Flon-Snow Lake Greenstone Belt. The Watts River deposit is a stratabound accumulation of sulphides that precipitated at or near the seafloor in association with contemporaneous volcanism. The depositional environment is similar to the base metal deposits of the Snow Lake mining camp. The deposit appears to be distal with no associated alteration pipe recognized to date. The sulphide mineralization is interpreted as three lenses comprised of coarse-grained sulphides with a trend of azimuth of 020° and dip between 65 and 75° to the east.

Snow Lake Gold Deposits

The Amisk mafic and felsic volcanic hosted deposits lie to the South of the Missi group hosted deposits of Squall Lake. The McLeod Road Thrust (MRT) is one of the main structural features of the area. The deposits are bound to the west by the MRT which is responsible for the emplacement of ca. 1.9-1.89 Ga volcanic and volcanoclastic rocks of the Snow Lake arc assemblage (Bailes and Galley, 2007) on top of the ca. 1.85 Ga Burntwood group metasedimentary rocks (Figure 7-2).

The southern part of the property comprises a moderately north-dipping and dominantly north facing bimodal sequence of coarsely re-crystallized and deformed mafic and felsic volcanic and volcanoclastic, epiclastic rocks and gabbro. The Birch Lake Thrust fault to the east was emplaced ca. 1845 Ma and thrust the Missi Group metasediments above the volcanic and volcanoclastic rocks.

The southern volcanic sequence is intruded by pre and syn ultramafic and mafic intrusions, and syn to post-tectonic granitoids. The metamorphic assemblages are of lower to mid-amphibolite facies. Imbrication of the sequence bound by the MRT and BLT occurred during south-directed transport and resulted in the formation of a strong regional northeast plunging stretching lineation, a regional foliation and the Nor-Acme anticline. These structures were subsequently folded around the northeast-trending Threehouse synform. (Rubingh et al. 2013).

Mineralization of the lode-gold vein-type deposits hosted in the Amisk group mafic and felsic volcanic rocks are structurally controlled and associated with shear zones, faults, fold hinges and axial planes that host simple to complex vein systems. The mineralization is associated with lithological contacts of contrasting properties in the sequence of interlayered volcanic and volcanoclastic rocks. Alteration in the mineralized zones consists of quartz-carbonate-mica, with arsenopyrite as the primary sulphide accompanying the gold.

The Squall Lake area (4 km northwest of the Town of Snow Lake) is comprised of several mineralized zones that are hosted in Missi group rocks which are the product of fluvial and alluvial sedimentation. In Snow Lake, the Missi Sequence is composed of cross bedded arenites and arkose. The Missi group is commonly intruded by discontinuous gabbroic sills and dykes that are medium to coarse grained. The Missi group arenites were thrust over the garnet staurolite schist of the Burntwood group (Beilhartz, 2006). The lowermost sill, a 20 - 40-metre-thick amphibolitic unit, is spatially associated with the known

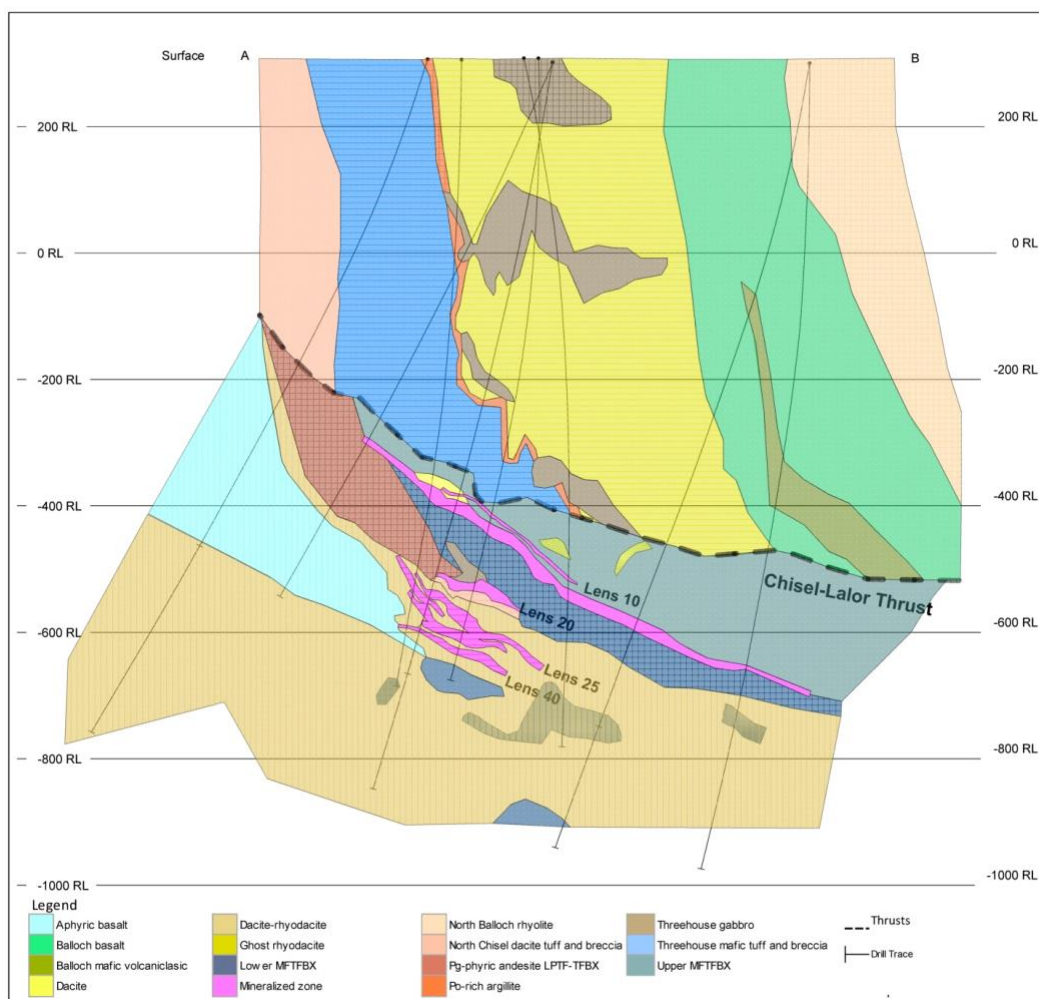
gold mineralization of the Squall lake area. This epigenetic gold mineralization is believed to be associated with structures generated by over-thrusting related to the McLeod Road Thrust Fault.

7.2. PROPERTY GEOLOGY

Lalor

The Lalor deposit is similar to other massive sulphide bodies in the Chisel sequence (Chisel Lake, Ghost Lake, Chisel North, and Photo Lake), and lies along the same stratigraphic horizon as the Chisel Lake and Chisel North deposits. It is interpreted that the top of the zone is near a decollement contact with the overturned hanging wall rocks, the Chisel-Lalor Thrust (Figure 7-3). The most common dyke intrusion throughout these rocks is a fine-grained feldspar-phyric gabbro to diorite.

FIGURE 7-3: LALOR MINE CROSS SECTION WITH DRILL HOLE TRACES



The extensive hydrothermal alteration and metamorphic recrystallization of the footwall rocks has produced some exotic aluminous mineral assemblages. These assemblages include chlorite and sericite dominant schists and cordierite + anthophyllite gneisses. Other minerals indicative of hydrothermal alteration that occur extensively throughout these rock assemblages include quartz, feldspar, kyanite, biotite, garnet, staurolite, hornblende, and carbonate. Clinopyroxene, gahnite and anhydrite also occur locally. These

assemblages are typical of metamorphosed footwall hydrothermal alteration commonly associated with VMS deposits and are similar to that at the other massive sulphide deposits in the Chisel Lake area.

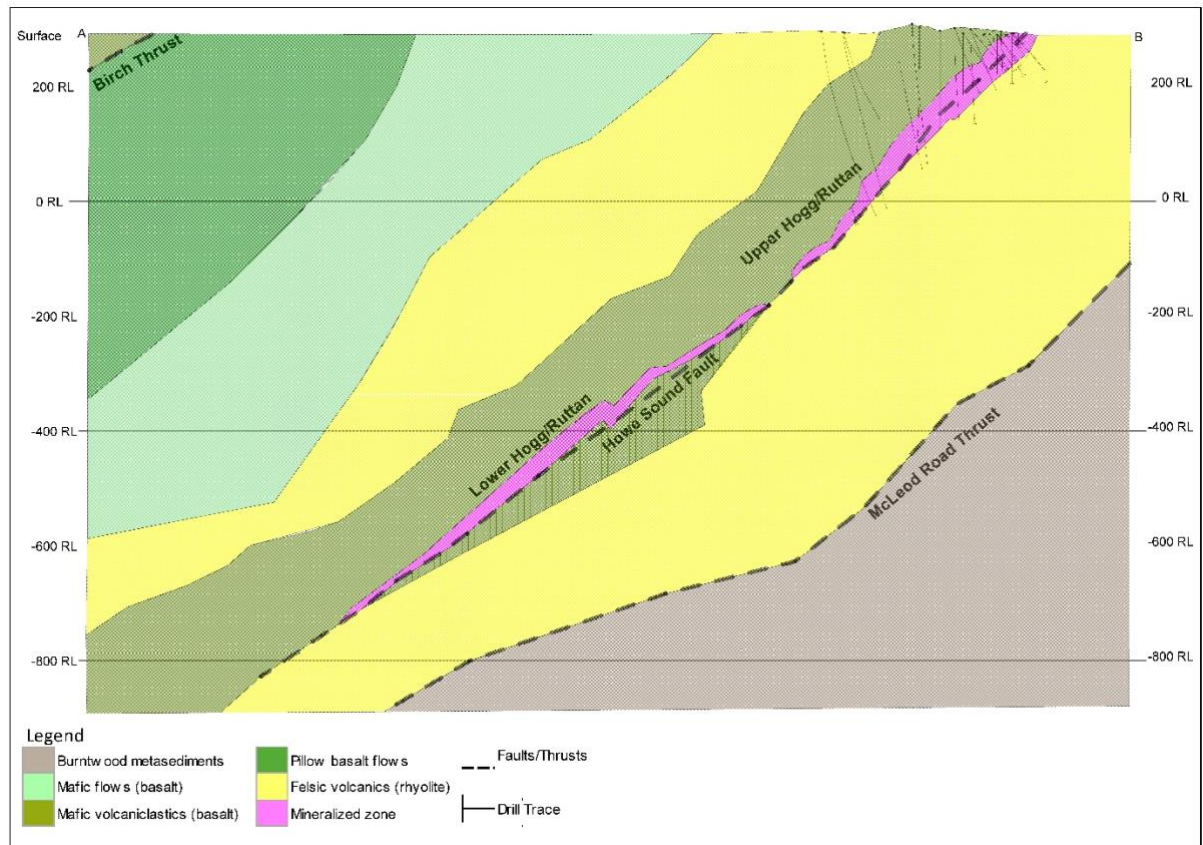
Snow Lake Gold Properties

The Snow Lake gold properties consists of the New Britannia Mine, No. 3 Zone, Birch Zone, Boundary Zone and Squall Lake deposits including the K1, K5, K7, K10, K11, Margaret (North and South), Margaret Extension, South, F1, Moon Gertie Zones, Bay and North Zones.

The New Britannia mine is a classical lode gold, quartz-carbonate vein type deposit. The mineralization is associated with shear zones, faults, and simple to complex vein systems. The auriferous zones (Toots, Dick, Ruttan and Hogg) lie along a curvilinear shear zone named the Howe Sound fault also referred to as the Nor-Acme fault (Figure 7-4). The main mineralized zones consist of sheared or “mylonitic” quartz-carbonate-mica alteration with arsenopyrite mineralization accompanying the gold emplaced in a simple intercalated sequence of altered felsic and mafic volcanics and pyroclastics of the Amisk Group. The host rocks are altered and include varying proportions of quartz and carbonate.

The gold bearing horizons in the No. 3 Zone are fault-hosted vein occurrences, consisting of a main shear vein and numerous extensional ladder veins. Gold mineralization is located within a 50 m wide fault zone at the folded contact between the coarse pyroxene mafic volcanoclastic rocks and the plagioclase-phyrlic pillowed basalt flows.

FIGURE 7-4: NEW BRITANNIA REPRESENTATIVE CROSS SECTION WITH DRILL HOLE TRACES



The Birch Zone is interpreted as hosted in a silicified shear zone and several parallel and sub parallel shears cut the mafic volcanic, mafic volcanoclastic and metasedimentary rocks (including some graphitic argillite) which were intruded by gabbroic dykes. Alteration and mineralization at the Birch Zone is similar to the No. 3 Zone.

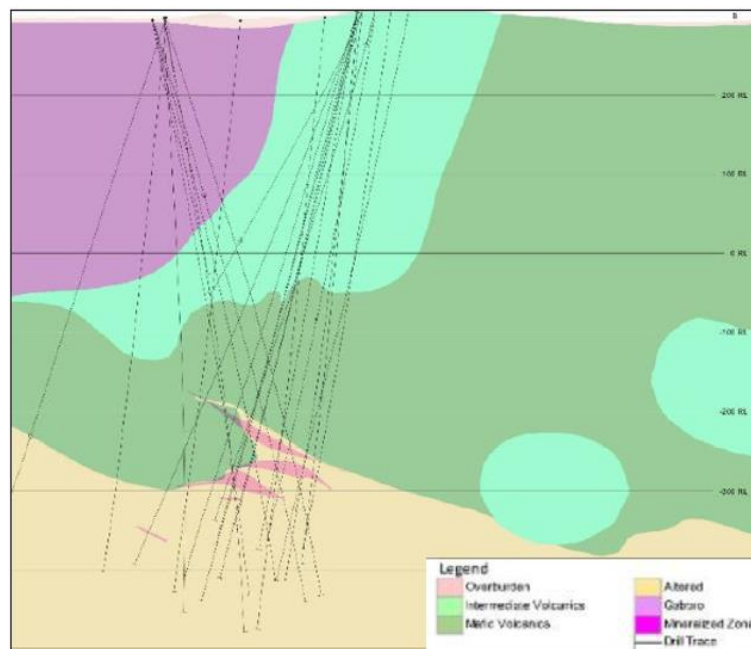
The Boundary Zone is located 0.5 km south of the No. 3 Zone and approximately 1 km north of the New Britannia Mine, and in the hanging wall of the McLeod Road Thrust fault. The lode-type gold deposit is located along the Nor-Acme anticline at the contact between mafic and felsic volcanics.

The Squall Lake area is host to several mineralized zones in Missi group rocks. Epigenetic gold mineralization on the property is believed to be associated with structures generated by over-thrusting related to the McLeod Road Thrust Fault (Beilhartz, 2006). Two economically significant gold mineralized sets of structures exist on the property. The first type of structure was generated by an initial shearing event at the contact of differing lithologies due to contrasting competencies. This type of structure accounts for the Lower Silicified Zone at the Margaret Extension and Margaret South and occur as a series of sub-parallel en-echelon mineralized lenses striking 050° to 070°. The second type of mineralized structure is a transcurrent type of shearing which developed large scale, sub-vertical shears across the property. (Beilhartz, D. 2006)

1901 Deposit

The 1901 Deposit is hosted in a shallowly NNE dipping package of highly altered volcanic rocks of calc-alkaline geochemistry (Figure 7-5). The volcanic rocks consist of altered mafic, intermediate and felsic volcanics, similar to other VMS deposits in the Chisel-Lalor area. A major flat lying fault in the hanging wall of the 1901 deposit separate the deposit from a hanging wall package of unaltered, steep to moderately dipping southwest facing open folded tholeiitic rocks of the Threehouse and North Chisel formations. The 1901 deposit is hosted in highly altered intermediate “gneiss” that structurally overlies a sequence of felsic volcanic rocks (felsic “gneiss”). The deposit occurs where the intermediate “gneiss” is thinned, and the 1901 VMS type mineralization is within 30m of the felsic “gneiss”. The 1901 deposit is characterized by a large zone of metamorphically recrystallized and hydrothermally altered rocks that are composed of sericite- and chlorite-rich rocks with abundant porphyroblasts of kyanite-silimanite, biotite, staurolite, anthophyllite, cordierite and garnet. The primary textures are rarely preserved, and the protolith is no longer recognizable except through (immobile) trace element geochemistry.

FIGURE 7-5 1901 REPRESENTATIVE CROSS SECTION WITH DRILL HOLE TRACES

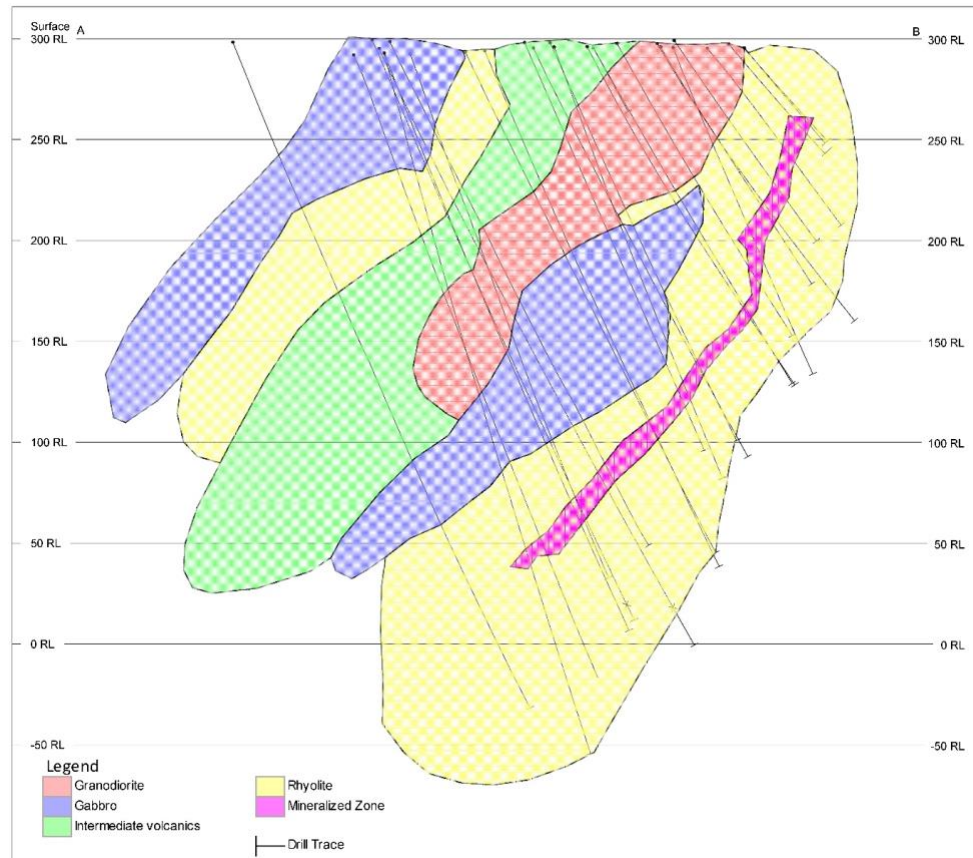


Pen II Deposit

The Pen II deposit is located in the Snow Lake arc assemblage (Figure 7-6). The volcanic stratigraphy that hosts the Pen II deposit is on the eastern margin of the younger Ham Lake intrusion. The stratigraphy

strikes north - northeast and dips moderately to steeply west. The deposit has extensive granitic to gabbro dykes throughout the hanging wall and numerous felsic to mafic volcanic flows in the footwall of the deposit. The mineralization is generally hosted within felsic volcanic or volcanoclastics or at a contact between felsic and mafic volcanics. All the rocks have undergone lower to middle amphibolite facies metamorphism and primary volcanic features have been obliterated.

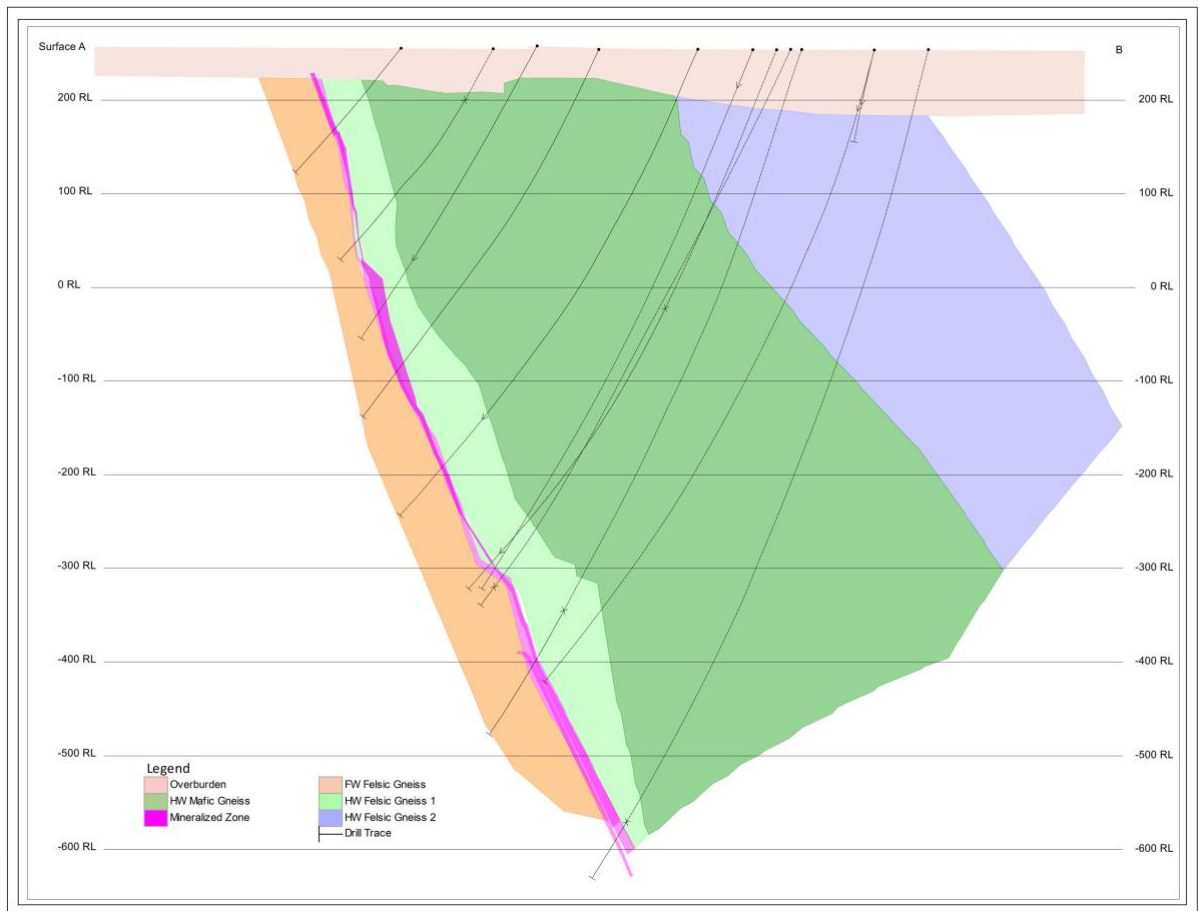
FIGURE 7-6 PEN II REPRESENTATIVE CROSS SECTION WITH DRILL HOLE TRACES



Watts Deposit

The Watts River deposit is believed to have formed in an arc-rift setting. The deposit is a stratabound accumulation of sulphide minerals that were likely precipitated in association with contemporaneous volcanism. The massive sulphide zone was interpreted to be "distal" as no alteration pipe associated with the base metal mineralization has been identified to date. The rocks hosting the Watts River deposit are coarsely recrystallized to upper amphibolite to perhaps granulite facies mineral assemblages. They are overprinted by a strong gneissic fabric and intruded by numerous small veins of granitic (quartzofeldspathic gneiss) material. The mineralization occurs in three main lenses that extend from the bedrock-surface contact to a depth of 900m below surface (Figure 7-7).

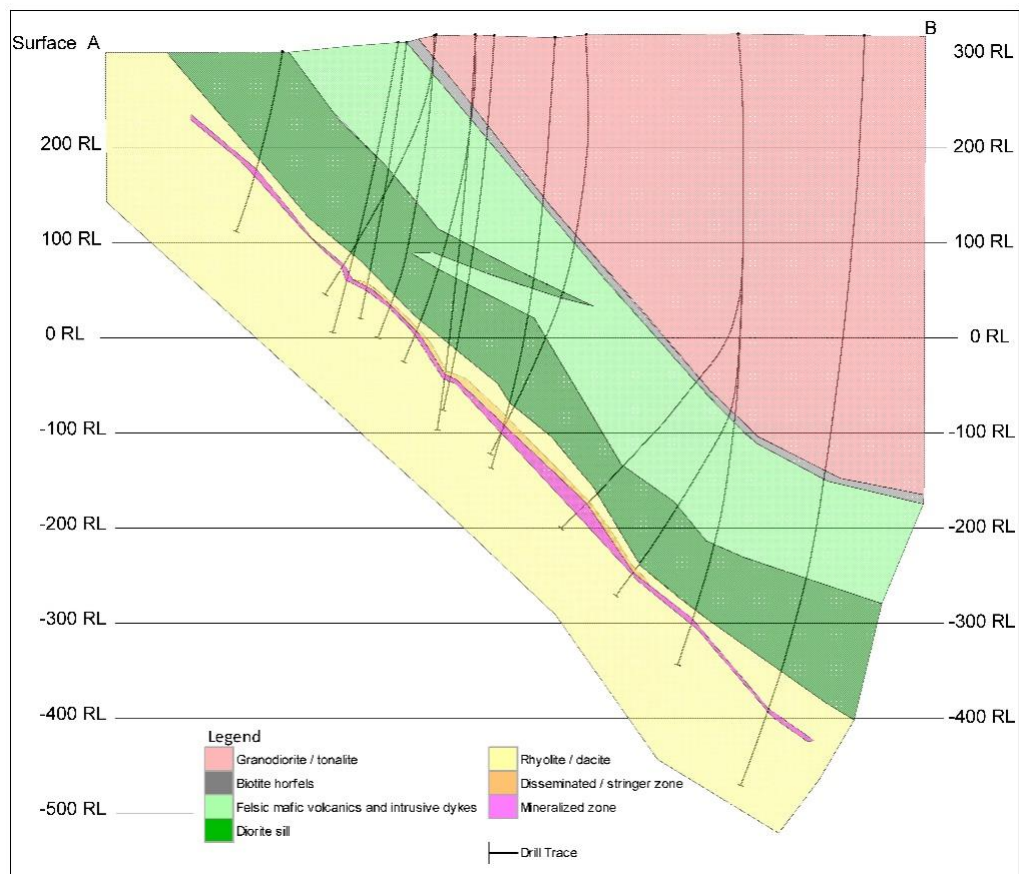
FIGURE 7-7 WATTS REPRESENTATIVE CROSS SECTION WITH DRILL HOLE TRACES



Wim Deposit

The Wim deposit is hosted within a northwest-striking, northeast-dipping package of mafic and felsic metavolcanic gneisses along the southern rim of the Herblet Lake Gneiss Dome Complex (Figure 7-8).

FIGURE 7-8: WIM DEPOSIT CROSS SECTION WITH DRILL HOLE TRACES



The meta-volcanic gneisses are interpreted to be derived from the early Proterozoic subaqueous and subaerial volcanics of the Amisk Group that is locally overlain by marine turbidites of the Burntwood Group and terrestrial sediments of the Missi Group and intruded by various dykes. Tectonic foliations in the vicinity of the Wim deposit trend northwest and dip moderately to steeply towards the northeast.

Several alteration types associated with the sulphide mineralization have been documented in outcrop and drill core. Other alteration types encountered in outcrop and drill core include medium to coarse-grained, cordierite-anthophyllite, sieve textured garnets with chalcopyrite inclusions, muscovite-sillimanite and garnet zones. There is a general coarsening of grain size with proximity to mineralization due to the metamorphic recrystallization of hydrothermally altered rock and the growth of strongly porphyroblastic minerals such as garnet, anthophyllite, cordierite and sillimanite. Altered rhyolites occur to the northwest and southeast of the deposit along the geophysical trend that coincides with the Wim deposit mineralization.

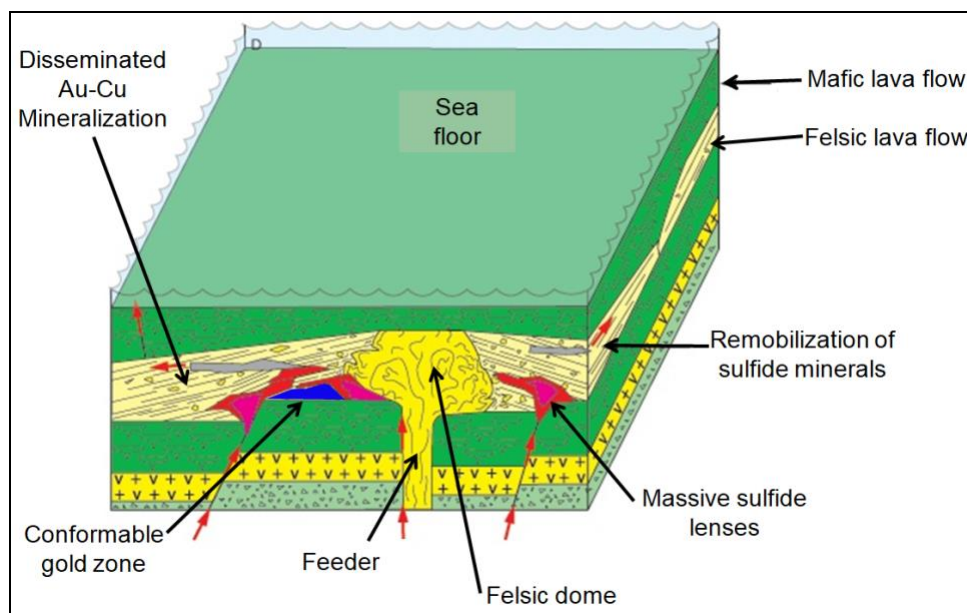
8. DEPOSIT TYPES

8.1. VMS DEPOSITS

The deposits of interest in this Technical Report are classified as VMS (volcanogenic massive sulphides). VMS deposits are an important source of copper, zinc, lead, gold and silver (Cu, Zn, Pb, Au and Ag) and form at or near the seafloor where circulating hydrothermal fluids driven by magmatic heat that have quenched through mixing with bottom waters or porewaters in near-seafloor lithologies (Figure 8-1).

FIGURE 8-1: CONCEPTUAL MODEL OF A VOLCANOGENIC MASSIVE SULFIDE (VMS) DEPOSIT

(Modified After S. Lavoie, 2003)



Massive sulfide lenses vary widely in shape and size and may be podlike or sheetlike. They are generally stratiform and may occur as multiple lenses from small pods of less than a ton (which are commonly scattered through prospective terrains) to giant accumulations like the Rio Tinto mine (Spain), Kholodrina (Russia), Windy Craggy (Canada), Brunswick No. 12 (Canada) or Ducktown (United States). Volcanogenic massive sulfide deposits are actively forming in extensional settings on the seafloor, especially mid-ocean ridges, island arcs, and back-arc spreading basins.

The location of VMS deposits is often controlled by synvolcanic faults and fissures, which permit a focused discharge of hydrothermal fluids. As illustrated on Figure 8 1, a typical VMS deposit will include the massive mineralization located proximal to the active hydrothermal vent, footwall stockwork mineralization, and distal products, which are typically thin but extensive. Footwall, and less commonly, hanging wall semi-conformable alteration zones are produced by high temperature water-rock interactions (Franklin et al., 2005).

Massive sulphide ore in VMS deposits consists of usually more than 40 percent sulfides, dominated by pyrite, pyrrhotite, chalcopyrite, sphalerite, and galena. Non-sulfide gangue typically consists of quartz, barite, anhydrite, iron (Fe) oxides, chlorite, sericite, talc, and their metamorphosed equivalents. Sulphide ore composition may vary from Pb-Zn- to Cu-Zn-, or Pb-Cu-Zn-dominated, and some deposits are zoned vertically and laterally. VMS deposits have stringer or feeder zones beneath the massive zone that consist of crosscutting veins and veinlets of sulfides in a matrix of pervasively altered host rock.

Lalor Mine and 1901 Deposit

The depositional environment for the mineralization at Lalor and the 1901 deposit are similar to present and past producing base metal deposits in felsic to mafic volcanic and volcanoclastic rocks in the Snow Lake mining camp. The deposits appear to have an extensive associated regional hydrothermal alteration.

The mineralized envelopes are shallow dipping, with zinc mineralization defined to date beginning at approximately 600 m from surface and extending to a depth of approximately 1,100 m. The mineralization trends about N320° to N340° azimuth and dips between 30° and 45° to the north. It has a lateral extent of about 1,400 m in the north-south direction and 800 m in the east-west direction.

Sulphide mineralization is dominated by pyrite and sphalerite. In the near solid (semi-massive) to solid (massive) sulphide sections, pyrite occurs as fine to coarse grained crystals ranging one to six millimetres and averages two to three millimetres in size. Sphalerite occurs interstitial to the pyrite. A crude bedding or lamination is locally discernable between these two sulphide minerals. Near solid coarse-grained sphalerite zones occur locally as bands or boudins that strongly suggest that remobilization took place during metamorphism.

Disseminated blebs and stringers of pyrrhotite and chalcopyrite occur locally within the massive sulphides, adjacent to and generally in the footwall of the massive sulphides. The hydrothermally altered rocks in the footwall commonly contain some very low concentrations of sulphide minerals. Some sections of massive pyrrhotite occur, but these tend to give way to pyrite-sphalerite-dominant zones.

The top two lenses of the stacked base metal zones at Lalor (referred to as Zone 10 and 11) and lenses 1 and 3 at the 1901 deposit have higher grade zinc and iron content. Lenses located lower in the stratigraphy and coded as Zones 20, 30, 31, 32, 40 and 42 at Lalor have moderate to high zinc grades hosted in near solid sulphides containing higher grade gold and locally appreciable amounts of copper.

The footwall gold mineralization is typical of any VMS footwall feeder zone with copper-rich, disseminated and vein style mineralization overlain by a massive, zinc-rich lens. The fact that the footwall zone is strongly enriched in gold suggests a copper-gold association which is comparable to other gold enriched VMS camps and deposits (Mercier-Langevin, 2009). Some of the footwall zones tend to be associated with silicification and the presence of gahnite. These zones are often characterized by copper-gold association and are currently interpreted as being associated with higher temperature fluids below a zone of lower temperature base-metal accumulations.

Gold and silver enriched zones occur near the margins of the zinc rich sulphide lenses and as lenses in local silicified alteration. Remobilization is illustrated in some of the gold-rich zones by late veining that is restricted to the massive lenses.

Some of the footwall zones tend to be associated with silicification and the presence of gahnite. These zones are often characterized by copper-gold association and are currently interpreted as being associated with higher temperature fluids below a zone of lower temperature base-metal accumulations.

General observations of the known gold zones indicate areas which are coarse-grained and porphyroblastic in nature are gold poor, while fine grain siliceous ($\pm$ veins $\pm$ sulphide traces) and strained looking stratigraphy tend to be gold rich. However, the intensity and style of alteration can vary strongly over short distances and may suggest that the alteration was forming discordant stockwork like zones that are now strongly transposed in the main foliation (Mercier-Langevin, 2009).

Watts Deposit

The Watts River deposit trends 020° azimuth and dips between 65° and 75° to the east. The sulphide intersections range from less than 0.3m up to 23m in core length, with a lateral extent of approximately 1,200m. Diamond drilling has intersected mineralization at depths of 850m below surface. Mineralization was intersected and interpreted as three lenses; Main Lens, Main Footwall Lens, and East Lens. The Main lens mineralization is comprised of coarse-grained pyrite, pyrrhotite, chalcopyrite, sphalerite, and minor galena. Mineralization was projected from the surface bedrock contact to a depth of 900m below surface.

as massive sulphides, fine disseminations and clots. The Main Lens has been divided into Main A, B, and C representing zones along the lens where appreciable copper and zinc grades and widths have been identified. The Main Footwall Lens lies in the immediate footwall of the Main Lens at depths of 600 to 880m below surface and tends to be slightly richer in sphalerite. The East Lens mineralization lies 400m into the hanging wall of the Main lens and extends from the surface bedrock contact to 280m below surface. Mineralization consists of narrow massive sulphides, disseminated, blobs, splashes, stringers of coarse-grained pyrite, pyrrhotite, chalcopyrite and lesser sphalerite. The massive sulphides exhibit a durchbeweget texture, characterized by the presence of rounded clasts of clear cherty quartz and amphibole inclusions within the sulphides. This texture suggests structural milling of the wall rock and the deposition of wall rock fragments within the recrystallized sulphides. The sulphides have generally been recrystallized to a coarse grain size, but sections of finer grained sulphides do occur.

Pen II Deposit

The Pen II deposit comprises a stratabound, semi-massive to massive sulphide lens with an adjacent stringer/disseminated sulphide zone.

Mineralization is characterized by disseminated to massive, recrystallized and medium to coarse-grained sphalerite, pyrite, pyrrhotite and minor chalcopyrite. Based on the drilling completed by Hudbay and Callinan Mines Ltd. the mineralization extends from surface to 500 m below surface. The current strike length of the deposit is 400 m with an average thickness of 4 m. The deposit is conformable to stratigraphy, trends to the northeast at a N40° azimuth, a 45-65° dip towards the northwest.

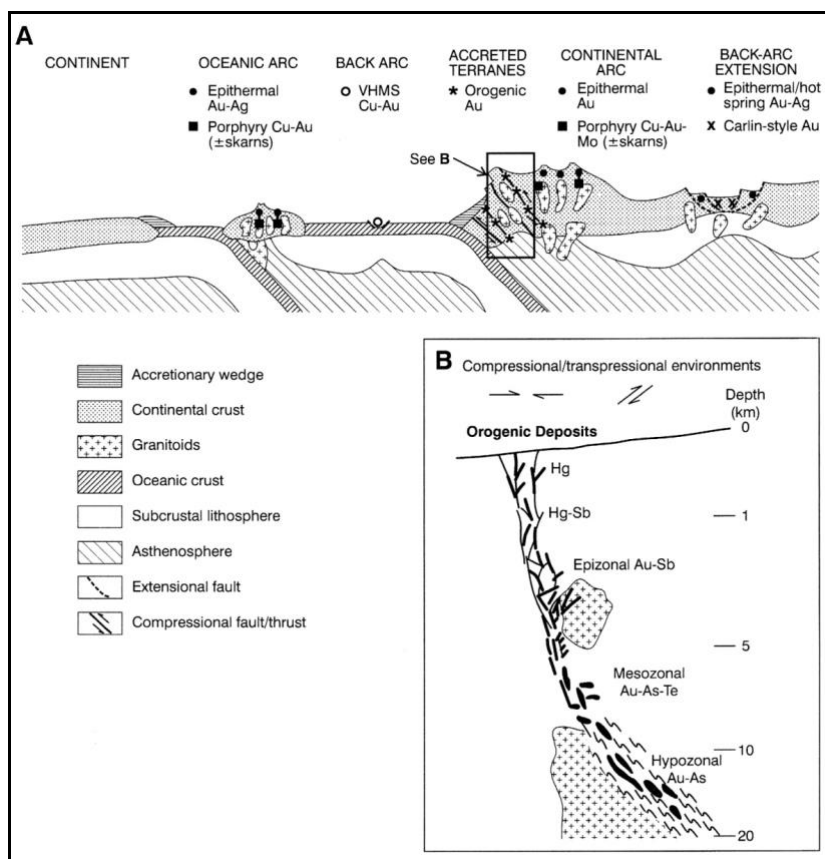
Wim Deposit

The Wim deposit comprises a stratabound, semi-massive to massive sulphide lens with an adjacent stringer/disseminated sulphide zone. Mineralization is characterized by disseminated to massive, recrystallized and medium to coarse grained pyrite, pyrrhotite, chalcopyrite and minor sphalerite. Based on the drilling completed by Murgor, the VMS mineralization extends from surface to 720 m below surface. The current strike length of the deposit is 725 m with an average thickness of 10 m. The Wim deposit is conformable to stratigraphy, trends to the northwest at a N310° azimuth, a 40-45° dip towards the northeast and a plunge of 40° to the north. In addition, disseminated and stringer sulphides occur in the stratigraphic footwall of the Wim sulphide lens and may represent the feeder zone of the stratabound sulphide lens.

8.2. SNOW LAKE GOLD PROPERTIES

The Snow Lake Gold Properties (NBM, No. 3 Zone, Boundary Zone, Birch Zone and Squall Lake Area) are considered to belong to the quartz-carbonate vein gold subtype of orogenic lode gold deposits. This subtype of gold deposits consists of simple to complex quartz carbonate vein systems associated with brittle-ductile rock behaviour, corresponding to intermediate depths within the crust, and compressive tectonic settings (Figure 8-2). Deposits of this type have been commonly referred to as mesothermal gold quartz vein deposits, but they encompass both mesothermal and hypothermal classes as initially defined by Lindgren (1933).

FIGURE 8-2: SCHEMATIC DIAGRAM SHOWING THE SETTING AND NATURE OF OROGENIC LODGE-GOLD DEPOSITS



Note: The plate tectonic environments of formation of orogenic lode-gold deposits and other gold-rich deposit styles, adapted from Goldfarb et al. (2005) and Groves et al. (2000). (B) Depth profile of orogenic lode-gold deposits, adapted from Groves et al. (1998)

At the regional scale, the quartz-carbonate vein gold deposits occur in two contrasting geological environments: deformed clastic sedimentary terranes and deformed volcano-plutonic terranes containing diverse volcanic assemblages of island-arc and oceanic affinities. Despite lithological and structural differences, these two types of environment share the following characteristics: greenschist to locally lower amphibolite metamorphic facies, brittle-ductile nature of deformation, and geological structures recording compressional to transpressional tectonic settings. Quartz-carbonate vein gold deposits in these environments tend to occur in clusters, or districts. Both types of environments are present in several districts, in which they are separated by major fault zones. However, in such cases auriferous quartz-carbonate veins preferentially occur in the volcano-plutonic domains.

At the New Britannia mine the mineralization is associated with shear zones, faults, and simple to complex vein systems. The auriferous zones (Toots, Dick, Ruttan and Hogg from west to east respectively) within the mine lie along a curvilinear shear (fault) zone named the Howe Sound fault also referred to as the Nor-Acme fault. One or more "slip planes" accompanied by a variety of altered mylonitic zones mark the location of the fault. The mylonitic zones (quartz-carbonate-mica) are predominantly less than 0.3 m thick but can thicken locally to between 3 m to 6 m. The gold bearing quartz-carbonate rocks are almost always situated next to or astride the fault. The main mineralized zone consists of quartz-carbonate alteration zones with arsenopyrite as the main sulphide accompanying the gold, emplaced in a simple intercalated sequence of altered felsic and mafic volcanic and pyroclastic. The contacts of the mineralized zones are often gradational; however, there have been sharp shear and fault contacts noted within some parts of the deposit, and a biotite alteration halo occurs within a few feet of the mineralized zone.

Mineralization in No. 3 Zone, Boundary and Birch Zones is hosted within sheared biotite-quartz and carbonate altered volcanic and volcanoclastic rocks that host quartz-albite-iron carbonate veins with fine to coarse gold, and is associated with acicular arsenopyrite within the veins, wallrock and wallrock fragments (Galley et al., 1991). Observations of the mineralization from underground at the No. 3 Zone noted that the quartz veins varied in true width from 0-1.5m and that the mineralization was not confined to the veins and can occur in up to 2 m from the main vein system.

Ten zones of gold-arsenopyrite mineralization have been identified in the Squall Lake area through previous exploration programs from 1943 to 1987. These zones occur within a 40-60 m stratigraphic interval above the staurolite schist unit and are aligned over a 10 km strike length along the northwest flank of the McLeod Lake Synform. These zones have been named K1, K5, K7, K10, K11, Margaret (North and South), Margaret Extension, South, F1 and Moon Gertie Zones. Two other similar zones to the South are the Bay and North Zones. The gold-arsenopyrite mineralization is directly associated with tabular quartz veins that are rimmed by fine-grained arsenopyrite, sub-parallel to the enclosing stratigraphy, or as tightly folded with a near vertical attitude. (Beilhartz, 2006).

9. EXPLORATION

9.1. LALOR MINE

In 2003, a Crone Geophysics high power time-domain electromagnetic (EM) system experimental survey was conducted over the deepest portion (approximately 600 m vertical depth) of the Chisel North Mine. The survey was designed and interpreted by Hudbay and was conducted by Koop Geotechnical Services Inc. The survey provided conclusive evidence that the system could detect conductive bodies at depths greater than 500 m. It was decided to extend the survey coverage further down-dip and down-plunge of the known mineralized lenses. A double-wired transmitter loop measuring 2 km by 2 km was used to maximize the EM field strength. The survey results were interpreted using three-dimensional (3D) computer modeling. The model indicated a highly conductive, shallow-dipping zone at a vertical depth of 800 m at Lalor.

Since the last NI 43-101 Technical Report in 2019, exploration drilling at Lalor has both focused on adding and converting inferred mineral resource estimates with a strong emphasis on confirming the continuity of the gold mineralization. Due to the low angle and stacking nature of the mineralization at Lalor, holes were extended beyond the base metal target depths to explore for feeder style mineralization rich in copper – gold, which led to increases in mineral resource inventory. Approximately 30,000 m of exploration underground drilling has been conducted per year at Lalor since 2019.

9.2. 1901 DEPOSIT

In the mid 1990's Airborne EM/MAG surveys (SPECTREM) were flown and highlighted numerous anomalies in the Chisel Basin, which were subsequently followed up with gridding, ground HLEM / magnetometer surveys and diamond drilling. Airborne VTEM geophysical surveys were also flown in 2007.

In January 2019, a winter drill program was initiated to follow-up untested DPEM and BHPEM targets. The initial hole CH1901, was testing and area known as the South Bull's eye and identified VMS type sulphide mineralization. After the completion of the 1901 discovery drillhole and additional 53 drillholes were completed in 2019 yielding a total of 35,138m of NQ core. Borehole EM was conducted on several drillholes and helped to direct the drilling. In 2020, a follow-up drill program was conducted on the 1901 deposit. A total of 39 NQ drillholes were completed for a total of 26,275m.

9.3. SNOW LAKE GOLD PROPERTIES

Exploration programs and exploration history on the Snow Lake Gold Properties (SLGP) are described in the previously disclosed NI 43-101 prepared for Alexis Minerals Corp. by Genivar, in 2010. A brief outline of the most recent exploration work on these properties is provided below.

In 2011, Bailes Geoscience was contracted by Alexis Minerals Corp. to perform a geochemical study of the lithological units in this area (McLeod Road-Birch Lake Thrust (MB)) sequence. The evaluation was conducted over 649 high-quality ICP-MS litho-geochemical samples collected from 35 exploration drill holes. The highlights of the results were:

- The McLeod Road-Birch Lake volcanic rocks are a thrust repetition of the Snow Lake arc assemblage to the south.
- Gabbro intrusions are geochemically identical to specific supracrustal units and therefore likely to be synvolcanic feeders for the rocks.
- Rhyolite flows in the MB are all very similar in geochemistry and indistinguishable from one another. They display arc tholeiite type FII rhyolite geochemistry.
- Alteration within the MB is very subtle compared to rocks associated with VMS. No gold-associated major element alteration trends were identified.

In 2017, Hudbay re-initiated a systematic exploration program for lode gold deposits in the Snow Lake area and completed an Induced Polarization (IP) survey and an unmanned aerial vehicle (UAV) magnetic survey over the northern part of the (SLGP) encompassing the Birch, No. 3 Zone and the Boundary deposit areas.

9.4. PEN II DEPOSIT

The PEN II zone was discovered in 1986 by HBED. A ground HLEM and magnetometer grid in the area had been extended to the northeast of claim CB146 and identified a previously unknown EM anomaly. The anomaly was drilled and intersected favourable mineralization, so the drilling campaign was continued for a total of 15 holes. This drilling campaign marked the discovery of what is known as the PEN II zone. Follow-up drilling was conducted by HBED between 1987 and 1991.

Between 1999 and 2000, Callinex optioned the property and drilled an additional three holes and two wedges and completed an airborne EM survey over the zone.

In 2007, HBED drilled 6 holes to determine the geological continuity of the PEN II zone and tested geophysical targets in the area. In 2017, Hudbay (previously HBED) completed an additional seventeen drillholes on the property for a total of 3,120 m.

In 2018, 23 holes were drilled, totaling 6,374 m to upgrade the majority of the mineral resource estimates at the Pen II deposit to an indicated category. During this program, two new small high-grade satellite zones were discovered within the top 400 m.

In 2019, the drilling at Pen II focused on confirming the continuity of the mineralization in the two satellite zones, helped to define the exact extent of the mineralization towards surface, and were used to collect geotechnical data and samples for metallurgical sampling.

9.5. WIM DEPOSIT

Exploration programs and exploration history at Wim is described in the previously disclosed NI 43 101 prepared for Alexandria resources by Golder, in 2015. Since acquiring the property at the end of 2018, Hudbay has initiated a drilling program to collect geotechnical and metallurgical samples in order to test the flowsheet of the future New Britannia mill. Results have confirmed that the mineralization at Wim could attain processing recoveries in copper, silver and gold similar to the ones estimated for the Lalor copper-gold zones.

9.6. WATTS

The property was first staked in 1995 by HBED to cover anomalies located by a 1993 SPECTREM airborne EM survey. Diamond drilling of a horizontal loop electromagnetic (HLEM) anomaly in 1995 by HBED resulted in the discovery of the Watts River Deposit.

A total of 19 holes from the early 1995 to 1997 drilling were borehole pulse EM surveyed. The surveys determined that the region of highest conductivity appeared to be in the vicinity of drill holes KUS223, 229, and 283, which lie within the Main Lens A. Five of the seven holes drilled during the 2000 to 2001 campaign were borehole pulse EM surveyed, defining mineralization in the East Lens as predominantly disseminated and minor massive sulphides.

In 2000 – 2001 the property was optioned to Fort Knox – Berland Resources, which completed eight (8) holes focusing on the newly discovered east lens which is approximately 400m in the hangingwall of the main deposit.

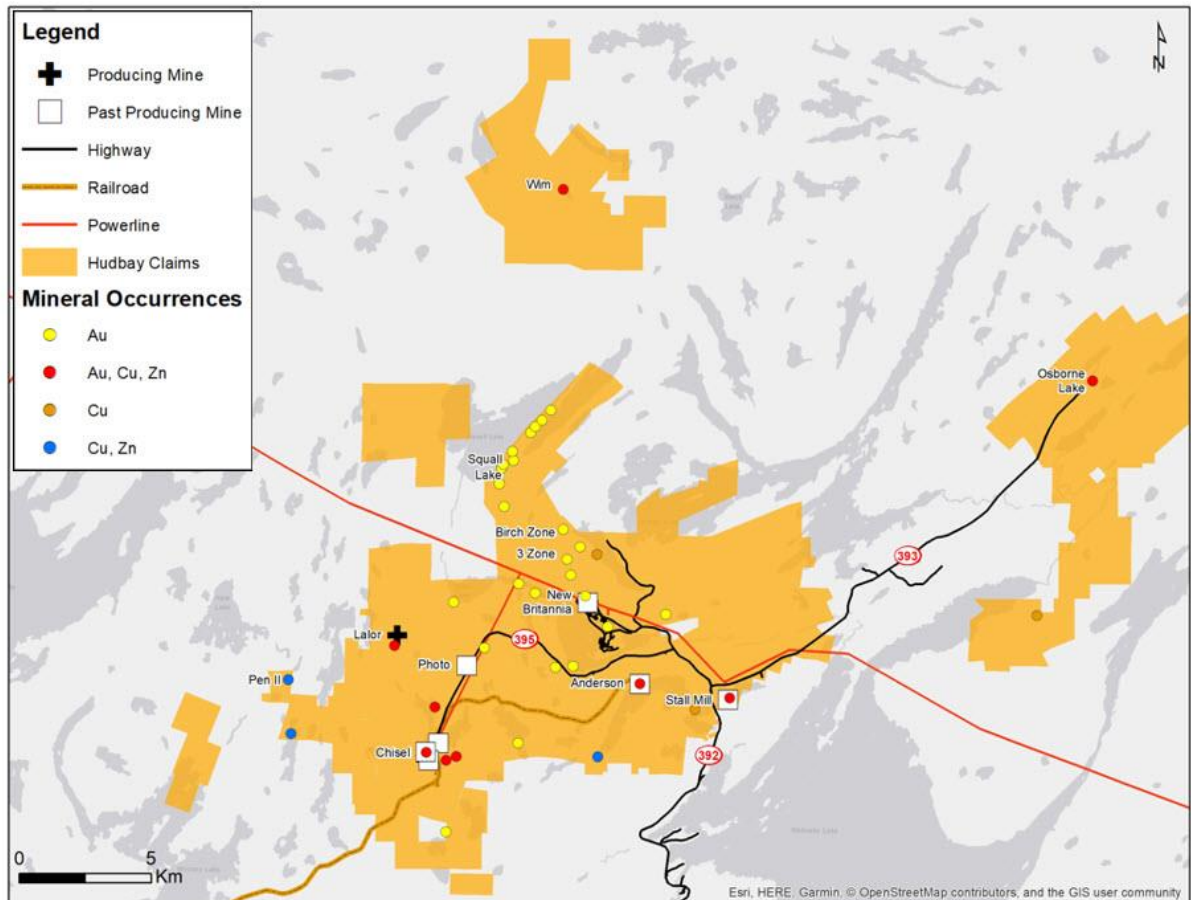
Four of the five holes encountered an in-hole response and these lie within the East Lens. The borehole EM surveys during the 2005 to 2007 drilling did not reveal any off-hole anomalies.

In 2019, Hudbay completed a diamond drill program on the Watts River deposit. Six diamond drillholes were completed for a total of 2,673m. The focus of the program was to look at possible expansion of the known mineralization.

9.7. REGION EXPLORATION

Hudbay has an extensive land package in the Snow Lake belt of 17,517 Ha that provides significant additional upside for further gold and base metals exploration (Figure 9-1).

FIGURE 9-1: MINERAL OCCURRENCES IN THE SNOW LAKE CAMP



Since 2018, Hudbay spent close to C\$10 million annually on major airborne and ground geophysical surveys as well as on surface exploration drilling in the Snow Lake area. This work was instrumental in identifying several base metal and gold targets that are to be followed up in the near future with comparable annual exploration budgets.

It is the QP's opinion, that sampling to date is both representative and adequate to develop a sound interpretation of the geology and the grade distribution for all the deposits covered by this Technical Report.

10. DRILLING

There are no known factors that could materially impact the accuracy and reliability of the drill results reported in this section.

10.1. LALOR MINE

Surface Drilling

As of January 1, 2021, a total of 305,487 m of surface drilling was completed at Lalor. Table 10-1 provides a summary by year.

**TABLE 10-1: SURFACE DRILLING PROGRAMS BY YEAR
(IN METERS DRILLED)**

Year	Hole Type	Drilling Company	Core Size	Number of Holes	Length (m)
2007	Surface drilling	Major Drilling Ltd.	NQ/BQ	33	37,431
2008	Surface drilling	Major Drilling Ltd.	NQ/AQ	75	78,470
2009	Surface drilling	Major Drilling Ltd.	NQ/AQ	78	90,676
2010	Surface drilling	Major Drilling Ltd.	NQ/AQ	31	41,680
2011	Surface drilling	Major Drilling Ltd./ Boart Longyear Inc.	NQ/AQ	22	28,891
2012	Surface drilling	Major Drilling Ltd.	NQ	3	4,722
2015	Surface drilling	Major Drilling Ltd.	NQ	2	2,956
2017	Surface drilling	Rodren Drilling Ltd.	NQ	4	4,869
2018	Surface drilling	Rodren Drilling Ltd.	NQ	13	13,979
2019	Surface drilling	Rodren Drilling Ltd.	NQ	2	1,813
Total				263	305,487

All surface holes have been surveyed using reflex downhole tools (Ez-Shot or Ez-Trac) systems at a spacing of 30 metres down the hole. The survey instruments measure magnetic azimuth, dip, temperature, earth's magnetic field and roll face of tool for wedging operations. This data is then copied onto reflex survey instruments by the drillers and provide to the geologist daily. Readings are then inputted into the drilling database and a correction is applied to the magnetic azimuth to calculate true azimuth based on yearly diurnal drift of the magnetic north pole. All down hole survey results are validated by a geologist before being entered into the drill hole database.

Underground Drilling

Underground drill holes are drilled at all dips and azimuths needed to provide adequate coverage of the orebody for interpretation and mining purposes (

Table 10-2). Holes with dips steeper than +70° are preferably avoided due to poor ergonomics and the increased risk for the drill crews.

Underground drilling at Lalor can be divided into four different categories based on the primary planned purpose of the hole:

- Exploration holes for targets outside known lenses or in areas of low drill density.
- Definition holes are drilled into known lenses to upgrade inferred resources to a higher category and to identify mineralization contacts for preliminary mine design purposes. Typical hole spacing is about 15 to 20 m.
- Delineation holes within indicated resources establish exact ore contacts for detailed mine planning and stope design purposes. Typical hole spacing is 10 to 15 m.

- Engineering holes are drilled for mine infrastructure purposes such as drain holes, holes for electrical cables and service holes for break through rounds.

**TABLE 10-2: UNDERGROUND DRILLING PROGRAMS BY YEAR
(IN METERS DRILLED)**

Year	Drilling Company	Core Size	Number of Holes	Length (m)
2012	Major Drilling Ltd.	NQ/BQ	50	7,519
2013	Major Drilling Ltd.	NQ/BQ	301	22,728
2014	Major Drilling Ltd.	NQ/BQ	396	35,669
2015	Major Drilling Ltd.	NQ/BQ	519	46,418
2016	Major Drilling Ltd.	NQ/BQ	336	34,070
2017	Major Drilling Ltd.	NQ/BQ	610	44,809
2018	Major Drilling Ltd.	NQ/BQ	744	73,691
2019	Major Drilling Ltd.	NQ/BQ	114	11,392
Total			3,070	276,295

AS shown in Table 10-3, 4,042 drill holed totaling approximately 641,000 m were included in the Lalor database to support the mineral resource estimate. Amongst these samples approximately 397,300 m were analyzed for Zn, Cu, Au and Ag and density was measured for 84,142 samples.

TABLE 10-3: DRILLING DATA BY YEAR

Year	Drill Holes	Metres Drilled
2007	39	44,905
2008	69	73,377
2009	76	89,749
2010	28	38,656
2011	19	24,723
2012	82	17,651
2013	299	22,650
2014	381	33,771
2015	515	48,981
2016	325	26,328
2017	587	44,118
2018	711	83,290
2019	614	65,241
2020	297	27,875
Total	4,042	641,315

The drill hole database was directly imported in MineSight® from acQuire® with a cut-off date for mineral resource estimate purposes of September 8th, 2020. In parallel, the drill database was also imported in Leapfrog® for additional and independent comparison and validation. No significant differences in the projection of the drill hole traces were found between the two datasets. Additional validation conducted in the drill hole database are described in section 12.

All underground surveying at Lalor mine is conducted in Universal Transverse Mercator projection using NAD83 datum Zone 14. The mine survey was tied into the Canadian Spatial Reference System grid through the use of the Global Navigation Satellite System (GNSS) and an initial reference point at surface. Using the initial reference point, a reference azimuth was established by the use of a gyro compass. Based on the initial GNSS point and gyro azimuth a traverse was conducted to carry the survey down the ramp to the 810 m level. Once the main underground development workings were completed, another traverse was

done to bring the survey down to 955 m level. All surveys into the individual levels are based on the control points created by the initial survey located at 810 m level and 955 m level. All surveying is done using electronic theodolites using the resection/free station technique.

Diamond drill lines are marked up according to layouts issued by the Mine's Geology Department. An electronic version of the layout is created for viewing on the electronic theodolite. In the field, the electronic theodolite is set up using resection or free station technique. Reflectorless Electronic Distance Measurement (EDM) is used to locate the planned drill azimuth line location according to the electronic drill hole layout. Collar location and a drill azimuth back sight is spray painted on the walls of the drift to function as front and back sights for diamond drillers to line up the drill. An anchor is installed on the painted lines to provide a permanent reference line. Once drilling is completed the collar survey is recorded using the resection setup method. Collar location is surveyed using reflectorless EDM surveying and collar location stored as single point data.

Downhole surveys were completed using a Reflex EZ-Shot®, EZ-A® or EZ-Trac® (Reflex) instrument. Surveys were completed at regular intervals of 30 m down the hole.

A down hole gyro survey was conducted at the Lalor Mine in June and July of 2016 to check the accuracy of the magnetic Reflex readings. The survey was conducted by company technicians using a Reflex TN14 gyro compass and a MEMS down hole gyro probe. Thirty-eight underground holes were surveyed by gyro method for a total of 11,358 m. No significant discrepancies were identified between azimuth values measured by magnetic Reflex survey methods and azimuth values measured by the gyro survey methods.

10.2. 1901 DEPOSIT

In January 2019, a winter drill program was initiated to follow-up untested DPEM and BHPDM targets. The initial hole CH1901, was testing and area known as the South Bull's eye and identified VMS type mineralization. After the completion of the 1901 discovery drillhole an additional 53 drillholes were drilled in the deposit area in 2019. An additional 39 drill holes were completed in 2020 for a total of 61,413m of NQ drill core between 2019 and 2020 (Table 10-4). All drill hole collars have been surveyed with a differential global positioning system (DGPS) with centimeter accuracy.

TABLE 10-4: SUMMARY OF THE 1901 DEPOSIT SURFACE DIAMOND DRILLING

1901 Deposit					
Year	Area	Hole Type	Operator	Number of holes	Total Metres Drilled
2019	1901	Surface	Hudbay	54	35,138
2020	1901	Surface	Hudbay	39	26,275

All drilling was carried out following CIM Exploration Best Practices. Each surface Exploration drill hole was lined up with a Reflex TN14 Gyrocompass, the drill hole was surveyed during drilling with a Reflex EZ-Trac at 30m intervals, and subsequently surveyed with a Reflex EZ-Gyro at the end of the hole. The Reflex data is downloaded directly from the IMDEXHUB-IQ into the acQuire database to ensure that there are no transcription errors.

The Reflex EZ-Trac and Gyro surveys were compared for several holes and no significant discrepancies were identified between azimuth values measured by magnetic Reflex survey methods and azimuth values measured by the gyro survey methods.

10.3. SNOW LAKE GOLD PROPERTIES

Table 10-5 lists the historical surface and underground drilling on the Snow Lake Gold Properties since the 1940s.

**TABLE 10-5: SUMMARY OF SURFACE AND UNDERGROUND DIAMOND DRILL HOLES
(AS OF JANUARY 1, 2021)**

Year	Area	Hole Type	Operator	Number of Holes	Total Meters Drilled
1945-1946	Squall	Surface	Squall Lake Gold Mines Ltd.	363	15,250
1950	Squall	Surface	Wekusko Consolidated Ltd.	8	503
1971-1972	Squall	Surface	Stall lake Mines Ltd.	6	367
1979	Squall	Surface	Corporate Oil and Gas	NA	NA
1980	Squall	Surface	W.B. Kobar	1	315
1980-1981	Squall	Surface	Camflo Mines Ltd.	35	3,223
1984	Squall	Surface	Barrick Resources Corp. (formerly Camflo Mines Ltd.)	114	8,559
1987	Squall	Surface	Zenco Resources	30	3,272
1990	Squall	Surface	Graham Gold Mining Corp.	11	1,067
2003	Squall	Surface	Coniagas/Trienergy/ MBMI	19	1,448
Total				587	34,004

1948-1956	NB	Surface & underground	Howe Sound Exploration Co. Ltd. & Britannia Mining & Smelting Co. Ltd.	1,629	92,766
1988-1989	NB	Underground	High River Gold Mines Ltd.	88	9,809
1988-1989	NB	Surface & underground	Inco Gold Co.	86	17,741
1994-2002	NB	Surface & underground	TVX Gold Inc. & High River Gold Mines	7,107	494,750
2003-2004	NB	Surface & underground	Kinross Gold Corp. & High River Gold	1,127	67,886
2007-2008	NB	Surface	Garson Gold Corp.	10	3,618
2009	NB	Surface	Garson Gold Corp.	3	913
2010-2011	NB	Surface	Alexis Minerals Corporation	146	50,872
Total				10,196	738,355

Alexis Minerals Corp. was the last operator to conduct drilling on the property. The 2010-2011 Alexis drilling campaigns focused on collecting information in areas with the potential to provide mineral resources to the mining plan and additional surface targets in the Snow Lake Gold area. the drill programs concentrated on areas around the Boundary, Birch, Mine East and near the New Britannia Mine.

This most recent drilling completed by Garson Gold Corp. and Alexis Minerals Corp was NQ diameter core utilizing a drilling company based in Alida, Saskatchewan called Dig-It Exploration. The location of all the drill holes were surveyed and tied into the established grid on the property. Down hole surveys were completed using Fordia Flexit Smart-tool System. The first reading was taken approximately 10 m down the hole with subsequent readings being taken every 50 to 60 metres down the hole.

Orientated core has been performed on some of the holes drilled at No. 3 Zone and at Boundary using the Ball Mark Orientation System. The orientated core was logged for structural measurements on veins and rock fabric.

In 2019, a drill program was completed in the 3 Zone deposit area. A winter program of 3 drill holes were completed for a total of 1,173 metres and a summer/fall metallurgical drilling program of 4 drill holes for a total of 1,160m (**Error! Reference source not found.**). Each drill hole was lined up with a Reflex TN14 Gyrocompass and the drill hole was surveyed with a Reflex EZ-Trac at 30m intervals downhole.

10.4. WATTS

A total of 19 holes from the early 1995 to 1997 drilling were borehole pulse EM surveyed. The surveys determined that the region of highest conductivity appeared to be in the vicinity of drill holes KUS223, 229, and 283, which lie within the Main Lens A. Five of the seven holes drilled during the 2000 to 2001 campaign were borehole pulse EM surveyed, defining mineralization in the East Lens as predominantly disseminated and minor massive sulphides.

Four of the five holes encountered an in-hole response and these lie within the East Lens. The borehole EM surveys during the 2005 to 2007 drilling did not reveal any off-hole anomalies.

In 2019, Hudbay completed a diamond drill program on the Watts River deposit. Six diamond drillholes were completed for a total of 2,673m. The focus of the program was to look at possible expansion of the known mineralization.

10.5. PEN II

Historical drilling at the Pen II deposit included 15 holes for a total of 2,029.26 metres, for which downhole surveys were completed sporadically using acid tests and tropari instrument. Since 2017, 9,531 metres of additional drilling have been collected with downhole survey utilizing reflex (Ez-Shot and Ez-Trac) systems. Readings were then inputted into the drilling database and a correction is applied to the magnetic azimuth to calculate true azimuth based on yearly diurnal drift of the magnetic north pole. All down hole survey results are validated by a geologist before being entered into the drill hole database. The drilling focused on improving the confidence of the mineralization between each drillhole. Hole collars were picked up utilizing a Differential Global Positioning System (DGPS) supplied by Hudbay's mine technical service group.

10.6. WIM

In total, 136 diamond drill holes and 43,285.36 m have been completed on the Wim deposit from 1962 to 2008. All drilling prior to Murgor's exploration campaigns on the Wim deposit was completed by HBED. HBED completed six drilling campaigns between 1962 and 1991, which included 75 diamond drill holes (AX, EX and BQ core sizes) for a total of 25,018.90 m of drilling of which 17 were wedge drill holes. In 2007 and 2008, Murgor completed a total of 61 NQ-sized diamond drill holes for a total of 18,266 m of drilling on the Wim deposit. Half core samples from the historical drilling programs (pre-2007) are stored at the Stall Lake Mill Complex located approximately 5 km from Snow Lake. All half core samples from Murgor's 2007 and 2008 drilling campaigns are permanently stored on the Wim project site or at the Manitoba government core storage at the past producing Centennial mine.

In winter 2019, a drill program of seven drill holes was completed on the WIM deposit. A total of 1,494.5 metres were drilled. The goal of the drill program was to complete several drill holes for metallurgical studies of the Wim deposit mineralization.

11. SAMPLE PREPARATION, ANALYSES & SECURITY

Sample preparation, analysis, and security procedures applied to Lalor and the Snow Lake satellite deposits were reviewed by Olivier Tavchandjian, P. Geo, the Qualified Person of this Technical Report. The sampling methodology, analyses and security measures have been documented in great detail in past published technical reports by Hudbay for its Manitoba operations and will only be summarized in this document. Generally, the same procedures were followed by the exploration team of Murgor Resources at Wim, who was under the supervision of a former chief geologist at Hudbay. Unless specified in the report, all the laboratories used for samples preparation, analysis and security are independent of Hudbay. The procedures followed at the New Britannia mine and other Snow Lake gold properties are summarized at the end of this section and are described in greater details in the Technical Report produced by Genivar for Alexis Resources in 2011 and available on SEDAR.

11.1. LALOR, 1901 & PEN II

Sample Preparation

At the Lalor mine, drill core is logged directly into the acQuire drill hole database using dedicated laptop computers. Lithologies, alteration, structures, mineralization core quality and recovery are described in detail throughout the core length. Sample intervals are clearly marked on the core.

The core is photographed before samples are cut and bagged for shipment. Photographing the core is the last step of the logging process to assure exact locations of all contacts, sample locations and numbers are captured. A standard setup as well as a single camera type with standardized settings are used to ensure photographs are of consistent quality. Once photographs have been captured, they are saved on the company server and linked to the acQuire drill hole database for easy network access.

Prior to cutting the core, the database manager prints a sample list for each drill hole that includes the sample identification number, hole name, sample type and the start and end footage of each sample. The core is cut along the cut line drawn by the geologist. In gouge and rubble intervals, an aluminium sampling scoop is used to separate the gouge into two halves in the core boxes. Completed sample bags are closed using zip ties. All saws and sampling buckets are rinsed with water after cutting each sample to prevent cross-contamination.

Once logging is completed, all data from sample books including QAQC samples are entered into Hudbay's acQuire drill hole database by hole number. Before dispatch, QAQC samples are inserted into the sample stream. Hudbay's practice involves insertion of the following every 100 samples:

- 1) Two blanks
- 2) Five duplicates
- 3) Five base metal standards
- 4) Two gold standards

Sample preparation and analysis for Lalor has been conducted at three different laboratories over time. Since September 2016, nearly all samples are prepared and assayed at Bureau Veritas. All drill core samples analyzed since the last technical report (NI 43-101 published on March 30, 2019) have been sent for analysis at Bureau Veritas. SGS was used as the umpire laboratory for quality control purposes.

ACME/Bureau Veritas

All samples arriving at ACME (Bureau Veritas) were checked against the custody chain form and prepped according to the lab protocols. The sample preparation includes weighing the sample, crushing 1 kg to minimum 80% passing 2 mm and a 250 g split crushed to minimum 85% passing 75 µm.

Base metals and other elements were analyzed with aqua regia digestions coupled with ICP-ES/MS (method AQ270 and AQ370). Assays returning "over the range" values were re-assayed with higher reporting methods (GC820 for copper, GC816 for zinc, GC817 for lead and MA404 for copper and zinc when values were above 20%).

All samples were analyzed for gold using a fire assay method coupled with atomic absorption (FA430). A gravimetric finish (FA530) was used when samples return gold values above 10 PPM.

Bureau Veritas is currently registered with ISO 9001 and ISO/IEC 17025 accreditations. Fine sample pulps are kept in secure storage at the laboratory after analysis. Pulps are only released after all data is validated.

SGS

All samples arriving at SGS are checked against the custody chain form. No sample preparation was conducted at SGS since the laboratory was only used to check assays done on pulps.

Base metals and other elements were analyzed with aqua regia digestions coupled with ICP-ES (method ICP14B). Assays returning "over the range" values were re-assayed with higher reporting methods (ICP13B when values were above 1% for zinc, copper and lead).

All samples were analyzed for gold using a fire assay method coupled with atomic absorption (FAA313). A gravimetric finish (FAG303) was used when samples return gold values above 10 PPM.

SGS is currently registered with ISO 9001 and ISO/IEC 17025 accreditations. Fine sample pulps are kept in secure storage at the laboratory after analysis. Pulps are only released after all data is validated.

Specific Gravity via gas pycnometer (Bureau Veritas, Vancouver)

A total of 16,559 density measurements were completed at Lalor since the last NI 43-101. At 1901, density measurements were performed on 13,664 samples. As for Pen II, a total of 1,306 density measurements were collected by Hudbay.

Quality Control and Quality Assurance

Data verification procedures and results since the previous technical report are included in this section based on information collected and reviewed from 2018 to 2020. During this period, all samples were assayed at the Acme/Bureau Veritas laboratory. For data verification procedures and results prior to 2018, refer to previous technical report dated March 30, 2019.

As part of Hudbay QAQC program, QAQC samples were systematically introduced in the sample stream to assess sub-sampling procedures, potential cross-contamination, precision and accuracy. Hudbay commonly includes 5% certified reference materials (CRM), 2% certified blanks, and 5% coarse duplicates. Blanks and CRMs were prepared by Ore Research and Exploration (OREAS) and Analytical Solutions Ltd. (ASL). High-grade gold standards are from Rocklabs. All QAQC samples were analyzed following the same analytical procedures as those used for the drill core samples.

Standards

A failure rate that is less than 5% is considered good, while a failure rate between 5% and 10% is considered reasonable.

At Lalor, from 2018 to 2020, a total of 8,663 OREAS and Rocklabs CRMs were analyzed at Bureau Veritas representing 5.4% of the sample stream. A failure rate that is greater than 10% would trigger additional investigations. In most cases, the analytical bias is considered adequate for resource estimate purposes. High positive bias for silver in a low grade standard A6 is attributed to the certified value of the CRM being close to detection limit of the assay method used.

At 1901, between 2019 to 2020 a total of 797 OREAS standards were analyzed at Bureau Veritas, representing 5.8% of the sample stream. The analytical bias for all metals at 1901 is considered more than adequate for resource estimate purposes. Some failures in low grade standards A6 & B6 were noted for zinc and copper. Failures were corrected by re-analysing part of the assays batch, including the standard, until all the QAQC results cleared the built-in checks in the acquire database.

At Pen II, between 2017 to 2018 a total of 77 OREAS were analyzed at Bureau Veritas, representing 4.9% of the sample stream. In most cases, the analytical bias on zinc is considered adequate for resource estimate purposes. High failure rates mostly impact copper and silver for standard values which are rarely found at Pen II. In any case, when failure of a CRM occurred, part of the assays batch including the standard was re-assayed until all the QAQC results cleared the built-in checks in the acquire database.

Blanks

A failure rate that is less than three times the detection limit (3xDL) 90% of the time is considered acceptable. A blank failure either equates to samples swap or cross contamination unless the expected value is close to the reporting limit. In any case, if a blank fails, as part of Hudbay's QAQC protocols, the full batch is re-assayed.

At Lalor, a total of 3,466 certified OREAS and ASL blanks (about 2.2% of the samples stream), were inserted into the sample stream to monitor potential cross-contamination or sample swaps at Bureau Veritas. In all cases aside from copper in the blank F7 analysed at Bureau Veritas, the cross-contamination or sample swaps are considered adequate for resource estimate purposes. The failure of copper in F7 is

linked to its certified value being equal to the 3xDL criteria. In this case, 98% of the results came back within ten times the detection limit (0.01%), which is a more appropriate performance gate, given its certified value. In any case, when failure of a blank CRM occurred, part of the assays batch, including the blank, was re-assayed until all the QAQC results cleared the database acQuire built-in checks.

At 1901, a total of 806 blanks, sourced from a certified ASL blank (F8) and a non-certified internal geological blank (GQC-granite) that has been historically used by Hudbay, were inserted into the sample stream to monitor potential cross-contamination or sample swaps at Bureau Veritas. The level of cross-contamination and/or sample swaps are very low and/or non-existent to consider data collected during the 2019 – 2020 drilling campaign and no significant issues were observed; however, a significant number of samples returned zinc, copper, and silver results that are higher than 3x detection limits. These failures are attributed to certified value of blanks used being significantly higher than the detection limits of zinc, copper, and silver in the ultra-trace AQ251-EXT method used in exploration projects. The maximum reported results for zinc, copper, and silver in the blanks are 0.02%, 0.006%, & 0.2 g/t, suggesting least cross-contamination and no sample swaps occurred at 1901.

At Pen II, a total of 82 certified OREAS and ASL blanks, or 5.3% of the samples stream, were inserted into the sample stream to monitor potential cross-contamination or sample swaps at Bureau Veritas. In all cases aside from copper in the blank F7, the cross-contamination or sample swaps are considered adequate for resource estimate purposes. The failure of copper in F7 is linked to its certified value being equal to the 3xDL criteria. In this case, 100% of the results came back within five times the detection limit, which is a more appropriate performance gate given its certified value. In any case, when failure of a blank CRM occurred, part of the assays batch including the blank was re-assayed until all the QAQC results cleared the acQuire database built-in check checks.

Coarse Duplicates

At Lalor, a total of 8,540 coarse duplicates were analyzed at Bureau Veritas representing 5.3% of the sample stream. Based on the paired analysis results, the laboratory appeared to be precise.

At 1901, a total of 804 coarse duplicates were analyzed at Bureau Veritas representing 5.9% of the sample stream. Based on the paired analysis results, the laboratory appeared to be precise.

At Pen II, a total of 43 coarse duplicates were analyzed at Bureau Veritas representing 2.7% of the sample stream. Based on the paired analysis results, the laboratory appeared to be precise.

Umpire Laboratory

At Lalor, a total of 8,273 pulp rejects were sent to and analyzed at SGS to validate the reproducibility of Bureau Veritas results. These external check samples represent 5.2% of the sample stream. Based on the paired analysis results, assay data from Bureau Veritas appeared to be reproducible.

At 1901, a total of 787 pulp rejects were sent to and analyzed at SGS to validate the reproducibility of Bureau Veritas results. This external check represents 5.8% of the sample stream. Based on the paired analysis results, the Bureau Veritas results appeared to be reproducible.

At Pen II, a total of 67 pulps rejects were sent to and analyzed at SGS to validate the reproducibility of Bureau Veritas results. This external check represents 4.3% of the sample stream. Based on the paired analysis results, the Bureau Veritas results appeared to be reproducible.

11.2. WATTS DEPOSIT

Sample Preparation

The sample preparation methods performed on samples collected at the Watts deposit during the 2019 drill program are the same as the ones used for Lalor mine and the 1901 deposit. Table 11-1 presents the number of samples prepared and analyzed by year.

TABLE 11-1: WATTS SAMPLES PREPARATION AND ANALYSIS BY YEAR

Year	Number of Samples Dispatched	Flin Flon Laboratory	TSL	Acme / Bureau Veritas Laboratory
1995	555	555	0	0
1996	412	412	0	0
1997	297	297	0	0
2000	47	0	47	0
2001	87	0	87	0
2005	275	275	0	0
2006	356	356	0	0
2007	341	341	0	0
2019	344	0	0	344
Total	2714	2236	134	344

Specific Gravity

Density measurements were performed for 699 samples at Hudbay's logging facility for samples collected between 1995 and 2007, and Bureau Veritas laboratory in Vancouver for samples collected in 2019. Density measurements were completed using a non-wax-sealed immersion technique to measure the weight of each sample in air and in water for the 1995 – 2007 samples. Density measurements at Bureau Veritas were completed using a gas pycnometer (method SPG04). Table 11-2 presents the number of density measurements by year

TABLE 11-2: DENSITY MEASUREMENTS AT WATTS BY YEAR

Year	Number of Density Measurements	Hudbay Logging Facility	Acme / Bureau Veritas Laboratory
1995	99	99	0
1996	101	101	0
1997	88	88	0
2000	26	26	0
2006	41	41	0
2019	344	0	344
Total	699	355	344

Sample Analysis

The laboratory analysis for samples collected at the Watts deposit between 1995 and 2007 were Hudbay's internal laboratory in Flin Flon and TSL laboratory in Saskatoon. All samples collected during the 2019 program were analysed at the Bureau Veritas laboratory in Vancouver with SGS laboratory conducting external checks.

Hudbay's Flin Flon Laboratory

Base metals and silver analysis are completed by aqua-regia digestion and read by inductively coupled plasma (ICP-AES). Gold analysis are completed on each sample by atomic absorption spectrometry (AAS) after fire assay lead collection. Gravimetric method is used for gold if the range is above the upper limit (7 PPM) of the AAS.

Bureau Veritas

The methods used by Bureau Veritas on samples collected at the Watts deposit during the 2019 drill program are the same as the ones used for Lalor, 1901 and Pen II deposits.

SGS

The methods used by SGS on pulp rejects sent by Bureau Veritas for samples collected at the Watts deposit during the 2019 drill program are the same as Lalor, 1901 and Pen II deposits.

Quality Assurance and Quality Control

Data verification procedures and results are based on information collected and reviewed from the 1995 – 2007, and 2019 drilling campaigns. QAQC samples, i.e., standards, blanks, and duplicates were introduced into the sampling stream for samples collected between 2005 and 2019. No QAQC samples were taken for samples collected between 1995 and 2001.

As part of Hudbay's QAQC program, for samples collected from 2005 to 2019, QAQC samples were systematically introduced into the sample stream to assess sub-sampling procedures, potential cross-contamination, precision and accuracy. Hudbay commonly includes 5% certified reference materials (CRM), 2% certified blanks, and 5% coarse duplicates in the sample stream. Blanks and CRMs were prepared by Ore Research and Exploration (OREAS) and Analytical Solutions Ltd. (ASL). All QAQC samples were analyzed following the same analytical procedures as those used for the drill core samples.

Standards

A total of 73 standards that represent 2.7% of total samples taken at Watts between 1995 and 2019 were introduced into the sampling stream between 2005 and 2019. A total of 53 OREAS standards A4, B4, C4, D4, & E4 were analysed at Hudbay's Flin Flon Laboratory in the 2005 – 2007 program, representing 5.5% of total samples taken during that time frame. In 2019, a total of 20 OREAS standards, A6, B6, C6, & D6 were analyzed at Bureau Veritas, representing 5.8% of the sample stream.

The analytical bias for all metals at Watts is considered more than adequate to validate the accuracy of the laboratory assay results. Quality control failures related to analytical results coming below lower limit in low to medium grade standards were noted for zinc and copper in the 2005 – 2007 program. Failures were corrected by re-analysing part of the assays batch, including the standard, until all the QAQC results cleared the built-in checks in the acquire database.

Blanks

A total of 78 blanks, sourced from a certified ASL blank (F8) and a non-certified internal geological blank (GQC-granite) that has been historically used by Hudbay, were inserted into the sample stream to monitor potential cross-contamination or sample swaps at Hudbay's Flin Flon lab and Bureau Veritas.

The level of cross-contamination and/or sample swaps are very low and/or non-existent to consider data collected during the 2005 – 2007 and 2019 drilling campaigns confirming the laboratory processed the samples in an adequate fashion; however, significant number of samples that returned zinc, copper, and silver results higher than 3 x detection limits were noted in the 2019 program. These failures are attributed to certified value of blanks used being significantly higher than the detection limits of zinc, copper, and silver in the ultra-trace AQ251-EXT method used in exploration projects. The maximum reported results for zinc, copper, and silver in the blanks are 0.005%, 0.006%, & 0.2 g/t, suggesting least cross-contamination and no sample swaps occurred at Watts.

Coarse Duplicates

Twenty coarse duplicates were analyzed from the 2019 program at Bureau Veritas representing 5.8% of the sample stream collected in 2019. Basic statistics, scatter plots, quantile-quantile plots, probability plots and absolute difference plots were used to assess the precision of the laboratory. Based on the paired analysis results, the laboratory appeared to be precise.

Umpire Laboratory

Twenty pulp rejects were sent to and analyzed at SGS to validate the reproducibility of Bureau Veritas results in the 2019 program. This external check represents 5.8% of the sample stream collected in 2019. Basic statistics, scatter plots, quantile-quantile plots, probability plots and absolute difference plots were used to assess the precision of the laboratory. Based on the paired analysis results, the Bureau Veritas results appeared to be reproducible.

11.3. WIM

Hudbay has not performed any exploration work on the Wim property since 1991. The sampling methodology, analyses and security measures used by the previous owners (i.e. Murgor Resources and Alexandria Minerals) have been documented in Alexandria Minerals' 2015 technical report and will only be summarised in this document.

Sample Preparation

The samples preparation methods performed on the Wim samples are the same as the ones used for the Lalor mine samples. The logs were entered in a Gemcom Access® database.

Specific Gravity

572 density measurements were collected by Murgor Resources in 2007 and 2008. These measurements were performed at TSL Laboratories in Saskatoon, using a non-wax-sealed immersion technique to measure the weight of each sample in air and in water.

Sample Analysis

"All drill cores assayed from the 2007 and 2008 drilling programs were completed by TSL Laboratories, using multi-acid digestion and atomic absorption spectrometry for the base metals and fire assay, atomic absorption and gravimetric finish for gold. All samples selected for analysis were prepared and analyzed for Cu, Zn, Au, Ag, Fe, As and Pb." TSL Laboratories is currently registered with ISO/IEC 17025 accreditation

Quality Assurance and Quality Control

Data verification procedures and results are based on information collected and reviewed from the 2007 and 2008 drilling campaign. During this period, all samples were assayed at the TSL laboratory in Saskatoon while whole rock geochemistry pulp samples were sent to ALS Laboratory in Vancouver. As part of Murgor's QAQC program, QAQC samples were systematically introduced in the sample stream to assess sub-sampling procedures, potential cross-contamination, precision, and accuracy, typically using a 10% insertion rate (i.e. one CRM, blank or duplicate every 10 samples). All QAQC samples were analyzed following the same analytical procedures as those used for the drill core samples at Wim.

The results of the review done by Golder indicated that Murgor's QAQC standard sampling program was working effectively, and the program met industry standards.

11.4. SNOW LAKE GOLD PROPERTIES

The sampling methodology, analyses and security measures used by the previous owners (i.e. Kinross and Alexis Resources) have been documented in the Technical Report produced by Genivar for Alexis Resources in 2011, available on SEDAR and will only be summarized in this document.

Limited confirmatory drilling was conducted by Hudbay in 2019 on the 3 Zone deposit. The sample preparation methods used for the samples collected during this confirmatory campaign are the same as the ones used for the Lalor mine, 1901 and Watts deposits.

Sample Preparation

Details of the sampling methods conducted by earlier operators of the New Britannia mine prior to 1995 have been lost. Between 1995 and 2003, the New Britannia sampling method consisted in core being logged on paper at the mine site by geologists. The written logs were later transferred into a Gemcom Access database. Technicians would lay out the core, split and sample the core according to the geologists' instructions. Chip samples were also collected during the mining operation by the geologist

either taken from the back of the drift if the muck pile had not been removed or taken 1.40 m to 1.52 m above the floor of the drift if the blast muck has been removed. The geologist was responsible for bringing the samples to the preparation area of the mill laboratory and recording the number of each sample into a record book.

In the 2019 drill program at the 3 Zone deposit, drill core is logged directly into the acQuire drill hole database using dedicated laptop computers. Lithologies, alteration, structures, mineralization core quality and recovery are described in detail throughout the core length. Sample intervals are clearly marked on the core.

The core is photographed before samples are cut and bagged for shipment. Photographing the core is the last step of the logging process to assure exact locations of all contacts, sample locations and numbers are captured. A standard setup as well as a single camera type with standardized settings is used to ensure photographs are of consistent quality. Once photographs have been captured, they are saved on the company server and linked to the acQuire drill hole database for easy network access.

Prior to cutting the core, the database manager prints a sample list for each drill hole that includes the sample identification number, hole name, sample type and the start and end footage of each sample. The core is cut along the cut line drawn by the geologist. In gouge and rubble intervals, an aluminium sampling scoop is used to separate the gouge into two halves in the core boxes. Completed sample bags are closed and secured at the neck using two zip ties. All saws and sampling buckets are rinsed with water after cutting each sample to prevent cross-contamination.

Once logging is completed, all data from sample books including QAQC samples are entered into Hudbay's acQuire drill hole database by hole number. Before dispatch, QAQC samples are inserted into the sample stream. Hudbay's practice involved insertion of the following quality control samples for every 100 samples:

- 1) Five blanks
- 2) Five duplicates
- 3) Five standards (gold + base metals)

Sample Analysis

Most of the drill cores and chips assays from 1995 to 2003 from the New Britannia mine were completed at the on-site mill laboratory using a fire assay/atomic absorption finish (FA/AA) method.

At the other Snow Lake properties, sample analysis, including determination of gold and arsenic on all samples and specific gravity and multi-element geochemistry on selected samples, was completed at TSL Laboratories Inc. (TSL) in Saskatoon, SK. TSL was established in 1981 and certified as ISO/IEC 17025. At TSL, the samples are dried and crushed and a 250 g riffle split of the crushed sample is pulverized to 95% passing 150 mesh (106 microns). Gold concentration is determined on all samples from a 30-g split of the pulverized sample by fire assay with atomic absorption finish. All samples with a gold concentration greater than 1,000 ppb (1 gram/tonne) were re-assayed using fire assay with a gravimetric finish. Specific gravity and ICP-MS multi-element analyses were completed on selected samples as requested by Garson project geologists.

The laboratory analysis for the 2019 drill campaign on the 3 Zone deposit was conducted at the Bureau Veritas Laboratory in Vancouver with SGS laboratory conducting external checks.

Bureau Veritas

All samples arriving at ACME (Bureau Veritas) were checked against the custody chain form and prepped according to the lab protocols. The sample preparation includes weighing the sample, crushing 1 kg to minimum 80% passing 2 mm and a 250 g split crushed to minimum 85% passing 75 µm.

All samples were analyzed for gold using a fire assay method coupled with atomic absorption (FA430). A gravimetric finish (FA530) was used when samples return gold values above 10 PPM.

Base metals and other elements were analyzed using the Ultra-trace by ICP-MS method (AQ251-EXT). Assays returning “over the range” values were re-assayed with higher reporting methods (AR404 for silver, copper, lead and zinc). If the range is above the upper limits of AR404, additional packages (i.e. GC816, GC817, and GC820) were used for zinc, lead, and copper respectively.

Bureau Veritas is currently registered with ISO 9001 and ISO/IEC 17025 accreditations. Fine sample pulps are kept in secure storage at the laboratory after analysis. Pulps are only released after all data is validated.

SGS

All samples arriving at SGS are checked against the custody chain form. No sample preparation was conducted at SGS since the laboratory was only used to check assays done on pulps.

All samples were analyzed for gold using a fire assay method coupled with atomic absorption (GE FAA313/GE FAA30V5). A gravimetric finish (GO FAG303/GO FAG30V) was used when samples return gold values above 10 PPM.

Base metals and other elements were analyzed with aqua regia digestions coupled with ICP-ES/MS (method GE ICP14B/GE ICM21B20). Assays returning “over the range” values were re-assayed with higher reporting methods (GO ICP13B/GO ICP21B100 when values were above 1% for zinc, copper and lead).

SGS is currently registered with ISO 9001 and ISO/IEC 17025 accreditations. Fine sample pulps are kept in secure storage at the laboratory after analysis. Pulps are only released after all data is validated.

Specific Gravity

Density measurements were performed for 827 samples from the 3Zone deposit at Bureau Veritas laboratory in Vancouver using a gas pycnometer (method SPG04) in 2019.

Quality Assurance and Quality Control

Prior to 2014, standard, blank and duplicate assay samples were added to each batch of 21 samples for drill core and to each batch of 24 samples for chip samples.

The sampling methods and approaches and concluded that the sampling and analytical procedures conformed to the industry standards at the time, and these were adequate to ensure a representative determination for the type of gold mineralization identified on the property. The QAQC data documented in the Genivar Technical Report (2010) is not available for the QP to review. The QP is confident that the audits done by Micon in 2005 and 2009 are sufficient for the purpose of reporting inferred mineral resource estimates for the Snow Lake gold properties. The QAQC, data verification and results from the 2019 confirmatory drill program at 3 Zone were collected and reviewed and no issues was observed. During this period, all samples were assayed at the Bureau Veritas laboratory in Vancouver, British Columbia and checks conducted at SGS laboratory.

Standards

In 2019, a total of 41 OREAS standards were analyzed at Bureau Veritas, representing 5% of the sample stream. The analytical bias for gold at 3 Zone is considered more than adequate for resource estimate purposes.

Blanks

In 2019, a total of 50 blanks non-certified internal geological blanks (GQC-granite) that has been historically used by Hudbay, were inserted into the sample stream to monitor potential cross-contamination or sample swaps at Bureau Veritas. The level of cross-contamination and/or sample swaps are very low and/or non-existent in the data collected during the 2019 drilling campaign. No significant issues were observed; however, a single blank returned a value greater than 5X the detection limit. The blank result was investigated and determined to be a result of contamination in the processing stage from a sample with visible gold (VG).

Coarse Duplicates

A total of 47 coarse duplicates were analyzed at Bureau Veritas representing 5.6% of the sample stream. Basic statistics, scatter plots, quantile-quantile plots, probability plots and absolute difference plots were used to assess the precision of the laboratory. Based on the paired analysis results, the laboratory appeared to be precise.

Umpire Laboratory

A total of 46 pulp rejects were sent to and analyzed at SGS to validate the reproducibility of Bureau Veritas results. This external check represents 5.5% of the sample stream. Basic statistics, scatter plots, quantile-quantile plots, probability plots and absolute difference plots were used to assess the precision of the laboratory. Based on the paired analysis results, the Bureau Veritas results appeared to be reproducible.

Based on the technical review of samples Preparation, Analysis and Security, the QP is satisfied that both the accuracy and precision of the laboratories used met the industry norms and formed a good basis upon which the mineral resource estimates can be reported.

12. DATA VERIFICATION

12.1. LALOR MINE, 1901 WATTS, 3 ZONE, WIM AND PEN II DEPOSITS

Site Visits

Frequent site visits have been conducted by the Qualified Person since September 2017. These visits included the core logging facilities and the underground mine to review the geology and mineralization of the different mineralized zones. Visits of the exploration and the mine technical services offices to meet with the geologists were also part of the process. The Flin Flon Hudbay's laboratory and the Bureau Veritas laboratory in Vancouver were also inspected by the Qualified Person.

Database Management

As described in the previous sections of this Technical Report (10 and 11), drilling, logging, sampling data and assays results are captured and stored in an acQuire database. The acQuire platform provides robust data security and long-term data storage solutions. The assay database is administered by the database manager with working copies kept on the local drive of a secure computer and backups placed on a secure location on a Hudbay server. Any requests for edits to the database are made to the database manager who updates all the copies. All .pdf copies of the assay certificates and logs are available on the Hudbay's internal SharePoint website with restricted access.

Database Review

The drill holes samples used for the mineral Resource estimate included Zn, Cu, Au, Ag, Fe, Pb, As, Ni, and bulk density (SG). The drill database was directly imported into MineSight. Under the direct supervision of the Qualified Person, reviews of the drill holes to look for obvious errors were performed along with random checks of the acQuire database to ensure honouring of the original data such as the collar location, the downhole survey data and the assays results. Minor errors in the downhole surveys data of a few holes were detected and immediately corrected. It is the QP's opinion that the data is adequate and acceptable for use in the estimation of mineral resources.

12.2. SNOW LAKE GOLD PROPERTIES

The data verification used by the previous owners are summarized in this section, but more details can be found in the report from Technical Report produced by Genivar for Alexis Resources in 2011 and available on SEDAR.

Site Visits

Todd McCracken, P. Geo, for WSP, conducted a site visit in November 2018. During the site visit, the following validation tasks were completed:

- Transfer of all available digital records to WSP;
- Review of the state of geological/mineralization knowledge;
- Review of the drilling quality assurance/control (QAQC) procedures;
- Review of survey techniques for collar positions and down-hole surveys;
- Review of the core handling and sampling procedures;
- Review of the SG/bulk density determination procedures;
- Review of database integrity and storage procedures;
- Inspection of the core logging and core storage facilities.

Database Review

WSP reviewed the process completed by Micon (Lewis, 2009) to verify completeness and reliability of electronic database files; Micon found no errors. Consequently, it can be reasonably assumed that the database is relatively free of errors.

Based on the technical review of data collection, quality control procedures, and inspection of previous studies, the QP is satisfied that both the integrity and quality of the mineral resource database met the industry norms and formed a good basis upon which the inferred mineral resource estimates can be reported.

13. MINERAL PROCESSING AND METALLURGICAL TESTING

It is believed that all metallurgical test work presented in this section was conducted on samples that are representative of the various types and styles of mineralization and as a whole, the Lalor mine, 1901 deposit, WIM and 3 Zone deposits.

13.1. STALL CONCENTRATOR

The Stall Concentrator located 16km from the Lalor mine was commissioned in 1979. Since then, it has processed ore from many of the Snow Lake area mines. With the introduction of Lalor ore to the Stall Concentrator, an increase in the copper and precious metal content warranted the refurbishment of the concentrator to enable the production of a copper and precious metal concentrate alongside the zinc concentrate.

The Stall concentrator began processing ore from Lalor in August 2012, initially producing only a zinc concentrate and by October 2012 a copper concentrate. As Lalor increase ore production, the mill underwent an initial expansion and by the summer of 2014 was capable of processing ore at a 2,800 tpd throughput rate. Continuous improvement and debottlenecking led by the operations team has lifted the processing capacity of the Stall concentrator to above 3,800 tpd and instantaneous processing rates above 200 tph.

Historical Metallurgical Performance

The Stall Concentrator metallurgical performance processing Lalor Base Metal Ore from October 2012 to December 2020 is shown in Table 13-1 and actual concentrate produced is shown in Table 13-2.

The Stall Concentrator currently is producing a copper concentrate grading 18 to 20% copper depending on the levels of lead and talc present in the feed ore and a zinc concentrate grading 49 to 51% zinc. Metal recovery to the copper concentrate ranges from 85 to 88% and 55 to 62% for copper and precious metals, respectively. Metal recovery to the zinc concentrate ranges from 90 to 93% depending on the levels of gahnite, a zinc oxide mineral that is not responsive to flotation, in the feed ore and the amount of zinc recovered to the upstream processes.

TABLE 13-1: HISTORICAL STALL CONCENTRATOR METALLURGICAL PERFORMANCE

Year	Tonnes	Feed Assays				Metal Recovery			
		Au (g/t)	Ag (g/t)	Cu (%)	Zn (%)	Au (g/t)	Ag (g/t)	Cu (%)	Zn (%)
2012	59,787	1.33	15.31	0.60	9.23	47.60	55.50	68.00	94.00
2013	422,287	1.51	29.38	0.60	8.91	59.60	56.50	77.70	94.90
2014	526,015	2.31	24.05	0.89	8.49	53.30	50.80	71.40	93.70
2015	928,501	2.53	21.29	0.71	8.21	55.80	54.75	84.47	90.81
2016	1,089,530	2.25	21.68	0.63	7.03	57.39	56.42	82.17	92.82
2017	1,102,035	1.91	22.85	0.65	7.76	56.26	56.24	82.46	92.44
2018	1,201,467	2.15	25.27	0.72	6.38	57.54	59.20	85.69	92.84
2019	1,290,300	2.13	25.48	0.73	6.39	56.81	60.43	85.84	91.11
2020	1,412,752	2.55	25.37	0.73	5.76	60.04	60.44	86.18	91.86

TABLE 13-2: HISTORICAL STALL CONCENTRATOR CONCENTRATE PRODUCTION

Year	Zinc Concentrate		Copper Concentrate			
	Tonnes	Zn (%)	Tonnes	Au (g/t)	Ag (g/t)	Cu (%)
2012	10,260	56.40	1,450	28.00	383.30	18.50
2013	67,346	51.50	9,784	39.20	386.90	20.20
2014	81,946	51.10	16,518	39.20	388.40	20.10
2015	133,808	51.76	26,847	48.75	403.14	20.68
2016	138,056	51.48	27,299	51.57	488.19	20.65
2017	152,767	51.73	29,361	40.31	482.32	20.18
2018	139,269	51.10	37,047	40.05	485.19	20.04
2019	147,227	51.04	40,855	38.27	486.28	19.86
2020	147,861	50.57	47,681	45.44	454.30	18.74

Historical Metallurgical Testwork (2009 to 2018)

Historical metallurgical testwork results from the discovery of Lalor through to 2018 are outlined and can be viewed in the previous technical reports dated March 29th 2012, March 30th 2017 and March 28th 2019.

Metallurgical Testwork (2019 to 2020) – Lalor Base Metal Ore

In Q1 2019 a study program was begun to determine the recovery entitlement of the copper, gold, silver and gold within the Lalor Base Metal ore, the gap between plant performance and recovery entitlement and the main diluents in the final mineral concentrates. The study programs included:

- Detailed flotation surveys (Blue Coast Research)
- Detailed base metal mineralogy (ActLabs)
- Detailed base metal and gold mineralogy (Advance Mineral Technology Laboratory, AMTEL)
- Batch flotation tests on future ore composites (Blue Coast Research)
- Jameson cell pilot plant testwork

In April 2019, Blue Coast Research with the assistance of site personnel conducted a multiple flotation circuit surveys. The flotation flowsheet and the material balance are outlined in Figure 13-1 and Table 13-3 respectively.

FIGURE 13-1: STALL CONCENTRATOR FLOTATION FLOWSHEET APRIL 2019

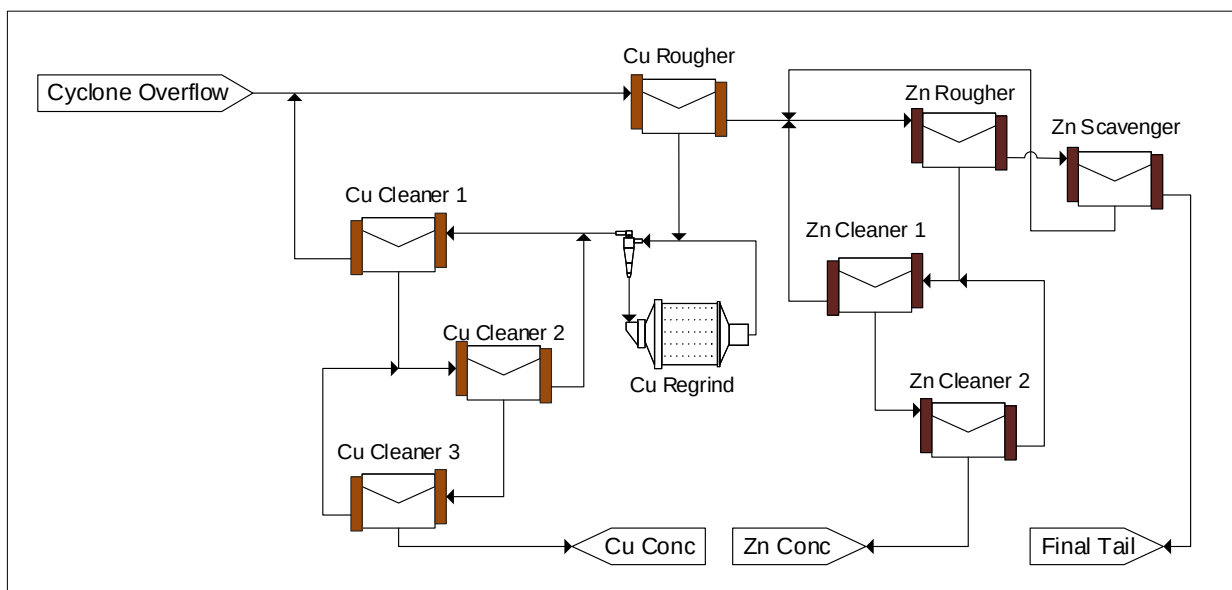


TABLE 13-3: FLOTATION CIRCUIT SURVEY MASS BALANCE

	Stream	Units	Cu Circuit			Zn Circuit		
			Feed	Conc	Tail	Feed	Conc	Tail
Mass Flow	Solids	Tph	184.3	4.3	180.0	180.0	18.3	161.7
Assays	Au	g/t	2.68	48.30	1.59	1.59	3.89	1.33
	Ag	g/t	23.70	595.37	10.07	10.07	42.45	6.40
	Cu	%	0.60	20.98	0.12	0.12	0.66	0.06
	Pb	%	0.27	7.09	0.10	0.10	0.54	0.06
	Zn	%	5.27	4.94	5.28	5.28	49.65	0.25
	Fe	%	14.63	22.77	14.43	14.43	10.41	14.89
	S	%	14.81	27.22	14.51	14.51	32.87	12.44
	MgO	%	6.88	4.33	6.94	6.94	0.63	7.65
Distribution	Mass	%	100.0	2.33	97.67	97.67	9.93	87.74
	Au	%	100.0	41.99	58.01	58.01	14.44	43.57
	Ag	%	100.0	58.52	41.48	41.48	17.79	23.69
	Cu	%	100.0	81.11	18.89	18.89	10.88	8.01
	Pb	%	100.0	61.92	38.08	38.08	19.93	18.15
	Zn	%	100.0	2.18	97.82	97.82	93.58	4.24
	Fe	%	100.0	3.63	96.37	96.37	7.07	89.31
	S	%	100.0	4.28	95.72	95.72	22.05	73.67
	MgO	%	100.0	1.47	98.53	98.53	0.92	97.62

In addition to the survey, Blue Coast Research conducted batch flotation tests on the cyclone overflow, copper regrind overflow and copper rougher tails to determine the recovery entitlement for the copper and zinc circuits. The flotation tests confirmed that the zinc circuit was operating at entitlement recovery, with the survey recording 93.58% zinc recovery (95.6% zinc circuit) against a laboratory results of 96.5% zinc circuit recovery. The flotation test also confirmed that the copper circuit was not operating at entitlement recovery. The laboratory tests returned a gold recovery entitlement of 67.7% (70.4% rougher recovery and 96.3% cleaner recovery) and a copper recovery entitlement of 92.3% (94.1% rougher recovery and 98.2% cleaner recovery). The absolute gap between the copper circuit recovery entitlement and the actual measured was 25.8% and 11.3% for gold and copper, respectively.

The recovery gap in the copper circuit formed the basis of the recovery improvement testwork program and all the mineralogy investigations focused on streams within the copper circuit.

Mineralogical Investigation

AMTEL and ActLabs conducted detailed mineralogy focusing assessment on select samples from the copper circuit collected throughout the flotation survey campaign. The stream analyzed were primary cyclone overflow, rougher concentrate, rougher tails, cleaner 1 tails and cleaner 3 concentrate. Pyrite was the principal sulphide mineral in the cyclone overflow followed by sphalerite, pyrrhotite, chalcopyrite and galena. Chalcopyrite is the dominate copper bearing mineral present accounting for greater than 99% of the copper mineralisation.

The target mineral and the most abundance in the cleaner 3 concentrate is chalcopyrite. With principal diluent minerals being talc, galena, and sphalerite.

In the cyclone overflow, the chalcopyrite liberations are 89.4% with the remainder composited with sphalerite (2.83%) and pyrrhotite (2.84%). There are no measurable levels of association between galena and talc with chalcopyrite, indicating these minerals are recovered to the cleaner 3 concentrate via mechanical or chemical processes and not insufficient liberation.

Of the chalcopyrite lost to the rougher tails, 51.7% is in the free/liberated highlighting that most of the chalcopyrite that is not recovered to the copper concentrate is recoverable. A further 17.5% is associated with pyrrhotite is loss to the rougher tails and is potentially recoverable. This indicates that mechanical or chemical processes are the principal causes for the mineral loss and that the size reduction processes are operating as designed. Additional flotation capacity or the application of pneumatic flotation devices would be beneficial for the recovery of these liberation classes.

Amtel conducted additional mineralogy assessment on the primary cyclone overflow, rougher tails, cleaner 1 tails and cleaner 3 concentrate focusing on the gold deportment.

In the cyclone overflow 36.7% of the gold is present as free/liberated particles accounting for 0.91 g/t of the gold in feed. Of note is the 13.5% of gold in feed that is above 40µm in size and is amenable to gravity recovery processes. Free/liberated gold accounts for 22.2% and 52.4% of the gold present in the rougher and cleaner 1 tails, respectively. This indicates that mechanical or chemical processes are the principal drivers of recovery loss in this liberation class as these particles are highly responsive to flotation. Additional flotation capacity or the application of pneumatic flotation devices would be beneficial for the recovery of these liberation classes.

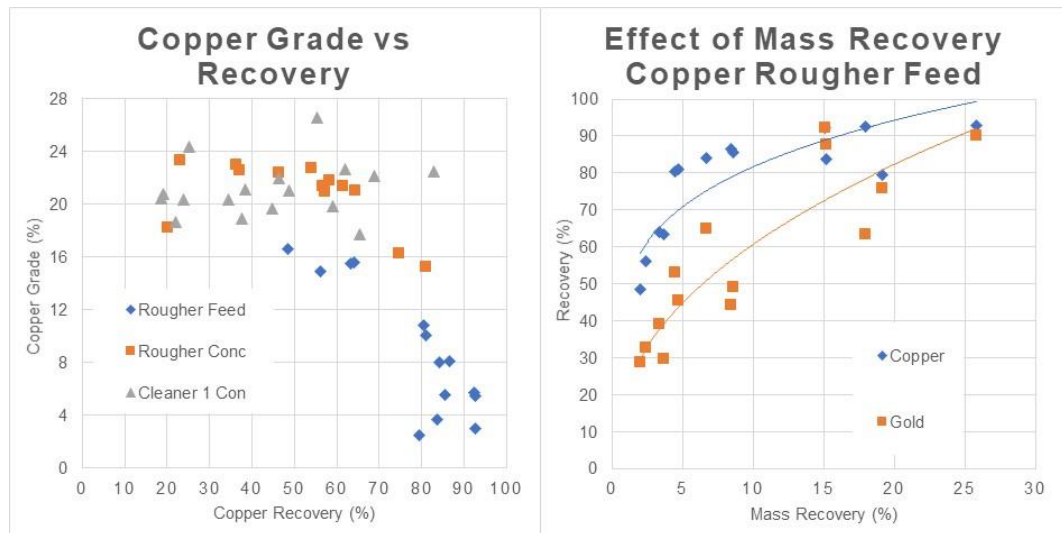
Sulphide association accounts for 29.3%, 22.9% and 35.2% of the gold in the cyclone overflow, rougher tails and cleaner 1 tails, respectively. The recovery of these gold particles is dependant on which sulphide mineral they are associated with. Gold not associated chalcopyrite has a high probability of being rejected. Sub-microscopic does occur within this fraction and is a contributor to the loss of gold to tails. Of the 0.61 g/t gold in the rougher tails, 0.17 g/t or 28% of the gold in the associated sulphide classes is sub-microscopic gold is not recoverable without diluting the copper concentrate.

Jameson Cell Pilot Plant Testwork

Based on the mineralogical investigation, Glencore Technology was contacted to supply a L500 Jameson Cell pilot plant for onsite metallurgical testing. The pilot plant testing was conducted on three streams in the copper circuit and three stream in the zinc circuit to determine if this flotation technology was suitable. The streams tested were copper rougher feed, copper rougher concentrate (post regrind), copper cleaner 1 tail, zinc rougher feed, zinc rougher concentrate and zinc cleaner 1 concentrate. The results for the copper streams tested are outlined in Figure 13-2 and in Figure 13-3 for the zinc stream tested.

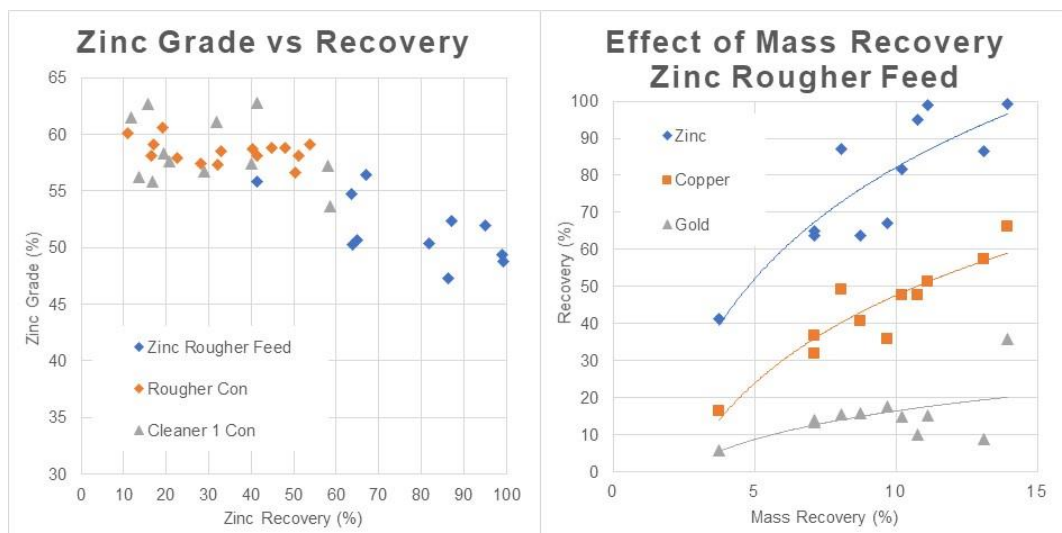
In the single pilot Jameson cell treating the copper rougher feed, copper and gold recoveries tracked a typical mass recovery curve, with +90% copper and +70% gold recoveries achieved at 15% mass recovery indicating the Jameson cell is suitable for the rougher application. When treating the reground copper rougher concentrate or copper cleaner 1 concentrate, final concentrate grade (20% Cu) was achieved up to 60% recovery which is typical for Jameson Cell applications in base metal operations. This indicates that the Jameson cell is suitable for cleaning duties.

FIGURE 13-2: JAMESON CELL PILOT TESTWORK RESULTS FOR THE COPPER CIRCUIT



When treating the zinc circuit streams it was evident that the sphalerite highly liberated and there were minimal competing minerals in the system. Final concentrate grade zinc concentrate (+50% Zn) was achieved with all stream tested with the zinc rougher feed achieving final concentrate grade at +80% recovery.

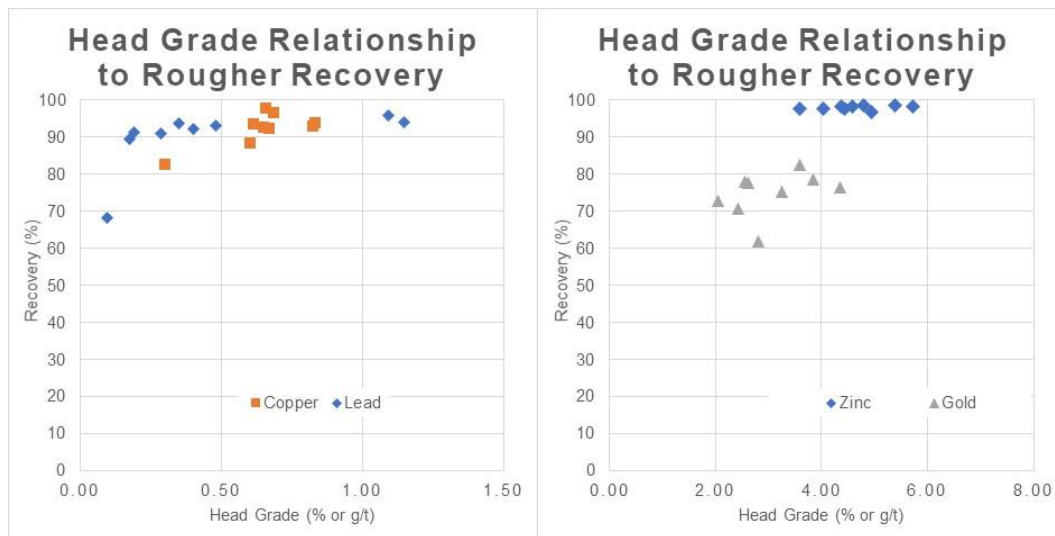
FIGURE 13-3: JAMESON PILOT TESTWORK RESULTS FOR THE ZINC CIRCUIT



Future Ore Testwork

Blue Coast Research conducted flotation testwork on nine composite samples created to simulate the future Lalor Base Metal ore from 2024 to 2030. The testwork was focused on understanding the flotation response in the ores to be treated on recovery of copper, gold, lead, and zinc. The results are outlined in Figure 13-4. The recovery of zinc in the zinc circuit is consistent with past performance. The recovery of copper and gold in the copper circuit highlights that recoveries greater than historical performances are achievable with 90 to 95% copper and 60 to 80% gold recovery possible.

FIGURE 13-4: HEAD GRADE RECOVERY RELATIONSHIP FOR COPPER, GOLD AND ZINC IN FUTURE LALOR BASE METAL ORES



Further work was undertaken on the composites to determine the effect of a finer grind on the copper, gold, and zinc recovery. Copper and zinc recovery was not materially impacted by the decrease in primary grind, and this is reflection of the highly liberated state these metals achieved at +100 μ m. Gold recovery is affected by a reduction in primary grind size, with the recovery improving on average by 1.2% per 10 μ m reduction in primary grind size.

Mineralogy analysis on the each of the composite samples, indicates a general upward trend in the proportion of pyrite/pyrrhotite in feed. Of concern is the level of galena and talc, which will have a significant impact on the copper flotation circuit without appropriate management strategies. Additional testwork has been initiated to map the lead grade profile across the LOM and the viability of a separate lead flotation stage in the flowsheet. A talc preflotation stage has already been designed to handle the variable levels of talc in feed.

13.2. NEW BRITANNIA CONCENTRATOR

In March 1949, the Nor-Acme mine and gold concentrator was opened. Production continued until 1958, with the Nor-Acme mill recovering approximately 610,000 oz of gold during this period. Between 1958 and the early 1990s, various companies worked on the property with the intention to bring the mine back into production. These efforts were unsuccessful.

In 1994, TVX and High River formed a joint venture to reopen the mine. Using ore from secondary mining areas, the mill was brought into production while completing construction of the main shaft. When production commenced in 1995, the name Nor-Acme was changed to New Britannia. Through various transactions, Kinross became the operator of the New Britannia mine-mill complex. Production mining ceased at the end of September 2004 and the mill was put on care and maintenance in 2005 due to a low gold price environment after producing 1.6 million oz of gold. In 2015, Hudbay acquired the New Britannia mill. It is currently being refurbished to process gold and copper-gold zone ores from the Lalor Mine.

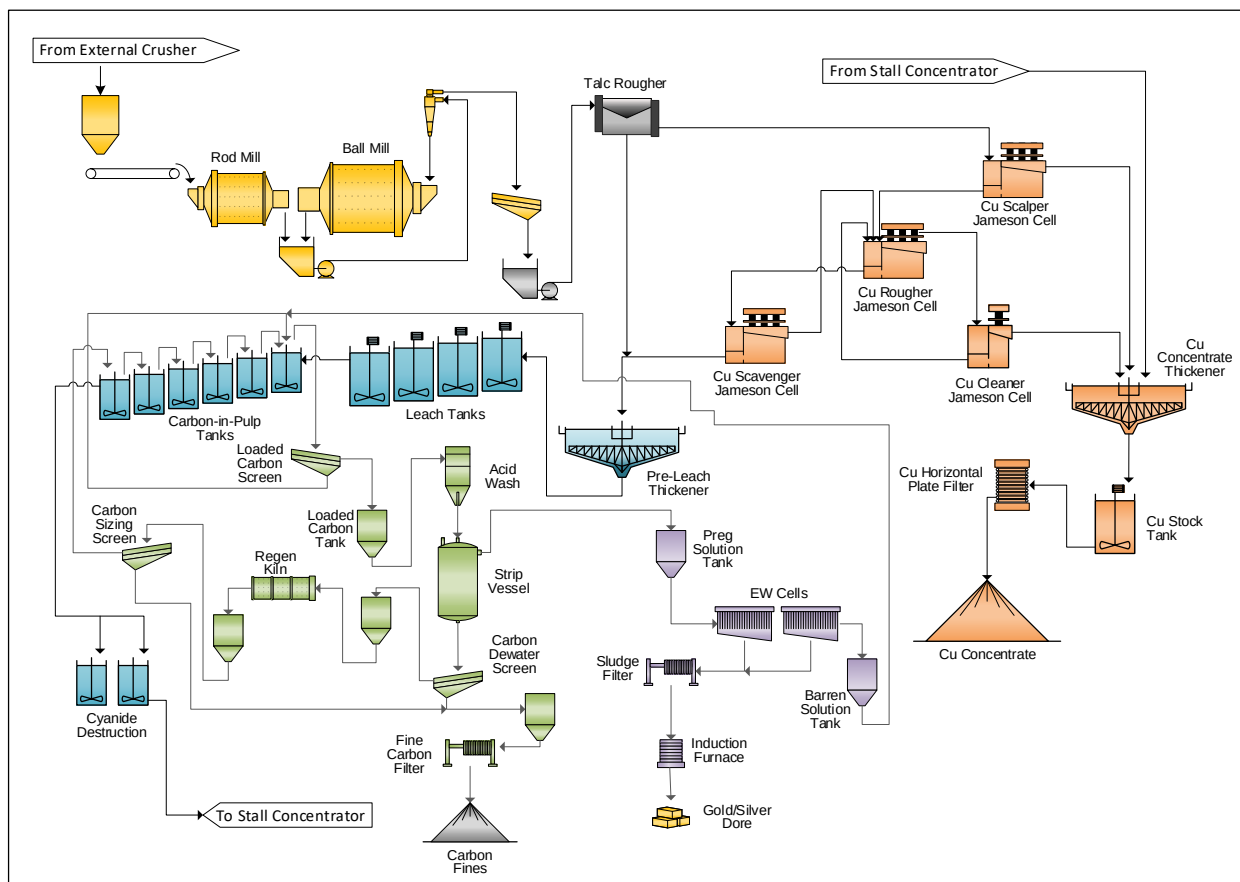
Two significant metallurgical test work programs were undertaken between 2015 and 2020 by SGS Lakefield and Blue Coast Research which form the basis of the concentrator refurbishment project.

The New Britannia is scheduled to achieve commercial production in H2 2021. The concentrator flowsheet is outlined in Figure 13-5. The LOM average metallurgical performance forecasts based on testwork completed and mine production grades are outlined in Table 13-4.

TABLE 13-4: LOM FORECAST METAL RECOVERY RANGE

Year	Flotation Recovery			Leach Recovery		Total Recovery		
	Au (%)	Ag (%)	Cu (%)	Au (%)	Ag (%)	Au (%)	Ag (%)	Cu (%)
LOM Average	60.0 to 65.0	51.0 to 59.0	91.0 to 95.0	26.0 to 32.0	20.0 to 25.0	91.0 to 94.0	76.0 to 80.0	91.0 to 95.0

FIGURE 13-5: NEW BRITANNIA PROCESS FLOWSHEET



Metallurgical Testwork – SGS Lakefield (2015)

In 2015, SGS Lakefield conducted a metallurgical test program on nine composite samples from within the Lalor gold zones with the objective of evaluating the processing options for the gold zone ores. The testwork included program mineralogy, comminution, gravity separation, flotation, cyanidation, CIP model and cyanide destruction.

Sample Selection

The 2015 test program used 395 kg of core from zone 25 and 315 kg from zone 27. The zone 25 samples were segregated into four variability composite samples and a single overall composite including of waste dilution mass. The zone 27 samples were segregated into three variability composites, a single overall composite excluding waste and an overall composite including waste. A final zone 25 and zone 27 blend composition sample was generated at a ratio of 2/3 and 1/3 zone 25 to zone 27, respectively. The head assays for the test composites are outline in Table 13-5.

TABLE 13-5: SGS LAKEFIELD COMPOSITE SAMPLE HEAD GRADES

Zones	Composite ID	Head Assays					
		Au (g/t)	Ag (g/t)	Cu (%)	Zn (%)	Fe (%)	S (%)
25	Composite A	1.93	15.7	0.04	0.09	7.20	0.55
	Composite B	6.88	51.3	0.17	0.31	8.70	1.37
	Composite C	7.90	78.0	0.32	0.60	5.27	1.90
	Composite D	8.83	32.1	0.69	0.63	7.70	2.52
	Zone 25 Overall Diluted	5.11	40.3	0.22	0.27	6.52	1.4
27	Main Central	11.8	28.3	6.28	0.82	14.9	8.25
	Upper	9.15	10.1	1.44	0.18	9.70	2.31
	Main North	10.7	28.8	6.60	0.38	12.4	10.4
	Zone 27 Overall	11.0	29.1	5.38	0.69	14.0	7.17
	Zone 27 Overall Diluted	8.41	20.5	4.28	0.57	12.0	6.02
25 and 27	Blend Composite	5.87	29.0	1.64	0.36	8.32	3.00

Mineralogical Investigation

A subsample of zone 25 overall and zone 27 overall composite was ground to a p80 of 153um for zone 25 and 101um for zone 27 with the results outlined in Table 13-6. The two composites varied significantly in both non-sulphide gangue and sulphide content. The zone 25 overall composite was composed predominately of quartz, amphibole/pyroxene, and feldspar with the sulphides accounting for less than 5% of the total mass. The zone 27 overall composite was composed predominately of quartz, amphibole/pyroxene, chalcopyrite, and garnets with the sulphides accounting for more than 20% of the mass.

TABLE 13-6: OVERALL COMPOSITE MINERALOGY

Mineral	Zone 25 Overall		Zone 27 Overall	
	Abundance (%)	Mean Graind Size (um)	Abundance (%)	Mean Graind Size (um)
Sphalerite	0.20	28.00	0.60	21.00
Chalcopyrite	0.60	24.00	15.70	33.00
Cubanite	0.00	10.00	1.70	25.00
Galena	0.30	20.00	0.00	8.00
Pyrite	1.00	68.00	0.00	17.00
Pyrrhotite	2.00	62.00	4.30	46.00
Arsenopyrite	0.00	0.00	0.00	0.00
Talc	1.40	33.00	0.00	16.00
Quartz	33.90	44.00	35.10	38.00
Feldspar	10.00	36.00	0.40	10.00
Garnets	3.00	9.00	13.30	32.00
Mg-Al Silicates	7.10	37.00	0.00	
Chlorites	6.30	31.00	3.30	25.00
Mica/Clays	4.40	31.00	5.70	26.00
Amphibole/Pyroxene	27.00	36.00	16.90	30.00
Other	2.80		3.00	

Communication Testwork

RWi and BWi tests were completed on several the composites sample with the results outline in Table 13-7 The RWi values range from 9.3 to 13.3 kWh/t for the material tested. Compared to the base metal ore zones (6.0 to 10.0 kWh/t) the material tested from zone 25 and zone 27 is 30 to 50% harder. The BWi

values range from 14.5 to 21.0 kWh/t for the material tested. Compared to the base metal ore zones (10.0 to 14.0 kWh/t) the material tested from zone 25 and zone 27 is 40 to 60% harder.

TABLE 13-7: COMMUNICATION TESTWORK RESULTS

Sample Name	Work Indices (kWh/t)		AI (g)
	RWI	BWI	
Comp A	-	21.0	-
Comp B	13.3	17.7	-
Comp C	11.4	15.4	-
Comp D	-	16.1	-
Gangue / Dilution	11.8	17.1	-
Main Central	9.3	14.5	0.359
LX0001-M	10.2	18.0	0.353
Statistics	Work Indices (kWh/t)		AI (g)
	RWI	BWI	
Average	11.2	18.0	0.356
Std. Dev.	1.6	2.8	0.005
Rel. Std. Dev.	13.9	15.5	1.312
Minimum	9.3	15.4	0.353
10th Percentile	9.6	15.9	
25th Percentile	10.2	16.6	
Median	11.4	17.7	0.356
75th Percentile	11.8	19.3	
90th Percentile	12.7	20.3	
Maximum	13.3	21.0	0.359

Flotation Testwork

The flotation testwork conducted by SGS Lakefield focused on the removal of copper prior to tailings cyanidation to recovery the residual gold and silver. Rougher, cleaner, and locked cycle tests were completed on the composite samples. The results from the rougher and cleaner tests are outlined in Table 13-8.

High recoveries of copper to the copper concentrate (+90%) and high concentrate grades were achieved where the lead head grade does not exceed the copper head grade. The gold and silver recovery ranged from 40.8 to 91.9%. The test work indicates that further work is required to investigate the potential to scalp a high grade concentrate from the rougher flotation stage.

TABLE 13-8: ROUGHER AND CLEANER FLOTATION TEAMWORK RESULTS

Test #	Feed	Calculated Head						Copper Cleaner Concentrate										Copper Rougher Concentrate									
		Assays (g/t, %)						Assays (g/t, %)					Distribution (%)					Assays (g/t, %)					Distribution (%)				
		Au	Ag	Cu	Pb	Fe	Zn	Au	Ag	Cu	Pb	Zn	Au	Ag	Cu	Pb	Zn	Au	Ag	Cu	Pb	Zn	Au	Ag	Cu	Pb	Zn
F8	Blend Comp	5.10	30.4	1.57	0.19	0.19	0.18	62.4	370	26.0	1.78	1.25	68.9	68.7	93.4	53.4	50.2	308	19.4	2.13	1.14	75.4	77.2	94.8	87.6		
F20	Blend Comp	5.42	29.6	1.51	0.19	0.19	0.18	71.6	370	28.3	0.65	0.84	65.1	61.6	92.4	17.0	58.3	316	20.6	2.28	0.79	74.8	74.3	94.8	84.2		
F22	Blend Comp	5.40	28.3	1.46	0.20	0.20	0.19	73.7	365	29.6	0.43	0.79	59.9	56.6	89.0	9.5	-	-	-	-	-	-	-	-	-	-	
F23	Blend Comp	5.92	29.1	1.50	0.17	0.17	0.17	78.2	285	27.5	0.44	1.01	65.2	48.3	90.7	13.0	61.9	258	20.5	1.64	0.98	72.3	61.4	94.6	67.2		
F25	75% Zone 25 / 25% Zone 27	5.94	35.0	1.16	0.24	0.24	0.33	88.5	516	27.6	0.69	0.88	60.2	57.0	91.1	11.5	67.3	429	18.9	2.47	1.12	70.4	71.2	93.8	61.4		
F26	50% Zone 25 / 50% Zone 27	6.78	30.1	2.10	0.16	0.16	0.40	54.9	251	29.4	0.35	0.95	57.3	50.4	87.4	12.0	49.8	254	22.8	1.71	1.11	72.8	71.3	94.9	81.5		
F27	25% Zone 25 / 75% Zone 27	7.59	25.3	3.04	0.09	0.09	0.46	50.8	161	30.3	0.18	0.97	65.3	53.3	90.7	17.9	47.9	170	25.6	0.88	1.10	76.9	70.4	95.7	82.6		
F30	50% Blend / 50% Comp 1	3.80	22.6	1.07	0.19	0.19	3.79	27.9	153	11.0	0.46	4.79	67.1	51.9	87.2	17.2	14.4	86.9	5.23	0.78	6.06	76.0	64.7	90.9	63.7		
F9	Zone 25 Overall Diluted	5.03	40.5	0.22	0.33	0.33	0.13	254	1992	14.8	14.3	1.23	69.0	67.3	90.1	59.3	89.6	718	4.77	5.91	0.67	77.8	77.6	92.8	78.3		
F18	Zone 25 Overall Diluted	5.02	40.6	0.20	0.29	0.29	0.13	388	2631	21.2	5.27	1.01	61.1	54.0	87.9	15.3	132	1040	6.51	7.26	0.70	75.6	73.7	93.2	72.9		
F11	Zone 25 Comp C	7.72	73.4	0.33	0.77	0.77	0.19	312	2767	16.6	12.1	1.40	66.6	62.1	83.5	25.9	134	1229	6.62	10.7	0.88	74.2	71.3	88.0	59.0		
F19	Zone 25 Comp C	7.76	70.2	0.29	0.58	0.58	0.18	406	3736	21.4	4.21	1.22	58.8	59.9	82.3	8.2	106	995	4.30	8.50	0.72	83.3	86.1	89.4	89.0		
F12	Zone 25 Comp D	7.54	29.2	0.65	0.15	0.15	0.47	184	633	23.3	2.74	2.04	61.8	55.0	91.3	45.3	79.0	287	8.80	1.61	1.36	73.1	68.7	95.0	73.3		
F16	Zone 25 Comp D	7.47	26.7	0.63	0.16	0.16	0.47	195	498	24.9	1.21	0.89	57.2	40.8	85.8	16.3	99.1	299	10.4	1.96	2.17	72.8	61.4	90.2	66.3		
F10	Zone 27 Overall Diluted	6.55	20.5	3.69	0.02	0.02	0.27	35.8	101	27.8	0.04	1.31	69.2	62.6	95.5	35.8	31.3	93.0	22.7	0.04	1.17	75.3	71.6	96.8	44.7		
F24	Zone 27 Overall Diluted	7.18	22.0	4.44	-	-	-	-	-	-	-	-	-	-	-	-	27.6	84.3	26.1	-	-	63.5	63.5	97.4	-		
F28	Zone 27 Overall Diluted	7.26	21.3	3.88	0.01	0.01	0.29	34.4	100	26.1	0.04	1.59	69.4	68.9	99.1	36.5	31.3	91.0	21.6	0.03	1.34	76.6	76.0	99.3	41.4		
F29	Zone 27 Overall Diluted	6.40	19.4	3.61	0.01	0.01	0.26	35.2	101	27.4	0.03	1.19	70.2	66.4	96.8	29.1	29.6	86.3	22.3	0.03	1.02	72.9	70.1	97.4	33.8		
F13	Zone 27 Main Central	10.7	35.0	6.35	0.02	0.02	0.45	37.4	102	30.7	0.03	1.27	66.5	55.4	92.0	34.4	35.6	109	27.7	0.04	1.26	72.4	67.4	94.9	51.2		
F17	Zone 27 Main Central	12.0	34.3	6.07	0.02	0.02	0.42	-	-	-	-	-	-	-	-	-	40.0	101	24.4	0.03	1.20	81.6	72.5	98.4	50.9		
F14	Zone 27 Upper	3.16	8.8	1.16	0.02	0.02	0.10	46.9	111	26.1	0.16	0.91	61.6	52.5	93.0	39.2	33.9	87.9	16.9	0.12	0.71	70.4	65.9	95.2	44.7		
F15	Zone 27 Main North	8.72	26.0	6.79	0.03	0.03	0.40	47.2	99	29.5	0.06	1.01	91.9	64.9	73.7	38.3	40.3	87.7	25.4	0.08	0.96	65.1	69.4	76.9	56.6		
F21	Zone 27 Main North	10.8	33.4	6.97	0.03	0.03	0.34	-	-	-	-	-	-	-	-	-	28.9	87.1	21.1	0.08	0.78	66.6	64.7	98.4	79.1		

The lock cycle tests conducted all produced copper concentrates at saleable grade and with metal recoveries consistent with the batch rougher and cleaner flotation tests. Detailed analysis of the concentrates indicates that concentrates produced by the zone 27 material is low in delirious minor elements such as F, As and Hg. The zone 25 concentrates however display elevated levels of Hg and Sb that are in the payable penalty range however as this ore is projected to be blended with zone 27 and the subsequently concentrates produced blended with Stall Concentrator copper concentrates, the impact of these penalty elements will be minimised.

Leach Testwork

Cyanidation leach tests were conducted on the tailings from the lock cycle and rougher tests. The test results are outline in Table 13-9.

From initial sighter tests the standard conditions of 40% solids, 8 hours pre-aeration, 50 g/t lead nitrate and 0.5 g/L cyanide concentrate were selected. The gold extraction values range from 72.2 to 85.6% after 48 hours with most of the extraction occurring within the first 24 hours period. Cyanide consumption to maintain a 0.5 g/L pulp concentration through the test ranged from 0.65 to 2.98 kg/t. As most of the gold extraction occurs within the first 24 hours further optimization testwork is required to leach time to reduce the cyanide consumption.

TABLE 13-9: FLOTATION TAILS CYANIDATION LEACH TESTWORK RESULTS

Test No.	Sample	Extraction (%) Time (hours)		Gold			Extraction (%) Time (hours)		Silver		
				Residue	Assay (g/t) Head				Residue	Assay (g/t) Head	
		24	48		Calc	Direct	24	48		Calc	Direct
CN-13	F10-Ro Tail & CI Tail	68.0	74.3	0.59	2.29	2.31	39.9	50.7	4.25	8.62	8.77
CN-14	F14-Ro Tail & CI Tail	75.4	78.2	0.28	1.26	1.26	59.9	65.3	1.45	4.18	4.34
CN-15	F17-Ro Tail	68.9	72.3	0.75	2.71	2.55	52.4	64.2	3.25	9.09	11.1
CN-16	F21-Ro Tail	64.6	72.9	0.28	1.03	2.05	32.9	51.0	3.25	6.6	6.70
CN-17	LCT-1 Comb Tail	80.6	80.7	0.46	2.35	2.51	62.5	65.2	4.50	12.9	11.6
CN-18	LCT-1 Comb Tail	79.8	78.9	0.48	2.27	2.51	60.2	60.3	4.90	12.3	11.6
CN-19	LCT-1 Comb Tail	80.3	79.1	0.44*	2.24	2.51	n/d	n/d	n/d	n/d	11.6
CN-20	LCT-1 Comb Tail	n/d	n/d	n/d	n/d	n/d	n/d	n/d	n/d	n/d	11.6
CN-21	LCT-1 Comb Tail	n/d	81.5	0.46	2.49	2.51	n/d	n/d	n/d	n/d	11.6
CN-25	LCT-2 - Comb Tail	85.3	85.0	0.27	1.76	1.94	63.5	69.2	4.50	14.6	21.9
CN-26	LCT-3 - Comb Tail	86.8	83.9	0.39	2.43	2.29	53.7	56.4	11.8	27.1	26.8
CN-27	LCT-4 - Comb Tail	82.3	81.9	0.48	2.65	3.21	75.1	79.7	2.50	12.3	13.6
CN-28	LCT-5 - Comb Tail	81.0	@ 48 h	0.40	2.08	1.91	64.4	@ 48 h	4.40	12.4	11.9
CN-29	LCT-5 - Comb Tail	80.2	@ 36 h	0.39	1.95	1.91	63.4	@ 36 h	4.20	11.5	11.9
CN-30	LCT-5 - Comb Tail	78.3	@ 24 h	0.42	1.94	1.91	62.5	@ 24 h	4.30	11.5	11.9
CN-31	LCT-5 - Comb Tail	76.1	@ 16 h	0.47	1.95	1.91	57.8	@ 16 h	4.80	11.4	11.9
CN-32	LCT-5 - Comb Tail	74.1	@ 8 h	0.52	2.01	1.91	56.0	@ 8 h	4.90	11.1	11.9
CN-33	F-22 - Comb Tail	79.6	-	0.37	1.79	1.87	62.4	-	4.20	11.2	10.9
CN-34	F-22 - Comb Tail	79.9	-	0.36	1.76	1.87	63.2	-	4.00	10.9	10.9
CN-35	F-24 - Comb Tail	-	74.0	0.79	3.02	3.14	-	62.4	3.60	9.6	9.60
CN-36	F-25 - Comb Tail	-	85.6	0.35	2.43	2.38	-	69.0	4.90	15.8	16.2
CN-37	F-26 - Comb Tail	-	78.2	0.53	2.43	2.79	-	60.8	5.40	13.8	16.8
CN-38	F-27 - Comb Tail	-	72.2	0.75	2.70	2.94	-	43.3	7.40	13.1	15.3
CN-41	F-29 - Comb Tail	74.2	74.1	0.54	2.07	2.19	62.1	66.0	2.20	5.81	7.48
CN-42	F-29 - Comb Tail	71.9	72.9	0.60	2.13	2.19	59.9	63.3	2.40	5.98	7.48
CN-43	LCT-6 - Comb Tail	78.8	78.5	0.36	1.67	1.85	67.0	65.3	3.40	10.3	12.1
CN-44	LCT-6 - Comb Tail	78.4	78.3	0.38	1.76	1.85	66.4	64.6	3.60	10.7	12.1

Cyanide Destruction Testwork

Cyanide destruction testwork using the SO₂/Air Process was conducted on the on the leach tails from tests CN 20, 21, 42 and 44. The reagent sources of used for the tests was sodium metabisulphite (SMBS) for the SO₂ and copper sulphate for the Cu²⁺. The results are outlined in Table 13-17. The CNT and CNWAD values in the leach tails ranged from 232 to 466 mg/L.

For the batch tests conducted on all the samples, <1.00 mg/L CNWAD was achieved with SO₂ additions ranging from 7.28 to 17.25 g/g CNWAD and Cu additions ranging from 0.05 to 0.12 g/g CNWAD. In CN-42 continuous tests, the removal of copper sulphate was attempted. The two tests achieved 210.00 and 38.20 CNWAD mg/L after 90min and 120min indicating that copper sulphate is required to catalyze the reaction even though 285 mg/L Cu was present in the feed solution. With the addition of 0.11 g/g copper sulphate to CNWAD and additional SO₂ the CNWAD was reduced to <1.00 mg/L.

The cyanide destruction tests were successful at reducing the CNWAD to <1.00 mg/L in all tests and further optimization is required around retention time, SO₂ and copper sulphate additions.

Metallurgical Testwork – Blue Coast Research (2019 to 2020)

From 2019 to 2020, Blue Coast Research conducted a metallurgical test program on thirteen composite samples from within the Lalor gold zones to confirm the optimum processing options for the gold zone ores. The testwork program included mineralogy, flotation, cyanidation, and cyanide destruction.

Sample Selection

The samples selected for the were sourced primarily from zone 27 with a single master composite and nine variability samples. A single zone 25 sample was also included to test individually and to enable the generation of two zone 25 and zone 27 blend composites (50/50 and 75/25 zone 25 to zone 27 respectively). The head assays for the test composites are outline in Table 13-10

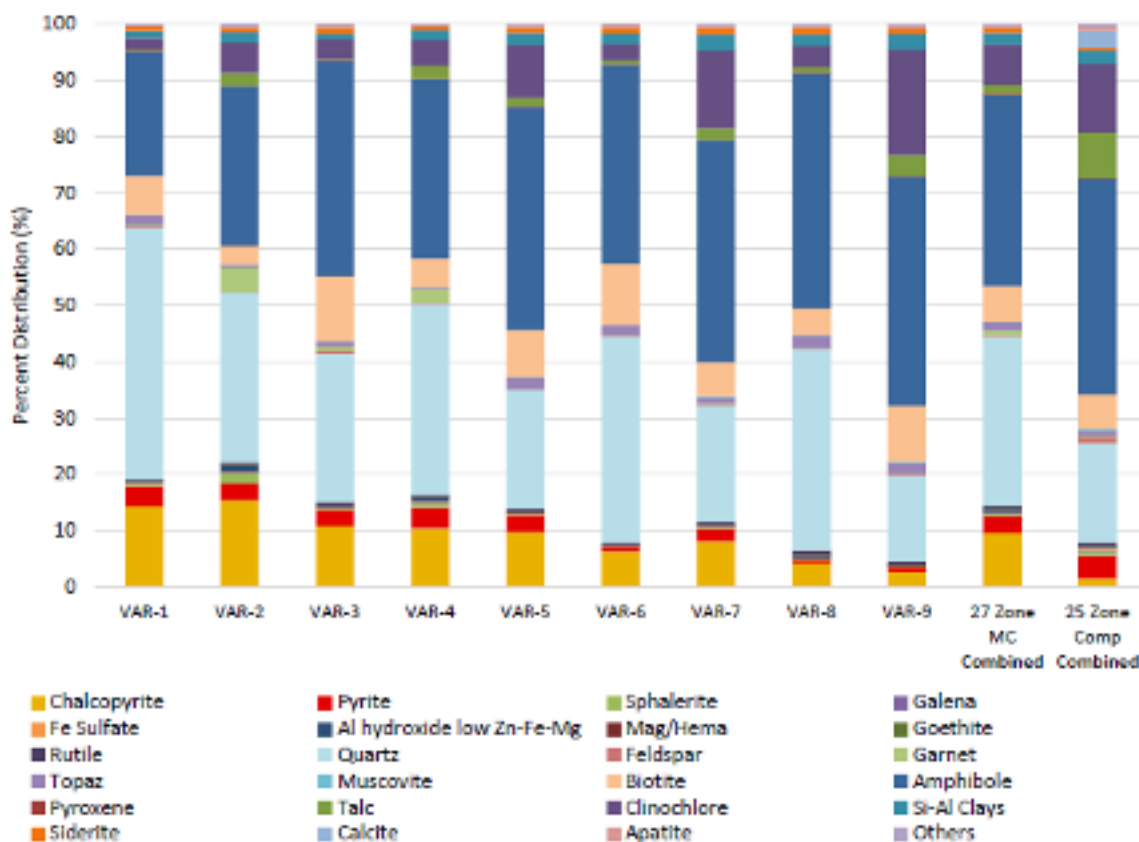
TABLE 13-10: BLUE COAST RESEARCH COMPOSITE SAMPLE HEAD GRADES

Comp ID	Au (g/t) FA-AA	Ag (g/t) 4AD-AA	Cu (%) 4AD-AA	Pb (%) 4AD-AA	Zn (%) 4AD-AA	Fe (%) 4AD-AA	MgO (%) 4AD-AA	S _{tot} (%) ELTRA
27 Zone Master Comp	6.17	18.3	2.95	0.01	0.31	12.5	7.96	4.02
Zone 25 Composite	5.02	36.4	0.53	0.18	0.44	8.24	13.2	2.35
50/50 Blend Composite	5.59	27.4	1.74	0.10	0.37	10.4	10.6	3.18
75/25 Blend Composite	5.31	31.9	1.13	0.14	0.40	9.30	11.9	2.77
VAR-1	10.1	19.0	4.60	0.02	0.28	11.7	4.04	6.11
VAR-2	16.2	46.0	4.18	0.01	0.87	12.6	7.27	5.99
VAR-3	3.36	14.5	3.61	0.02	0.25	15.8	5.99	4.79
VAR-4	6.98	28.0	3.39	0.02	0.41	11.8	7.30	5.18
VAR-5	7.47	23.5	3.19	0.01	0.19	14.2	9.12	4.52
VAR-6	5.74	16.0	2.04	0.01	0.11	11.6	6.32	2.42
VAR-7	8.12	24.5	2.66	0.01	0.17	12.5	10.4	3.51
VAR-8	1.70	11.5	1.45	0.01	0.17	10.9	7.20	1.80
VAR-9	2.02	3.00	0.89	0.01	0.07	11.2	12.6	1.22

Mineralogical Investigation

A subsample of each of the composite samples was ground to 90um and submitted for mineralogical analysis with the result outlined in Figure 13-6. The dominant mineral species across all the composite samples were quartz, amphibole, clinocllore and chalcopryrite. The mineralogy of the composite sample was consistent with the 2015 SGS program except for Talc. The level of talc present in each of the composites, particularly the zone 25 composite are significantly higher than in the 2015 program. In the 2015 program the talc content of the zone 25 composite was 1.5%. When compared with the zone 25 composite in the Blue Coast program at ~10%, the significant increase in talc would have an impact on the flotation circuit.

FIGURE 13-6: COMPOSITE SAMPLE MINERALOGY



Additional focused mineralogy was conducted on the zone 25 and zone 27 master composite to determine the liberation state of the chalcopyrite. Results showed that most of the chalcopyrite in both zone 25 and zone 27 is free/liberated across all size fractions with only minor quantities associated with sphalerite, pyrite, and other non-sulphide minerals. This indicates that there is little benefit to the recovery of copper at primary grind sizes below 90µm.

Flotation Testwork

The flotation testwork conducted by Blue Coast Research focused on the removal of copper prior to tailings cyanidation to recover the residual gold and silver. Rougher flotation tests were completed on the zone 25 (SKT 8, 9, 11, 34) and zone 27 master composite sample with the results outlined in Table 13-11.

TABLE 13-11: ROUGHER FLOTATION TESTWORK RESULTS

Test ID	Mass Pull (%)	Grade to Conc			Recoveries to Conc (%)		
		Au (g/t)	Ag (g/t)	Cu (%)	Au	Ag	Cu
SKT-1	27.2	23.1	58	10.7	89.6	82.7	99.0
SKT-2	23.6	25.9	59	12.5	88.0	81.2	99.0
SKT-3	17.2	39.0	85	16.8	87.8	79.5	97.4
SKT-4	16.6	33.6	86	17.1	80.1	74.7	96.0
SKT-5	25.7	27.1	58	11.5	90.0	84.1	99.4
SKT-6	22.3	25.7	64	13.2	85.9	77.0	98.5
SKT-7	25.5	24.2	53	12.3	89.3	89.6	99.4
SKT-8	25.7	17.2	98	2.0	90.0	95.5	96.9
SKT-9	20.3	21.2	129	2.5	88.8	92.1	96.8
SKT-10	22.4	31.8	64	13.4	87.7	84.6	99.4
SKT-11	24.1	18.4	121	2.1	90.7	92.5	98.7
SKT-12	20.1	29.6	79	14.6	86.2	88.3	99.6
SKT-13	21.0	31.2	73	14.3	88.5	89.5	99.6
SKT-34	21.9	17.9	115	2.1	86.0	84.8	96.9

The rougher flotation tests confirmed that high recoveries are achievable for copper, gold, and silver in the rougher flotation stage. The tests on zone 25 master composite averaged a rougher concentrate copper grade of 2.0 to 2.5% Cu compared to the zone 27 composite rougher concentrate grade of 10.7 to 17.1%. The principal driver for the depressed concentrate grade in the zone 25 rougher tests was the presence of high levels of Talc in the feed. The MgO grade in the zone 25 test was 12 to 14% indicating the bulk of the mass in concentrate was Talc.

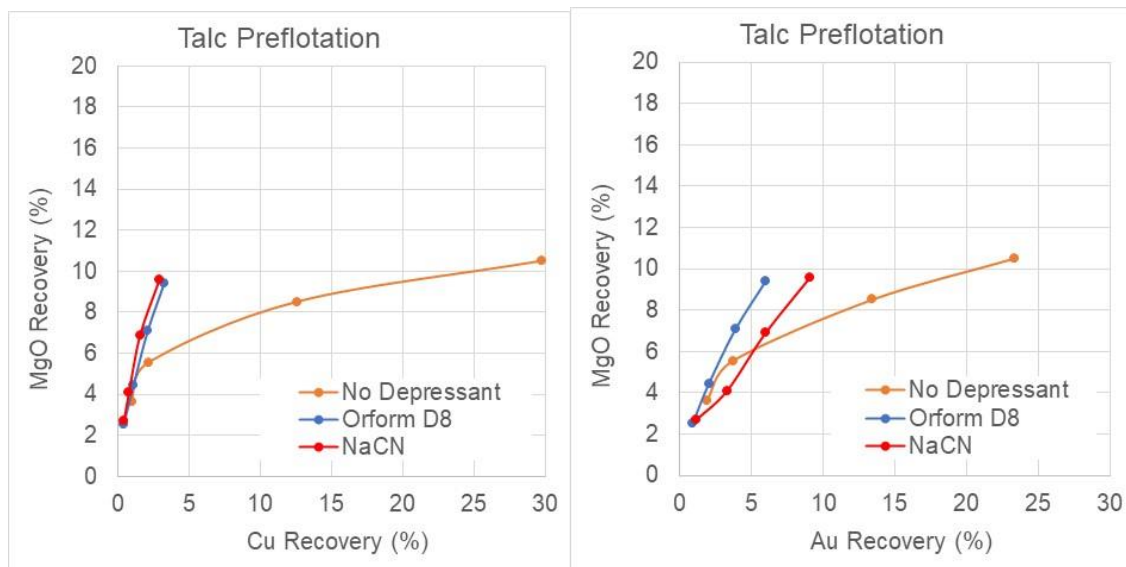
The final stage of the rougher flotation testwork was to test the zone 25 and zone 27 blend composites (Table 13-27). The blend of the high talc zone 25 ore with the cleaner zone 27 ore did not impact recovery however the concentrate grade was significantly impacted. The greater the proportion of zone 25 ore in the blend the lower the achievable concentrate grade dropped. In the 50/50 the concentrate MgO ranged from 9 to 11% with the 75/25 blend ranging from 12 to 14%.

Cleaner flotation testwork was conducted on a bulk concentrate sample generated from the zone 27 composite. The zone 27 ore displayed exceptional flotation response with recovery of copper, gold and silver all exceeding 96% and at a saleable concentrate grade.

Cleaner flotation testwork was also conducted on the blend composites. As found in the rougher flotation tests, the recovery was not adversely affected by the blending on the zone 25 and zone 27 ores. The effect of blending these ores was the depression of the concentrate grade due to the presence of talc. In the 50/50 blend the MgO in the concentrate was 8 to 10% and in the 75/25 blend it was 11 to 12%. The addition of CMC did not significantly impact the concentrate quality.

The presence of talc at these levels indicates that a dedicated preflotation stage is required to selectively recovery the talc directly to flotation tailings ahead of the copper flotation circuit to ensure a saleable concentrate grade can be produced. Additional flotation tests were conducted on the efficacy of a preflotation stage with the results outline in Figure 13-7. In the presence of only MIBC and at pH 9.5 the talc was recovered to the concentrate in a single rougher stage. The subsequent recovery of copper and gold to the talc concentrate was unacceptably high and required additional optimization. The addition of a sulphide depressant (Orform D8 or NaCN) significantly reduced the recovery of copper and halved the recovery of gold to the talc concentrate. As the flotation tailings in the New Britannia circuit reports to the leach circuit the loss of this level of gold is acceptable as it will be recovered to Doré. Additional optimization is required including investigating the effectiveness of a talc cleaning stage as is typical in base metal concentrators operating a talc preflotation circuit.

FIGURE 13-7: TALC PREFLOTATION ROUGHER TESTWORK RESULTS



Two lock cycle tests were conducted on 50/50 blend composite without a talc preflotation stage. The major changes between each of the lock cycle test was the removal of the cleaner scavenger stage, the change in pH and the increase in CMC addition.

The results for the lock cycle tests are outline in Table 13-12. LCT-1 was operated aggressive generating an exceptional recovery at the expense of final concentrate grade. It is important to note that for LCT-1 the effort to control talc recovery was minimal with the resulting final concentrate grade of 15.4% Cu and +6.0% MgO. LCT-2 increased the effort to remove MgO from the concentrate with the CMC increase in the rougher from 150 g/t to 400 g/t and in the cleaners from 25 g/t to 50 g/t. LCT-2 achieved a 25.9% Cu and 1.0% MgO concentrate grade however at a significant discount to copper and gold recovery.

TABLE 13-12: LOCK CYCLE TEST RESULTS

Test ID	Grade to Final Con			Recovery to Final Con (%)		
	Au (g/t)	Ag (g/t)	Cu (%)	Au	Ag	Cu
LCT-1	46.1	185.8	15.4	84.2	83.2	97.7
LCT-2	67.7	226.0	25.9	68.2	57.3	86.4
Average	56.9	205.9	20.7	76.2	70.2	92.1

13.3. FLOTATION TAILS LEACH TESTWORK

Cyanidation leach tests were conducted on the tailings from the lock cycle and rougher tests (Table 13-13). The flotation tailings from the batch flotation tests, bulk flotation tests (cleaner testwork feed generation) and lock cycle tests were used as the feed stock for the test work. The base conditions for the flotation tails leach testwork were 40% solids, 48-hour leach duration, 0.5 g/L cyanide concentration and pH at 10.5 to 11.0.

CN-1 and CN-2 were grind size variability flotation tests with a flotation feed p80 of 90µm and 120µm, respectively. CN-1 and CN-2 show that at a p80 grind of 90µm 74% gold extraction is achieved verses 68.4% gold extraction at 120µm. A 90µm grind was carried forward for all the subsequent leach tests. CN-3 to CN-5 assessed the impact of pre-aeration on the gold extraction. Minimal difference between no pre-aeration and 4 hours pre-aeration was found and unless stated, pre-aeration was not utilized in future tests.

TABLE 13-13: FLOTATION TAILS LEACH TESTWORK GOLD EXTRACTION RESULTS

Test ID	Final Solution Grade			Final Residue Grade			Recovery (%)		
	Au (g/t)	Cu (ppm)	Ag (g/t)	Au (g/t)	Cu (ppm)	Ag (g/t)	Au	Cu	Ag
CN-1	0.58	99	-	0.31	295	-	74.0	33.7	-
CN-2	0.64	47	-	0.45	281	-	68.4	20.4	-
CN-3	1.06	76	-	0.41	995	-	79.9	10.3	-
CN-4	1.30	74	-	0.38	985	-	83.7	10.0	-
CN-5	1.25	75	-	0.42	1080	-	82.2	9.5	-
CN-6	1.95	110	-	0.48	945	-	80.7	10.6	-
CN-7	1.92	93	-	0.44	1035	-	82.0	8.5	-
CN-8	1.93	95	-	0.42	1035	-	82.7	8.6	-
CN-9	1.78	132	-	0.43	1005	-	81.2	11.9	-
CN-10	1.55	154	-	0.41	965	-	79.9	14.2	-
CN-13	0.78	85	2.59	0.23	345	1.98	78.7	20.4	56.4
CN-14	1.50	283	6.11	0.41	1845	4.33	79.8	13.7	58.6

All tests achieved low cyanide consumptions except for CN-14. CN-14 feed material was the tailings from LCT-2 which reported higher residue copper in the flotation tailings at 0.18% versus the residual copper in tailings for CN-1 to CN-13 between 0.03 to 0.10%. With the correct operation of the flotation circuit ahead of the leach stage, cyanide consumption below 1.0 kg/t is achievable.

To minimize the loading of copper onto carbon in the CIP stage the CNFree/Cu molar ratio in leach liquor must be maintained above 4.0. Most tests exceeded the required CNFree/Cu molar ratio low limit. CN-14 low CNFree/Cu molar ratio was a direct result of the elevated residual copper grade in the flotation tailings. CN-9 and CN-10 were the dosage tests for lead nitrate which indicates the addition of lead nitrate enhanced the leach rate of copper. The leach liquor copper tenor was 95 ppm without the addition of lead nitrate and rose to 154 ppm with its addition. The addition of lead nitrate does not improve the extraction of gold and appears to enhance the extraction rate of copper potentially causing downstream issues with carbon loading.

13.4. 1901 BASE METAL ORE METALLURGICAL TESTWORK

1901 is a satellite deposit located between the Chisel and Lalor deposit to the east side of the decline. This deposit consists of four distinct lens, two high zinc base metal lenses and two high gold bearing lenses. reportable level.

In 2020 a metallurgical testwork program was initiated with Blue Coast Research to cover the four identified ore lenses. At the time of writing, only testwork for the two base metal lenses, Lens 1 and 3 has been completed to a

Sample Selection

Three composite samples were created for each of base metal lens. As the base metal lenses are zinc dominant the composite sample were select to represent a low, average, and high zinc grade associated with the lens. The distribution of the drill core selected to create the metallurgical testwork composites are shown in Figure 13-8 and the grades in Table 13-14.

FIGURE 13-8: METALLURGICAL TESTWORK COMPOSITE SAMPLE DRILL CORE LOCATION

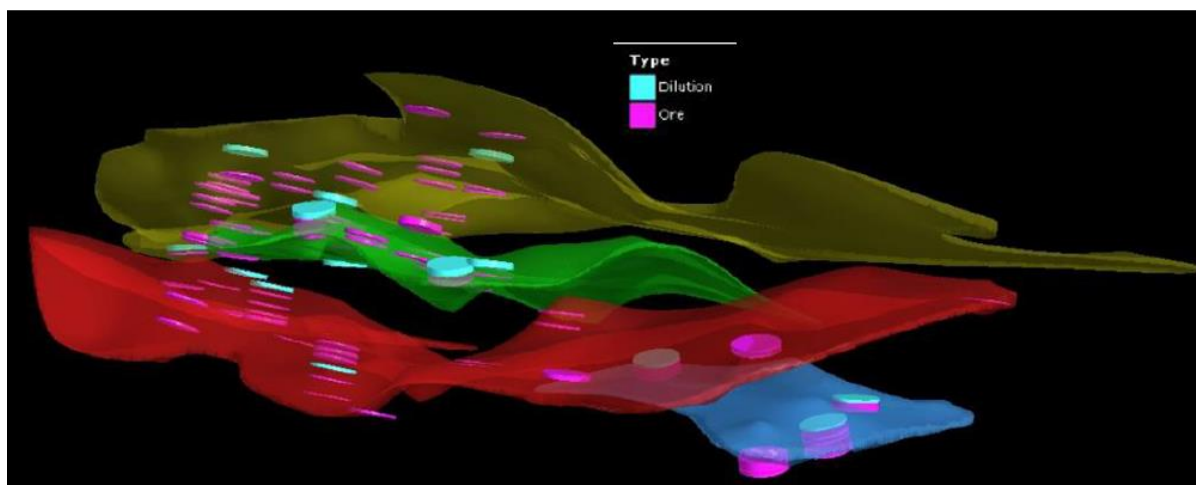


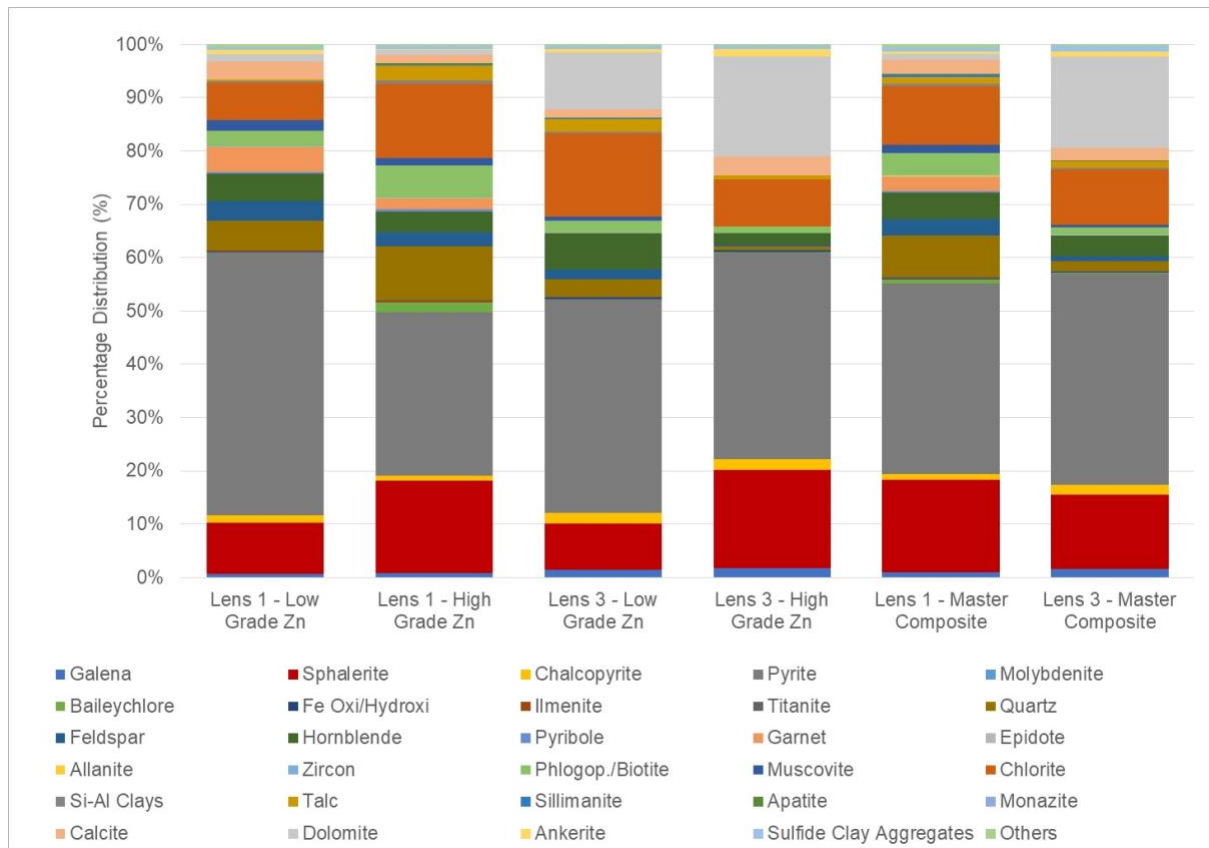
TABLE 13-14: 1901 BASE METAL LENSE COMPOSITE ASSAYS

Composite	Assays						
	Au (g/t)	Ag (g/t)	Cu (%)	Pb (%)	Zn (%)	Fe (%)	MgO (%)
Lens 1 - Low Grade Zinc	2.55	38.40	0.39	0.43	4.61	26.90	3.60
Lens 1 - Master Composite	2.12	43.00	0.30	0.72	9.57	21.40	4.90
Lens 1 - High Grade Zinc	0.69	34.00	0.26	0.56	12.22	18.00	5.70
Lens 3 - Low Grade Zinc	2.44	32.40	0.55	0.87	4.77	22.10	8.70
Lens 3 - Master Composite	1.69	26.80	0.55	0.92	4.65	22.30	8.10
Lens 3 - High Grade Zinc	0.81	22.30	0.60	0.97	9.09	22.80	7.60

Mineralogical Investigation

A subsample of each of the composite samples was ground to a p80 of 100µm and submitted for mineralogical analysis with the result outlined in Figure 13-9. The dominant mineral species across all the composite samples is pyrite, chlorite, sphalerite, and dolomite. Chalcopyrite and galena are the principal minerals for copper and lead, respectively.

FIGURE 13-9: 1901 COMPOSITE SAMPLE MINERALOGY



Flotation Testwork

The flotation testwork conducted by Blue Coast Research on the 1901 composite samples used a sequential copper-zinc flotation circuit approach. Flotation tests were completed on each of the six composites with the bulk of the optimization testwork carried out on the master composite sample from both Lens 1 and Lens 3. The feed grades for each of the composite samples shows that the lead assay is higher than the copper assay indicating that lead contamination of the copper concentrate is likely.

TABLE 13-15: 1901 FLOATATION TESTWORK COPPER CIRCUIT RESULTS

Test ID	Cu Rougher Grade					Cu Rougher Recovery (%)				
	Au (g/t)	Ag (g/t)	Cu (%)	Pb (%)	Zn (%)	Au	Ag	Cu	Pb	Zn
FKT-1	19.2	352	2.61	6.89	8.82	82.3	82.6	90.2	95.4	9.3
FKT-2	26.5	348	6.71	3.85	10.1	45.5	32.8	83.5	22.5	4.2
FKT-3	11.0	250	2.01	4.84	5.52	75.4	78.5	88.7	94.1	7.6
FKT-4	10.1	201	1.55	3.91	5.52	81.3	81.9	90.6	96.0	9.9
FKT-5	12.3	254	2.08	5.24	6.59	73.4	80.4	82.6	94.3	9.7
FKT-6	15.2	273	2.22	5.14	5.34	80.7	79.7	88.2	92.5	7.4
FKT-7	8.68	130	3.32	5.87	4.54	76.5	84.3	96.8	95.3	10.3
FKT-8	10.48	206	2.22	2.73	3.16	81.6	77.2	88.7	91.4	10.4
FKT-9	4.07	180	1.50	3.77	7.42	72.2	76.1	84.7	93.8	9.2
FKT-10	17.99	178	3.29	5.53	3.21	81.1	83.6	90.9	94.6	10.2
FKT-11	4.94	129	3.96	6.70	7.19	70.7	80.4	93.6	95.8	10.7
FKT-12	10.14	143	3.30	5.82	5.76	78.4	81.7	90.6	94.5	11.3
FKT-13	12.99	173	4.00	7.07	4.81	83.3	81.6	91.3	93.8	8.0
FKT-14	18.79	328	3.83	4.56	4.36	79.7	69.4	83.3	85.5	7.9
FKT-15	14.21	237	2.67	3.19	3.43	77.7	73.5	84.7	88.5	9.2
FKT-16	5.22	197	1.72	4.13	8.72	70.3	73.3	84.8	93.6	8.6

To examine the potential to selectively recovery copper and gold from lead, FKT-2 was run utilizing the collector TNC 312. The use of TNC 312 decreased the lead recovery from 95.2% to 22.5% and an improvement in the rougher concentrate grade from 2.6% Cu to 6.7% Cu. The cost of this improvement was a reduction in copper and gold recovery to 83.5% and 45.5% respectively which is an unacceptable reduction.

At the time of writing, further testwork is underway to examine the potential to include the same lead recovery stage into the sequential flotation circuit already considered for Lalor's future production. This testwork will operate a copper-lead-zinc sequential flowsheet utilizing TNC 312 as the copper circuit collector and 3418A as the lead circuit collector.

Zinc circuit flotation tests are consistent with testwork and historical performance from the Lalor Base metal ores, achieving +97% zinc circuit rougher recovery.

13.5. WIM METALLURGICAL TESTWORK

WIM is a satellite deposit located north of the Lalor deposit. This deposit consists of one ore zone that host of high-grade copper and moderate gold grade ore. In 2020 a metallurgical testwork program was completed by Blue Coast Research with the focus being the performance of the WIM ore through the New Britannia mill. No tailings cyanidation testwork was completed in the testwork campaign.

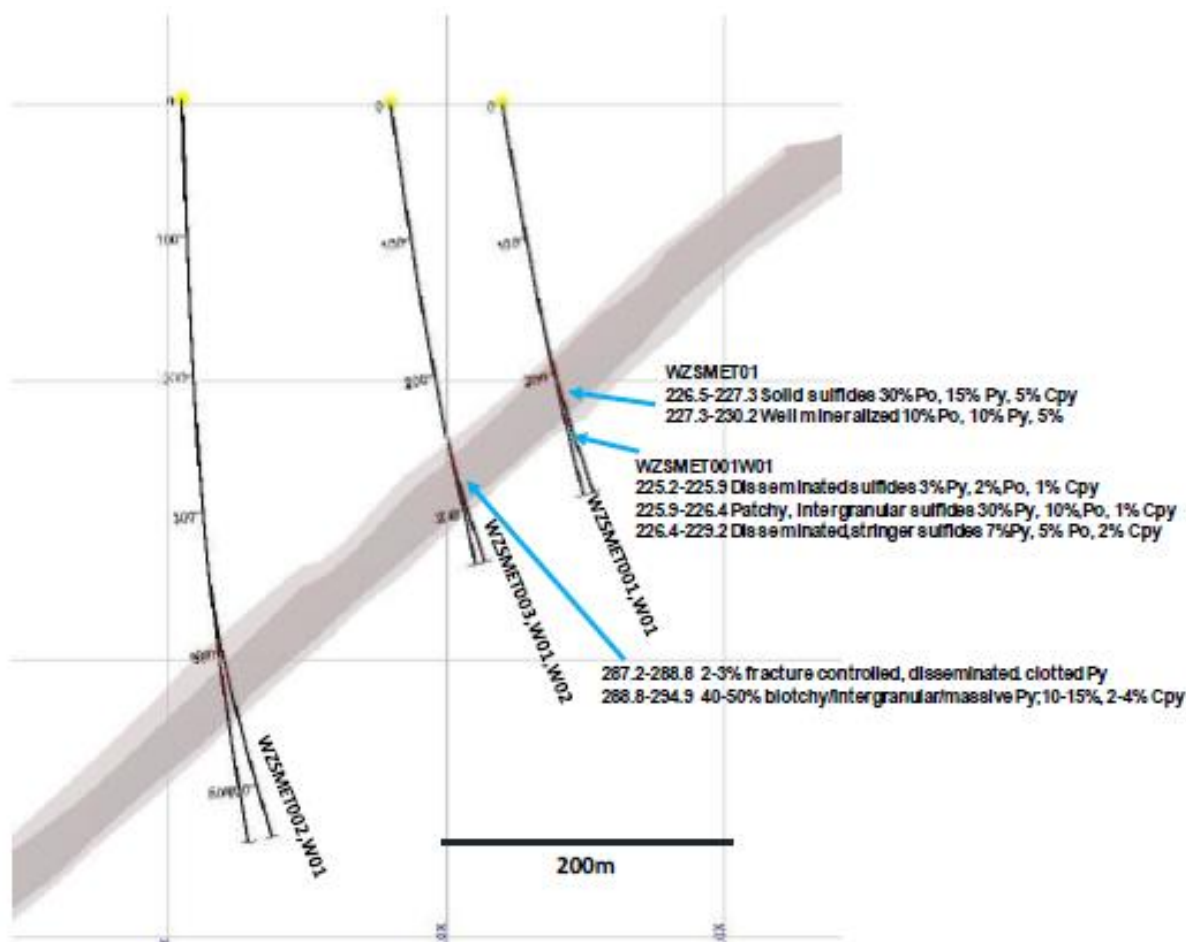
Sample Selection

Four composite samples were created for the WIM ore zone. Three variability composite samples were created based on the spatial distribution of the drill core locations outlined in Figure 13-10. A single master composite was also created from subsamples of the three variability samples. The head grades for all four composite samples are outline in Table 13-16.

TABLE 13-16: WIM COMPOSITE SAMPLE HEAD GRADES

Comp ID	Au (g/t) FA-AA	Ag (g/t) 4AD-AA	Cu (%) 4AD-AA	Pb (ppm) 4AD-AA	Zn (%) 4AD-AA	Fe (%) 4AD-AA	MgO (%) 4AD-AA	S (%) ELTRA	C _{org} ELTRA	C _{tot} ELTRA
WIM MC	1.55	5.60	1.81	0.003	0.19	24.2	3.61	19.0	0.02	0.05
MET-1	1.41	6.70	1.73	0.005	0.13	18.2	2.11	14.4	0.07	0.02
MET-2	2.16	6.40	2.02	0.003	0.22	24.7	4.82	18.8	0.04	0.02
MET-3	1.00	3.10	1.18	0.004	0.12	23.4	1.81	19.9	0.05	0.02

FIGURE 13-10: WIM COMPOSITE SAMPLE DRILL HOLE LOCATIONS

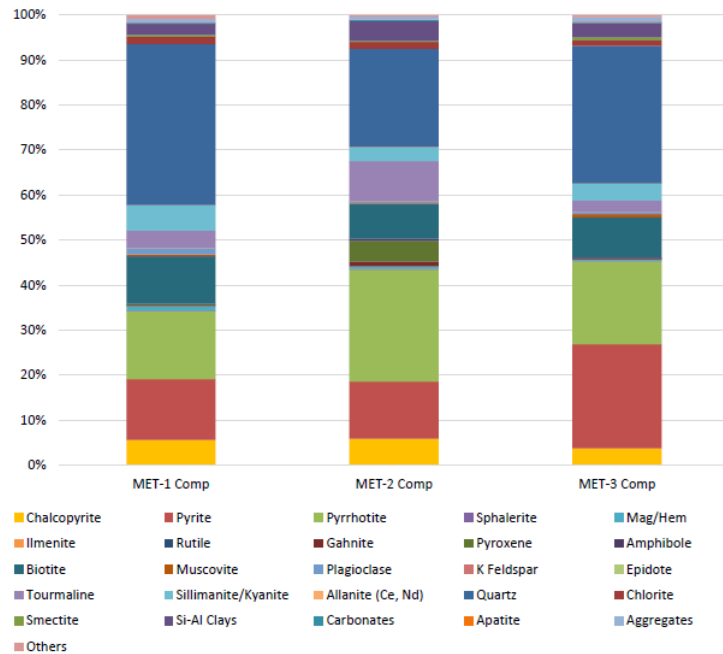


Mineralogy

A subsample of each of the composite samples was ground to a p80 of 90µm and submitted for mineralogical analysis with the result outlined in

Figure 13-11. The dominant mineral species across all the composite samples is pyrite, pyrrhotite, quartz and chalcopyrite. The level of pyrrhotite in the feed sample is significantly higher than comparable ore in the Lalor zone 25 or zone 27.

FIGURE 13-11: WIM COMPOSITE SAMPLE MINEROLOGY



Comminution Testwork

Standard Bond ball mill work index tests were completed on each of the composite samples with a closing screen size of 106µm. The BWi for the samples ranged from 17.9 to 18.5kWh/t. This is comparable to the Lalor gold zone ores.

Flotation Testwork

The flotation testwork conducted by Blue Coast Research on the WIM composite samples used a single copper flotation circuit configuration. Flotation tests were completed on each of the four composites with the bulk of the optimization testwork carried out on the master composite sample to determine the optimal conditions for the variable composites.

TABLE 13-17: WIM MASTER COMPOSITE ROUGHER FLOATATION TESTWORK RESULTS

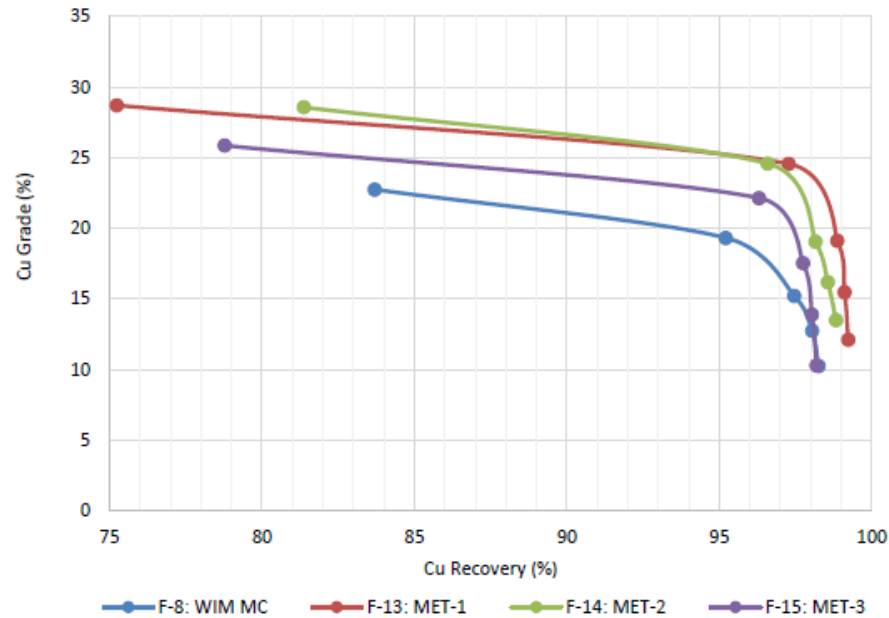
Test ID	Grade to Rougher Conc								Recoveries to Rougher Conc (%)							
	Au (g/t)	Ag (g/t)	Cu (%)	Pb (ppm)	Zn (%)	Fe (%)	MgO (%)	S (%)	Au	Ag	Cu	Pb	Zn	Fe	MgO	S
F-1	3.86	14	5.65	69	0.24	30.6	3.60	30.1	76.3	74.3	96.1	38.0	39.7	40.7	29.8	52.8
F-2	5.78	17	5.98	66	0.33	25.3	4.42	23.9	85.3	79.5	98.3	46.2	50.9	32.2	35.8	38.7
F-3	5.12	18	7.01	66	0.36	28.1	4.30	27.2	78.1	76.2	97.9	41.3	50.4	30.3	28.1	37.3
F-4	4.27	15	5.97	78	0.31	24.5	4.63	22.9	80.1	75.9	97.5	50.1	48.0	29.8	34.2	36.0
F-5	3.87	14	5.50	61	0.28	31.8	3.65	30.6	80.2	78.0	98.0	52.6	47.0	42.5	31.7	53.2
F-6	5.28	19	7.22	69	0.38	27.6	4.30	25.8	82.9	82.9	98.7	38.7	47.6	28.2	28.8	34.5
F-7	6.17	22	8.95	50	0.42	20.2	5.31	17.9	73.5	68.7	98.6	28.2	41.4	16.5	28.8	18.6

For tests F-1 through F-6 Aerophine 3418A was the only collector with F-7 utilizing only TNC 312. All tests achieved exceptional copper and gold recoveries at +96% and +73% respectively. The conditions from F-7 (pH 11.5, TNC 312) were selected as the optimal rougher baseline conditions due to the improved selectivity against iron sulfides when compared to the 3418A based tests (18.6% S recovery verses +34%

S recovery). Additionally, as the WIM ore would be processed through the New Britannia mill, flotation gold recoveries are not as critical due to the tails leach operation.

Following the selection of the baseline rougher conditions, two stage cleaner flotation tests were carried out without a regrind step. Final concentrate grade was achieved (+20% copper) at +96% copper recovery indicating this ore is amenable to the New Britannia flotation circuit.

FIGURE 13-12: WIM COMPOSITE SAMPLE CLEANER FLOTATION TESTWORK GRADE RECOVERY CURVES



The conditions identified in the master composite flotation testwork were repeated in for the three variability composites. Each of the variability samples achieved +99% copper and +65% gold recovery in the rougher stage. Outlined in Figure 13-12, each of the composite samples were able to achieve +20% copper concentrate grade at +95% recovery.

13.6. 3 ZONE METALLURGICAL TESTWORK

3 Zone is a satellite deposit located north of the Lalor deposit and is part of the historical New Britannia mine. This deposit consists of one ore zone that host of high-grade copper and moderate gold grade ore. In 2020 a metallurgical testwork program was completed by Blue Coast Research with the focus being the performance of the ore through the New Britannia mill. This ore is a primary gold ore with minimal sulfur content, to that extend only gravity concentration and cyanidation testwork was performed on the selected samples.

Sample Selection

Four composite samples were created for the 3 Zone ore. Three variability composite samples were created based on the spatial distribution. A single master composite was also created from subsamples of the three variability samples. The head grades for all four composite samples are outline in Table 13-18.

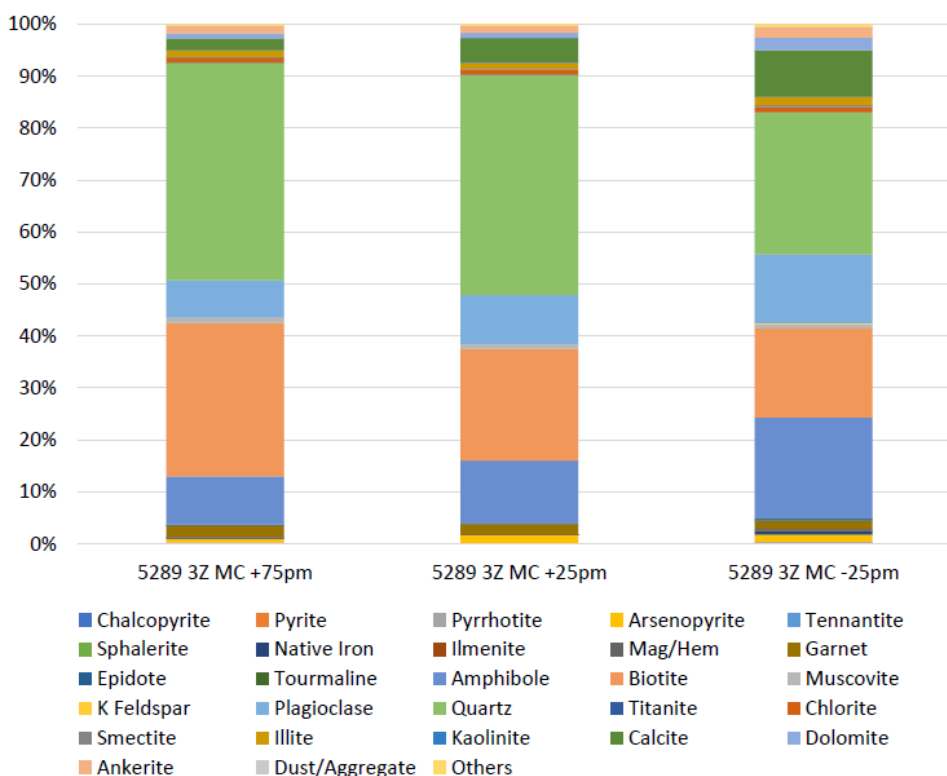
TABLE 13-18: 3 ZONE COMPOSITE SAMPLE HEAD GRADES

Composite ID	Au (g/t) FA-AA	Ag (g/t) AR-AA	As (%) AR-AA	Cu (%) AR-AA	Fe (%) AR-AA	S _{tot} (%) Eltra	C _{tot} (%) Eltra	C _{org} (%) Eltra
3Z-MC	5.33	0.73	0.52	0.01	4.53	0.34	1.28	0.02
VAR-H1	6.53	0.60	0.67	0.01	4.22	0.44	1.33	0.02
VAR-H3	3.34	0.70	0.41	0.02	4.52	0.29	0.97	0.01
VAR-H4	7.20	1.50	0.43	0.02	5.34	0.34	1.57	0.02

Mineralogy

A subsample of the master composite samples was ground to a p80 of 75µm and submitted for sized mineralogical analysis with the result outlined in. The dominant mineral species were quartz, amphibole, biotite, and plagioclase. The principal sulfide mineral is arsenopyrite.

FIGURE 13-13: 3 ZONE MASTER COMPOSITE SAMPLES SIZED MINEROLOGY



Leach Testwork

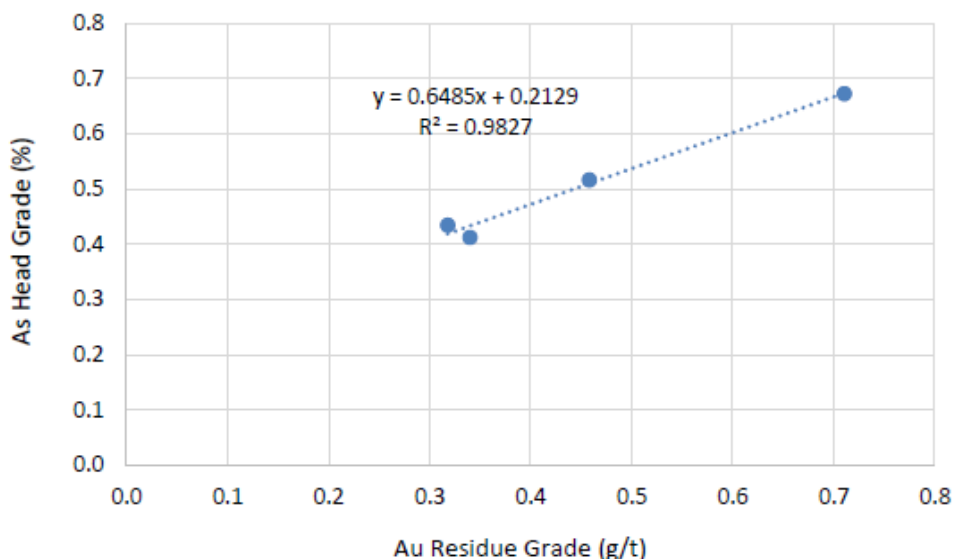
Leach tests were completed on each of the four composites with the bulk of the optimization testwork carried out on the master composite sample to determine the optimal conditions for the variable composites. The results of the initial seven optimization tests are outline in Table 13-19. All tests exceeded 89% gold recovery with exceptionally low consumption of cyanide indicating amenability to direct cyanidation. CN-3 conditions were carried through as the baseline test conditions.

TABLE 13-19: 3 ZONE MASTER COMPOSITE LEACH TESTWORK RESULTS

Test ID	Purpose	Final Solution Grade	Final Residue Grade	Recovery (%)	Reagent Consumption (kg/t)		Measured p80
		Au (g/t)	Au (g/t)	Au	CN	CaO	(µm)
CN-1	Grind sensitivity: 120 µm	4.06	0.50	89.5	0.10	0.46	128
CN-2	Grind sensitivity: 90 µm	4.31	0.49	90.1	0.16	0.49	92
CN-3	Grind sensitivity: 75 µm	4.65	0.46	91.3	0.15	0.59	75
CN-4	Pre-aeration 8 hours; 0 g/t PbNO ₃	3.82	0.44	90.0	0.10	0.60	73
CN-5	Pre-aeration 8 hours; 50 g/t PbNO ₃	4.54	0.44	91.4	0.07	0.72	75
CN-6	Pre-aeration 8 hours; 200 g/t PbNO ₃	3.66	0.45	89.5	0.12	0.74	73
CN-7	Grind sensitivity: 50 µm	3.81	0.40	90.7	0.74	0.50	50

The tests were repeated for the three variability composite samples with gold recovery and cyanide consumption rates 85 to 95% and 0.25 to 0.62kg/t, respectively. The higher cyanide consumption and lower recoveries positively correlate to the arsenic feed grade indicating that the gold in the 3 Zone ore is refractory in the arsenopyrite and that further work is required to identify the process route to recovery the fraction of gold unrecoverable by the current New Britannia leach circuit.

FIGURE 13-14: 3 ZONE ARSENIC HEAD GRADE CORRELATION WITH LEACH RESIDUE GOLD GRADE



Gravity Concentration Testwork

Gravity recovery amenability testwork was carried out on the 3 Zone master composite. The ground sample was initial upgraded through a Knelson concentrator. The Knelson concentrator achieved 78.6% gold recovery at a 1.2% mass recovery. This indicates that the 3 Zone ore present in the master composite is amenable to gravity concentration methods and that further testwork is required to map the gravity recoverable gold potential through the 3 zone orebody.

TABLE 13-20: 3 ZONE KNELSON CONCENTRATOR TESTWORK RESULTS

Product	Weight		Grade (g/t)		Recovery (%)	
	g	%	Au	Ag	Au	Ag
Knelson Con	116.3	1.17	351	42.3	78.6	50.7
Knelson Tail	9857.4	98.83	1.13	0.49	21.4	49.3
Total	9973.7	100.0	5.20	0.97	100.0	100.0
ERD Head	10000.0		5.33	0.73		
Reconciliation	99.7		97.7	132.6		

13.7. PEN II METALLURGICAL TESTWORK

Metallurgical test work conducted on composites representative of the Pen II deposit support the assumption that the zinc rich inferred mineral resource estimates could be processed at the Stall concentrator with similar performance as observed from Lalor and 1901 mineral reserve estimate. The reader is referred to the March 2019 Lalor and Snow Lake Technical report for more details on the test work conducted and results obtained.

14. MINERAL RESOURCE ESTIMATES

No known factors exist which could materially affect the mineral resource estimates.

14.1. LALOR MINE

Hudbay prepared an update of the Lalor mine resource model using Leapfrog® version 5.1.1 and MineSight® version 15.60, two industry standard commercial geological and mining software. The construction of this 3D resource model and the estimation of mineral resources were performed by Hudbay personnel following Hudbay procedures in compliance with best industry standards and the CIM guidelines. The work was supervised and validated by Olivier Tavchandjian, P.Geo., Vice President, Exploration and Geology and Qualified Person for the purpose of the present Technical Report.

Modeling Database

4,118 drill holes totaling approximately 650,000 m were included in the Lalor database to support the mineral resource estimate. Amongst these samples approximately 397,300 m were analyzed for Zn, Cu, Au and Ag.

The drill hole database was directly imported in MineSight® from acquire® with a cut-off date for mineral resource estimate purposes of September 8th, 2020. In parallel, the drill hole database was also imported in Leapfrog® for additional and independent comparison and validation. No significant differences in the projection of the drill hole traces were found between the two datasets.

Modeling of the Mineralized Envelopes

The Lalor mineralized envelopes trend along an azimuth of approximately N320° with a general dip of approximately 37° to the north east. Economic mineralization delineated so far at Lalor occurs in nine semi-massive to massive sulfide lenses typical of VMS deposits (lenses 10, 11, 20, 30, 31, 32, 33, 40 and 42), three conformable gold-rich disseminated to stringer zones (lenses 21, 23 and 25), two copper-gold rich feeders (lenses 17 and 27) and as small footwall and hanging wall gold rich zones associated to lens 10 (lenses 9, 13 and 16). The base metal and gold rich zones are stratigraphically stacked and are mostly aligned to the main regional deformation, averaging N320°/37°. The Lalor mine is continuous along a strike length of approximately 1,400 m in north-south direction, approximately 780 m in an east-west direction and with a vertical extent of approximately 760 m.

Geological units of sulphides horizons were created in 3D using core logging data and assay results. These broad geological envelopes were then refined to create base metal and gold rich mineralized envelopes. The distribution of gold mineralization follows two styles: the first one, which is synchronous to the base metal deposition, is directly associated and contained within the base metal mineralized envelopes, typically on their footwall. The second one, posterior to the base metal deposition, is contained outside the sulphides horizons and is typically sulphides poor with the exception of the Au-Cu rich mineralized envelopes 17 and 27 which are Au-Cu copper stringer zones.

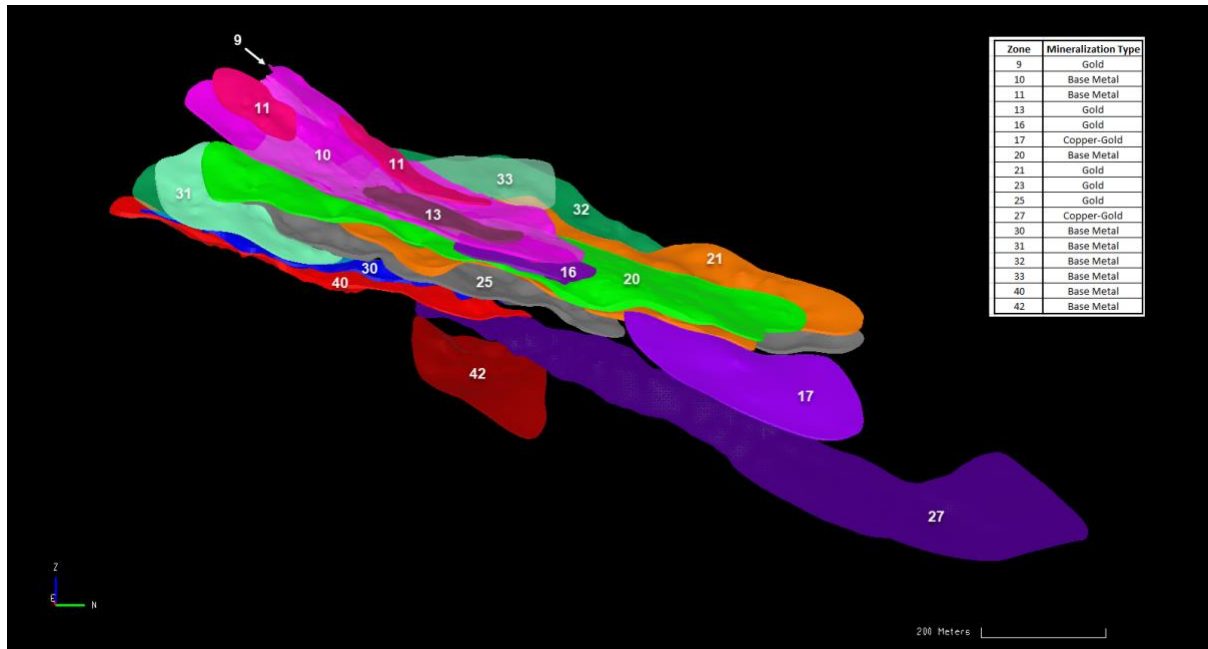
The construction of the mineralized envelopes was based on the type of mineralization intersected. Smooth and continuous 3D solids were built using 4% zinc or 1% copper cut-off to guide the construction of the base metal mineralized envelopes, with the inclusion of any nearby gold intersections, while a 2.0 to 2.5 g/t cut-off was used to guide the construction of the gold rich mineralized envelopes.

In a final validation step, these envelopes were loaded into MineSight® in order to ensure proper tagging of the solids to actual drill hole locations and that drillhole traces matched both in Leapfrog and Minesight. The mineral envelopes were used as hard boundaries in all cases for grade interpolation purposes so as to correctly prevent spreading of sulfide mineralization into the gold zones and vice-versa. Wireframes of the mineralized envelopes are shown in Figure 14-1.

Density

Density values generated from multi regression formulas are based on 25,182 assay measurements. The predicted density values were validated through comparisons against the measured density.

FIGURE 14-1: 3D VIEW OF MINERAL ENVELOPE WIREFRAMES, LOOKING WEST



Note: Lens 23 is located under lens 32 and not shown in this view.

Compositing

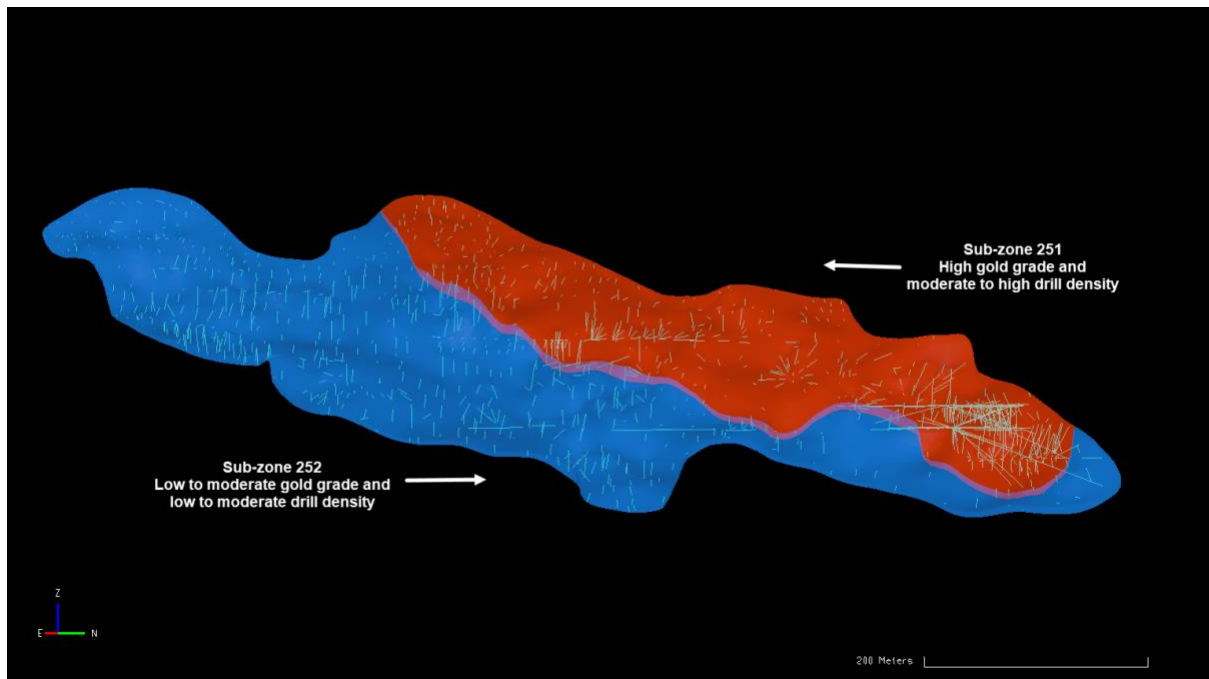
Assay intervals were regularized by compositing drill hole data within each interpreted mineralized envelope. Although most of the drill holes were assayed on 1m interval, a composite length of 2.5 m was selected as more appropriate to conduct interpolation into the 5 m x 5 m x 5 m block size selected by the mine engineering team as the optimum selective mining unit (SMU).

Grade compositing was weighted by specific gravity as there is a significant positive correlation between some of the metal grade and density. The compositing process was validated by comparing total length, density and length weighted average grade for each metal of the 2.5 m composites to the original assays.

Exploratory Data Analysis

Exploratory data analysis (EDA) includes basic statistical evaluation of the composites for gold (Au), silver (Ag), copper (Cu), zinc (Zn), and density (SG). The EDA was conducted separately for each mineralized envelope and also aimed to identify sub-domains that did not support the use of hard contacts for grade interpolation purposes but that justified being used for the purpose of block model validation, smoothing assessment and correction and resource classification. Capping analysis and smoothing assessment were also conducted separately in each sub-domain. An example of sub-domaining is illustrated for gold zone 25 on Figure 14-2 where the high-grade core of the lens is subdivided based on drilling coverage and also from its lower grade halo as defined by Leapfrog Geo interpolations.

FIGURE 14-2: SUB-DOMAIN IN MINERALIZED ENVELOPE 25



Note: drill hole traces represented by the lines in grey

The 2.5 m composite statistics for Au, Ag, Cu, Zn and calculated SG are summarized in Table 14-5 in the grade estimation validation section. As expected, zinc shows significantly lower CV and skewness values in the base metal lenses while the opposite is observed for gold.

Grade Capping

Because the composites of the four metals of economic potential show highly skewed histograms in most of the mineralized envelopes, the deciles² analysis (Parish) method was used to define high-grade outliers and to confirm the need for grade capping. This analysis was conducted on the 2.5 m composites for each sub-domain independently.

This method considers capping when the last decile of the population contains more than 40% of the metal and that the last percent contains more than 10% of the metal. Capping values were selected to limit the weight of the high-grade outliers on the overall population. Table 14-1 summarizes the capping applied to all the mineralized envelopes for gold and silver and the proportion of metal removed through this process. In all cases, it was found that no capping would be required for zinc or copper.

² I.S. Parrish (1997) Mining Engineering Journal, Geologist's Gordon Knot: to cut or not to cut (pp. 45-49)

TABLE 14-1: 2021 CAPPING THRESHOLDS BY LENS OF GOLD AND SILVER (G/T)

Mineralized Zone	Total number of composites	Au capping value (gpt)	Number of composites capped	Metal removed	Ag capping value (gpt)	Number of composites capped	Metal removed
Z09	174	20	4	6.0%	400	3	3.8%
Z10	5325	50	8	13.0%	250	42	4.5%
Z11	371	8	3	4.0%	275	6	3.3%
Z13	211	15	2	2.0%	350	3	2.4%
Z16	276	30	2	8.0%	80	4	4.0%
Z17	101	25	2	26.0%	60	2	2.9%
Z20	6218	40	15	3.1%	260	24	1.8%
Z21 HGO	4024	95	15	5.0%	400	2	0.7%
Z21 LGO	1491	30	4	3.0%	300	4	0.7%
Z23	982	50	8	5.0%	300	3	0.9%
Z25 HGO	3493	145	9	5.1%	375	10	2.6%
Z25 LGO	3239	60	11	6.9%	220	12	2.0%
Z27	1971	65	10	1.0%	250	8	1.6%
Z30	860	25	3	0.8%	200	3	1.2%
Z31	2105	50	6	4.5%	330	6	0.4%
Z32 HGO	2802	80	15	5.0%	400	2	1.0%
Z32 LGO	1612	25	5	2.0%	300	1	0.3%
Z33	358	15	3	14.0%	200	5	5.0%
Z40	1496	35	5	9.0%	300	6	2.5%
Z42	569	25	1	14.0%	-	-	-

Note: HGO = high grade and LGO = low grade

The results obtained from the Parrish method were then compared to the natural breaks in the cumulative frequency distribution in the last decile. The metal at risk was also assessed in the population of outliers by comparing the difference in metal using the median and mean values of these outliers. As mentioned in our previous NI43-101, an independent third-party review was conducted in November 2018 by Mr. Dominique Francois-Bongarçon (Agoratek) who concurred with the use of the Parrish as a valid approach for gold capping in the gold zones.

Variography

Down-hole and directional variograms for Zn, Au, Cu, Ag, and SG were created for each individual mineral envelope using the MineSight data analysis software. Pairwise relative variogram were preferred due to the skewed nature of metal distribution in most lenses. The major, semi-major and minor axis were built in order to fit the attitude of each mineral envelope. A linear combination of a nugget and two nested spherical models were adjusted in all cases. Once generated, a systematic visual check was conducted to ensure that the search ellipsoid would be correctly oriented with respect to the geometry of the mineral envelopes.

Grade Estimation and Interpolation Methods

The block model consists of regular blocks (5 m along strike by 5 m across strike by 5 m vertically). The block dimensions were selected to match the smallest mining unit (SMU) at Lalor mine. Where a block was intersected by two lenses, the mineral envelopes were used to assign the percentage of the block that belongs to each lens.

Both nearest neighbour (NN) and ordinary kriging (OK) grade interpolations were completed on the uncapped and capped grades, using a strict composite and block matching code by lens and three passes with increasing minimum information requirements (Table 14-2). The search passes were selected to

ensure best local estimates recognizing that OK has a smoothing effect but making no attempt during interpolation to reduce this smoothing as it would negatively impact the quality of the local estimates. Over-smoothing is addressed through the post-processing of the model described in sub section “smoothing assessment”.

Grade Estimation Validation

The grade estimation process was validated for each envelope to ensure appropriate honouring of the input data and subsequent unbiased resource reporting and use of the model for reserve estimation.

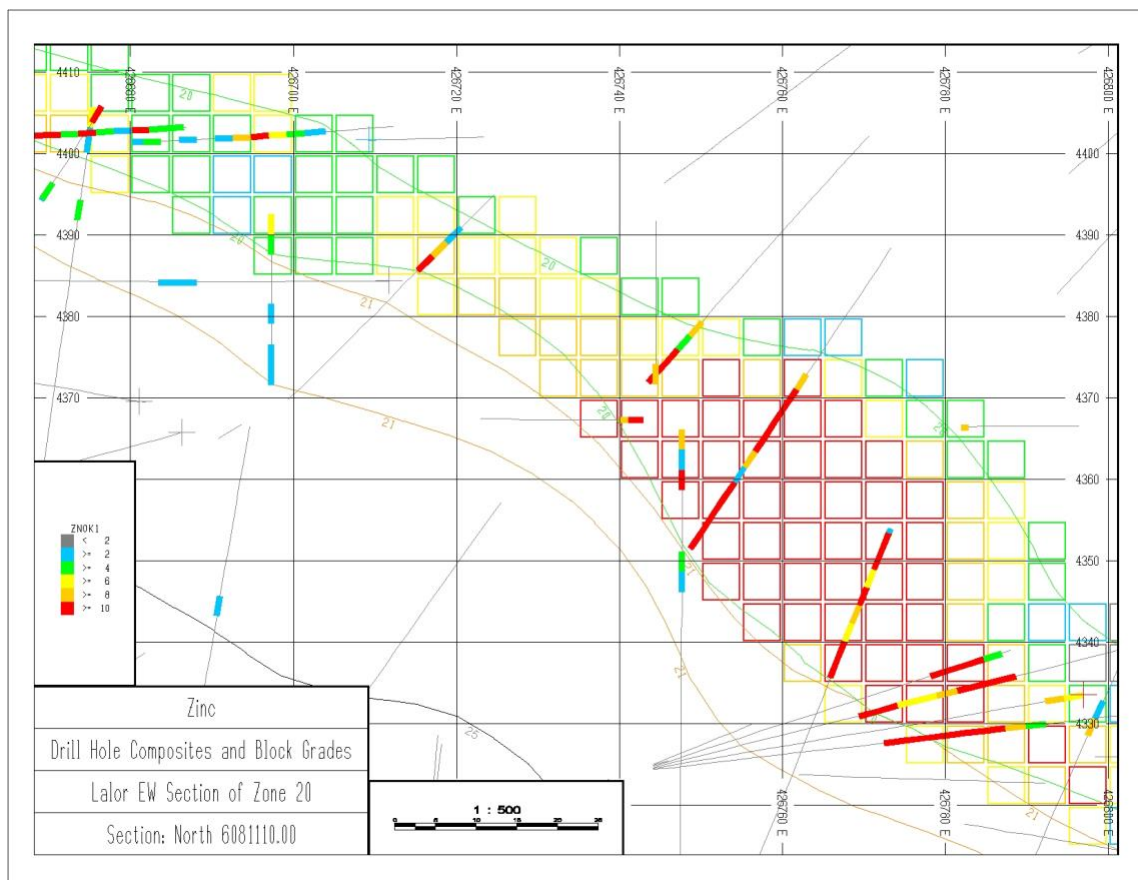
TABLE 14-2: SEARCH ELLIPSE PARAMETERS

	Min # of Comps	Max # of Comps	max # of Comps per hole	Declustering	max Comps per Quadrant	Search ellipse
Pass #1	1	32	6	no	-	150% of variogram ranges
Pass #2	16	32	6	yes	8	75% of variogram ranges
Pass #3	16	32	6	yes	8	50% of variogram ranges

Visual Inspection

Visual inspection of block grade versus composited data was systematically conducted in section and plan views. This check confirmed a good reproduction of the data by the model. An example is illustrated in Figure 14-3 for the zinc grade in zone 20.

FIGURE 14-3: VISUAL CHECK ON MODEL FOR ZN GRADE IN LENS 20



Global Bias Checks

This validation step consists in comparing the global average grade of each element (after capping for gold and silver) between the original composites, the kriged block estimates and the nearest neighbour estimates. This investigation was conducted not only by envelope but also for each individual sub-domain. The average grade obtained from the nearest neighbor interpolation is a useful benchmark but not a perfect one as it fails to incorporate the nugget effect measured by the variogram. A global check was performed to verify that the kriged mean block estimate was located between the mean of the composites and the mean of the nearest neighbour model.

Smoothing Assessment

The larger number of composites used for grade estimation in the block model significantly improves the individual block grade estimates but, at the same time, results in a much smoother model requiring a careful assessment and, in many cases, a post-processing of the OK estimates. The extent of grade 'over-smoothing' in the model was investigated separately by sub domains based on material differences in grade distribution and/or drilling density. The mean and variance of the kriged estimates were compared to the variance of the composites after declustering (obtained from a nearest neighbour interpolation). The expected true variance between SMUs can be calculated from the variogram models.

Smoothing Correction

An indirect log-normal correction was used to perform a change of support on the kriged models in order to obtain unbiased grade tonnage curves for the selected mining block size of 5m x 5m x 5m. This correction is only valid globally and provides poorer local estimates than the smoothed OK model but does not materially alter the global average grade within each zone and provides the correct grade-tonnage curve for the variogram models fitted on the drill hole data. This correction was applied by domain with consistent statistical properties and uniform drilling coverage. The targeted variance was reached within acceptable limits in most cases, as illustrated in Table 14-3.

TABLE 14-3: SUMMARY OF THE APPLIED SMOOTHING CORRECTION

Lens	Sub Zone	OK Mean	NN model variance	OK model variance	Smoothing Ratio	Expected Variance	Achieved variance
9	<i>Au global</i>	3.18	18.48	4.60	2.07	9.54	8.33
10	Zn HG	7.62	67.84	23.10	1.56	36.01	33.7
	Zn LG	1.30	4.08	1.47	1.47	2.17	1.96
11	<i>Zn global</i>	7.78	55.33	22.20	1.32	29.37	27.49
13	<i>Au global</i>	2.38	11.84	1.91	3.20	6.11	4.85
16	<i>Au global</i>	3.32	32.25	7.96	2.09	16.66	15.71
17	<i>Au global</i>	2.55	10.66	0.65	7.88	5.1	4.92
20	Zn HG	5.89	44.21	17.30	0.81	14.02	17.3
	Zn LG	2.51	13.53	3.57	1.20	4.29	3.57
	Zn LG	2.20	9.32	1.20	2.46	2.96	2.47
21	Au LG	2.70	27.97	2.89	2.91	8.41	7.07
	Au HG	4.82	117.30	16.14	2.19	35.27	31.85
23	Au HDD	4.75	51.52	5.86	2.89	16.96	16.1
	Au LDD	5.07	69.01	7.90	2.88	22.72	18.58
25	Au HG	6.54	220.90	26.15	2.80	73.17	69.53
	Au LG	2.64	13.63	1.51	2.99	4.52	4
27	Au HG & HDD	5.21	91.94	10.63	3.56	37.82	29.1
	Au HG & LDD	3.91	26.78	2.38	4.63	11.02	7.47
30	Zn HG	5.83	25.50	3.22	2.01	6.47	6.33
	Zn LG	2.99	12.59	1.46	2.19	3.19	3.19
31	Zn HDD	3.63	16.92	5.80	1.05	6.08	5.8
	Zn LDD	3.48	16.89	5.29	1.15	6.07	5.29
32	Zn HG	5.99	47.55	16.79	0.99	16.64	16.79
	Zn LG	2.87	9.77	2.57	1.33	3.42	3.23
33	<i>Zn global</i>	2.84	12.19	1.85	2.31	4.26	3.56
40	Zn HG	5.72	16.69	3.86	1.64	6.31	5.88
	Zn LG	4.97	15.67	3.55	1.67	5.93	5.24
42	Zn LG & HDD	4.07	28.64	7.87	1.16	9.1	8.97
	Zn HG & LDD	6.27	24.95	6.17	1.28	7.92	7.72

Note: HG = high grade, LG = low grade, HDD = high drilling density & LDD = lower drilling density

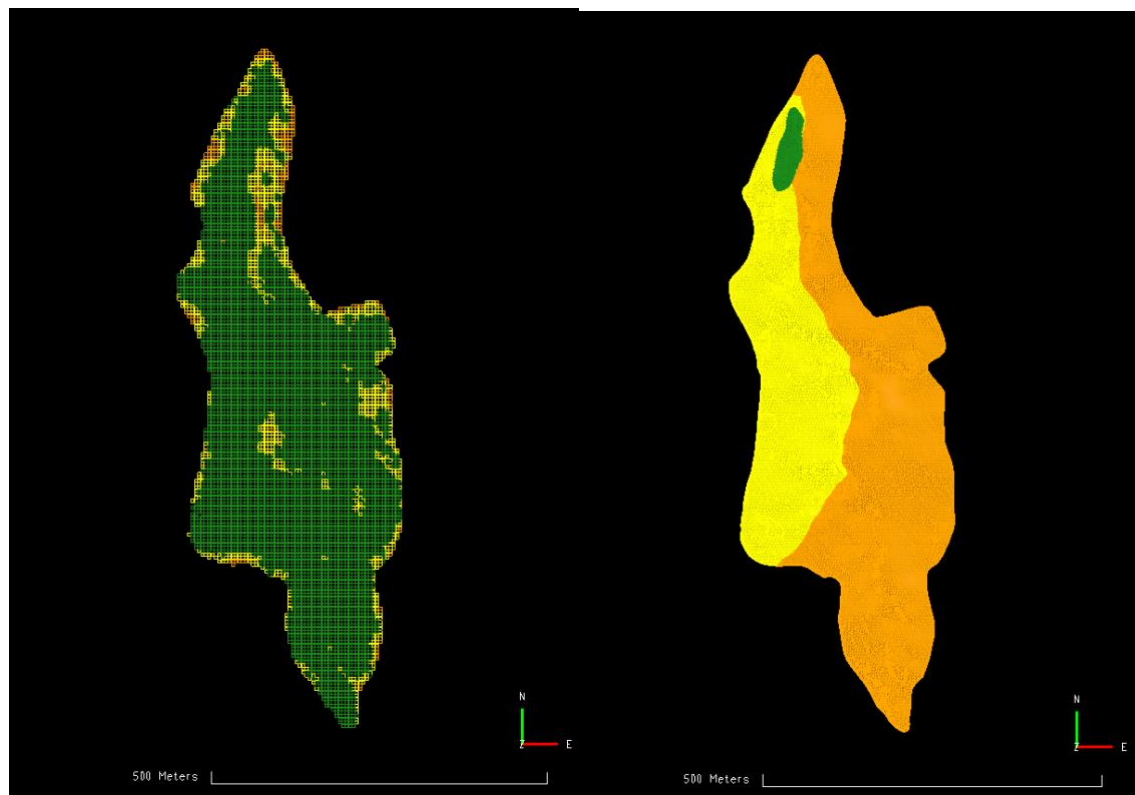
Classification of Mineral Resources

During the interpolation process, several control parameters were recorded for each block, e.g., number of samples, number of holes, the distance to the nearest sample and the average distance to all the samples used for the interpolation as well as the number of quadrants with samples, the kriging variance and the regression slope of kriging for each individual block estimate.

The regression slope values obtained from the kriging of zinc (in the case of base metals lenses) or gold (in the case of gold zones) grade estimates was used as the primary criteria for resource classification with 80% and 60% regression slope thresholds used respectively to separate measured from indicated and from inferred resources. From detailed reserves to mill reconciliations exercises conducted by Hudbay at its operating mines, including Lalor, this criteria was found to be a reliable first pass measure of quarterly and annual performance in tonnes and grade prediction.

The block by block coding assignment was then smoothed to remove isolated blocks of one category within another. Proportions of measured, indicated and inferred category blocks were not changed significantly through this process. In assigning the final categorization, careful consideration was also given to the proportion of mineralization grading above the anticipated cut-off grade. Gold lenses followed a more prudent approach in which some categories were downgraded and measured was only applied in areas of high drill density or which was silled through with development. Figure 14-4 illustrates the classification process followed for Lens 25.

FIGURE 14-4: RESOURCE CLASSIFICATION IN MINERALIZED ENVELOPE 25



Note: the insert on the left presents the block by block resource classification (Measured in green, indicated in yellow and Inferred on orange) based on the regression slope of gold in mineralized envelope 25 while the insert on the right present the final classification. Confidence was down graded based on the proportion of blocks above cut off.

Reasonable Prospects of Economies Extraction and Mineral Resource Estimates

Since 2019, Hudbay has implemented a stringent approach to resource reporting for underground deposits. With this approach, the potential for economic extraction of the mineral resource estimates at Lalor are reported within the constraint of a mineable stope-shape, created using Stope Shape Optimizer (SSO). The SSO application mimics what a mine planner would do, generating strings on sections, linking these to create a wireframe shape and then evaluating the wireframes against a block model. The SSO provides a stope-shape that maximizes recovered resource value above a cut-off while also catering for practical mining parameters (such as minimum and maximum mining width, anticipated wall dilutions, minimum and maximum wall angles, minimum separation distances between parallel and sub-parallel stopes, minimum and maximum stope heights and widths, etc.). The mining parameters used to build the mineable shapes are detailed in Section 15.

NSR cut-off values are used at Hudbay in the SSO process, whereby a calculation of value per tonne is applied to the block model based on anticipated metal recoveries, metal prices and exchange rate, and concentrate transportation, smelting, and refining costs. The cut-off value input parameter in SSO considers operating and capital costs, as based on the current Lalor mine and mill budgets, to ensure that each potential stope created would cover all its associated operating mining and milling costs to be deemed profitable if mined. This process ensures that reported mineral resource estimates exclude small isolated individual resource blocks that may meet economic cut-off criteria on an individual basis but could not be aggregated into mineable shapes and include 'geological dilution' (representing small blocks grading below cut-off) in the economic shape to maintain minimum mining criteria.

The stope optimization process was run on all resource categories combined and as a result, a small amount of one resource category could be included in a stope shape dominated by another resource category. This usually occurred on the fringes of the zones defined for resource classification. Resource classification is based on broad smoothing of interpolation parameters used in the construction of the resource model and visual inspections confirmed that minor modifications to the resource categories were warranted so that all stope shapes would be entirely assigned to one resource category.

The resulting mineral resource estimates, as of January 1, 2021, are presented in Table 14-4. There are no measured or indicated mineral resource estimates in addition (exclusive) to those that were converted to mineral reserve estimates. All measured and indicated mineral resource estimates were either converted to mineral reserve estimates, as documented in section 15, or deemed non-economic after applying the appropriate resource to reserve conversion factors.

TABLE 14-4: LALOR MINERAL RESOURCE ESTIMATES

Lalor Mineral Resource Estimates (Exclusive of Mineral Reserves) – January 1, 2021						
		Tonnes	Cu (%)	Zn (%)	Au (g/t)	Ag (g/t)
Base Metal	Inferred	590,000	0.31	3.48	2.8	55.0
Gold	Inferred	5,610,000	1.17	0.35	4.6	26.0
Base Metal and Gold	Inferred	6,200,000	1.09	0.65	4.5	28.7

Notes:

1. Totals may not add up correctly due to rounding.
2. Mineral resources that are not mineral reserves do not have demonstrated economic viability.
3. Mineral resources in the above tables do not include mining dilution or recovery factors.
4. Base metal mineral resources are estimated based on the assumption that they would be processed at the Stall concentrator while gold mineral resources are estimated based on the assumption that they would be processed at the New Britannia concentrator, which is currently being refurbished.
5. Metal prices of US\$1.10 per pound zinc, US\$1,500 per ounce gold, US\$3.10 per pound copper, and US\$18.00 per ounce silver with an exchange rate of 1.30 C\$/US\$ were used to estimate mineral resources.
6. Mineral resources are estimated using a minimum NSR cut-off of \$105 per tonne for waste filled mining areas and a minimum of \$116 per tonne for paste filled mining areas.

14.2. 1901 DEPOSIT

Hudbay prepared resource model using Leapfrog® version 5.0.3 and MineSight® version 15, two industry standard commercial geological and mining softwares. The construction of this 3D resource model and the estimation of mineral resources were performed by Hudbay personnel following Hudbay procedures in compliance with best industry standards and the CIM guidelines. The work was supervised and validated by Olivier Tavchandjian, P.Geo., Vice President, Exploration and Geology.

Modeling Database

Drill holes totaling approximately 80,600 m were included in the 1901 database to support the mineral resource estimate update. Amongst these, approximately 18,300 were analyzed for Zn, Cu, Au and Ag while density was measured for 15,150m.

The drill hole database was directly imported in Leapfrog from acQuire® with a cut-off date for mineral resource estimate purposes of June 9th, 2020. The drill hole database was then exported from Leapfrog and imported into MineSight via .csv files. The drillhole traces in MineSight were validated against the drillhole traces in Leapfrog to ensure no errors were introduced during the import process.

Modeling of the Mineralized Envelopes

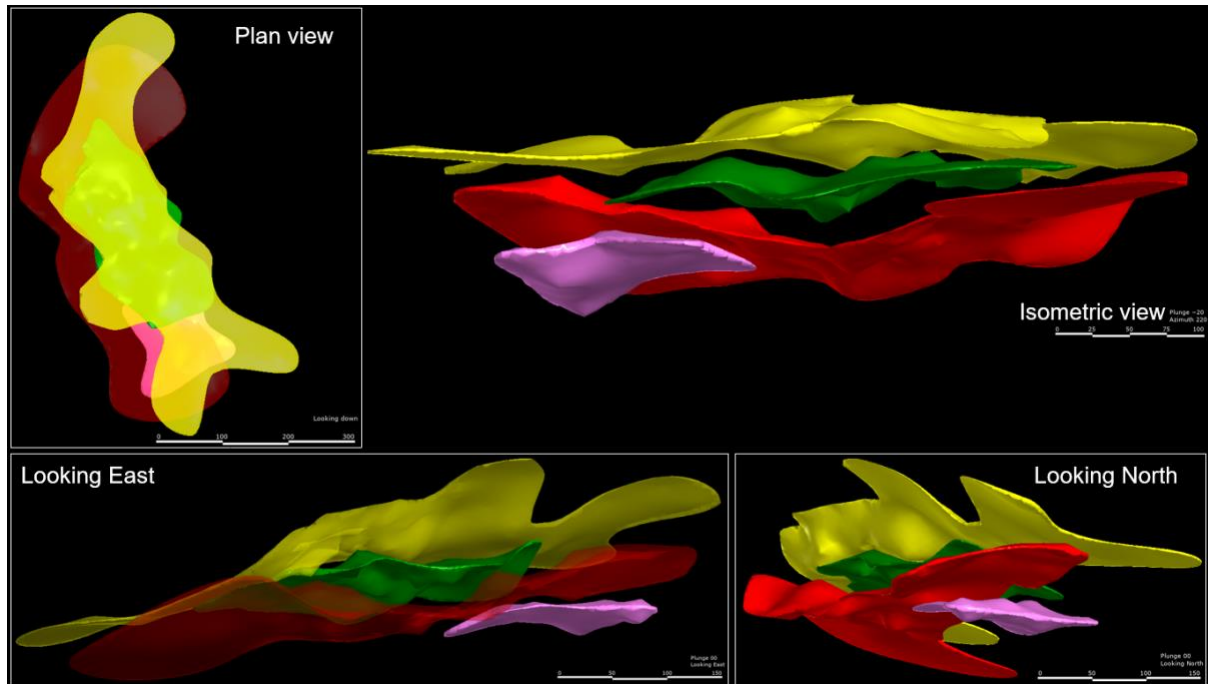
The 1901 mineralized envelopes trend along an azimuth of approximately 340° with a general dip of approximately 25° to the east. Economic mineralization delineated so far occurs in two semi-massive to massive sulfide lenses typical of VMS deposits (lens 1 & 3) and two conformable gold-rich disseminated to stringer zones (lens 2 & 4). The base metal and gold rich zones are stratigraphically stacked and are mostly aligned to the main regional deformation. The mineralization is continuous along a strike length of approximately 550 m in north-south direction, approximately 230 m in an east-west direction and with a vertical extent of approximately 80 m.

Geological units of sulphides horizons were created in 3D using core logging data and assay results. These broad geological envelopes were then refined to create base metal and gold rich mineralized envelopes. The distribution of gold mineralization appears follows two styles: the first one, which is synchronous to the base metal deposition, is directly associated and contained within the base metal mineralized envelopes, typically on their footwall. The second one, posterior to the base metal deposition, is contained outside the sulphides horizons and is typically sulphides poor.

The construction of the mineralized envelopes was solely based on the type of mineralization intersected. To this effect, assay intervals with high Au grade but no zinc or copper were excluded from base metal lenses and vice versa. Instead, smooth and continuous 3D solids were built using 3% zinc or 1% copper cut-off to guide the construction of the base metal mineralized envelopes while a 2 g/t cut-off was used to guide the construction of the gold rich mineralized envelopes.

In a final validation step, the envelopes were loaded into MineSight in order to ensure proper tagging of the solids to actual drillhole locations. The mineral envelopes were used as hard boundaries in all cases for grade interpolation purposes so as to correctly prevent spreading of sulfide mineralization into the gold zones and vice-versa. Wireframes of the mineralized envelopes are shown in Figure 14-5.

FIGURE 14-5: VIEW OF MINERAL ENVELOPE WIREFRAMES



NOTE: Lens 1 in yellow, 2 in green, 3 in red and 4 in fuchsia

Density

Since almost all the assay intervals contained within the different mineralized lenses had density measurements (only two samples with missing density in lens 3), density was only predicted for the barren material surrounding the mineralized zones. As for the two missing density measurements in lens 3, they were set to the average density measured in the lens (i.e. 3.52). Density values generated from a multi regression formula is based on 14,408 density measurements. The predicted density values were validated through comparisons against the measured density.

Compositing

Assay intervals were regularized by compositing drillhole data within each interpreted mineralized envelopes. Although most of the drillholes were assayed on 1m interval (Figure 7 4), a composite length of 2.5 m was selected as more appropriate to conduct interpolation into the 5 m x 5 m x 5 m block size selected by the project evaluation team as the optimum selective mining unit (SMU). Grade compositing was weighted by specific gravity as there is a positive correlation between some of the metal grade and density. The compositing process was validated by comparing total length, density and length weighted average grade for each metal of the 2.5m composites to the original assays.

Exploratory Data Analysis

Exploratory data analysis (EDA) includes basic statistical evaluation of the composites for zinc (Zn), gold (Au), copper (Cu), silver (Ag) and density (SG). The EDA was conducted separately for each mineralized envelope. Within each mineral envelope, the EDA also aimed to identify sub-domains that did not support the use of hard contacts for grade interpolation purposes but that justified being used for the purpose of block model validation, smoothing assessment and correction and resource classification.

The 2.5 m composite statistics for Zn, Au Cu, Ag and calculated SG are summarized in Table 7 5 in the grade estimation validation section. Overall, the statistics of the composites indicate that the kriging interpolation will as expected introduce some level of smoothing and that a thorough smoothing

assessment and post-processing of the initial interpolation is warranted to produce unbiased grade-tonnage curves and mineral resource and mineral reserve estimates.

Grade Capping

The deciles analysis (Parish) method was used to define high-grade outliers and to assess the need for grade capping. This analysis was conducted on the 2.5m composites for each sub-domain independently.

This method considers capping when the last decile of the population contains more than 40% of the metal and that the last percent contains more than 10% of the metal. Based on this method, gold was capped at 14 PPM in lens 1, 75 PPM in lens 2 and 20 PPM in lens 3, while silver was capped at 375 PPM in lens 2 and 160 PPM in lens 3. Capping was not applied to lens 4 composites given their limited numbers. Overall, capping removed the gold metal content by 2.7% in lens 1, 19.4% in lens 2 and 10.6% in lens 3, while capping removed the silver metal content by 15% in lens 2 and 2% in lens 3.

Variography

Down-hole and directional variograms for Zn, Au, Cu, Ag, Fe, Pb, As and SG were created for lens 1, 2 and 3 using the MineSight data analysis software. Given the limited number of composites and lens 4, the variogram parameters of lens 2 were adapted to match the nugget of lens 4 and its overall strike and dip.

The major, semi-major and minor axis were built in order to fit the attitude of each mineral envelope. A linear combination of a nugget and two nested spherical models were adjusted in all cases. Once generated, a systematic visual check was conducted to ensure that the search ellipsoid would be correctly oriented with respect to the geometry of the mineral envelopes.

Grade Estimation and Interpolation Methods

The block model consists of regular blocks (5 m along strike by 5 m across strike by 5 m vertically). Grade interpolation used a strict composite and block matching system based on the mineralized envelopes codes. For example, only composites coded as lens 1 were used to estimate the grade of the block that was assigned to lens 1. Both nearest neighbour (NN) and ordinary kriging (OK) grade interpolations were completed, using three passes with increasing minimum information requirements (Table 7 4). The search passes were selected to ensure best local estimates recognizing that OK has a smoothing effect but making no attempt during interpolation to reduce this smoothing as it would negatively impact the quality of the local estimates.

TABLE 14-5: SEARCH ELLIPSE PARAMETERS

	Min # of Comps	Max # of Comps	max # of Comps per hole	Declustering	max Comps per Quadrant	Search ellipse
Pass #1	1	32	6	no	-	150% of variogram ranges
Pass #2	16	32	6	yes	8	75% of variogram ranges
Pass #3	16	32	6	yes	8	50% of variogram ranges

The NN interpolation is solely used to validate that there is no global bias and to assess the level of grade smoothing in the OK interpolation for the selected SMU dimension.

Grade Estimation Validation

The grade estimation process was validated for each envelope to ensure appropriate honouring of the input data and subsequent unbiased resource reporting and use of the model for reserve estimation through the following steps:

- Visual checks of appropriate honouring of the input data but acknowledging that some natural smoothing should occur between samples as the grade of a sample in the middle of a block is not the average grade of the block;

- Absence of global bias by comparing the mean grade estimated by kriging to the original composite average grade and to a declustered grade obtained from a nearest neighbour interpolation.
- Change of support assessment and correction.

Visual Inspection

Visual inspection of block grade versus composited data was systematically conducted in section and plan views. This check confirmed a good reproduction of the data by the model.

Global Bias Checks

This validation step consists in comparing the global average grade of each element between the original composites, the kriged block estimates and the nearest neighbour estimates. This investigation was conducted by envelope.

A nearest neighbour interpolation is equivalent to the declustered statistics of the composites based on weighting each composite by its polygon of influence. The average grade obtained from this method is a useful benchmark but not a perfect one as it fails to incorporate the nugget effect measured by the variogram.

A global check was performed to verify that the kriged mean block estimate was located between the mean of the composites and the mean of the nearest neighbour model. Differences between the 2.5m composites, the NN and OK grades are acceptable in all the sub-domains within each mineralized envelope.

Smoothing Correction

An indirect log-normal correction was used to perform a change of support on the kriged models in order to obtain unbiased grade tonnage curves. For some of the elements, the correction did not fully attain the targeted variance reflecting that the log-normal model does not perfectly fit these elements. However, the targeted variance was reached within very close limits in most cases (Table 14-6).

TABLE 14-6: SUMMARY OF SMOOTHING CORRECTIONS – 1901 DEPOSIT

	Lens	OK Mean	NN model variance	OK model variance	Smoothing Ratio	Expected Variance	Achieved variance
Zinc	Lens 1	7.29	46.74	12.55	1.54	19.32	19.84
	Lens 2	0.70	0.81	0.20	1.97	0.40	0.32
	Lens 3	5.50	20.78	4.39	2.78	12.18	8.87
	Lens 4	0.09	0.02	0.01	0.72	0.01	0.01
Copper	Lens 1	0.20	0.026	0.007	1.22	0.009	0.009
	Lens 2 HG	1.73	4.088	0.506	5.04	2.555	1.323
	Lens 2 LG	0.35	0.172	0.084	1.28	0.107	0.103
	Lens 3	0.58	1.113	0.468	1.53	0.716	0.553
	Lens 4	0.65	0.405	0.042	6.07	0.253	0.214
Gold	Lens 1	1.24	3.67	0.26	4.96	1.29	1.26
	Lens 2 HG	10.81	152.51	13.96	4.40	61.46	48.26
	Lens 2 LG	2.84	10.79	0.74	5.87	4.35	5.03
	Lens 3	1.71	6.93	1.13	3.48	3.93	4.02
	Lens 4	1.94	3.55	0.15	9.52	1.42	1.51
Silver	Lens 1	31.64	1521.60	182.52	2.95	538.58	571.40
	Lens 2 HG	56.61	7212.99	608.83	6.19	3771.36	2271.84
	Lens 2 LG	17.90	393.61	64.39	3.20	205.80	211.40
	Lens 3	24.49	674.92	200.42	2.19	439.74	419.52
	Lens 4	3.58	13.02	1.36	5.01	6.80	6.60

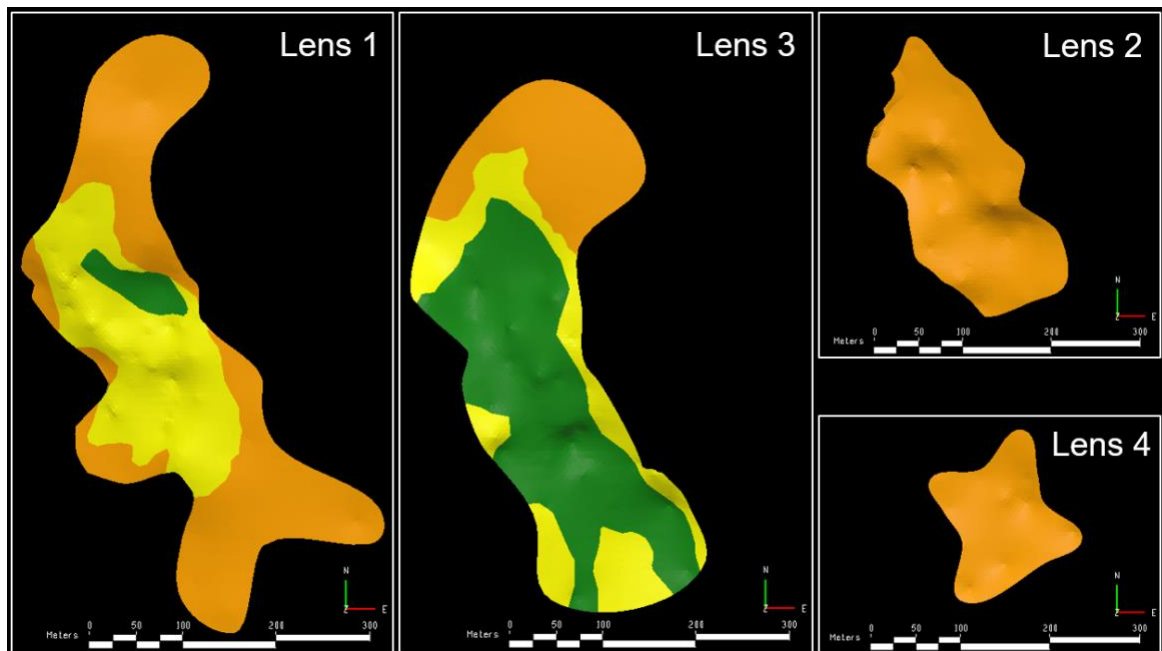
Note: HG = high grade & LG = low grade

Classification of Mineral Resource

As for Lalor, the regression slope values obtained from the kriging of zinc (for lens 1 and 3) grade estimates was used as the primary criteria for resource classification. Blocks from the second and third interpolation pass combined with a 60% regression slope thresholds were separate indicated from inferred resources. The block by block coding assignment was then smoothed to remove isolated blocks of indicated within areas of mostly inferred category or isolated inferred blocks within areas of mostly indicated category blocks. Proportions of indicated and inferred category blocks were not changed significantly by this process. Figure 14-6 presents a plan view of the resource categories.

Given the lower level of confidence caused by the highly skewed distribution of gold and limited amount of drilling, all blocks contained within lens 2 and 4 were classified as inferred.

FIGURE 14-6: PLAN VIEW OF THE RESOURCE CLASSIFICATION



NOTE: MEASURED IN GREEN, INDICATED IN yellow and Inferred in orange

As for Lalor, the mineral resource estimates for 1901 are reported within the constraint of an optimized stope-shape. The parameters used as input to define the mineable shape covers all the relevant technical and economic constraints including minimum stope and waste pillar dimensions and the NSR value calculation are very similar to those used at Lalor for the base metal and gold lenses assuming beneficiation at the Stall and New Britannia mills. The mineral resource estimates were reported to ensure that each potential stope would cover all its associated operating and capital costs. The resulting mineral resource estimates are presented in Table 14-7.

TABLE 14-7: 1901 MINERAL RESOURCE ESTIMATES

1901 Mineral Resource Estimates (Exclusive of Mineral Reserves) – January 1, 2021						
		Tonnes	Cu (%)	Zn (%)	Au (g/t)	Ag (g/t)
Base Metal	Inferred	310,000	0.85	6.44	2.0	24.6
Gold	Inferred	480,000	0.72	0.55	6.7	37.3
Base Metal and Gold	Inferred	790,000	0.77	2.84	4.9	32.3

Notes:

1. Totals may not add up correctly due to rounding.
2. Mineral resources that are not mineral reserves do not have demonstrated economic viability.
3. Mineral resources in the above tables do not include mining dilution or recovery factors.
4. Base metal mineral resources are estimated based on the assumption that they would be processed at the Stall concentrator while gold mineral resources are estimated based on the assumption that they would be processed at the New Britannia concentrator, which is currently being refurbished.
5. Mineral resources were initially estimated using metal price assumptions that vary marginally over the assumptions used to estimate mineral resources at Lalor. In the QP's opinion, the combined impact of these small variations does not have any impact on the mineral resource estimates.
6. 1901 mineral resources are estimated using a minimum NSR cut-off of \$140 per tonne.

14.3. WATTS RIVER

Hudbay prepared resource model using Leapfrog® version 4.3 and MineSight® version 15, two industry standard commercial geological and mining software. The construction of this 3D resource model and the estimation of mineral resources were performed by Hudbay personnel following Hudbay procedures in compliance with best industry standards and the CIM guidelines. The work was supervised and validated by Olivier Tavchandjian, P.Geo., Vice President, Exploration and Geology.

Modeling Database

Drill holes totaling 24,757 m were included in the Watts River database to support the mineral resource estimate. Amongst these, 2,107 m were analyzed for, Cu, Au, Ag and Zn.

The drill hole database was imported in Leapfrog® and MineSight® from .csv files with a cut-off date for mineral resource estimate purposes of September 27, 2019.

Modeling of the Mineralized Envelope

Watts River mineralized envelope trend along an azimuth of approximately 20° with a general dip of approximately 65° to the east. Economic mineralization delineated so far occurs in two semi-massive to massive sulfide lens. These copper zones appear to be mostly aligned to the main regional deformation.

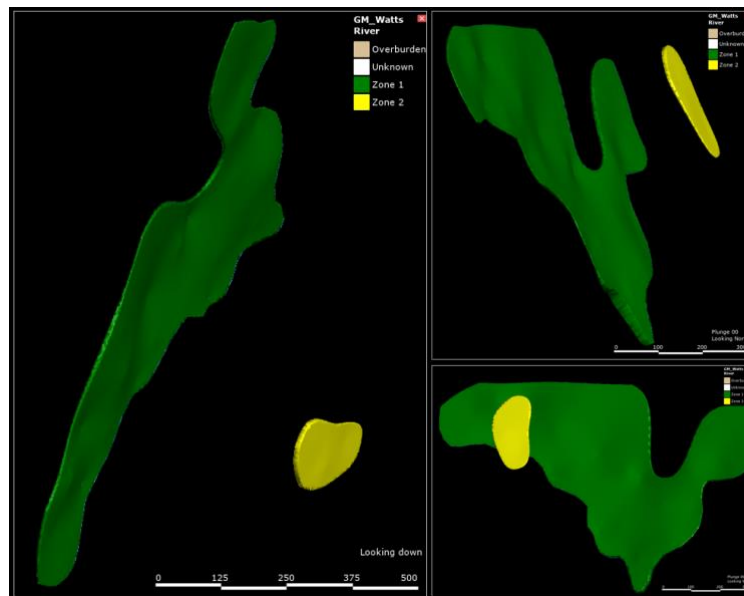
Units of overburden, barren and mineralized zones were created in 3D using core logging data and assay results. Two broad geological envelopes were then refined to create the mineralized envelopes.

The construction of the mineralized envelopes was based on the type of mineralization intersected. A smooth and continuous 3D solid was built using a 1 gram per tonne gold, a 0.5% copper or a 3% zinc cut-off to guide the construction of the mineralized envelope. Figure 14-7 present a plan view (main left), a looking North view (upper right) and a looking West view (lower right) of the mineralized envelopes.

In a final validation step, the envelope was loaded into MineSight® in order to ensure proper tagging of the solids to actual drillhole locations. The mineral envelope was used as hard boundary in all cases for grade

interpolation purposes to correctly prevent spreading of sulfide mineralization into the barren zone and vice-versa.

FIGURE 14-7: VIEW OF MINERAL ENVELOPE WIREFRAME



Density

Density values generated from two multi regression formulas are based on 197 density measurements contained within the main mineralized envelope: a formula for samples with iron assays results and a formula when samples do have iron assays results. The regression formulas were used to predict the density value where no actual measurement was taken.

In the case of the satellite mineralized envelope, only 10 samples had SG measurements. Nevertheless, a regression formula was developed to predict the density. Given the fact the coefficient of determination obtained was moderate (0.55 adjusted), it was deemed better to use the regression formula than use the average density for the zone. The predicted density values were validated through comparisons against the measured density.

Compositing

Assay intervals were regularized by compositing drillhole data within the interpreted mineralized envelope. Although most of the drillholes were assayed on lengths of less than 1 m interval (Reference Missing), a composite length of 2.5 m was selected as more appropriate to conduct interpolation into the 5m x 5m x 5m block size selected by the project evaluation team as the optimum selective mining unit (SMU). Grade compositing was weighted by specific gravity as there is a positive correlation between some of the metal grade and density. The compositing process was validated by comparing total length, density and length weighted average grade for each metal of the 2.5m composites to the original assays.

Exploratory Data Analysis

Exploratory data analysis (EDA) includes basic statistical evaluation of the composites for copper (Cu), gold (Au), zinc (Zn), silver (Ag) and density (SG). The EDA was conducted in the mineralized envelopes only. Within the mineral envelopes, the EDA also aimed to identify sub-domains that did not support the use of hard contacts for grade interpolation purposes but that justified being used for the purpose of block model validation, smoothing assessment and correction and resource classification.

Overall, the statistics of the composites indicate that the kriging interpolation will as expected introduce some level of smoothing and that a thorough smoothing assessment and post-processing of the initial interpolation is warranted to produce unbiased grade-tonnage curves and mineral resource and mineral reserve estimates.

Grade Capping

The deciles analysis (Parish) method was used to define high-grade outliers and to assess the need for grade capping. This analysis was conducted on the 2.5m composites in the mineralized envelopes.

This method considers capping when the last decile of the population contains more than 40% of the metal and that the last percent contains more than 10% of the metal. In all cases, it was found that capping would not be required for the two base metal zones.

Variography

Variogram analysis was performed with MineSight using a combination of nugget and two nested structures calculated along the envelope plan.

Grade Estimation and Interpolation Methods

The block model consists of regular blocks (5 m along strike by 5 m across strike by 5 m vertically) coupled with an ore percentage model. The mineral envelopes were used to assign the percentage of the block that belongs to each unit.

Grade interpolation used a strict composite and block matching system based on the mineralized envelopes codes. For example, only composites coded as LITHO=1 (i.e. the main mineralized envelope) were used to estimate the grade of the block that was assigned to ORE1=1 and ORE2=1. Both nearest neighbour (NN) and ordinary kriging (OK) grade interpolations were completed, using three passes with increasing minimum information requirements (Table 14-8). The search passes were selected to ensure best local estimates recognizing that OK has a smoothing effect but making no attempt during interpolation to reduce this smoothing as it would negatively impact the quality of the local estimates.

TABLE 14-8: SEARCH ELLIPSE PARAMETERS

	Min # of Comps	Max # of Comps	max # of Comps per hole	Declustering	max Comps per Quadrant	Search ellipse
Pass #1	1	32	6	<i>no</i>	-	150% of variogram ranges
Pass #2	16	32	6	yes	8	75% of variogram ranges
Pass #3	16	32	6	yes	8	50% of variogram ranges

The NN interpolation is solely used to validate that there is no global bias and to assess the level of grade smoothing in the OK interpolation for the selected SMU dimension.

Grade Estimation Validation

The grade estimation process was validated for each envelope to ensure appropriate honouring of the input data and subsequent unbiased resource reporting and use of the model for reserve estimation through the following steps:

- Visual checks of appropriate honouring of the input data but acknowledging that some natural smoothing should occur between samples as the grade of a sample in the middle of a block is not the average grade of the block;
- Absence of global bias by comparing the mean grade estimated by kriging to the original composite average grade and to a declustered grade obtained from a nearest neighbour interpolation;

- Volume variance correction to assess and correct if need be the smoothing effect of the grade interpolation

Visual Inspection

Visual inspection of block grade versus composited data was systematically conducted in section view. This check confirmed a good reproduction of the data by the model.

Global Bias Checks

This validation step consists in comparing the global average grade of each element between the original composites, the kriged block estimates and the nearest neighbour estimates.

A nearest neighbour interpolation is equivalent to the declustered statistics of the composites based on weighting each composite by its polygon of influence. The average grade obtained from this method is a useful benchmark but not a perfect one as it fails to incorporate the nugget effect measured by the variogram.

A global check was performed to verify that the kriged mean block estimate did not present any bias when compared to the composites and the nearest neighbour model. Differences between the 2.5m composites, the NN and OK grades are acceptable.

Smoothing Assessment

The visual validation conducted in sections confirmed that the block grade interpolation was consistent with the supporting composite data. The larger number of composites used for grade estimation in the block model significantly improves the individual block grade estimates but, at the same time results in a much smoother model requiring a careful assessment and, in many cases, a post-processing of the OK estimates.

The extent of grade 'over-smoothing' in the model was investigated in the main mineralized envelope based on material differences in grade distribution and/or drilling density. The mean and variance of the kriged estimates were compared to the variance of the composites after declustering. The expected true variance between SMUs was calculated from the variogram models. Smoothing assessment was not conducted on the satellite given the fact that the low number of composites did not generate valid variograms.

Smoothing Correction

An indirect log-normal correction was used to perform a change of support on the kriged models in order to obtain unbiased grade tonnage curves. This correction is only valid globally and provides poorer local estimates than the smoothed OK model but does not materially alter the global average grade within each zone and provides the correct grade-tonnage curve for the variogram models fitted on the drillhole data. It is an appropriate method to predict the recoverable tonnage and grade such as the volume mined over three months of production which should be a realistic aim for a long-term reserve model based on exploration drilling. The targeted variance was reached within very close limits in most cases (Table 14-9).

TABLE 14-9: SUMMARY OF SMOOTHING CORRECTIONS – WATTS RIVER

	OK Mean	NN model variance	OK model variance	Smoothing Ratio	Expected Variance	Achieved variance
Cu%	1.40	1.52	0.60	1.29	0.78	0.74
AuPPM	0.55	0.53	0.19	1.43	0.27	0.26
AgPPM	20.25	341.41	116.07	1.46	169.68	159.55
Zn%	2.06	6.30	1.00	3.04	3.03	2.40
Pb%	0.26	0.0013	0.0005	1.25	0.0006	0.0006

Classification of Mineral Resource

During the interpolation process, a number of control parameters were recorded for each block, e.g. number of samples, number of holes, the distance to the nearest sample and the average distance to all the samples used for the interpolation as well as the number of quadrants with samples, the kriging variance and the regression slope of kriging for each individual block estimate.

The regression slope values obtained from the kriging of copper grade estimates was used as the primary criteria for resource classification with 80% and 60% regression slope thresholds used respectively to separate measured from indicated and from inferred resources. From Hudbay's experience, this criteria was found to be a reliable first pass measure of quarterly and annual performance in tonnes and grade prediction.

Usually, this block by block coding assignment would have been smoothed to remove isolated blocks of one category within another. However, only a limited amount of blocks within ORE1 located at the immediate vicinity of drillhole intervals had regression slopes higher than 0.6.

Given the relative lower level of confidence caused by the limited amount of drilling, all blocks contained within the mineralized envelope were classified as inferred.

Reasonable Prospects of Economics Extraction and Mineral Resource Estimates

As for Lalor, the mineral resource estimates for Watts are reported within the constraint of an optimized stope-shape. The parameters used as input to define the mineable shape covers all the relevant technical and economic constraints including minimum stope and waste pillar dimensions and the NSR value calculation are very similar to those used at Lalor for the base metal lenses assuming beneficiation at the Stall mill. The mineral resource estimates were reported to ensure that each potential stope would cover all its associated operating and capital costs. The resulting mineral resource estimates are presented in Table 14-10.

TABLE 14-10: WATTS MINERAL RESOURCE ESTIMATES

Watts Mineral Resource Estimates (Exclusive of Mineral Reserves) – January 1, 2021						
		Tonnes	Cu (%)	Zn (%)	Au (g/t)	Ag (g/t)
Base Metal	Inferred	3,150,000	2.34	2.58	1.0	31.0

Notes:

1. Totals may not add up correctly due to rounding.
2. Mineral resources that are not mineral reserves do not have demonstrated economic viability.
3. Mineral resources in the above tables do not include mining dilution or recovery factors.
4. Base metal mineral resources are estimated based on the assumption that they would be processed at the Stall concentrator.

5. Mineral resources were initially estimated using metal price assumptions that vary marginally over the assumptions used to estimate mineral resources at Lalor. In the QP's opinion, the combined impact of these small variations does not have any impact on the mineral resource estimates.
6. Watts mineral resources are estimated using a minimum NSR cut-off of \$150 per tonne, assuming processing recoveries of 90% for copper, 80% for zinc, 70% for gold and 70% for silver.

14.4. PEN ZONE

The following summarize the method used for the Pen II mineral resource model developed by Hudbay's in 2018. A more detailed description on how the model was built was published in Hudbay 2019 Technical Report and is available in SEDAR.

Modeling Database

95 drill holes were used to support the mineral resource estimate with a cut-off date of August 29th, 2018. The files were imported as collar, downhole survey and assay data into MineSight® and Leapfrog® for independent comparison and validation. No significant differences in the projection of the drillhole traces were found between the two datasets.

Modeling of the Mineralized Envelope

The Pen zone mineralized envelope trends along an azimuth of approximately 40° with a general dip of approximately 40° to the north east. Economic mineralization delineated occurs in a semi-massive to massive sulfide lens typical of VMS deposits.

A mineralized envelope was created in 3D using core logging data and assay results. This broad envelope of sulfide occurrence was then refined to create a narrow smooth and continuous 3D solid including most of the zinc rich mineralization.

Density Assignment

Density values were calculated using a linear regression formula from the zinc and iron content based on 166 measurements and supported by a coefficient of correlation of 97%.

Compositing

Due to the narrow nature of the mineralization, the grade interpolation process was simplified to a pseudo-2D approach. Assay intervals were regularized to a single full length composite from hanging wall to footwall. This modeling approach was adopted to prevent any lateral selectivity in the mine design that is not deemed realistic at PEN II. Grade compositing was weighted by specific gravity as there is a significant positive correlation between some of the metal grade and density in the mineral envelope. After compositing, the total length, the density and length weighted average grade for each metal of the composites was compared to those of the assays before compositing to ensure they were similar.

Variography

Down-hole and directional variograms for Zn, Au, Cu, Ag, and SG were created using the MineSight data analysis software. The major, semi-major and minor axis were built in order to fit the attitude of the mineral envelope. A linear combination of a nugget and at two nested spherical models were adjusted. Once generated, a systematic visual check was conducted to ensure that the search ellipsoid would be correctly oriented with respect to the geometry of the mineral envelope.

Grade Estimation and Interpolation Methods

Both nearest neighbour (NN) and ordinary kriging (OK) grade interpolations were completed, using three passes with increasing minimum information requirements. The composite selection parameters for grade estimation in each domain were selected to ensure best local estimates. The percentage of the blocks contained within the mineralized envelope is recorded in the model.

The first interpolation pass uses 150% of the variogram ranges and is restricted to a minimum of one composite and a maximum of eight composites. The second interpolation pass uses 75% of the variogram ranges and is restricted to a minimum of three composites and a maximum of eight composites. The third interpolation pass uses similar constraints to the second pass but with a reduced 50% of the variogram ranges. At PEN II, all the composites have a different length and as a result the interpolation was both length and density weighted.

Grade Estimation Validation

The grade estimation process was validated to ensure appropriate honouring of the input data and unbiased resource reporting and use of the model for reserve estimation.

Visual inspection of block grade versus composited data was systematically conducted in section and plan views. This check confirmed a good reproduction of the data by the model.

A global check was performed to verify that the kriged mean block estimate was located between the mean of the composites and the mean of the nearest neighbour model.

Classification of Mineral Resources

As for Lalor, the regression slope values obtained from the kriging of zinc grade estimates was used as the primary criteria for resource classification. Blocks from the second and third interpolation pass combined with a 60% regression slope thresholds were used to separate indicated from inferred resources. There are no measured resource estimates reported at PEN II. The block by block coding assignment was then smoothed to remove isolated blocks of indicated resources within areas of mostly inferred resources and vice versa.

Reasonable Prospects of Economics Extraction and Mineral Resource Estimates

As for Lalor, the mineral resource estimates for PEN II are reported within the constraint of an optimized stope-shape. The parameters used as input to define the mineable shape covers all the relevant technical and economic constraints including realistic minimum stope and waste pillar dimensions. The NSR value calculation is based on the one used at Lalor for the base metal lenses assuming beneficiation at the Stall mill. The mineral resource estimates were reported to ensure that each potential stope would cover all its associated operating and capital costs. The resulting mineral resource estimates are presented in Table 14-11. There are no mineral reserve estimates reported for the PEN as the required engineering work has not been completed yet.

TABLE 14-11: PEN II MINERAL RESOURCE ESTIMATES

PEN II Mineral Resource Estimates (Exclusive of Mineral Reserves) – January 1, 2021						
		Tonnes	Cu (%)	Zn (%)	Au (g/t)	Ag (g/t)
Base Metal	Indicated	470,000	0.49	8.89	0.3	6.8
	Inferred	130,000	0.37	9.81	0.3	6.8

Notes:

1. Totals may not add up correctly due to rounding.
2. Mineral resources that are not mineral reserves do not have demonstrated economic viability.
3. Mineral resources in the above tables do not include mining dilution or recovery factors.

4. Mineral resources were initially estimated using metal price assumptions that vary marginally over the assumptions used to estimate mineral resources at Lalor. In the QP's opinion, the combined impact of these small variations does not have any impact on the mineral resource estimates.
5. Pen II mineral resources are estimated using a minimum NSR cut-off of \$75 per tonne and assume that the mineral resources would be amenable to processing at the Stall mill.

14.5. WIM & 3ZONE

In May 2015, Golder Associates ("Golder") prepared a resource model for the Wim property using Datamine® version 2.1.1547.0, an industry standard commercial geological and mining software. The same modeling approach and software were used by Golder in 2019 to construct a resource model for the 3 Zone gold deposit, a satellite ore body to the former producing New Britannia mine.

At both deposits, a smooth and continuous 3D mineralized envelope was built using a copper equivalency cut-off grade in the case of WIM and a gold cut-off grade in the case of 3 Zone to guide the construction of the mineralized envelope while including material below cut-off where necessary to maintain the continuity of the envelope. Scatter plots were used to identify assay outliers for grade capping. In the case of WIM, missing density values were generated using a linear regression formula from either the Fe grade or from the Cu grade. A composite length of 0.9 m in the case of WIM and 1m in the case of 3 Zone were selected to conduct interpolation into the blocks. Compositing was weighted by density at WIM but not at 3 Zone due to the absence of correlation in the latter case between density and gold grade.

A block size of 15 m east-west (strike), 3 m north-south (thickness), and 5 m elevation was selected at WIM while a block size of 5 m east-west, 5 m north-south, and 2m elevation was selected in the case of 3 Zone. At WIM, grade for Cu, Zn, Au, Ag and density (SG) was estimated using an inverse distance interpolant (IDW2) in the unfolded space and weighting by density while only gold grade was interpolated for 3 Zone using ordinary kriging (OK). In both cases, a nearest neighbor (NN) interpolation was also performed to provide declustered sample grade for block model validation. The search ellipsoids were defined from pairwise variograms modeled by a linear combination of a nugget and nested spherical structures.

The information provided by Golder was validated to ensure the built quality matched Hudbay technical standards.

Resource classification was assigned based on the search parameters used for grade interpolation. The block by block coding assignment was then smoothed to remove isolated blocks of indicated within areas of mostly inferred category and vice versa. At both WIM and 3 Zone, the resource models have supported pre-Feasibility studies and no resource estimates are reported in addition to those that been included in a viable mine plan and converted to reserve estimates.

14.6. NEW BRITANNIA MINE AND BIRCH DEPOSIT

The mineral resource estimate performed by Kinross followed a conventional and industry standard approach. The approach and general methods of resource estimation are summarized as follows:

- 1) Sectional interpretation of mineralization zones at a cutoff grade of 0.097 oz. per ton and a minimum true width of 7 ft.
- 2) All assays greater than 0.50 oz. per ton were cut to 0.500 oz. per ton.
- 3) The sectional interpretation of the mineralization was digitized and wireframed, using Gemcom Software.
- 4) The assay data was composited into 3-ft. intervals and basic statistics were completed on the assay and composited assay data.
- 5) Geostatistical analysis was completed and variogram models established.

- 6) A block model with dimensions 15 ft. x 15 ft. x 15 ft. was used for grade estimation.
- 7) Grade interpolation was completed using OK using a number of interpolation passes.
- 8) Validation of the block model using graphical and statistical techniques.
- 9) Tonnage was established using the percent model utility in Gemcom and tabulated by rock type using the percentage of the block within the mineralized wireframe.
- 10) The blocks were composited, using Gemcom, into 50-ft. levels and 50-ft. sections for tabulation and reporting of tons and grade.
- 11) Tabulation and reporting of the resources was completed at US\$400, US\$450, and US\$500 gold prices corresponding to cutoff grades of 0.097 oz. per ton, 0.086 oz. per ton, and 0.077 oz. per ton.

The review of the reports and the two procedures verify that the Kinross resource estimate, (audited and reported in the Micon International Ltd. NI-43-101 technical report dated October 2006), is a consistent representation of the drill data.

The cutoff grades for the resource have been estimated over a 6-ft. minimum true width and the parameters described in Table 14-12. The formula to calculate the cutoff grade is:

$$\text{costs}/(\text{gold price} \times \text{recovery}) \times 31.103 / \text{exchange rate}$$

TABLE 14-12: COST AND RECOVERY PRICE ASSUMPTION

Parameter	Units	Value
Estimated Mining Cost	C\$/tonne	43 – 115
Estimated Milling Cost	C\$/tonne	35
Estimated G&A	C\$/tonne	20
Total Operating Cost	C\$/tonne	98 - 170
Assumed Gold Price	US\$/Tr.Oz	1,250
Assumed Exchange Rate	C\$/US\$	1.3
Gold Recovery	%	93

Table 14-13 presents the resource estimate for the New Britannia mine and Birch property, as of January 1, 2021, using a variable cut off grade. Given its historical nature, the entire resource is classified as inferred. Mineral resources that are not mineral reserves do not have demonstrated economic viability.

TABLE 14-13: MINERAL RESOURCE ESTIMATE FOR THE NEW BRITANNIA MINE AND BIRCH DEPOSIT ⁽¹⁾⁽²⁾⁽³⁾

Zone	Location	Cut-off (Au g/t)	Tonnes	Grade
Lower Dick NW	1780 -	3.3	7,000	4.76
Main Mine 1780 +	1780 +	3.5	1,142,000	4.55
Lower Dick NW	780 -	3.3	213,000	4.55
Lower Dick EW	1780 -	2	823,000	4.59
Lower Ruttan	1780 -	2	568,000	4.28
Birch	Birch	3.3	569,000	4.42
Total Inferred			3,322,000	4.49

1. Totals may not add up correctly due to rounding.
2. Mineral resources that are not mineral reserves do not have demonstrated economic viability.
3. Mineral resources in the above tables do not include mining dilution or recovery factors.

WSP reviewed all the available data, modeling procedures, block models and technical reports and more specifically, audited the geological model, mineralization wireframes, domain statistical validation, and visual trend slice validation. Available historical data and technical reports were also examined. Based on

the independent checks conducted by WSP, the QP is satisfied that the procedures followed by Kinross and Alexis Minerals were adequate to support the declaration of inferred mineral resource estimates.

15. MINERAL RESERVE ESTIMATES

No mineral reserve estimates are reported for the Pen II, Watts and New Britannia Mine deposits and as a result, this section documents how the joint LOM plan and mineral reserve estimates were developed for the Lalor mine, 1901, WIM and 3 Zone deposits only. The Lalor mine, 1901 deposit and its satellite deposits mineral reserves as of January 1, 2021 are summarized in Table 15-1.

TABLE 15-1: LALOR MINE AND SNOW LAKE DEPOSITS MINERAL RESERVE ESTIMATES (AS OF JANUARY 01, 2021)

Snow Lake Mineral Reserve Estimates – January 1, 2021							
			Tonnes	Cu (%)	Zn (%)	Au (g/t)	Ag (g/t)
Base Metal	Proven	Lalor	6,860,000	0.50	5.79	2.6	29.4
		1901	890,000	0.40	6.61	2.0	28.1
		Subtotal	7,740,000	0.49	5.88	2.5	29.2
	Probable	Lalor	1,190,000	0.64	4.32	3.2	31.9
		1901	690,000	0.25	9.49	1.4	29.9
		Subtotal	1,880,000	0.50	6.23	2.6	31.1
Gold	Proven	Lalor	3,950,000	0.60	1.03	5.2	27.7
		Subtotal	3,950,000	0.60	1.03	5.2	27.7
	Probable	Lalor	3,630,000	1.16	0.53	5.7	27.9
		WIM	2,450,000	1.63	0.25	1.6	6.3
		3 Zone	660,000	-	-	4.2	-
		Subtotal	6,740,000	1.22	0.38	4.0	17.3
Base Metal and Gold	Proven and Probable	Lalor	15,620,000	0.69	3.25	4.0	28.8
		1901	1,580,000	0.34	7.87	1.7	28.9
		WIM	2,450,000	1.63	0.25	1.6	6.3
		3 Zone	660,000	-	-	4.2	-
		Total	20,310,000	0.75	3.14	3.5	25.2

Notes:

1. Totals may not add up correctly due to rounding.
2. Metal prices of US\$1.10 per pound zinc, US\$1,500 per ounce gold, US\$3.10 per pound copper, and US\$18.00 per ounce silver with an exchange rate of 1.30 C\$/US\$ were used to confirm the economic viability of the mineral reserve estimates. Mineral reserves are estimated using a minimum NSR cut-off of \$105 per tonne for waste filled mining areas and a minimum of \$116 per tonne for paste filled mining areas.
3. For Lalor, individual stope gold grades were capped at 10 g/t, as a prudent estimate until reserves to mill reconciliations can establish that the high-grade gold can indeed be entirely recovered. This capping method resulted in the reduction of the global gold reserve grade by approximately 3%.
4. WIM mineral reserves are estimated assuming processing recoveries of 98% for copper, 88% for gold, and 70% for silver based on processing through New Britannia's flotation and tails leach circuits.
5. 3 Zone mineral reserves are estimated assuming processing recoveries of 85% for gold based on processing through New Britannia's leach circuit.

The Deswik mine design software was used to optimise the mine plan in all cases based on cut-off costs, stope geometry parameters, and mineral resource shapes from the resource model described in Section 14. Appropriate dilution and recovery factors were applied based on mining method with a combination of paste and unconsolidated waste backfill material.

The mineral reserves were sequenced and scheduled into an integrated LOM schedule using Deswik interactive scheduling software and exported to spreadsheets for financial analysis.

The author considers that the mineral reserves as classified and reported comply with all disclosure in accordance with requirements and CIM definitions. The author is not aware of any mining, metallurgical, infrastructure, permitting or other relevant factors that could materially affect the mineral reserve estimate.

There are no legal, political, environmental or other risks that could materially affect the potential developments of the reserves in the report.

15.1. MINING STRATEGY AND RESERVE ESTIMATIONS PROCESS

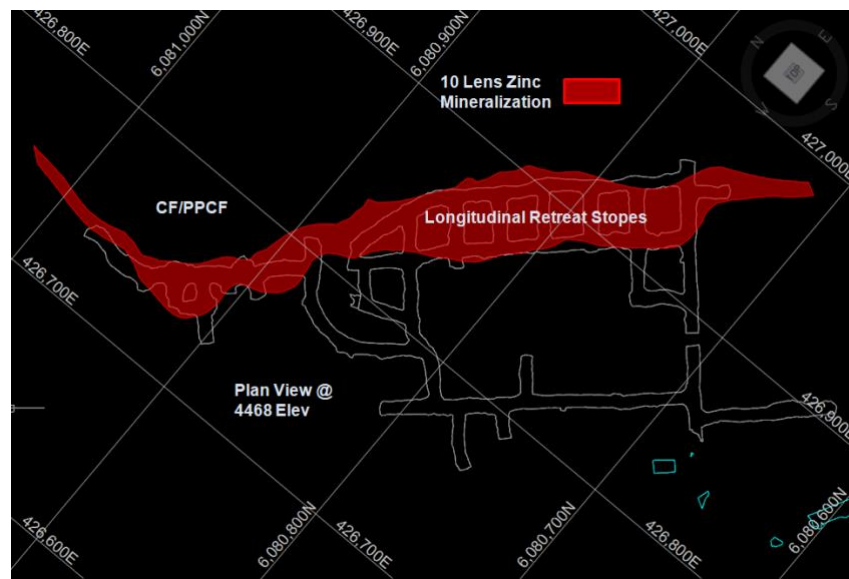
Mining Method Selection & Mining Strategy

Lalor Mine has been in production continuously since mid 2012 and has produced 8.8 million tonnes by the end of 2020. Mining methods are the major determinants of dilution, recovery, economics, and production scheduling, which are used to determine mineral reserves. Production at Lalor is planned to be done using predominantly transverse longhole mining with minor longitudinal retreat and cut and fill mining.

Early in the mine life, the complexity of stacked zinc and gold zones within the deposit was realized. In the zinc rich 10 and 11 Lenses, where there is no lens stacking, longhole retreat, transverse longhole and post pillar cut and fill mining were used successfully. See Figure 15-1 for typical mining arrangement in 10 Lens.

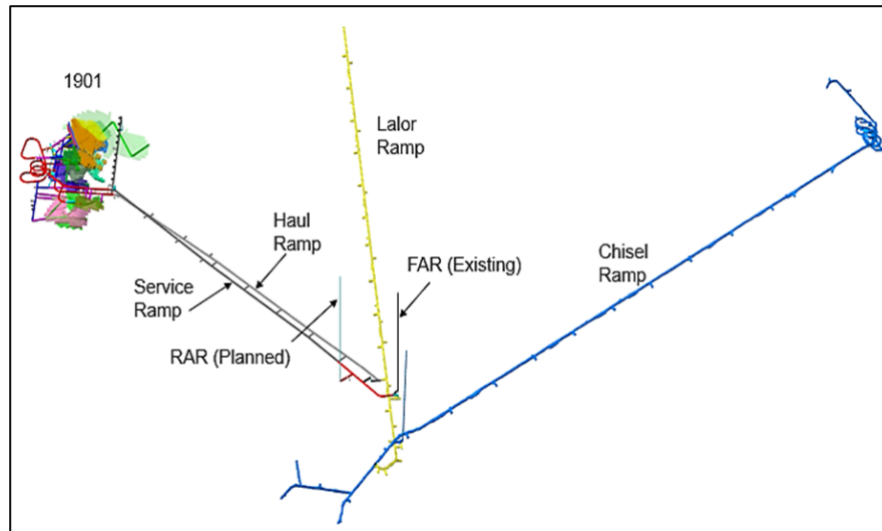
Selective post pillar cut, and fill mining was initially used in 20, 21, 25 and 32 lenses as shown in Figure 15-2. Post pillar mining allowed geological evaluation of each round, leading to a better understanding of the ore. This helped to determine: 1) the possibility of mining the 20 and 32 lens zinc ores separately from 21 and 25 lens gold ores and 2) the ability to visually identify and segregate gold ores from surrounding waste. The result was that most zinc ore stopes can be designed and mined separately from the gold, but that the microscopic nature and distribution of the gold in gold zones required additional diamond drilling, sampling and assaying information than required for the zinc stopes to determine production grades.

FIGURE 15-1: 840 LEVEL 10 LENS PLAN VIEW AT LALOR



Updated geological interpretations and the decision to construct a paste backfill plant supported the decision to convert some areas of the mine similar to those in Figure 15-2 from post pillar cut and fill to longhole mining across all lenses. For narrow and steeply dipping stopes typical of 31 lens, longitudinal retreat mining was preferred, while for wide stopes crossing multiple lenses, typical of 20, 32 and 21 lenses shown in Figure 15-2, transverse longhole mining using footwall drawpoints was preferred.

FIGURE 15-3: ACCESS TO 1901 DEPOSIT ISOMETRIC VIEW



1901 is expected to produce 1,000 tonnes per day of ore at steady state and has been incorporated into the Lalor production strategy as an extension to the existing mine. Personnel and equipment from 777 and Lalor are to be redeployed to 1901 in future years, to sustain peak production levels at the mine and mills.

At WIM and 3 Zone, longitudinal longhole mining with unconsolidated rock fill is the chosen method for the underground mines. To eliminate the use of cemented back fill, pillars in ore will be left for stability purposes. Each deposit will be mined as a ramp access mine with limited infrastructure, with 3 Zone using the existing portal and ramp development. The existing ramp at 3 Zone will be mucked out and refurbished and new ramp development started once the bottom of the decline has been reached. To keep capital costs low, execution of 3 Zone will begin once WIM begins to ramp down, thus allowing some infrastructure to be moved and installed at 3 Zone (pumps, temporary fans, electrical, main fans, heaters, portable refuge stations and equipment).

The production rates over the LOM are 840 tonnes per day (8 years) for WIM, and 450 tonnes per day (4 years) for 3 Zone, with the ore being processed at the New Britannia Mill once Lalor ore is depleted.

Reserve Methodology

The following steps were followed in developing the reserve estimates:

- Calculate two payable (NSR) values for each individual block in the resource model depending on whether processing would occur at the Stall concentrator or at the New Britannia concentrator, using long-term metal prices, concentrator recoveries, metal payability and downstream smelter treatment and refining costs assumptions. Supporting test work, actual performance and forecasts supporting these assumptions are discussed in details in Sections 13, 17, 19 and 21 of this Technical Report.
- Design stopes in the Deswik Stope Optimizer, considering depleted mineral resources, existing workings, resource categories and mine and mill operations costs. Dilution and recovery are estimated and applied at this step. Stopes are designed for both the Stall concentrator option and the New Britannia concentrator option
- Establish stope economics using a secondary NSR calculation where, along with mine and mill operations costs, mine capital, waste development and offsite administration costs are applied to each stope. Assign whether stopes can be upgraded to mineral reserves based on resource classification.

- Design ore development required for mining the reserves. Deplete development from the stopes. Interrogate grades of designed development for inclusion in mineral reserves.
- Sequence and schedule development and stope production for input to a financial Life of Mine (LOM) study to support mineral reserve economics.

The above methodology takes into consideration the three different ore types and the milling options for the future production from Lalor mine.

15.2. DILUTION AND RECOVERY

Planned dilution factors are input in the stope optimizer by mining method assuming those currently experienced at Lalor Mine for transverse longhole open stope, longitudinal retreat longhole and post pillar cut and fill mining.

Dilution

Dilution is included in the conversion of mineral resource to mineral reserves and can be internal (planned) or external. Dilution is reported as the ratio of total waste divided by mineral resource.

Internal dilution is primarily incurred due to the shallow dipping nature of the deposit and stacking of lenses which can result in multiple lenses being grouped together for mining purposes. The space between the lenses that are mined in the course of stope mining is considered as internal dilution. In the Deswik stope optimizer routines multiple lenses may be required to be extracted as a single mining unit based on stope mining parameters as shown in Table 15-2.

Internal dilution included in mineral reserves can also include low grade development required to achieve footwall dip angles in longhole stopes and may include material mined between lenses where multiple lenses are grouped together. Current grade control practices will continue to be used to assign material as ore or waste on a round by round basis to attempt to minimize dilution.

TABLE 15-2: STOPING PARAMETERS BY MINING METHOD

Stope Shape Parameters	Units	Lalor		1901		WM	3 Zone
		Longhole	Cut & Fill	Cut & Fill	Room & Pillar	Longhole	Longhole
Length Along Strike	Metres						
Minimum		16	20	7	7	15	10
Maximum		23	150	7	7	15	10
Width FW to HW	Metres						
Minimum		4	5	8	14	3	3
Maximum		50	-	-	14	20	25
Waste Pillar Width	Metres	5	0	4	0.25	0.01	0.01
Stope Height	Metres						
Minimum		10	5	5	4	20	15
Maximum		25	5	5	-	20	15
Hangingwall Dip	Degrees						
Minimum		30	90	90	-11	45	45
Maximum		90	90	90	11	130	130
Footwall Dip	Degrees						
Minimum		50	90	90	-11	45	45
Maximum		90	90	90	11	130	130
Dilution	Metres						
Hangingwall Fixed		1	0	0	0.25	0	0
Footwall Fixed		0	0	0	0.25	0	0

Note: Lalor Cut & Fill length along strike (min and max) values represent a collection of drifts whereas the 1901 values represent the size of each drift.

At Lalor, external dilution is set at a fixed distance of 1.0m into the hanging wall after the stope geometry shape is finalized. Internal dilution and external dilution are included as part of the optimized mining shape. If dilution occurs within the block model, it is assigned block model grades. Otherwise, it is set at zero grades and a bulk density of 2.8 t/m³. Dilution incurred in converting resources to reserves is summarized by mining method in Table 15-3.

TABLE 15-3: DILUTION INCURRED BY MINING METHOD

Average Dilution	Lalor (%)	1901 (%)	WIM (%)	3 Zone (%)
Development	19.9	13.9	9.7	79.2
PPCF, CF, & RP	18.3	7.8	-	-
Longhole	26.8	-	20.0	20.0
Total	24.6	8.0	17.8	31.7

Mining Recovery

Mining recovery is defined as the ratio between mineral resource tonnes delivered to the concentrator and in-situ mineral resource tonnes estimated in the model. Mining recovery assumptions used by type of stoping and backfilling method in the Deswik Scheduler are shown in Table 15-4. These assumptions were based on a combination of first principles and actual performance at the mine since the start of operations. Resultant recovery factors by mining method are shown in Table 15-5. Some of the mineral resources not recovered relate to: rib, post and sill pillars required to maintain rock stability, inefficiencies in mining represented by small blocks of ore along ore/waste contacts and underbreak and inefficiencies in mucking reflecting ore unable to be mucked from longhole stopes.

TABLE 15-4: RECOVERY FACTORS BY MINING METHOD

Average Recovery	Criteria	Backfill	Lalor (%)	1901 (%)	WIM (%)	3 Zone (%)
PPCF, CF, & RP	Standard 7m x 7m pillars	Waste	75	75	-	-
Longhole	Transverse & Retreat	Paste	89	-	-	-
	Transverse & Retreat	Waste	78	-	83	83

TABLE 15-5: AVERAGE RECOVERY FACTORS BY MINING METHOD

Average Recovery	Lalor (%)	1901 (%)	WIM (%)	3 Zone (%)
Development	100	100	100	100
PPCF, CF, & RP	75	75	-	-
Longhole	89	-	83	83
Total	87	76	87	86

15.3. DESWIK STOPE OPTIMIZER

Stopes were built in the Deswik stope optimizer considering measured, indicated, and inferred mineral inventories from the block model. Stope construction considered the NSR block values and appropriate dilution and recovery factors based on resource geometry, mining and backfill methods.

To reduce the possibility of including low grade material in stopes, cut-off costs attempt to replicate the steady state cost of mine and mill operations, and include ore extraction, onsite mine G&A and mill operating costs. This generates stopes that might be mined economically during the normal operation of the mine. Cut-off costs (\$/tonne) used to construct stopes in the stope optimizer are shown in Table 15-6.

TABLE 15-6: CUT-OFF COSTS

Mine Method Backfill Mill	Lalor Longhole Paste Stall	Lalor Longhole Paste New Brit	Lalor Longhole Waste Stall	Lalor Longhole Waste New Brit	Lalor PPCF Waste Stall	Lalor PPCF Waste New Brit	1901 PPCF Waste Stall	WIM Longhole Waste New Brit	3 Zone Longhole Waste New Brit
Mining	\$15.23	\$15.23	\$15.23	\$15.23	\$38.83	\$38.83	\$51.15	\$60.11	\$60.11
Ore Removal	\$20.65	\$20.65	\$20.65	\$20.65	\$20.65	\$20.65			
Backfill	\$13.28	\$13.28	\$2.32	\$2.32	\$2.32	\$2.32			
General Mine Expense	\$42.79	\$42.79	\$42.79	\$42.79	\$42.79	\$42.79	\$32.88	\$20.71	\$19.99
Subtotal	\$91.95	\$91.95	\$80.99	\$80.99	\$104.59	\$104.59	\$84.03	\$80.82	\$80.10
Mill	\$23.99	\$42.76	\$23.99	\$42.76	\$23.99	\$42.76	\$40.00	\$40.00	\$40.00
Sustaining Capital	-	-	-	-	-	-	\$14.82	\$33.56	\$33.56
Total	\$115.94	\$134.71	\$104.98	\$123.75	\$128.58	\$147.35	\$138.85	\$154.39	\$153.66
DSO Cutoff	\$115.94	\$134.71	\$104.98	\$123.75	\$128.58	\$147.35	\$140.00	\$150.00	\$150.00

Note: with Lalor being an operating mine, a second test is conducted to only retain the stopes that can cover the sustaining capital costs associated with their extraction, which is on average \$34.61/t for the Lalor mine.

Iterations were run considering mining methods and preferred milling location. At this stage, the multiple runs of the stope optimizer continued to consider measured, indicated, and inferred resources. The sequence used to build the preliminary Deswik stope inventory is:

- 1) Attempt to build longhole stopes in the stope optimizer using payability from the flotation only milling process (NSR Stall) assuming longhole mining and Stall milling cutoff costs.
- 2) Where longhole stopes were unable to be built by the stope optimizer, or where excessive dilution was designed into the stope, attempt to design post pillar cut and fill stopes using the stope optimizer, using higher cut and fill mining costs which incurs less planned dilution and the Stall milling cutoff cost.
- 3) For gold lenses that are distinctly separated from zinc lenses, attempt to build longhole stopes using payability and costs from the New Britannia concentrator. This assumes higher gold payability and higher mill operating costs.
- 4) For gold lenses that are distinctly separated from zinc lenses and for which longhole stopes cannot be designed for, or for areas incurring high dilution, attempt to design post pillar cut and fill stopes using payability from New Britannia concentrator and using higher cut and fill mining costs, which incurs less dilution.

At 1901, WIM, and 3 Zone, the in-situ mineral inventory values helped determine which mill the ores will be routed to, and so multiple optimization runs based on either the NSR ZN or NSR AU fields were not necessary for each lens or zone. For these satellite deposits, an elevated cut-off value was applied, with the intention of capturing the expected sustaining capital costs, to ensure that the resulting mineable shapes would pass further economic testing and remain profitable.

15.4. STOPE NET SMELTER RETURN CALCULATION

To determine if stopes could be included in the final mine plan and in the mineral reserve estimates, only those stopes supported by measured and indicated resource estimates were retained. Then, the tonnes and grades contained in the DSO stopes were subjected to a secondary financial test using 2021 budget unit costs. Each stope was assigned a mill circuit to determine stope revenues, and a mining method and backfill method to determine mining costs. Direct mining costs derived from mining and backfill method, onsite G&A costs, offsite administration costs, mine sustaining capital costs and stope specific waste development costs were applied on a stope by stope basis to determine final stope economics:

- Mining, milling and onsite G&A costs are based on the 2021 budget and are applied to each stope generated in the DSO as shown in Table 15-6
- Capital equipment costs were applied at \$8.83/t to all stopes generated in the DSO.

- Capital Major Installations were applied at \$4.11/t to all stopes generated in the DSO.
- Capital waste development includes ramp and level development metres required to access new mining areas:
- Ramp and ancillary development (remucks, sumps, etc.) metres were estimated and applied to mining areas at 2021 budget unit costs.
- Haulage drift metres (footwall drifts) were estimated based on the stope arrangement and were applied as required to access additional stopes along strike at 2021 budget unit costs.
- Stope specific operating waste development includes longhole stope cross cuts and cut and fill entrance cross cuts. Estimated metres were applied on a stope by stope basis at 2021 budget unit costs.
- Offsite administration costs were applied at 2021 budget unit costs and are based on copper and zinc concentrates produced from stopes.

For 1901, WIM, and 3 Zone, pre-feasibility study (PFS) level mine designs, development and production schedules, as well as capital and operational cost estimates were prepared by Stantec (1901) and BBA (WIM and 3 Zone). The key parameters established from these studies were used as input into a consolidated long-range business plan of the Manitoba Business Unit to further confirm the positive economic contribution of each.

15.5. ORE DEVELOPMENT

The final step to establishing mineral reserves was to create ore development shapes required to mine stopes. Planned 3D development shapes were built and removed from stopes in Deswik when creating Interactive Scheduler activities. The development shapes were interrogated and assigned as mineral reserves.

15.6. RECONCILIATION OF RESERVES TO PRODUCTION

Recent reconciliation between reserve estimates and past production at Lalor provide a useful insight into the reliability of the LOM plan to predict future production

The reconciliation results presented in this section encompass 100% of the mining production for the first three quarters of 2020 including long-hole mining, post-pillar cut, and fill and development mined as ore. The mine's production was reconciled to the mill and compared to the reserve estimates for the stopes and cuts entirely mined during this period.

Results presented in Table 15-7 below show that all tonnes and metal grade reconcile within 6%. The 6% difference in gold grade may actually be closer to 5% as 20 thousand tonnes of unaccounted high grade gold production reported to underground stockpiles in preparation for the New Britannia mill start-up in 2021 and is not credited to the 2020 mine's production. Although these results are considered very good overall, they also show that Lalor is still in the process of adapting its past grade control protocols which were previously focused towards maximising base metal production. The mill credited 10% more copper and zinc metal production than the mineral reserve estimates while gold was 2% lower. This is recognised by the mine with on-going initiatives aimed at improving and re-balancing the focus of the underground grade control program to ensure gold mining recovery is maximised going forward including closer drillhole spacing prior to extraction, assaying all blasthole samples to optimize blast design as well as systematic bucket sampling from production. With all these initiatives in place, we intend to maintain a very reliable reserve model and identify further opportunities to maximize gold extraction.

TABLE 15-7: RESERVE MODEL DEPLETION RECONCILED TO MINE'S CREDITED PRODUCTION BY THE MILL

2020 Mining reconciliation	Ktonnes	Au g/t	Ag g/t	%Cu	%Zn
2021 Reserve model depletion	1,383	2.59	24.97	0.67	5.63
Actuals	1,432	2.43	25.71	0.70	5.90
Delta	4%	-6%	3%	6%	5%

16. MINING METHODS

Geology, geotechnical information, orebody geometry, productivity and mining experience are major drivers to select the best suited mining methods including considerations for lateral development, production, backfilling and ore transportation.

The hanging wall and footwall rocks at Lalor are generally of good quality, allowing the use of mechanized drilling and blasting techniques. The mineralized lenses dip at an average of 30°, but locally varying from sub-horizontal to 55°. Mining methods currently in use include cut and fill and longhole open stope. Paste backfill is used to increase recovery and accelerate the mining cycle. Low grade areas will be filled with rock from waste development.

Ore is mucked using Load Haul and Dump (LHD) loaders which are operated remotely in inaccessible areas. Ore is loaded into underground haul trucks or ore passes and transported to the ore handling system at the production shaft for hoisting to surface. Ore delivered to the production shaft is sized to less than 0.55m by one of two rock breakers. Ore is hoisted from the mine by two 16 tonne capacity bottom dump skips in balance. On surface, ore is truck hauled to a primary crusher at the Chisel North mine site, crushed to less than 0.15m, and then trucked to either the Stall or New Britannia concentrator for processing.

For 1901, two mining methods will be used. The primary method will be Post-Pillar Mechanized Cut and Fill (PPMCAF) used where the resource has sufficient vertical thickness to mine multiple consecutive cuts. Where there is limited vertical thickness and the resource can be mined in a single cut, Inclined Room and Pillar will be used. The PPMCAF stopes will be backfilled with unconsolidated waste rock.

At WIM and 3 Zone, longhole retreat mining methods were selected to excavate the orebodies. Utilizing this method will require a bottom-up mining sequence with mining fronts progressing in from the orebody extents towards the center of the orebody. The orebodies have been divided into various mining blocks to allow multiple levels to be extracted simultaneously.

16.1. LATERAL DEVELOPMENT

At Lalor, lateral development is horizontal or near horizontal tunnelling, travelable by rubber tired mechanized equipment. Ramps are typically mined at up to +/- 15% grade to provide access between levels. Rounds are 4m long and typical dimensions are 6m wide by 5m high. Standard ground support, consists of resin grouted rebar and welded wire mesh across the back and to within 1.8m of the sill on the walls and face. Mine services, including compressed air, process water and discharge water pipelines, paste backfill pipeline, power cables, leaky feeder communications antenna and ventilation duct, are installed in main levels and stope entrances.

For 1901, access has been started from the existing Lalor ramp and will approach the deposit from the hangingwall side. The main development profile will be 5 m wide x 5 m high, with dimensions established to allow for the required ventilation flow velocity as well as to meet other criteria like minimum equipment clearances. All ramp and lateral excavations will be developed using traditional drill and blast methods and diesel-powered mobile equipment.

At WIM and 3 Zone, all lateral ore and waste development is designed to accommodate Sandvik LH514 and LH517 LHDs and Sandvik TH550 trucks (or equipment of equivalent size). Levels generally consist of sills in the ore with level access off the main ramp to the sill. Sumps, as well as the access drifts for the return air and fresh air raises, will be off the level access.

16.2. VERTICAL DEVELOPMENT

At Lalor, main ventilation raises and ore pass raises are developed by a mining contractor using a raisebore and/or Alimak climber. Ground support and ladder ways are installed as required. Longhole slot raises, transfer ore passes, and auxiliary ventilation raises are limited to approximately 35m long and are

developed using longhole conventional drop raise techniques. Drain, paste backfill and electrical cable holes are drilled using longhole or raisebore drills and are reamed to designed diameter.

For 1901, major vertical development will consist of an existing 3m diameter fresh air raise (FAR) as well as a planned 3.5m diameter return air raise (RAR) that is to be raisebored roughly 500m in length. Additional internal raises will be used for ventilation and egress and they are planned to be developed with an Alimak method.

The requirements for vertical mine development at WIM and 3 Zone include FARs and secondary egresses. The main ventilation raises are sized based on ventilation requirements and will also be used as a second means of egress. It is envisioned that the raises will be developed using an Alimak method and slashed to the appropriate dimensions as the raise is supported and manway installed.

16.3. STOPE MINING

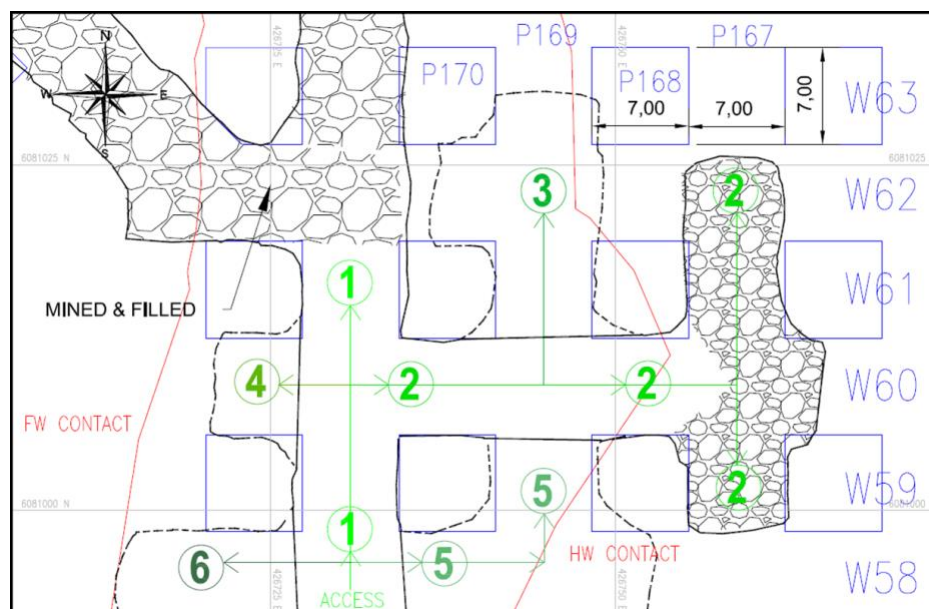
The two mining methods used at Lalor mine, 1901 deposit and the Snow Lake satellite deposits are variations of cut and fill mining or longhole open stope mining. In general, where the dip exceeds 35° and the orebody is of sufficient thickness, longhole open stope mining is preferred, whereas cut and fill mining methods are used in narrower, flatter areas. At Lalor, longhole mining represents 87% of total stope mining, and stope mining accounts for 92% of all mining, with development accounting for the other 8%.

Cut and Fill Mining

Single pass overhand mechanized cut and fill mining is the preferred method where the ore is flatter than 35° and horizontal widths are less than 10m. The ore is accessed from a footwall drift by a crosscut developed at approximately -15%. Ore is mined in 5m high horizontal cuts. Unconsolidated backfill is placed tight to the back and the entrance crosscut is then back slashed to provide access to the next cut.

Where ore widths are greater than 14m wide, mining is conducted by overhand post pillar cut and fill. Post pillars provide ground support to allow for selective mining of wide stopes. Backfill is placed to within 1.8m of the back and the entrance crosscut is back slashed to provide access to the next cut. Drifts and crosscuts in stopes are typically 7m wide x 5m high, with 7m x 7m post pillars. Mining retreats towards the access to reduce risks associated with poor ground conditions. A typical mining sequence is shown in Figure 16-1 as a plan view.

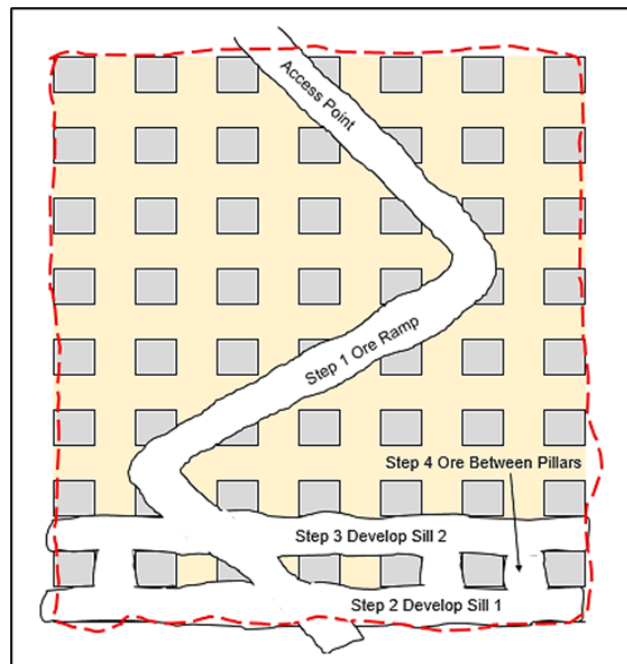
FIGURE 16-1: TYPICAL POST PILLAR CUT AND FILL MINING PLAN VIEW



At 1901, in areas that undulate and are typically narrower, an inclined room and pillar method will be used. This method does not require backfill, as there is only one cut per level. A schematic demonstrating this method is shown in Figure 16-2, with the following sequence of steps followed for stope production:

- Step 1 – develop the Ore Ramp from the Access Point to the extent of the resource. The Ore Ramp will traverse the ore at a gradient appropriate to keep the floor of the ramp at the FW contact.
- Step 2 – develop Sill 1 along strike at the stope limit.
- Step 3 – develop Sill 2 along strike.
- Step 4 – mine the resource between pillars between Sill 1 and Sill 2, retreating from the stope extents to the Ore Ramp. The configuration of the pillars adjacent to the Ore Ramp will be adjusted to the geometry, while maintaining the planned extraction ratio.
- Repeat Steps 2-4 for subsequent sills until mining has retreated to the initial access point.

FIGURE 16-2: TYPICAL INCLINED ROM AND PILLAR MINING PLAN VIEW



Longhole Open Stope Mining

Benefits of longhole mining over cut and fill methods include reducing exposure to risk as a non-entry method and higher productivity/lower cost than cut and fill methods.

The minimum design height of a longhole stope is 17m. In shallower dipping areas of the ore the interval is typically 17 to 20m, and in steeper dipping areas the interval is 20 to 25m. Stope widths assume a minimum mining width of 5m. Maximum open stope length and width is determined considering the rock quality and hydraulic radius of the combined exposed hanging wall and back span of the stope. A 5m rib pillar is added when unconsolidated rock is used as backfill.

FIGURE 16-3: TYPICAL ISOMETRIC VIEW – TRANSVERSE LONGHOLE OPEN STOPPING

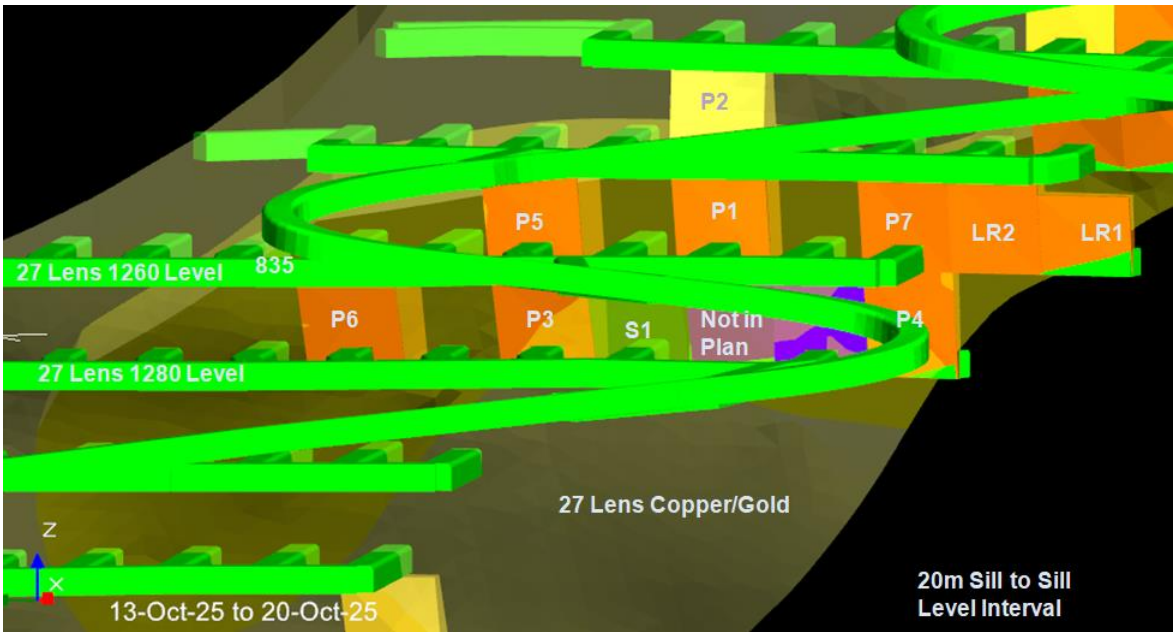
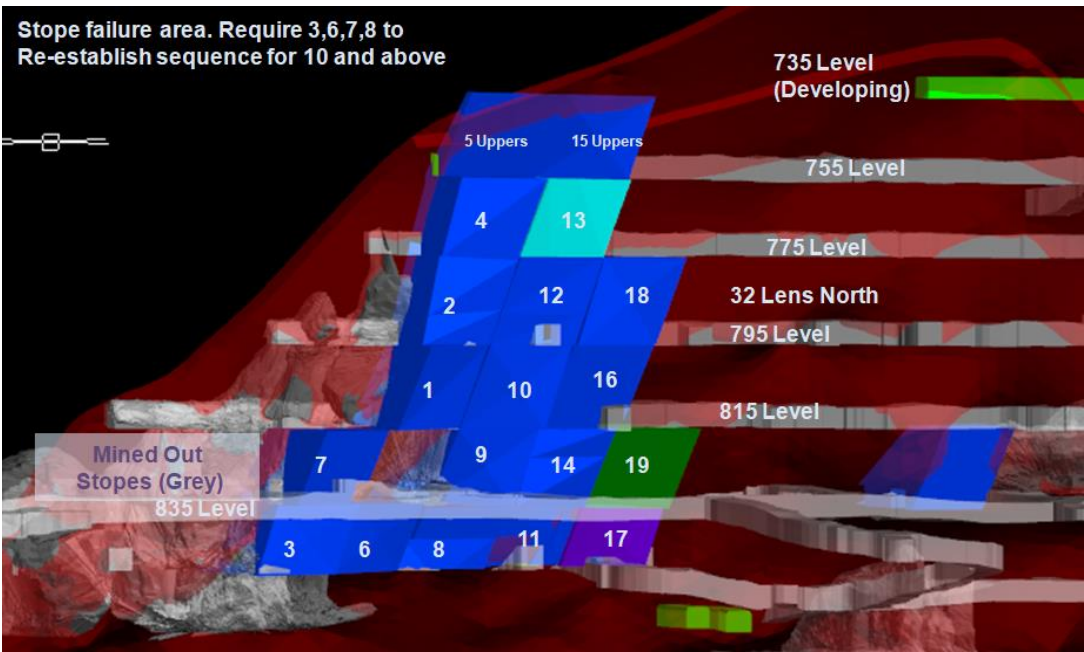


FIGURE 16-4: TYPICAL LONG SECTION – LONGITUDINAL RETREAT LH OPEN STOPPING



Teleremote and Autonomous Mucking

Teleremote and autonomous mucking allows stope mucking to and orepass to continue through shift change and blast clearing times. Lalor Mine uses Epiroc's Autonomous Mucking System which includes an operator's station on surface allowing control of up to three LHD's concurrently. The surface operator drives the LHD remotely to record a route and then switches to autonomous mode. The LHD travels the recorded route without operator input. Safety devices are installed on the mucking level to restrict access by mine workers and equipment during the operation. Future plans are to isolate truck routes from the stoping areas allowing teleremote loading of trucks as the mine expands below shaft bottom.

16.4. BACKFILL

All stopes at Lalor mine except uppers require backfill to maintain long term stability and to provide a floor to work from for subsequent mining. Backfill is either unconsolidated waste rock backfill (URF) or consolidated paste backfill. URF is used in stopes where pillar or wall confinement is not required and the value of the adjacent pillars does not warrant the added expenditure of paste backfill. Paste backfill is an engineered product comprised of mill tailings and a binder (3-5% cement by weight), mixed with water to provide a thickened paste that is delivered by borehole and pipes to stopes. Paste backfill presents several advantages over unconsolidated fill such as slurried mill tailings or loose waste rock including increased stability, better ventilation controls, higher mining recovery and lower mining dilution and it also eliminates potential flows of re-liquefied unconsolidated tails.

At WIM and 3 Zone, URF backfill material will come from underground development. The waste produced before stoping begins will be stockpiled on surface and hauled underground as required to replace removed ore. Once stoping begins, waste will be stockpiled underground for use as fill.

16.5. ORE HANDLING

Ore is mucked by LHD, loaded to underground haul trucks and hauled to one of the two rock breaker/grizzlies on 910m level for sizing to less than 0.55m. A 40m bin below the grizzly provides approximately 1,200 tonnes of coarse ore storage. A chute at the bottom of the bin at 955m level feeds ore to a conveyor that loads a measuring flask with approximately 14 to 16 tonnes of ore. Ore is skipped to surface by two 16 tonne capacity bottom dump skips in balance. The ore enters the headframe chute from the skips and is deposited into the surface ore bin or to the exterior concrete bunker via gravity. From the surface bin or bunker, ore is truck hauled to a primary crusher at the Chisel North mine site, crushed to minus150mm, and then trucked to a concentrator to process.

Ore from 1901 will be truck hauled and dumped onto an ore pad on surface near the Chisel North Portal. A trucking simulation was completed using Simio to determine the number of trucks required to meet the 1,000 tonnes per day maximum production rate envisaged. The simulation also includes backhauling 300 tonnes per day of waste rock in the ore trucks for use as backfill. The simulation was run with a 14.0 trucking-hour day with day and night shifts split evenly (7.0 hours of truck seat time per shift). The most favourable truck scenario to achieve 1,000 tonnes per day of ore haulage and backhaul 300 tonnes per day of waste rock is using two MT65 trucks.

At WIM and 3 Zone, all primary production mucking will be completed by LHD, with ore moving to a remuck bay located near the entrance of the level where it will be rehandled and side loaded onto trucks and hauled to surface. The ore will be stockpiled and hauled to New Britannia mill via surface haul trucks operated by a contractor.

16.6. GEOTECHNICAL DESIGN

A geotechnical database has been established from drill core logs and geotechnical core sample laboratory tests. Geotechnical data is continuously collected following standard procedures. This is done by the Lalor Geology department and data is shared with the Ground Control team. Adequate core samples are collected for all the ore zones to ensure we have sufficient geotechnical data from hanging wall, orebody and footwall to further test rock properties. Ground monitoring instruments (dCABLES) are

used for ground movement monitoring and more advanced ground control instrumentation will be considered going forward.

1901 rock mass characterization studies for geomechanical purposes and mine design have been based on basic geotechnical logging of resource holes, together with limited geomechanics laboratory testing, point load testing of full and half core, and evaluation of structural orientations from acoustic televiewer surveys. Rock mass conditions for the 1901 deposit can generally be described as very competent, massive to slightly jointed, with strong (50-100 MPa) to very strong (100-250 MPa) intact rock strengths. For WIM and 3 Zone, typical stope sizes and dimensions from existing Hudbay operations were considered practical for their respective studies. Although diamond drill holes were reviewed and indicate RQD values of High, future work is required to further identify structure and lithology of the mineralized zones and to characterize the rock mass, conduct rock mass behaviour analysis, and confirm ground support requirements.

16.7. GROUND SUPPORT SYSTEMS

Except when using cut and fill mining methods, drifts have arched backs to increase stability and factor of safety. Ground support is classified as either primary support or secondary support.

Primary support refers to reinforcement of the rock mass immediately following excavation to ensure safe working conditions before re-entry. Secondary support is additional support applied after the installation of primary support to provide further support in large spans, long term infrastructure excavations and structurally controlled areas where wedge failures may be a concern.

With 1901 currently in preliminary stages of project development, typical Lalor support for production levels and access were evaluated based on Q ratings (Barton, N.). For PFS level studies, it was confirmed that the Lalor standards for ground support are adequate for 1901.

Primary Support

The standard for Lalor is #6 or #7 2.3m (7'4") resin rebar installed on a 1.2m x 1.2m (4' x 4') square pattern with a same size rebar in the center of the pattern. Screen (#6 gauge welded wire mesh) and bolts are installed on the walls, back and face prior to re-entry. At drift intersections, 3.6m (12') resin rebar in a 1.2m x 1.2m pattern with a center #7 7'4" rebar is used for primary back ground support.

16.8. UNDERGROUND DEVELOPMENT

To access the 1901 deposit, the service ramp and haulage ramp (twin ramps) will provide two working faces for a single jumbo, with additional ancillary faces available (connections between ramps, remucks, safety bays, electrical cut-outs, etc.). The initial priority will be to establish the ventilation and infrastructure at the start of the access, before proceeding toward 1901. Once development reaches 1901, the priority will be to establish a ventilation loop and open additional faces for a second jumbo crew to access the first production areas.

At WIM, a new ramp will be developed to access the underground workings. The ramp will extend from surface to the bottom of the mine, roughly 700m vertical meters below surface. The ramp is typically driven 5.3m wide x 5.5m high with an arched back and a 15% maximum gradient to accommodate the fleet.

The existing ramp at 3 Zone will be rehabilitated and used as an access and then extended down to the bottom of the mine. According to the provided wireframes it currently extends down roughly 130 vertical meters below surface, and the new ramp development will begin around 100 vertical meters below surface. The existing ramp portal is currently filled with rock fill, but the amount of fill used is unknown at this time. For the 3 Zone study, the assumption is that before development can begin, roughly 100m of the existing ramp will need to be mucked out and approximately 900m of rehabilitation will be required before development can begin.

Vertical Development

Raises

Vertical development (or raises) is planned at Lalor mine and the Snow Lake satellite deposits for establishing or extending the ventilation and secondary egress systems. Drop raises used as slots in longhole stopes are included in stope drilling and blasting. Raises less than 35m can be mined as drop raises. Longer raises are evaluated to choose between raise boring and Alimak considering the conditions, end use and cost of both methods.

Paste Backfill, Drain and Service Holes

At Lalor, paste backfill is distributed to stopes via a series of steel cased vertical holes between levels and pipelines installed in footwall drifts. Paste backfill holes are drilled using a raisebore pilot bit. Service holes are holes drilled between levels for electrical cable distribution throughout the mine.

For 1901, WIM, and 3 Zone, allowances have been included in the cost estimates for drain holes and service boreholes.

16.9. DIAMOND DRILLING

Diamond drilling is done to define the extents of mineralization and is required for forecasting grades from stopes. The delineation holes are typically drilled from a drift located in the footwall or hanging wall. Stope definition holes are drilled from stope drawpoint stubs or undercut drifts for grade planning (which could vary sequence in some cases).

16.10. MINE DRAINAGE AND DEWATERING

Lalor mine is a relatively dry underground mine with no significant hydrological concerns. The drainage for Lalor mine and the past producing Photo Lake and Chisel North mines is interconnected. Mine water from all mine sources is pumped to the Chisel Open Pit, where it is pumped to a water treatment plant and then released to the environment. Main collection areas feeding the water treatment plant are: 140m level pump station at Photo Lake, a pumping station at Chisel North 540m level (referred to as the 990 sump) and satellite sumps at Lalor mine.

1901's mine drainage is to be integrated with Lalor systems. Service water from mining processes and inflows from the rock mass will flow in drainage ditches along the side of drifts and will be collected at each level sump, typically located near a ramp where possible. A main dewatering station on 580L will be outfitted with two filter weir sumps to filter slimes from the water and a clean water retention sump, which will pump to the water reservoir at Remuck No. 3 (in Lalor ramp). This main dewatering station will be established early so as the mine expands, all water from future workings will be collected and filtered prior to dewatering. Water will be directed to the main dewatering station via boreholes or staged through piping from levels below using submersible pumps.

Like Lalor and 1901, the WIM and 3 Zone dewatering systems are based on dirty water pumping achieved through a network of interconnected sumps, trenches, drain holes, and slurry pumps. Portable, standardized, skid mounted equipment to maximize ease of installation, flexibility, and reusability will be used. The pumping equipment includes pump stations, booster stations and clarifiers, and will be reused at 3 Zone once no longer required at WIM.

16.11. MINING OPERATIONS

Mining Fleet

Lalor mine is a ramp and shaft accessible mine with production and development done by rubber tired underground mining equipment. The mine equipment fleet required to achieve 4,500 tpd is shown in Table 16-1. Lalor mine production will be approximately 20% jumbo and 80% longhole. Additional contractor

equipment not listed in Table 16-1 is planned to be on-site to complete the sustaining capital lateral development metre requirements in the LOM plan. An allowance for replacement of mine equipment has been included in the mine plan.

The 1901 mobile equipment fleet will include units common with Lalor mine. There will be one jumbo and three scissorlifts available from the 777 mine, as well as one jumbo, two bolters, and two 65 tonne haul trucks available from Lalor. Although some auxiliary equipment may also be shared with Lalor, the remaining units are planned to be purchased by Hudbay.

WIM's underground fleet will be purchased new from suppliers and transferred to 3 Zone once available.

TABLE 16-1: MINE MOBILE EQUIPMENT

Description	Lalor	1901	WIM	3 Zone
Underground Trucks 65 tonne	4	2	0	0
Underground Trucks 42 tonne	12	1	5	3
LHD 14 tonne	10	3	5	3
LHD 18 tonne	5	0	0	0
LHD - Service	3	1	0	0
Two Boom Jumbo	6	2	2	2
Roof Bolters	10	3	2	2
Longhole Drills	5	0	2	2
Blockholer	2	0	0	0
Powder Truck	5	1	1	1
Scissor Lift	12	3	3	3
Boom Truck	3	1	1	1
Grader	2	1	1	1
Backhoe	5	0	0	0
Cable Reeler	1	0	0	0
Forklift	15	0	0	0
General Service Truck	17	2	0	0
Skid Steers	2	0	0	0
Front End Loader	3	1	0	0
Personnel Carriers	33	3	2	2
Total Mine Equipment	155	24	24	20

Mine Power

Grid electricity is supplied by Manitoba Hydro, the provincial power utility. Manitoba Hydro's 115 kV power line terminates at the Chisel North mine site, approximately 3.5 road km from the Lalor mine site. This supplies power to the Hudbay owned main distribution substation consisting of two (2) 115-25 kV 24 MVA transformers.

At 1901, the underground electrical power distribution system is to be fed from Lalor's 825mL transformer.

The WIM and 3 Zone mine electrical and power distribution originates at the New Britannia Mill site. Two primary feeders have been proposed to supply the load of the underground mine plants, with both primary feeders being egressed out of the mine services substation via an underground concrete encased ductbank (or direct buried), to ensure that the risk of contact with electrical plant is minimized. The overhead distribution will be constructed alongside the existing gravel road and will consist of wood poles and wires.

Production Schedule

The Life of Mine (LOM) production schedule combining Lalor, 1901, WIM and 3 Zone is shown in Table 16-2: Mine Production Schedule to Table 16-5. The Deswik software was used to assist with the LOM sequencing and scheduling to generate the production schedule. Historic mining rates and systems, equipment or crews capabilities were applied and levelled. From this output, adjustments were made to

further balance the capabilities to create the final plan. Concentrates produced from Lalor and contained metal in concentrates are also shown in tables below.

TABLE 16-2: MINE PRODUCTION SCHEDULE

Year	Lalor					1901					WIM					3 Zone	
	Tonnes	Au (g/t)	Ag (g/t)	Cu (%)	Zn (%)	Tonnes	Au (g/t)	Ag (g/t)	Cu (%)	Zn (%)	Tonnes	Au (g/t)	Ag (g/t)	Cu (%)	Zn (%)	Tonnes	Au (g/t)
2021	1,627,500	3.54	26.43	0.70	4.24	-	-	-	-	-	-	-	-	-	-	-	-
2022	1,627,500	4.01	27.04	0.77	3.65	-	-	-	-	-	-	-	-	-	-	-	-
2023	1,855,000	4.09	26.07	0.74	3.02	-	-	-	-	-	-	-	-	-	-	-	-
2024	1,855,000	4.26	26.94	0.64	3.00	-	-	-	-	-	-	-	-	-	-	-	-
2025	1,855,000	4.59	27.19	0.91	2.49	-	-	-	-	-	-	-	-	-	-	-	-
2026	1,763,001	4.20	27.76	0.65	2.79	91,999	2.29	38.16	0.23	6.26	-	-	-	-	-	-	-
2027	1,475,329	4.13	31.71	0.83	3.13	334,187	2.04	42.67	0.27	6.16	-	-	-	-	-	-	-
2028	1,284,108	3.63	34.46	0.51	3.57	343,393	1.96	21.83	0.38	8.65	-	-	-	-	-	-	-
2029	1,273,696	3.40	32.42	0.48	3.84	353,804	1.43	25.41	0.32	9.09	-	-	-	-	-	-	-
2030	822,886	3.79	32.51	0.50	3.40	312,205	1.58	21.84	0.44	5.77	104,294	0.76	4.64	1.22	0.09	-	-
2031	183,027	5.24	33.68	0.58	2.06	145,381	1.14	31.52	0.29	12.53	413,553	1.24	6.01	1.62	0.18	-	-
2032	-	-	-	-	-	-	-	-	-	-	438,130	1.55	5.66	1.47	0.32	-	-
2033	-	-	-	-	-	-	-	-	-	-	438,020	1.74	6.51	1.72	0.42	-	-
2034	-	-	-	-	-	-	-	-	-	-	401,392	1.82	6.67	1.71	0.28	37,564	3.40
2035	-	-	-	-	-	-	-	-	-	-	316,455	1.82	6.92	1.71	0.17	218,824	4.17
2036	-	-	-	-	-	-	-	-	-	-	287,784	1.68	6.76	1.67	0.13	219,102	4.17
2037	-	-	-	-	-	-	-	-	-	-	48,783	1.87	6.86	1.70	0.13	186,565	4.46
Total	15,622,046	4.02	28.80	0.69	3.25	1,580,969	1.73	28.88	0.34	7.87	2,448,411	1.60	6.31	1.63	0.25	662,055	4.21

TABLE 16-3: MILL PRODUCTION SCHEDULE

Year	Stall										New Britannia												
	Lalor					1901					Lalor					WIM					3 Zone		
	Tonnes	Au (g/t)	Ag (g/t)	Cu (%)	Zn (%)	Tonnes	Au (g/t)	Ag (g/t)	Cu (%)	Zn (%)	Tonnes	Au (g/t)	Ag (g/t)	Cu (%)	Zn (%)	Tonnes	Au (g/t)	Ag (g/t)	Cu (%)	Zn (%)	Tonnes	Au (g/t)	
2021	1,362,841	3.31	26.41	0.71	4.87	-	-	-	-	-	213,750	4.99	26.55	0.64	0.22	-	-	-	-	-	-	-	
2022	1,087,500	2.99	27.55	0.82	5.13	-	-	-	-	-	540,000	6.08	26.00	0.67	0.68	-	-	-	-	-	-	-	
2023	1,315,000	3.40	26.39	0.86	3.90	-	-	-	-	-	540,000	5.76	25.28	0.45	0.88	-	-	-	-	-	-	-	
2024	1,315,000	3.78	26.71	0.72	3.80	-	-	-	-	-	540,000	5.42	27.49	0.45	1.06	-	-	-	-	-	-	-	
2025	1,315,000	4.41	26.98	1.10	3.02	-	-	-	-	-	540,000	5.02	27.70	0.44	1.22	-	-	-	-	-	-	-	
2026	1,223,001	3.62	28.59	0.75	3.65	91,999	2.29	38.16	0.23	6.26	540,000	5.52	25.88	0.42	0.84	-	-	-	-	-	-	-	
2027	935,329	3.36	32.75	0.69	4.40	334,187	2.04	42.67	0.27	6.16	540,000	5.47	29.90	1.07	0.93	-	-	-	-	-	-	-	
2028	744,108	2.49	32.96	0.34	5.43	343,393	1.96	21.83	0.38	8.65	540,000	5.19	36.53	0.75	1.02	-	-	-	-	-	-	-	
2029	821,585	2.43	27.32	0.41	5.38	353,804	1.43	25.41	0.32	9.09	452,111	5.17	41.68	0.59	1.06	-	-	-	-	-	-	-	
2030	486,584	2.27	36.15	0.56	5.60	312,205	1.58	21.84	0.44	5.77	387,210	5.70	27.93	0.43	0.63	104,294	0.76	4.64	1.22	0.09	-	-	
2031	69,080	3.24	38.06	0.60	3.90	145,381	1.14	31.52	0.29	12.53	113,947	6.45	31.02	0.56	0.95	413,553	1.24	6.01	1.62	0.18	-	-	
2032	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	438,130	1.55	5.66	1.47	0.32	-	-
2033	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	438,020	1.74	6.51	1.72	0.42	-	-
2034	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	401,392	1.82	6.67	1.71	0.28	37,564	3.40
2035	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	316,455	1.82	6.92	1.71	0.17	218,824	4.17
2036	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	287,784	1.68	6.76	1.67	0.13	219,102	4.17
2037	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	48,783	1.87	6.86	1.70	0.13	186,565	4.46
Total	10,675,027	3.35	28.48	0.74	4.34	1,580,969	1.73	28.88	0.34	7.87	4,947,019	5.48	29.55	0.59	0.90	2,448,411	1.60	6.31	1.63	0.25	662,055	4.21	

Note: 2021 Stall tonnage includes ore processed at the Flin Flon mill.

TABLE 16-4: MILL RECOVERY SCHEDULE

Year	Stall								New Britannia						
	Lalor				1901				Lalor			WIM			3 Zone
	Au	Ag	Cu	Zn	Au	Ag	Cu	Zn	Au	Ag	Cu	Au	Ag	Cu	Au
2021	59%	60%	86%	92%	-	-	-	-	90%	76%	93%	-	-	-	-
2022	61%	63%	88%	91%	-	-	-	-	91%	76%	92%	-	-	-	-
2023	69%	72%	93%	91%	-	-	-	-	91%	75%	91%	-	-	-	-
2024	66%	70%	93%	91%	-	-	-	-	91%	76%	91%	-	-	-	-
2025	69%	72%	93%	89%	-	-	-	-	90%	76%	91%	-	-	-	-
2026	67%	71%	93%	91%	42%	44%	81%	94%	91%	75%	91%	-	-	-	-
2027	65%	69%	93%	91%	46%	48%	84%	94%	91%	77%	93%	-	-	-	-
2028	50%	53%	87%	94%	53%	55%	88%	94%	91%	78%	92%	-	-	-	-
2029	54%	57%	89%	93%	49%	51%	86%	94%	91%	79%	92%	-	-	-	-
2030	60%	64%	93%	92%	56%	58%	90%	93%	91%	76%	91%	77%	64%	97%	-
2031	62%	65%	93%	91%	47%	49%	84%	97%	91%	76%	92%	84%	70%	98%	-
2032	-	-	-	-	-	-	-	-	-	-	-	87%	67%	97%	-
2033	-	-	-	-	-	-	-	-	-	-	-	89%	71%	98%	-
2034	-	-	-	-	-	-	-	-	-	-	-	89%	71%	98%	84%
2035	-	-	-	-	-	-	-	-	-	-	-	89%	71%	98%	85%
2036	-	-	-	-	-	-	-	-	-	-	-	88%	70%	98%	85%
2037	-	-	-	-	-	-	-	-	-	-	-	90%	71%	98%	85%
Total	64%	66%	91%	92%	50%	51%	87%	94%	91%	77%	92%	88%	70%	98%	85%

TABLE 16-5: METAL PRODUCTION SCHEDULE

Year	Lalor				1901				WIM			3 Zone
	Au ounces (000s)	Ag ounces (000s)	Cu tonnes (000s)	Zn tonnes (000s)	Au ounces (000s)	Ag ounces (000s)	Cu tonnes (000s)	Zn tonnes (000s)	Au ounces (000s)	Ag ounces (000s)	Cu tonnes (000s)	Au ounces (000s)
2021	115	824	10	61	-	-	-	-	-	-	-	-
2022	160	946	11	51	-	-	-	-	-	-	-	-
2023	190	1,134	13	46	-	-	-	-	-	-	-	-
2024	191	1,150	11	46	-	-	-	-	-	-	-	-
2025	208	1,188	16	35	-	-	-	-	-	-	-	-
2026	182	1,133	11	41	3	49	0	5	-	-	-	-
2027	152	1,079	11	38	10	219	1	19	-	-	-	-
2028	112	912	6	38	11	134	1	28	-	-	-	-
2029	103	889	5	41	8	149	1	30	-	-	-	-
2030	86	626	4	25	9	128	1	17	2	10	1	-
2031	26	142	1	2	3	72	0	18	14	56	7	-
2032	-	-	-	-	-	-	-	-	19	54	6	-
2033	-	-	-	-	-	-	-	-	22	65	7	-
2034	-	-	-	-	-	-	-	-	21	61	7	3
2035	-	-	-	-	-	-	-	-	17	50	5	25
2036	-	-	-	-	-	-	-	-	14	44	5	25
2037	-	-	-	-	-	-	-	-	3	8	1	23
Total	1,524	10,022	99	424	44	751	5	117	110	347	39	76

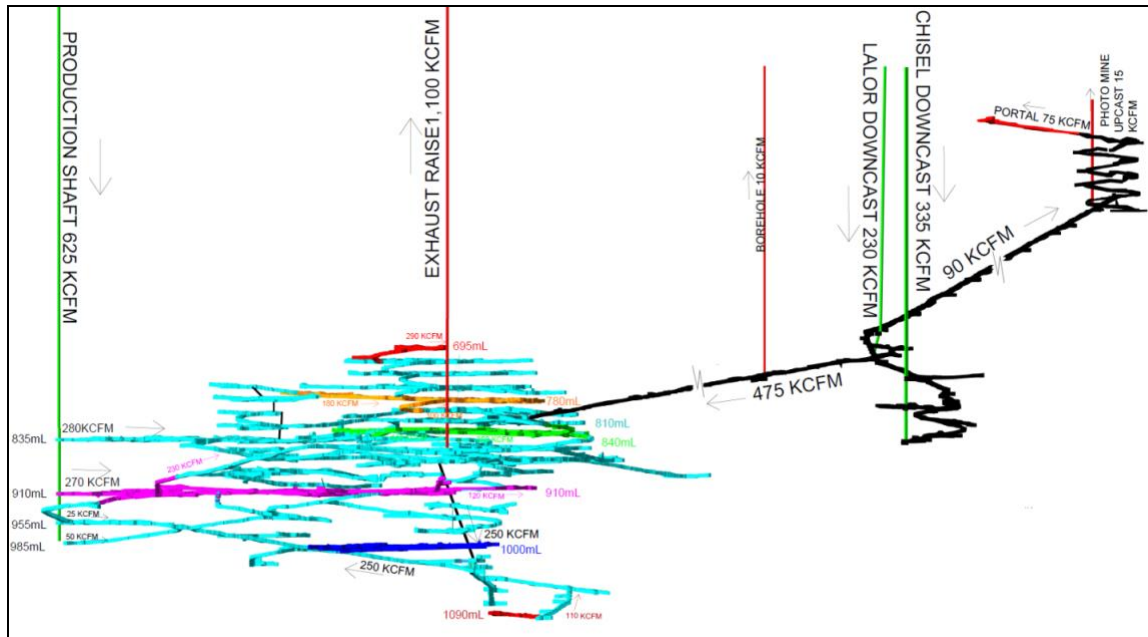
Note: Production includes metal contained in concentrate and doré. Totals may not add up due to rounding.

Mine Ventilation

Lalor mine is a “push-pull” system primarily driven by pull. The exhaust shaft is equipped with two fans drawing 1.1 million cfm from the mine (Figure 16-5). Fresh air is supplied into the mine from the Chisel and Lalor downcast raises (0.6 million cfm). The Chisel and Lalor downcast raises are equipped with 600HP and 400HP fans respectively. An additional 0.6 million cfm is downcast via the Lalor production shaft, equipped with two 250HP fans. In the upper mine, ventilation raises and an internal ramp between 795mL to 755mL will provide ventilation to expanding stope fronts. In the lower mine, an extension of the 810 to 1025mL raise is planned to connect 1025mL to 1090mL. An internal exhaust raise is planned from the 1000mL to the bottom of the exhaust shaft at 840mL. This will improve blast clearing times for the lower part of the mine. Individual mining faces are ventilated using 60 HP to 250 HP fans and 1.37 m ventilation duct. Three propane heaters heat air in the mine during the winter from Chisel, the access ramp and the production shaft.

Ventilation planning utilizes Ventsim modelling to create intermediate stages from the LOM Deswik schedule. The Ventsim model is continually updated as the mine continues to expand both upward and downward in multiple lenses.

FIGURE 16-5: CURRENT LALOR MINE VENTILATION LONG SECTION



1901's ventilation will be a "push-pull" system, with fresh air introduced underground via a new fan and heater plant installed on surface at the existing 10 ft diameter fresh air borehole, and exhaust air returned to surface through a new 13 ft diameter RAR with main fans underground. The fresh air will be heated as required with a direct fired propane heater.

The WIM and 3 Zone deposits will use an exhaust-air decline approach. This system will be a "push" type system. The main driver of the system will be two-fan intake installation. This installation will be located at the top of the FAR which will also serve as the secondary egress. The seasonally required, mine-air heater system will also be located at the top of the FAR.

16.12. WORKFORCE

The majority of operations, maintenance and technical personnel at Lalor work 11.5 hour shifts on a 5-5-4 day cycle or a 7-7 day cycle or 8 hour day shifts, 40 hours per week. The mine is operated under Collective Bargaining Agreements between Hudbay management and local unions. The mine operations workforce is comprised of Hudbay hourly operations and maintenance personnel as well as salaried supervision, mine administration and technical staff, plus contractor personnel for specialized work and workforce shortages. Personnel count will vary year to year. Steady state personnel requirements are shown in Table 16-6 .

At 1901, development, production, construction, and maintenance work will be completed by Hudbay personnel. A mining contractor will complete specialized work that includes: raises (raisebore and Alimak), diamond drilling, major construction, grubbing and site preparations for the RAR, and main ventilation plant installation.

At WIM and 3 Zone, all development is planned to be completed by contractors, with Hudbay staff carrying out production activities.

TABLE 16-6: MINE OPERATIONS WORKFORCE

Discipline	Lalor	1901
Direct Operations	325	56
Supervision and Administration	81	6
Health and Safety	9	1
Mine Maintenance	132	26
Mine Technical	47	4
Total Mine Personnel	594	93

16.13. MINE SAFETY AND HEALTH

All personnel are required to work under the applicable laws of the Province of Manitoba, Canada. All contractors working on site are required to have an approved health and safety program in place and have on site representation. Hudbay Plant Safety Rules and Regulations are used at Lalor mine including safety and health monitoring programs, dust, water and environmental monitoring, personal protective equipment and task analysis and job procedures completed by employees daily.

Refuge Stations

Refuge stations are required at Lalor mine as per mine regulations and Hudbay standards and are incorporated into the mine design. Hudbay's standard refuge station is excavated from rock and requires two ventilation bulkheads, and compressed air. Depending on the size of the refuge and the people working in the area a backup oxygen generator may be installed. Potable water, stretcher kit, first aid supplies and supplies to seal off the bulkheads are installed. In new development areas portable refuge stations replace excavated refuge station.

Three refuge stations will be constructed at 1901. There will be a refuge station in the access ramps and two at the 1901 workings on 580mL and 625mL. The refuge stations will be the same as at Lalor and will also serve as lunchrooms and can be used as a line-up room.

At WIM and 3 Zone, portable refuge stations and sanitary facilities will be strategically placed throughout the mine in active working areas.

Personal Evacuation Infrastructures

The primary route in and out of the Lalor mine is the production shaft equipped with a service cage. The shaft is equipped with a small auxiliary hoist and six person cage. In case of power failure, the auxiliary hoist can be operated by an emergency diesel generator to evacuate personnel from the mine. If the production shaft is not usable, the second egress from the mine is the main ramp to surface at Chisel North mine.

Secondary means of egress are incorporated into the designs of the Snow Lake satellite deposits. At 1901, the twin ramps and internal raises (with escapeways installed) provide alternate egresses back to the Lalor ramp, which is considered the primary entry and exit route. At WIM and 3 Zone, the main ventilation raises (FARs) will be supported and equipped with a manway so that they can be used as second egresses.

17. RECOVERY METHODS

17.1. INTRODUCTION

The Snow Lake operations consists of two facilities to process ore from the Lalor Mine: The Stall and New Britannia mills. The Stall mill employs conventional crushing, grinding and flotation operations. The nominal throughput rate is 3,800 tonnes per day and the mill operates 24 hours per day, 365 days per year, with scheduled downtime for maintenance as required. The mill produces a copper concentrate with gold and silver credits and a zinc concentrate. The New Britannia mill is currently being refurbished with an expected completion by the end of 2021. Conventional crushing, grinding, flotation and cyanide leach and recovery methods will be used to process ore from the Lalor mine. The nominal throughput rate will be 1,500 tonnes per day, operating 24 hours per day, 365 days per year, with scheduled downtime for maintenance.

17.2. STALL MILL

In 1979, the Snow Lake mill was constructed to process ore from six operating mines within a seven-mile radius of Snow Lake, which were previously being treated at the Flin Flon mill. The new mill had two independent flotation circuits, known as the Stall circuit and the Chisel circuit. Each circuit treated the output of three mines.

The Chisel circuit processed a blend of ore from the Chisel Lake, Lost Lake and Ghost Lake mines. The circuit's capacity was approximately 1,000 tonnes per day, on a five-day basis, producing separate copper, lead and zinc concentrates.

The Stall circuit treated a blend of ores from Anderson Lake, Osborne Lake and Stall Lake mines, at a mill throughput of roughly 2,800 tonnes per day, producing approximately 410 tonnes of copper concentrate and 162 tonnes of zinc concentrate per day. Ores were truck hauled to the new mill, and concentrates were rail-hauled to the Flin Flon metallurgical complex.

The mill operated continuously until shutdown in 1993, because of depletion of the Chisel Lake Open Pit and Stall Lake Mines. The mill was reopened in 1994 to process ore from the Photo Lake Mine and continued to process ore from the Chisel North Mine until 2009, when both the mine and mill were placed on care and maintenance due to depressed metal prices. The mine and mill were re-started in 2010, following recovery of metal prices.

The mill, now referred to as the Stall mill, processes ore from the Lalor Mine, which began commercial operations in 2012. Initial capacity was 2,800 tonnes per day and through modifications and improvements, milling performance has increased to an average rate of 3,800 tonnes per day, although it has been demonstrated to reach peak throughput rates of up 4,500 tonnes per day. This mill is also envisaged to process the ore produced from the 1901 deposit.

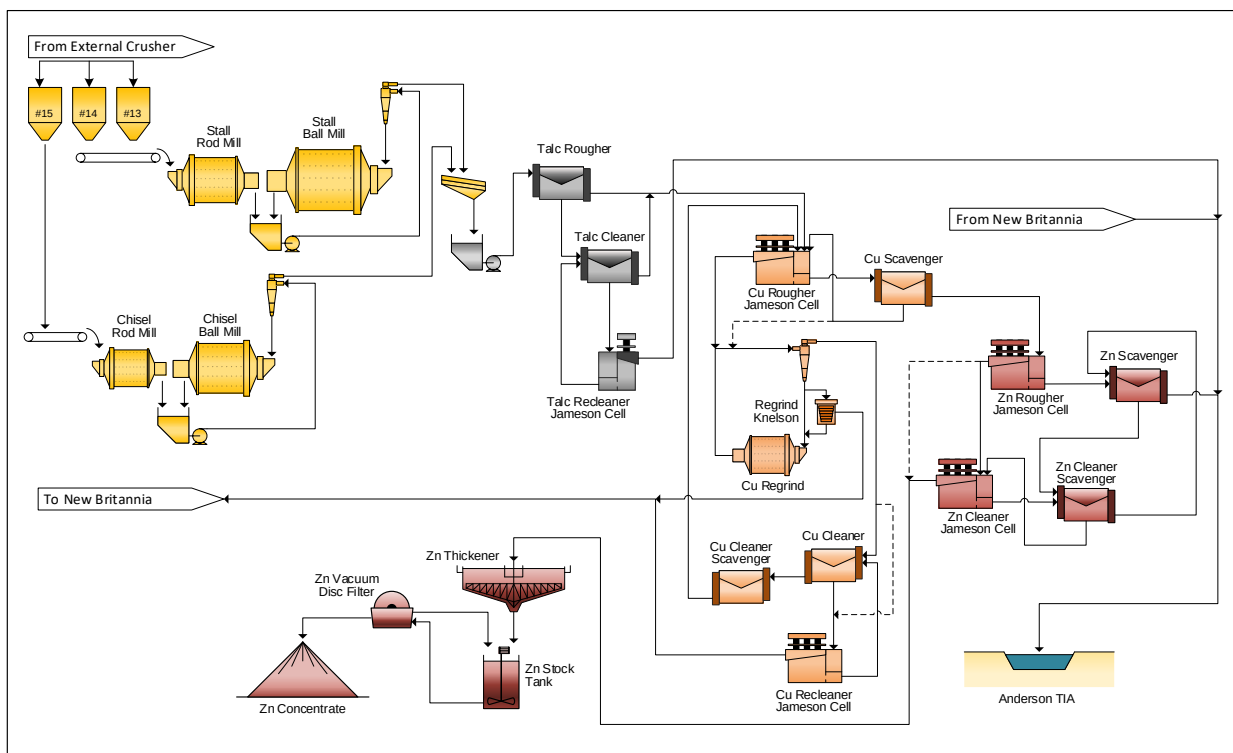
Flowsheet

The Stall mill consists of the following unit operations:

- Crushing:
 - Hewitt Robins 762mm x 1220mm Jaw Crusher
 - Tyroc 2.4m x 6.1m double deck screen
 - Symons Standard cone crusher
 - Ancillary conveying, dust collection, and other systems.
- Grinding
 - Three fine ore bins and associated belt feeders.
 - Two rod mills in primary grinding duty:

- 3.2m x 4.9m 600kW Stall Rod Mill
 - 2.1m x 3.0m 150kW Chisel Rod Mill
- Two ball mills, in secondary grinding duty:
 - 3.8m x 5.5m 1200kW Stall Ball Mill
 - 3.2m x 4.0m 600kW Chisel Ball Mill
- Ancillary pumps, a trash screen, hydrocyclones, and other systems.
- Talc flotation
 - Rougher flotation using Wemco 120 self-aspirated cells.
 - Two stages of cleaner flotation using Denver DR24 cells as the first cleaner and a Jameson Cell as the second cleaner.
- Copper flotation:
 - Rougher flotation using a Jameson Cell.
 - Scavenger flotation using Wemco 120 self-aspirated cells.
 - Regrinding of rougher and scavenger concentrate with a closed circuit 2.4m x 3.6m 400hp ball mill.
 - Gravity recovery of gold on the regrind hydrocyclone underflow using a Knelson mill.
 - Cleaner and cleaner scavenger flotation using Denver DR24 cells.
 - Recleaner flotation using a Jameson Cell.
- Zinc flotation:
 - Rougher flotation using a Jameson Cell.
 - Scavenger flotation using Wemco 120 self-aspirated cells.
 - Cleaner flotation using a Jameson Cell.
 - Cleaner scavenger flotation using Denver DR24 cells.
- Concentrate filtration and concentrate load-out:
 - Multistage centrifugal pump set for transfer of copper concentrate from the Stall mill to the New Britannia mill via a 125mm diameter pipeline.
 - Filtration of the zinc concentrate using two vacuum disc filters
 - Ancillary conveying, storage, and concentrate load-out systems.

FIGURE 17-1: STALL MILL FLOWSHEET



Crushing

Coarse ore from the mine is transported to a primary crushing plant located at the Chisel mine site and is operated by a third-party contractor. The product from the crushing plant is then trucked to the Stall mill and placed in coarse ore bins or a stockpile. Material from the bins is reclaimed via vibrating feeders and conveyed into a Hewitt Robins 762mm x 1220mm jaw crusher. The discharge is combined with that of the secondary Symons Standard cone crusher to feed a Tyroc 2.4m x 6.1m double deck vibrating screen. The undersize from the bottom deck is conveyed to the fine ore bins (FOB) while the oversize of the two decks is combined to feed the Symons cone crusher and then recirculated to the screen deck in a closed loop.

The described crushing circuit is in the process of being decommissioned and replaced with an external crushing circuit to service both the Stall and New Britannia mills. The new external crushing circuit start-up is aligned with the New Britannia mill commissioning schedule.

Grinding

From the fine ore bins, crushed ore is conveyed to the two independent Stall and Chisel circuits. The Stall circuit, which processes approximately two thirds of the mill feed, includes a 3.2m diameter x 4.9m rod mill and a 3.8m diameter x 5.5m ball mill. The Chisel circuit has a 2.1m diameter x 3.0m rod mill and a 3.2m diameter x 4.0m ball mill.

Each rod mill discharges into a dedicated hydrocyclone feed pump box where it is combined with the ball mill discharge and with dilution water. Each hydrocyclone underflow returns to their respective ball mill. The hydrocyclone overflow from each circuit combines and discharges into the hydrocyclone overflow pump box. The product reports to a flotation feed pump box through a trash screen. Lime is added to each rod mill to control the pH in the flotation feed.

Flotation

The flotation circuit outlined below is based on the Stall Mill Recovery Project Feasibility Study completed by Ausenco in Q3 2020. The recovery project envisions upgrades to the existing flotation circuit through the addition of flotation cells and instrumentation to improve the metal recovery and product quality. Detailed engineering is progressing through 2021 with completed and startup targeted for Q1 2023.

The upgraded flotation circuit consists of three sequential flotation circuits to enable the production of a copper-gold-silver and a zinc concentrate. These sequential flotation circuits are in order of process:

- Talc Flotation
- Copper Flotation
- Zinc Flotation

Talc Flotation

The talc flotation circuit concentrates and removes hydrophobic gangue. The trash screen undersize is pumped to a bank of five Wemco 8.5m³ cells for rougher flotation. The rougher concentrate undergoes two stages of cleaning via a bank of nine Denver DR24 cells, as the first cleaning stage, and a single Z1200/1 Jameson Cell, as the final cleaning stage. The talc concentrate is discarded to the final tailings. The final cleaning stage tails reports to the rougher concentrate pump box while the rougher and first cleaner tailings become the feed to the copper circuit.

MIBC is the only reagent added to the talc preflotation flotation circuit; although, there is infrastructure to add specialty sulfide depressants to the first and final stage of cleaning as required to minimize valuable mineral loss.

Copper Flotation

The talc flotation circuit tailings are pumped to the copper rougher. The copper rougher consists of a single E3432/8 Jameson cell. The rougher tailings will undergo scavenging in ten Wemco 8.5m³ cells, with the scavenger concentrate combined with the talc flotation circuit tailings and pumped to the rougher Jameson cell. The scavenger tailings is pumped to the zinc flotation circuit. The rougher concentrate is pumped to a 2.44m x 3.66m regrind ball mill with a 300kW motor. The regrind mill is operated in closed circuit with a target P80 of 30 to 40 µm. The option is available to direct the scavenger concentrate to the regrind mill when required.

The reground rougher concentrate is pumped to a bank of twelve Denver DR24 cells for the first stage of cleaning. The first cleaner concentrate undergoes a second stage of cleaner in a single E2514/3 Jameson cell. The final copper concentrate discharges via gravity to the copper concentrate holding tank prior to being pumped to the New Britannia mill. The second cleaner tailings are pumped to the head of the first cleaner. The tailings from the first cleaner are pumped to a bank of ten Denver DR24 cells for cleaner scavenging. The cleaner scavenger concentrate is combined with the second cleaner tailings for retreatment in the first cleaner. The cleaner scavenger report to the rougher Jameson cell, together with the talc circuit tailings and the scavenger concentrate.

The copper flotation circuit utilizes lime to maintain a pH across the rougher and cleaner stages of 10.0 to 10.8. The circuit employs Aerophine® 3418A as the sulfide mineral collectors, and MIBC as the frother.

Zinc Flotation

The copper circuit tailings are pumped to the zinc rougher, which consists of a single E3432/8 Jameson cell. The rougher tailings are scavenged in seven Wemco 8.5m³ cells with the rougher scavenger concentrate being pumped to the cleaner scavenger. The rougher scavenger tailings are the flotation final tailings and are pumped to the final tails pump box. The rougher concentrate can report to two

designations; when processing high grade zinc feed, the rougher concentrate is of sufficient grade to report directly to the zinc concentrate thickener; otherwise, the rougher concentrate is pumped to the zinc cleaner.

The zinc cleaner consists of a single E3432/8 Jameson Cell. The concentrate of the cleaner is at final concentrate grade and is pumping to the zinc concentrate thickener. The cleaner tail is combined with the scavenger concentrate and is pumped to a bank of twelve Denver DR24 cells, which act as a cleaner scavenger. The cleaner scavenger concentrate is pumped back to the cleaner Jameson cell. The cleaner scavenger tailings combine with the rougher tailings and are pumped to the scavengers.

The copper flotation circuit utilizes lime to maintain a pH across the rougher and cleaner stages of 11.0 to 11.5. The circuit employs Aero® 7279 as the sulfide mineral collectors and MIBC as the frother.

Concentrate Dewatering

Flocculated copper concentrate is pumped to a 3.4m diameter thickener. Underflow, at a target density of 65% solids, is pumped to an agitated stock tank. Currently, thickened copper concentrate is further dewatered to approximately 19% moisture on a disc filter, but after start-up of the New Britannia mill, the copper concentrate will be pumped via pipeline to New Britannia for dewatering and the existing copper dewatering section in the Stall mill will be decommissioned.

The flocculated zinc concentrate follows the same process as for copper but is further dewatered to approximately 11 % moisture using two disc filters.

Tailings

Plant tailings, which are composed of zinc rougher scavenger tailings, talc concentrate and certain sumps and flows from the dewatering circuits, are pumped out of the plant to the Anderson Tailings Impoundment Area (TIA) for final disposal. Tails required for paste are diverted to the Anderson booster pump station. The capacity of the pumping station ranges from 110 to 130tph of solids. Two pipelines have been installed between the Anderson booster pump station and the paste plant, located at Lalor mine site. Paste is delivered from a discharge hopper to underground via one of two nominal 8-inch diameter, hardened steel cased boreholes from surface to the 780 m level of the mine, the main distribution level to direct the paste to other levels above and below. Only one borehole is required during normal operation. The second borehole is available as a spare in the event of a plug or excessive wear on the primary hole. A network of underground lateral piping and level to level boreholes transfers the paste from the base of boreholes to the required underground locations.

Water

The Stall mill utilizes two sources of water, freshwater and reclaim water (recycled from the tailings facility).

Freshwater accounts for approximately 25% of the water usage within the mill and is withdrawn from Snow Lake via a 7km freshwater pipeline that feeds a holding tank at the Stall mill. It is then distributed to various areas of the process where clean, high-quality water is required. This includes, worker washroom and shower facilities, gland water, reagent mixing and instrument cleaning.

Reclaim water accounts for the remaining water usage within the mill and is withdrawn from the Anderson TIA. This water is deposited into the Anderson TIA as final process tailings. The water is then reclaimed for re-use within the mill via a 5km pipeline and is the primary source of water for the grinding and flotation processes.

Stall Mill Recovery Improvement Project

Based on the historical performance, the circuit flowsheet configuration and the metallurgical testwork the forecast metal recovery range for the Stall Concentrator treating Lalor Base Metal Ore is outlined Table 13-8. If changes are not made to the current flowsheet, e.g., installation of additional flotation capacity, the

recovery of recovery of copper, gold and zinc will be within the historical performance range. Hudbay plans to implement the upgrades to the Stall flowsheet based on the successes of the test program described in section 13 of this report. The upgraded flowsheet design calls for the installation of:

- Increase copper rougher and cleaner capacity with the addition of Jameson cells.
- Addition of talc preflotation circuit with the re-configuration of existing flotation cells and addition of a Jameson cell cleaner stage.
- Increase zinc circuit cleaning capacity and add froth washing through the addition of Jameson cells.
- Increase recovery of free gold through the addition of a Knelson gravity concentrator on the copper regrind cyclone underflow.

The comparison of the current and forecasted recovery and product grades increase as per the upgraded flowsheet are summarised in Table 17-1.

TABLE 17-1: FORECAST METAL RECOVERY FOR THE CURRENT FLOWSHEET AND AN UPGRADED FLOWSHEET

Year	Tonnes (ktpa)	Metal Recovery				Concentrate Grades	
		Au (%)	Ag (%)	Cu (%)	Zn (%)	Cu (%)	Zn (%)
Current Flowsheet	1,000 to 1,400	55.0 - 64.0	51.0 - 59.0	84.0 - 89.0	90.0 - 93.0	17.0 - 20.0	48.0 - 51.0
Upgraded Flowsheet	1,000 to 1,400	64.0 - 70.0	65.0 - 74.0	91.0 - 95.0	90.0 - 93.0	20.0 - 22.0	53.0 - 56.0

The Stall Recovery Improvement Project feasibility study was completed in 2020 by Ausenco with a Class 3 capital estimation of CAD\$24.5M excluding tax. The breakdown of direct, indirect and owners' costs inclusive of contingency is 57%, 33% and 10% respectively. The cost estimation benefited from benchmarking of labor and bulk rates with the New Britannia project.

TABLE 17-2: STALL RECOVERY PROJECT CLASS 3 CAPEX

DISC.	DISCIPLINE DESCRIPTION	Total Manhours	Total Labour (\$ CAD)	Total Mat (\$ CAD)	Total Equip (\$ CAD)	Total Freight (\$ CAD)	Total Subcon (\$ CAD)	Total Growth (\$ CAD)	Total Cost (\$ CAD)
B	B EARTHWORKS	-	-	-	-	-	-	-	-
C	C CONCRETE	775	62,556	143,472	-	-	-	42,455	325,485
S	D STEELWORK	6,362	289,419	442,940	-	45,789	65,831	58,301	1,270,115
F	E PLATEWORK	959	48,111	236,701	-	18,399	-	17,202	379,635
M	F MECHANICAL	15,164	954,286	-	5,812,897	433,381	-	397,093	8,772,326
P	G PIPING	4,177	157,588	1,098,138	-	54,907	-	144,025	1,639,184
E,L	H ELECTRICAL	2,400	112,762	51,292	690,191	21,087	-	60,233	1,072,112
I	J INSTRUMENTATION	1,877	88,189	-	574,322	19,824	-	38,465	827,592
A	M BUILDINGS / CONTAINERS	-	-	-	-	-	-	-	-
R	VENDOR REPS	-	-	-	-	-	-	-	-
	SUB-TOTAL DIRECT COSTS	31,713	1,712,910	1,972,543	7,077,409	593,388	65,831	757,773	14,286,449
O	OWNERS COSTS	-	-	-	-	-	1,428,645	-	1,428,645
U	CONTRACTOR INDIRECTS	-	-	-	-	-	1,955,472	97,774	2,053,245
T	COMMISSIONING	17,240	-	-	-	-	3,501,357	-	3,501,357
	FIRST FILLS	-	-	-	-	-	-	-	-
	COMMISSIONING SPARES	-	-	-	-	-	-	-	-
V	CAPITAL SPARES	-	-	-	-	-	234,933	-	234,933
	OPERATION SPARES	-	-	-	-	-	-	-	-
	SUB-TOTAL DIRECT COSTS	17,240	0	0	0	0	7,120,406	97,774	7,218,180
Y,D	CONTINGENCY + DEMOLITION	422	15,459	326,524	-	-	2,581,657	30,551	2,973,218
	PROJECT TOTAL	48,953	1,712,910	1,972,543	7,077,409	593,388	7,186,237	855,547	24,477,847

The project is scheduled to progress through detailed engineering in 2021 and construction in 2022 following the completion of the New Britannia Project. Completion and operation of the Stall Recovery Improvement Project is schedule for Q1 2023.

Future Stall Projects

As the Lalor mine matures and additional satellite ore sources are identified to supplement the feed blend to the Stall mill, metallurgical projects are actively advanced to match the configuration of the mill to the requirement of the ore. The following are projects that are being advanced through metallurgical testwork, early stage engineering or both.

- The current life of mine plan for Lalor shows an increasing Pb:Cu ratio from 0.3 to >1.0. In addition, the 1901 deposit has a Pb:Cu ratio of >1.5 indicating the need for a lead mitigation strategy. Preliminary results indicate that a traditional sequential Cu-Pb-Zn flotation circuit will mitigate the increasing lead in feed and provide additional lead concentrate revenue stream by capturing a significant portion of the copper, gold and silver that is currently lost to the zinc concentrate.
- Stall Tails Leach – The flotation recovery of gold is limited by the mineralogy and association of gold with iron sulfides. Due to the mineralogical differences between the gold zone and base metal zone ores, the flotation tailings from the base metal ores are not compatible with the same leaching conditions as the gold zone ore flotation tailings.
- Stall Fine Grinding – Along side the Tailings Leach project, a fine grinding project is being advanced to determine the improvements in flotation recoveries when the flotation feed size is reduced from the current p80 of 120µm to 60µm.

17.3. NEW BRITANNIA MILL

In March 1949, the Nor-Acme mine and gold mill was opened. Production continued until 1958, with the Nor-Acme mill recovering approximately 610,000 oz of gold during this period. Between 1958 and the early 1990s, various companies worked on the property with the intention to bring the mine back into production. These efforts were unsuccessful.

In 1994, TVX and High River formed a joint venture to reopen the mine. Using ore from secondary mining areas, the mill was brought into production while completing construction of the main shaft. When production commenced in 1995, the name Nor-Acme was changed to New Britannia. Through various transactions, Kinross became the operator of the New Britannia mine-mill complex. Production mining ceased at the end of September 2004 and the mill was put on care and maintenance in 2005 due to a low gold price environment after producing 1.6 million oz of gold. In 2015, Hudbay acquired the New Britannia mill. It is currently being refurbished to process ores from the Lalor gold and copper-gold zones.

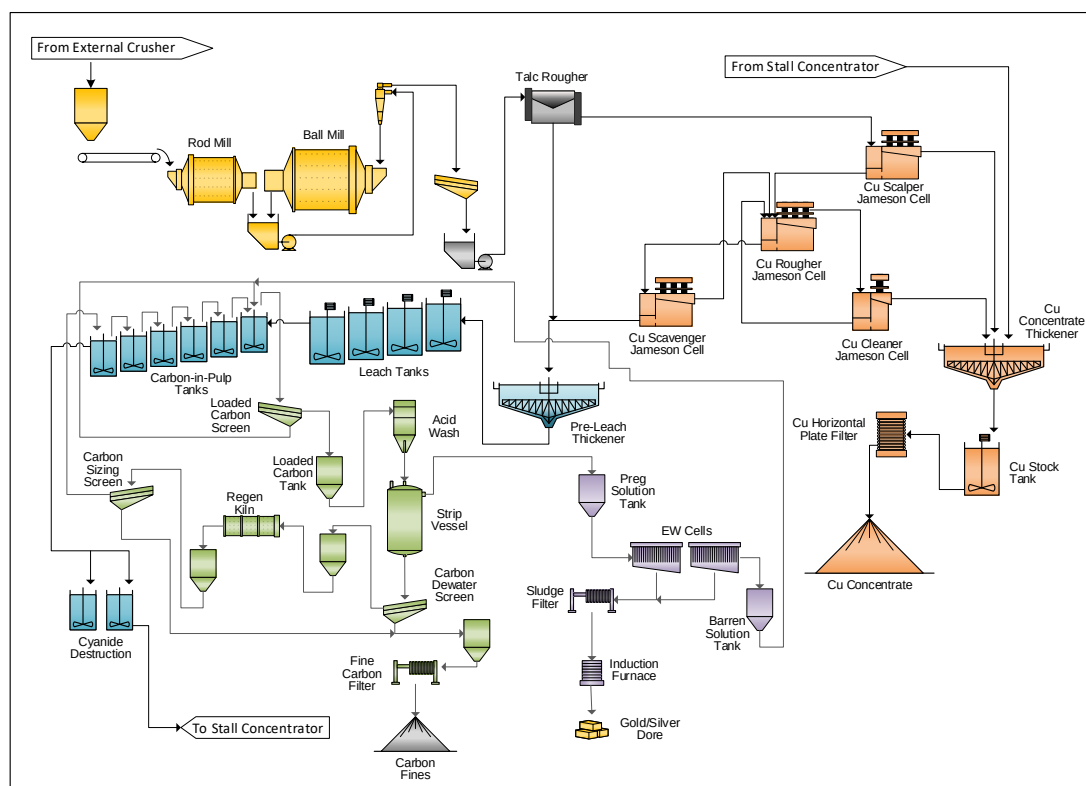
Flowsheet

The New Britannia mill consists of the following unit operations:

- Ore Receiving
- 150 tonne capacity dump pocket including a grizzly screen cover.
- Associated belt feeder and conveyor systems
- Single 2700 tonne capacity fine ore bin
- Grinding
- 3.5m x 4.5m 750kW Rod Mill
- 4.2m x 6.1m 1500kW Ball Mill
- Ancillary pumps, a trash screen, hydrocyclones, and other systems
- Talc flotation
- Rougher flotation using Direct Flotation Reactors
- Copper flotation:

- Rougher and cleaner flotation using Jameson Cells.
- Leaching, CIP, Elution and Refining
- Four leach tanks
- Six carbon-in-pulp tanks
- Loaded carbon, carbon dewatering and regenerated carbon screens.
- Fine carbon filter
- Carbon regeneration kiln
- Acid wash vessel
- Pressure strip vessel
- Two electrowinning cells
- EW sludge filter
- Induction furnace
- Concentrate filtration and concentrate load-out:
- 8m diameter high-rate thickener
- Horizontal plant pressure filter
- Tailings
- Two cyanide destruction tanks
- Multistage centrifugal pump sets for transfer of final tailings from the New Britannia mill to the Stall mill via a 200mm diameter pipeline.

FIGURE 17-2: NEW BRITANNIA MILL FLOWSHEET



Ore Receiving

Coarse gold and copper-gold ore from the mine is transported to the external crushing plant at the Stall Mill. The product from the crushing plant is delivered to the New Britannia site using 40 tonne trucks, where the material will be loaded into a dump pocket and conveyed to the fine ore bin (FOB).

Grinding

The crushed material is reclaimed from the FOB via conveyors into the rod mill and then discharged into a hydrocyclone feed pump box, where it is combined with the ball mill discharge and dilution water. A variable speed hydrocyclone feed pump feeds a the hydrocyclones cluster. The hydrocyclone underflow returns to the ball mill while the overflow flows via gravity to a trash screen and then is pumped to the flotation circuit.

Flotation

From the hydrocyclone overflow pump box, the slurry is pumped to the talc flotation circuit. Talc flotation consists of two 1.7m diameter Direct Flotation Reactors (DFR), operating as either a hydrophobic gangue removal stage ahead of the Cu flotation circuit to improve final concentrate quality, or as an additional Cu scalping stage in the event of above-design Cu feed grades.

The talc flotation tailings are sent to the flotation feed pump box and from here it is pumped to the first of three E1732/4 Jameson cells. The concentrate from the first Jameson cell, the Cu scalper, is at final product grade and reports to the Cu concentrate thickener. The tailings from the Cu scalper are pumped to the second Jameson cell, the Cu rougher, with the concentrate from this cell reporting to a single stage of cleaning in a Z1200/1 Jameson cell. The tailings from the Cu rougher is pumped to the third Jameson cell, the Cu Scavenger, with the concentrate reporting back to the Cu rougher along with the Cu cleaner tailings. The concentrate from the Cu cleaner reports along with the Cu scalper concentrate to the Cu

concentrate thickener. Note that the combination of two 1.7m DFRs in talc flotation duty, combined with three same-sized Jameson cell units in the copper flotation circuit allows for maximum circuit flexibility to changes in head grades or feed mineralogy.

Concentrate Dewatering

The combined concentrate from the Cu scalper and Cu Cleaner is mix with the copper concentrate from Stall Mill in the 8m diameter copper thickener. The concentrate is thickened, pressure-filtered, and stored in a concentrate shed, before being loaded onto trucks and transported offsite.

The Cu scavenger tails, depleted of copper, reports to the 12 m diameter pre-leach thickener for thickening prior to cyanidation. The underflow product of the thickener is approximately 50% solids.

Thickener overflow from the copper concentrate and pre-leach thickeners is returned to the process water tank.

Leaching

The thickened cleaner scavenger tailings and rougher tailings is fed to four leach tanks in series with aeration. The total retention time is at least 50 hours. Hydrated lime is added to ensure a pH above 11.5 and sodium cyanide solution is added to leach the gold. The slurry stream discharge from the final leach tank gravitates to six CIP leach tanks in series with a 1.2 hr retention time in each. Carbon is added to the sixth tank and is transferred counter-currently, adsorbing gold and silver as a cyanide-gold complex as it progresses upstream to the first CIP tank.

At the first tank, loaded carbon is transferred to a fine vibrating screen where it is separates the carbon from the slurry. The loaded carbon gravitates to a holding tank and is batch transferred periodically and washed with a mild acid solution to remove impurities. The washed carbon is then stripped using a hot caustic cyanide solution (pregnant solution). The pregnant solution proceeds to electrowinning. The sludge collected is dried and treated with fluxes and melted in the induction furnace to obtain the Doré. The waste slag from is collected, crushed, and added to the rod mill.

The stripped carbon, after elution, will retain a significant quantity of organic and inorganic contaminants (foulants) present in the cyanide slurry. Prior to returning to the process, the carbon will be regenerated by removing these contaminants in a kiln.

The final tailings from the sixth CIP tank gravitates to the carbon safety screen where fine carbon is collected and sent to the fine carbon tank. The screen undersize is followed by the cyanide destruction stage. The cyanide destruction is achieved via the INCO SO₂/Air process utilising sodium metabisulphite as the SO₂ source. The carbon safety screen undersize is pumped to a splitter box which evenly distributes the tailings slurry between two air-sparged cyanide destruction tanks operating in parallel each with 1hr retention time. The tailings slurry post cyanide destruction gravitates to the final tailings pump box.

Tailings

The final tailings from the New Britannia mill is pumped to the Stall mill via a 6.8km pipeline. This same pipeline corridor will house the copper concentrate pipe and the reclaim water (both with flow from Stall to New Britannia). Once the tailings line reaches the Stall mill tailings pump box, the paste plant has the option of directing the flow to the paste plant or to the Anderson TIA, depending on the needs for paste backfill.

Water

The New Britannia mill utilizes three sources of water, fresh water, process water and reclaim water (recycled from the tailings facility).

The fresh water is utilized in areas of the process where clean high-quality water is required, including the worker washroom and shower facilities, gland water, reagent mixing and instrument cleaning. The freshwater is sourced from Snow Lake to the New Britannia mill. It is fed into a holding tank within the mill building and distributed to the process as required.

Process water is used in the grinding and flotation sections. It is sourced from the concentrate and pre-leach thickener overflow streams.

Reclaim water accounts for the remaining water usage within the mill and is withdrawn from the Anderson TIA. This water is deposited into the Anderson TIA as part of final tailings. The water is then reclaimed for use within the New Britannia mill via a pipeline along the same corridor as the tailings and concentrate pipelines. The amount of reclaim water required by the New Britannia mill is proportional to the quantity of water deposited with the final tailings.

Future New Britannia Projects

As the Lalor mine matures and additional satellite ore sources are identified to supplement the feed blend to the New Britannia mill, metallurgical projects are actively advanced to match the configuration of the mill to the requirement of the ore. The following are projects that are being advanced through metallurgical testwork, early stage engineering or both.

- **Throughput Increase** – The previous operation of the New Britannia mill saw throughput rates of up to 2,000tpd based on the availability of ore from the underground operation. With the increase in the mineral reserves of the Lalor gold zones alongside additional the satellite deposits opportunity exists to increase the throughput rate. The study to date has identified the tailings pipeline, pre-leach thickener and various minor pumping systems require upgrade to accommodate the increased tonnage rate. All the other major systems within the New Britannia mill have sufficient capacity to accommodate the increased tonnage. In addition to the processing equipment upgrades the study includes the upgrade of the pipeline corridor road to accommodate ore haulage from Stall to New Britannia to remove traffic from the public access road.
- **1901 Gold Zones** – The 1901 deposit contains two gold-copper lenses which are suitable feed for the New Britannia mill and would supplement the Lalor gold zone ore. This project is advancing through metallurgical testwork.

18. PROJECT INFRASTRUCTURE

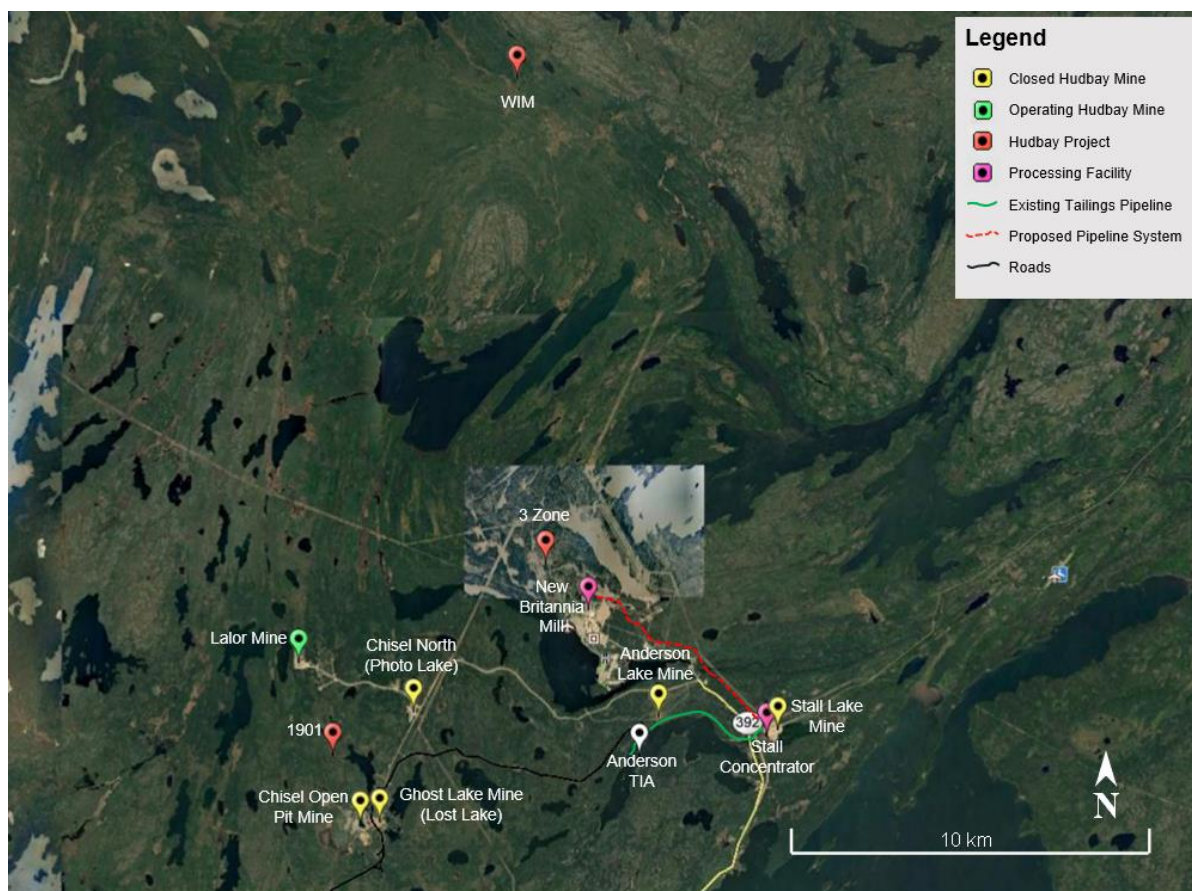
18.1. AREA SURFACE INFRASTRUCTURE

Lalor Mine is located 16 km by road from the Town of Snow Lake, Manitoba. General area infrastructure includes provincial roads, 115 kV Manitoba Hydro grid power within four km of Lalor and Manitoba Telecom (Bell) land line and cellular phone service.

The Town of Snow Lake is a full-service community with available housing, hospital, police, fire department, potable water system, restaurants and stores. The community is serviced by a 914m gravel airstrip to provide emergency medical evacuation. Hudbay owns a contractor operated 260 person camp in Snow Lake and the New Britannia mine, mill and office.

Figure 18 1 provides an overview of the various infrastructures and of the material flow from the mine to the tailings disposal facility in Snow Lake.

FIGURE 18-1: MAP OF SNOW LAKE MINING, PROCESSING AND TAILINGS FACILITIES



Source: Google Earth, 2021 image

Chisel North Mine Site

The Chisel North mine site is located 3.5 km from Lalor Mine. Infrastructure includes:

- Building complex with offices, changehouse and shop,

- Lalor Mine electrical substation with two (2) 115-25 kV 24 MVA transformers,
- Water pumping station (booster pump station) with holding tanks and pumps, and
- Contractor owned and operated ore crusher. Stockpile capacity at Chisel North is 15,000 tonnes.

South of the Chisel North mine site, along PR 395, there are:

- Two mine fresh air raises equipped with downcast fans, mine heaters and each with a 30,000 US gallon propane tank.

Chisel Lake Mine Site

The Chisel Lake mine site is a decommissioned mine located approximately 7.5 km from Lalor. Infrastructure includes:

- Mined out open pit used for potentially acid generating (PAG) waste rock disposal and storing mine discharge water and water treatment plant sludge,
- High density sludge process acidic water treatment plant that can treat up to 2,500 gpm, and
- Fresh (process) water pumps for Lalor Mine.

Anderson Tailings Impoundment Area Site

The permitted Hudbay Anderson TIA is located approximately 12 km from Lalor. Infrastructure includes:

- Slurried tails pump station,
- 13 km of 14" buried pipeline from Anderson TIA to Lalor to deliver slurried tailings to the Lalor paste backfill plant, and
- 13 km of 10" buried pipeline to return water from the paste plant to Anderson TIA.

Mine Access Corridor

The mine access corridor is the roadway between the PR395 at the Chisel North mine site to Lalor mine site. Infrastructure in the corridor includes:

- Four km of 25 kV overhead power lines,
- Buried and heat traced process water and mine discharge pipelines, and
- Buried slurried tailings and return water pipelines.

WIM and 3 Zone Project Sites

The WIM deposit is located approximately 15 km by road north of the New Britannia mill, 315 m above sea level. Access to WIM is currently via a winter road.

3 Zone is accessible via road and located approximately 3 km by road northwest of New Britannia Mill. From Lalor, it is located approximately 8 km northeast (12 km by road).

Required infrastructure at both sites includes:

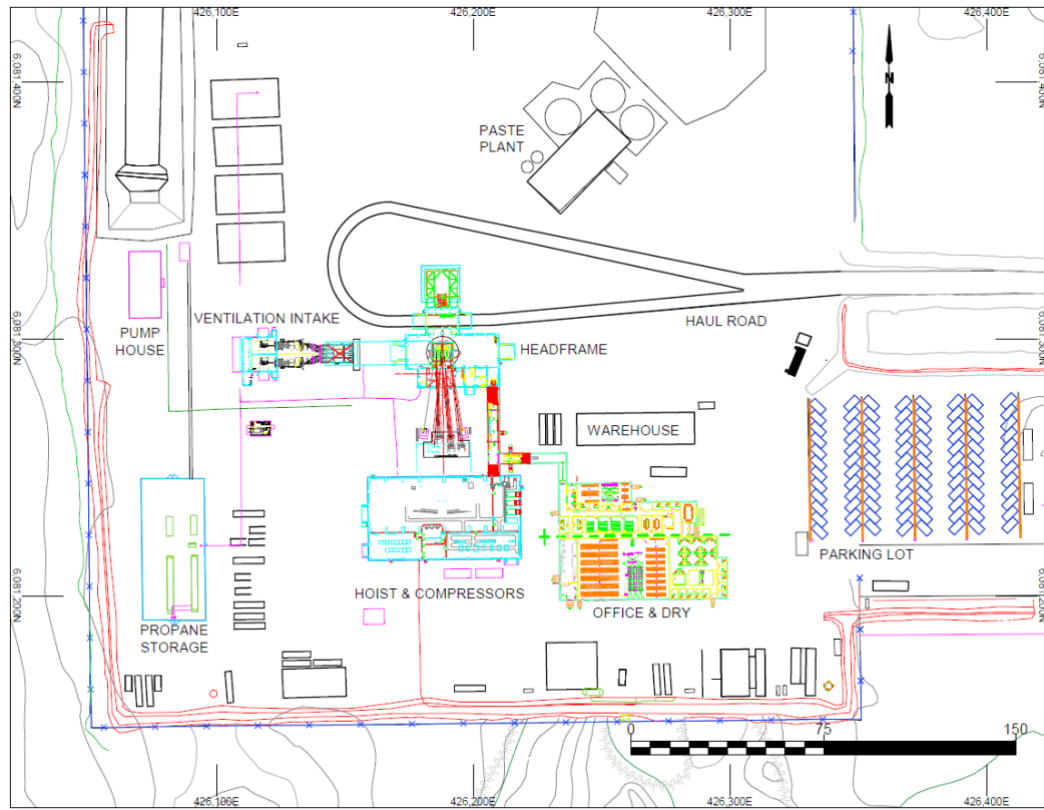
- A gravel access road to site from the New Britannia site. This road will require year-round maintenance.
- A trailer with washroom facilities and a small conference area.
- Powerlines along access road to site.

18.2. MINE SITE SURFACE INFRASTRUCTURE

A site drawing of the Lalor access road and services is shown in Figure 18-2. The Lalor mine surface infrastructure includes the following:

- Mine site services: This includes pole and buried electrical power lines, transformers and switchgear, and buried freshwater, mine discharge water and tailings pipelines.
- Hoist house: Contains electrical distribution for the site, hoist and communication control room, production hoist (Davy Markham, double drum, 4,828 kW), service hoist (Davy Markham, double drum, 2,414 kW), three GA 250 screw compressors (1,477 cfm @120 psi each).
- Headframe: Contains utility hoist (Davy Markham, single drum, 314 kW), bin house (approximate capacity 1,000 tonnes of ore), external bunker (capacity 1,200 tonnes of ore), two 250 hp downcast fans (313,000 cfm each) and mine air heater.
- Two exhaust fans (2,500 hp and 575,000 cfm each) located approximately 1.0 km from the production shaft.
- Main pump station: Includes holding tanks for discharge water, process water and potable water, PAL water system and pumps for discharge, potable, process and fire water.
- Paste backfill plant with tailings receiving and storage tanks, tailings thickener, tailings filters and cement storage silos.
- 341 person change house complex that also houses staff offices and 100 person and 50 person change houses adjacent to the main complex.
- Two 30,000 US gallon propane tanks.
- Office complex for health & safety, training and mine rescue.
- Fuel tanks and pumps for diesel and gas.
- Warehouse and shop.
- Multiple storage containers (sea containers) and contractors' office trailers.

FIGURE 18-2: LALOR MINE SITE GENERAL ARRANGEMENT



Site drawings of the WIM and 3 Zone access roads and services are shown in Figure 18-3 and Figure 18-4. The mines surface infrastructure includes the following:

- Maintenance facilities: the primary maintenance facility for surface and underground equipment constructed to service Lalor Mine is too distant to utilize and a small two bay facility has been costed at the site for the duration of operations.
- Warehouse: all purchasing is handled by the on-site staff, with regular freight movement between sites. The on-site warehouse space accommodates spares for mining as well as milling operations.
- Fuel and lube distribution: the fuel system includes a surface fuel farm and storage tanks with fuel distribution to the underground operations via a fuel and lube truck. This service will be provided by a contractor.
- Main administration offices: Management, supervisory and technical teams will be centralized in an office complex, located at either New Britannia mill or Hudbay's Lalor Mine site.
- Mine dry & conference space: A trailer will be placed at the WIM and 3 Zone locations to provide facilities for staff. The trailers will have areas for meetings and a lunchroom. The new dry facilities at New Britannia site be utilised and workers bused to the site.
- Electrical and power distribution: the above ground electrical distribution originates at the New Britannia Mill site and terminates at the WIM and 3 Zone portals. From there, the underground electrical distribution system will be supplied by two primary feeders from the mine services power distribution switchgear on the surface.

FIGURE 18-3: WIM SITE LAYOUT

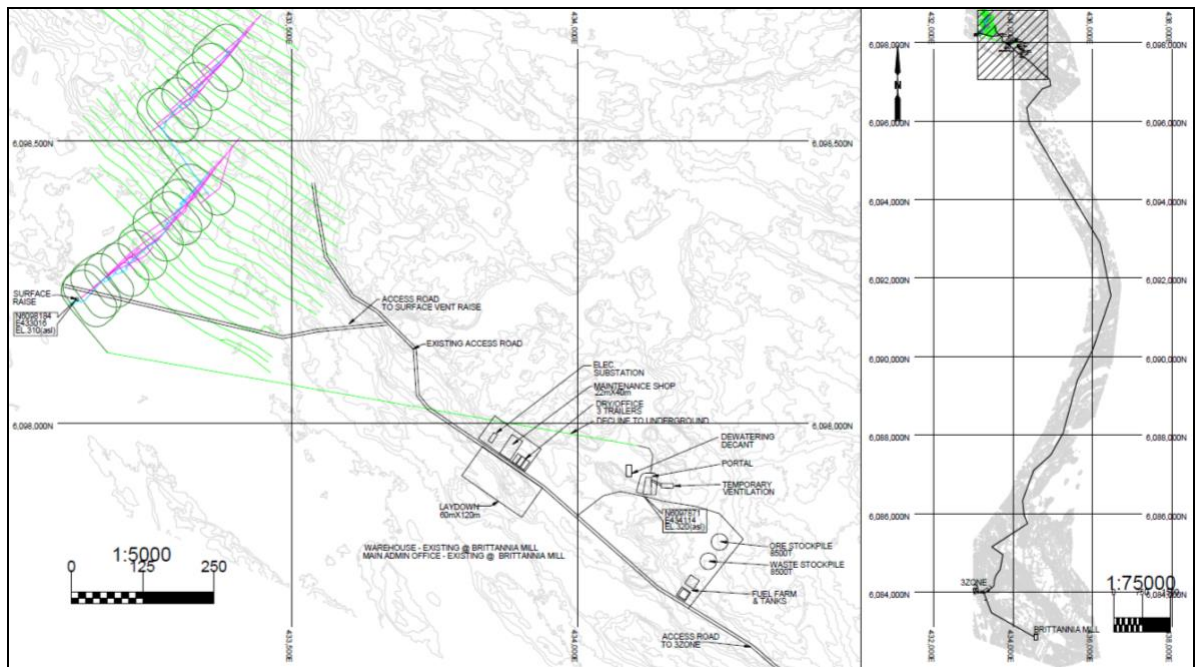
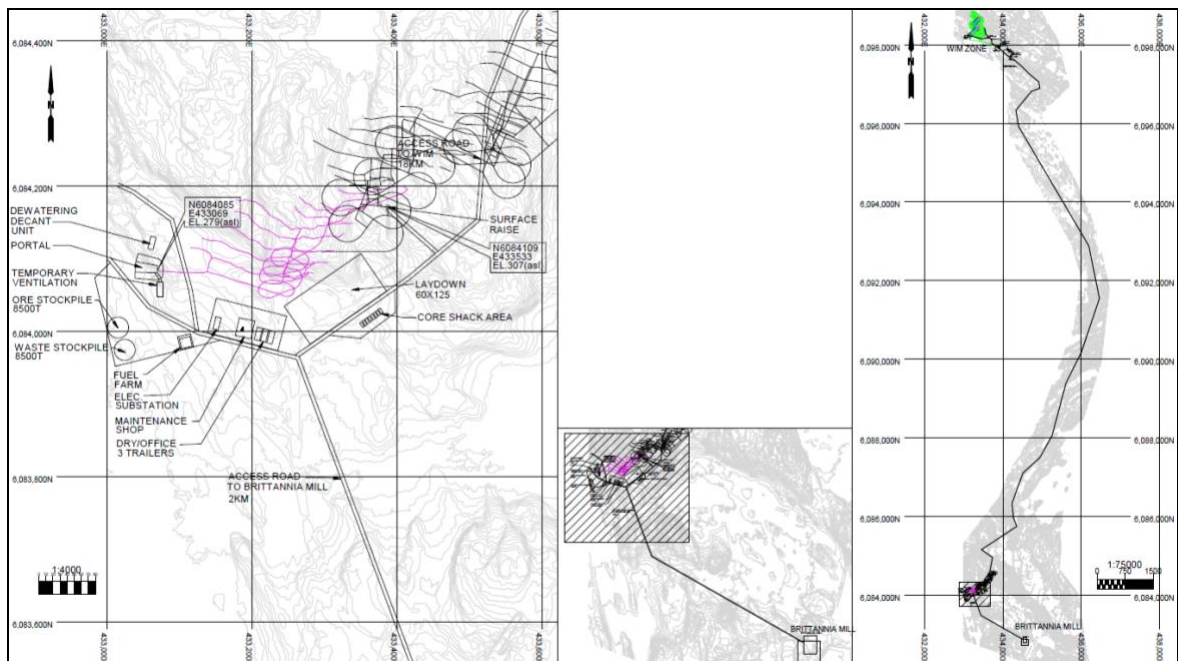


FIGURE 18-4: 3 ZONE SITE LAYOUT



18.3. MINE UNDERGROUND INFRASTRUCTURE

Mining infrastructure includes excavations and equipment normally used below the shaft collar. It includes for Lalor:

- Main Production shaft; 6.9m diameter concrete lined with five compartments equipped with:
- Two - 16 tonne skips
- One double deck cage (50 people per deck)
- Counter weight
- Utility conveyance (6 person cage)
- Three main shaft stations at 835m, 910m and 955m levels
- Lateral development, consisting of 6m x 5m ramps and level development connecting all mine workings.
- Ramp from surface to 810m level. Secondary egress from the mine. Total distance is approximately 6.0 km.
- Power Distribution consisting of:
- 2 x 25 kV power cables down the shaft
- 7.5 MVA 25 kV to 13.8 kV transformers at the 835m level and 910m level shaft stations
- Primary distribution throughout the mine is 13.8 kV with transformers to 600 V for local distribution
- Compressed air, process water and discharge water pipelines throughout the mine.
- Leaky Feeder underground wireless radio communication system.
- Fiber-optic backbone for data and video.
- Ore handling system consisting of two rock breakers and bins on 910m level feeding chutes and conveyor system on 955m level.
- Mobile Maintenance shop located at the Chisel North underground workings. Lalor maintenance shop located near the 910m level shaft station is under construction.
- Water Discharge system consisting of a series of drain holes and sumps with submersible pumps. Two settling cones on 910 m level. Clean water sumps on 955m level with 1,250 hp 10 stage Mather Platt pump (740 gpm).
- Internal ore passes to transfer ore from upper levels of the mine to 910m level.

Electrical Distribution and Equipment

Grid electricity is supplied by Manitoba Hydro, the provincial power utility. Manitoba Hydro's 115 kV power line terminates at the Chisel North Mine site, approximately 3.5 km by road from the Lalor mine site. This feeds power to the Hudbay owned main distribution substation consisting of two 115-25 kV 24 MVA transformers. The substation is completely equipped with an E-house complete with four GE Powervac circuit breakers and Tie breaker. From there, power is routed as follows:

- 1) Breaker 52-C1 provides power to:
 - (a) The Chisel North mine site
 - (b) The Chisel Lake mine site
 - (c) The Chisel North mine downcast fan substation
 - (d) Lalor mine ramp downcast fan site

- 2) Breaker 52-L1 – Lalor CCT1 provides power to the following areas:
 - (a) Lalor mine site pump house and water treatment plant
 - (b) Lalor mine exhaust fans
 - (c) Lalor paste plant
 - (d) Hoist house electrical room Bus 1 which distributes power to:
 - Mine underground Feeder A
 - VFD drives for production hoist
 - MCC for production hoist
 - Hoist house PDC (compressors)
 - MCC for Service hoist
 - Auxiliary hoist
- 3) Breaker 52-L2 – Lalor CCT2 provides power to the following areas:
 - Mine UG Feeder B which feeds
 - Headframe Complex
 - VFD Drives for Service hoist.
 - Intake fans
 - Lalor mine Office/Dry

Mine Underground Power Supply

Lalor mine underground mine electrical distribution to the mine workings consist 13.8 kV that is further stepped down to 600V. This power is used to power:

- (a) Auxiliary ventilation fans,
- (b) Auxiliary pumps,
- (c) Mobile electric jumbo drills and bolters, and
- (d) Underground fiber communication network.

Lalor mine underground mine power distribution equipment currently consists of:

- (a) 27 X 1 MVA 13.8-0.6 kV mine power centres (aka. portable sleds),
- (b) 16 X 0.75 MVA 13.8-0.6 kV mine power centres, and
- (c) 6 x 200A S&C load break switches.

The mine power centers are located in constructed underground substations.

Emergency Generators

Emergency generators are required to maintain water flows and to evacuate the mine in the case of power failure. Emergency generators are located at:

- Lalor mine pump house and treatment plant - Cummins 350 kW
- Booster pump station (Cummins 350 kW)
- Lalor auxiliary hoist (Cummins 1000 KVA)
- Paste backfill plant (Cummins 1000KVA)

- Anderson pump station (Cummins 1000 KVA)

All generators are equipped with automatic transfer switch that will in the event of a power failure transfer to generator power and vice versa.

1901 Underground Infrastructure

Mining infrastructure at 1901 is based on current practices at Lalor to maintain commonality with equipment, materials, operating procedures, and workforce skillsets. Mine services (compressed air, power, process water, drainage, etc.) are to be integrated with Lalor systems. Key elements include:

- Ventilation: push-pull ventilation system – 305 kcfm airflow requirement
- Fresh air down existing (unused) surface borehole, through new haul ramp and exhaust to surface via new service ramp and RAR
- Fresh air heated as required with a direct fired propane heater
- 3.5mØ RAR to be raisebored (495m length)
- 2 x 200 hp main fresh air fans and 2 x 250 hp return air fans
- Electrical: feed from Lalor 825mL transformer
- 9 x electrical sub-stations
- Service water (drilling, dust suppression) and drainage:
- Mining activity requires 350 L/min during shift average, with potential peak of 1,140 L/min
- Ditches, borehole collection sumps, level sumps, main pump station required
- Supply from / discharge to water reservoir in Remuck #3 in Lalor Ramp
- Other infrastructure such as 3 x refuge stations, SatStat fueling station, explosives & materials storage, escapeways also required

WIM and 3 Zone Underground Infrastructure

A new portal and ramp will be constructed at WIM, while the existing ramp and decline at 3 Zone will be refurbished for use during planned operations. Key elements of the underground infrastructure include:

- Dewatering equipment based on a system capacity of 500 GMP (114m³/h). All components are sized to handle that flow rate regardless of the elevation, using the following components:
- Pump stations: each portable pump station comprises of a tank with 5 minutes of retention time, two slurry pumps (one operating and one standby), and a power and control panel.
- Booster stations: each portable booster station comprises two slurry pumps (one operating and one standby), and a power and control panel.
- Clarifiers: each clarifier unit can treat up to 70 m³/h and uses lamella plates to separate solids
- Piping: all piping is to be 8" HDPE DR11 with Victaulic couplings. Mine communication and control systems: through leaky feeder and phone line systems installed in the trailer on site. The leaky feeder system is used through the primary development and level accesses to provide radio communication.
- Other infrastructure: portable refuge stations and sanitary facilities, SatStat fueling station, explosives & materials storage, and escapeways

18.4. PASTE PLANT AND ANDERSON PUMP STATION

Construction and commissioning of the paste plant and distribution system was completed in the third quarter of 2018. Paste is critical for the sustainability of the mine production plan. The paste plant is designed to fill voids left by mining of approximately 4,500 tpd. Taking into account waste generated from development in the LOM and the plan to not hoist waste from underground, the combined paste/waste backfilling capacity is approximately 6,000 tpd.

The paste plant is located northeast of the existing headframe complex at the Lalor mine site. Paste delivery capacity is 165 tph solids (tails) or 93 m³/hr paste. The paste plant is capable of varying the binder content in the paste to provide flexibility in the cure rate of the paste where higher and early strength may be required depending on mining method.

Tails are pumped from the Stall concentrator to the Anderson TIA. Tails required for paste are diverted to the Anderson booster pump station. The capacity of the pumping station ranges from 110 to 130 tph to allow for variation in the output of tailings from the concentrator. Tailings are directed into the Anderson TIA when not required for the paste plant.

Two pipelines have been installed between the Anderson booster pump station and the paste plant located at Lalor mine site, approximately a 13 km distance. The tails slurry pipeline is a nominal 14 inch diameter and the return water pipeline is a nominal 10 inch diameter. The main route of this pipeline is on top of the existing abandoned rail bed (property owned by Hudbay), then along the west side of Provincial Road #395 and finally along the south side of the Lalor mine access road to the paste plant.

Paste is delivered from a discharge hopper to underground via one of two nominal 8 inch diameter, hardened steel cased boreholes from surface to the 780m level of the Lalor mine. Only one borehole is required during normal operation. The second borehole is available as a spare in the event of a plug or excessive wear on the primary hole. The boreholes were drilled and cased in 2016.

A network of underground lateral piping and level to level boreholes transfers the paste from the base of boreholes to the required underground locations. The 780m level is the main distribution level to direct the paste to other levels above and below. Underground development connects the paste fill distribution system to mine workings via existing and planned development into stopes.

Operational improvement projects have been implemented to enhance the reliability of the tailings delivery system on surface. This includes the establishment of additional water recirculation, improved environmental containment, flexibility for water input due to tailings level expansion at Anderson tails area, and winterizing improvements at the Lalor Paste Plant. While some of these improvements are already in place, remaining construction on these improvements is slated for completion in the second quarter of 2019.

The paste plant utilizes the Lalor mine compressors. Paste plant and Anderson booster station infrastructure includes:

- Four slurry booster pumps
- Thickener – 14m x 5.5m
- Two filter feed tanks – 14m x 14.5m
- Disc filter – 9.4m x 4.8m x 4.5m
- Vacuum pump
- Paste hopper
- Continuous mixer
- Bag house dust collector and dust collector blower
- Two cement silos – 4.1m x 15.2m

- Two screw conveyors
- Two rotary valves and loss in Wt feeder
- One flocculent system
- Two MCC rooms – ABS and Paste Plant
- One crane
- Electric heating installations

18.5. STALL CONCENTRATOR AND ANDERSON TIA

Hudbay operates the Stall concentrator approximately 16 km from Lalor. The mill infrastructure consists mainly of two buildings. The crusher building houses the jaw and cone crushers, the screen deck and conveyors. The mill building hosts the grinding mills, flotation circuits, dewatering processes and concentrate shed areas. Three trailers are on site to provide office space. As of January 2019, the current Stall concentrator infrastructure includes:

- Two 350 tonnes coarse ore bins and 3 750 tonnes fine ore bins
- One Hewitt Robins 762mm x 1219.2mm jaw crusher
- One Symons Standard 7 ft cone crusher
- One Tyroc double deck vibrating screen – 2.4m by 6.1m screen frame, 50.8mm on top deck, 19.05mm bottom deck
- Two Rod mills – AC – (2.1m x 3.0m) and (3.2mx4.9m)
- Two Ball mills – AC – (3.2m x 4.0m) and (3.8m x 5.5m)
- One Regrind mill – AC – (2.4m x 3.7m)
- 17 - 8.5 m3 Wemco cells
- 22 Denver cells
- Two thickeners – 3.4m and a thickener/clarifier – 2.1m
- Two Zn Filters – 2.7m and 1.8m x 6 discs agitdisc - and one Cu Filter – 1.8m x 7 discs
- Eight cranes
- One lime silo
- Two MCC Rooms
- Three compressors and 3 blowers
- Utility pipeline – crusher and mill
- Two transformers on site to 4.16 kV and 6 Power Distribution Centers (PDCs)
- Two fuel tanks and pumps for diesel and gas respectively, 2 main propane tanks and auxiliary propane tanks for office trailers
- One backup generator (1 MW).
- Three x water pump stations including holding tanks, MCCs and generators
- Quonset and shed for part storage and muster point and 3 trailers for office space
- Dry space for approximately 90 personnel in total

Hudbay has submitted a Notice of Alteration to Manitoba Sustainable Development to expand the TIA within the existing limits to accommodate the future tailings produced through the entire Lalor mine operations. The commissioning of this expansion is anticipated to be completed in the fall of 2019.

The New Britannia site is a brownfield site previous operated from 1995 to 2004 and placed into care and maintenance in 2005. The infrastructure that will be required for processing operations including existing infrastructure will include (Figure 18-5):

- Administration building
- Dry facilities and warehouse
- Maintenance Workshops
- Ore receiving dump pocket
- Fine ore bin
- Mill building (includes mills, reagents mixing, CIP, elution, gold room, control room)
- Pre-leach thickener and leach tanks
- Flotation and concentrate loadout building
- Pipeline corridor (tailings to Stall, reclaim water and copper concentrate to Stall)
- Pipeline service road

[illegible]

The administration building will house nonproduction personnel which includes HR, accounting, safety and environmental. The building is pre-existing from the previous operation and has been refurbished. The mill dry facilities house the plant change rooms, maintenance offices and warehouse. The building is pre-existing from the previous operation and has been refurbished. The maintenance workshops have been relocated from the mill building to redundant preexisting buildings onsite includes the hoist building.

The mill building houses the grinding circuit, CIP, elution, reagent mixing, DFRs, gold room, process/reclaim water systems, lime slaking, cyanide destruction, plant control, assay laboratory and plant production offices. This building is pre-existing from the previous operation and is being extensively refurbished with the complete replacement of several subsystems including but not limited to the carbon regeneration kiln, acid wash vessel and carbon screens.

To the east and outside of the mill building is the pre-leach thickener, fine ore bin and leach tanks. These buildings are pre-existing from the previous operation and is being refurbished. Included in the refurbishment is the expansion and lining of the containment area around the leach tanks.

The flotation building is a new structure immediately to the north of the mill building. It contains the copper flotation circuit, concentrate dewatering system and concentrate loadout. The concentrate pipeline from Stall mill discharges in this building into the concentrate thickener.

The pipeline corridor between New Britannia and Stall mill consists of a 8" tailings pipeline carrying the New Britannia leach tailings after cyanide destruction to the Stall mill, a 5" concentrate pipeline carrying Stall copper concentrate to the dewatering circuit at New Britannia and a 6" reclaim water pipeline from Stall to New Britannia. The pipeline corridor construction included a light vehicle service road, leak containment system and a bridge to cross Snow Creek.

TABLE 18-1: NEW BRITANNIA MILL REFURBISHMENT – PROJECT DEVELOPMENT SCHEDULE

Development Capex	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Total
Engineering/Early Works	0.2	4.1	3.9	10.1	0.6								18.9
A/E CAD\$M					4.9	15.1	20.7	38.9	\$52.2	23.9	7.9	2.3	165.9
Project Timeline													
Environmental Permitting	→												
Preliminary Engineering		→											
Project Sanctioning				→									
Detailed Engineering				→									
Procurement				→									
Construction Early Mill				→									
Construction Flotation									→				
Commissioning and Ramp Up									→				
Commercial Production												→	

18.7. SNOW LAKE CAMP

The camp is located in the town of Snow Lake and services Hubbay employees and contractors for the mine and mill operations. The camp began operations in January 2011 and has an area of 13,124m². As of February 2021, the camp has seven dorms and three bunkers for a total of 260 rooms. There are also three dedicated trailers for non-dormitory purposes: a 640m² trailer provides a dining room for employees and kitchen for camp employees and the two remaining trailers are a recreation room and a gym.

Utilities are provided by the Town of Snow Lake, including potable water, sewage and electricity. Heating is electrical and propane. The camp also provides parking spots for employees. Snow Lake Camp

19. MARKET STUDIES AND CONTRACTS

19.1. METAL MARKETING

Lalor produces the majority of its zinc concentrate and copper concentrate with gold and silver credits from the Stall Mill. During 2021, a portion of Lalor's production will be delivered to and processed in the Flin Flon Mill.

The refurbished New Britannia mill will process run of mine from the Lalor gold zone to produce a copper concentrate and gold doré bar.

Hudbay does not use intermediate agencies to sell its products from Lalor. The production from Lalor is free of encumbrances and from any streaming agreement and there are no existing contracts that would materially impact the planned revenues in the demonstration of economic viability of the mineral reserve estimates.

Zinc Concentrates and Zinc Metal

Lalor zinc concentrate produced from its Stall Mill is delivered by truck to Hudbay's operations in Flin Flon. It is processed in the Flin Flon zinc plant into refined zinc metal and then sold and delivered to customers in Canada and the United States of America. Production of refined zinc metal will occur until the closure of the Flin Flon zinc plant which is expected during the first half of 2022.

Upon the closure of the Flin Flon zinc plant, Lalor's zinc concentrate will either be sold direct to third party zinc smelters located throughout the world or to merchants for delivery to zinc smelters located throughout the world. To date, no sales contracts have been established for the sale of Lalor zinc concentrate but it is anticipated that all sales contracts will be at standard industry terms commensurate with the quality of Lalor zinc concentrate.

Copper Concentrates

Lalor copper concentrate produced from its Stall Mill and the Flin Flon Mill is sold directly to a copper smelter in North America under a long term frame agreement. This agreement references the annual benchmark agreements between major concentrate producers and non-integrated smelters for the purpose of fixing the annual treatment and refining charge. The Lalor copper concentrate is delivered by rail.

Lalor copper concentrate expected to be produced from the refurbished New Britannia Mill have not been committed. It is anticipated that all sales contracts for the sale of copper concentrate produced from the refurbished New Britannia mill will be at standard industry terms commensurate with the quality of New Britannia copper concentrate.

Gold Doré

The refurbished New Britannia mill will commence production of gold doré in 2021. It is forecast to contain approximately 70% silver, 25% gold and 5% other elements. The doré bar will be delivered and sold to refineries using specialized third-party transportation at standard industry terms.

19.2. CONTRACTS

Engineering, supply and construction contracts are initiated, managed and administrated by Hudbay's Manitoba Business Unit. Hudbay follows a standard contracting-out process that specifies contractors' requirements to be eligible to be considered for work. Contractor selection criteria include ability to complete the work within the required time, safety record and programs, price, and proposed alternatives. The Lalor contracts that are in place have rates and charges that are within industry norms.

Hudbay has a marketing division that is responsible for establishing and maintaining all marketing and sales administrations of concentrates and metals. As well, Hudbay conducts ongoing research of metal prices and sales terms as part of normal business and long range planning process to achieve market terms. Contract terms used in the Lalor financial evaluation are based on this research and the author has reviewed these results and they support the assumptions made in this Technical Report.

In 2020, Hudbay entered into a gold forward sale and prepay arrangement and received an upfront payment of \$115 million to fund the refurbishment of the New Britannia gold mill. As a result, Hudbay is obligated to deliver a total of 79,954 gold ounces in 2022 and 2023 or approximately 3,331 gold ounces per month over a 24-month period from January 2022 to December 2023. The upfront payment was based on gold forward curve prices averaging approximately \$1,682 per ounce.

20. ENVIRONMENTAL STUDIES, PERMITTING AND SOCIAL OR COMMUNITY IMPACT

The permits required for the current Lalor operation, including the Lalor mine, Stall concentrator and Anderson tailings facility have all been issued and remain valid.

At this time, there are no known environmental concerns which could adversely affect Hudbay's ability to operate the Lalor mine. Since the mine site is nearby existing facilities in the Snow Lake area, the Lalor mine was able to utilize infrastructure, services, and previously disturbed land associated with permitted, pre-existing and current mining operations in the Snow Lake area. The Lalor mine and associated projects are designed to minimize the potential impact on the surrounding environment by keeping the footprint of the operations as small as possible and by using existing licensed facilities for the withdrawal of water and disposal of wastes.

Initial proposals for baseline work at Wim have been prepared by AECOM. Once complete these environmental studies will form the basis of the required approvals needed to advance this project should it be deemed viable.

3 Zone is part of the New Britannia site. Significant environmental studies of the area are available, and additional environmental assessments would be utilized to augment our understanding of property and any potential offsite impacts. Approvals to advance this project would be through Provincial regulators as part of an alteration of the existing Environment Act Licence for the property.

The 1901 deposit would leverage all existing surface and underground development near Lalor operations. Significant environmental baseline work has recently been conducted by AECOM and in conjunction with the significant amount past studies will be used to gain approvals for this development should it prove viable.

The re-commissioning of the New Britannia concentrator involves the placement of the Lalor gold ore tailings in the fully licensed Anderson tailings facility or the Lalor Paste fill plant via a pipeline connecting Stall concentrator to Lalor Mine. All environmental permits are in place for the refurbishment of the New Britannia mill and associated infrastructure.

The New Britannia site includes the Birch Lake Tailings Management Facility (BLTMF). The BLTMF is currently on care and maintenance since the closure of the mine in 2005. At this stage, and given Hudbay's history in the area, there is no indication that the approvals will not be obtained within the project schedule. Initial environmental reviews and assessments will need to be completed at Watts. Permitting efforts will be completed prior to the project advancement. Hudbay plans to engage qualified consulting firms to carry out baseline studies when there is a better understanding on when the Watts potential mine production would best fit into the integrated business plan of the Snow Lake operations of Hudbay.

Hudbay engaged AECOM to carry out baseline studies for the Pen II property. Once completed, these studies will comply with the regulatory requirements and will be part of the permitting submission process should the Pen II Zone project be advanced.

Based on Hudbay's long-term (more than 50 years) mining experience in the Snow Lake region, and baseline studies to date, there is no known First Nation or Aboriginal hunting, fishing, trapping or other traditional use of the land in the zone of potential influence for the Lalor mine and associated facilities. Post closure, all water quality and earthen structures will be monitored and inspected in order to ensure the sites' conditions meet the applicable regulatory requirements.

20.1. ENVIRONMENTAL STUDIES AND PLANNING

Lalor Mine Stall Mill and Paste Plant

Prior to commencing advanced exploration work at Lalor, environmental impact assessments including a baseline investigation were initiated in 2007 by AECOM, a consulting group that specializes in environmental service. This investigation included all the required terrestrial and aquatic field studies. The Lalor Advanced Exploration Project plan (AEP) was submitted in the first quarter of 2010 and approved in the second quarter of 2010. The Lalor mine Environment Act Licence (EAL) application was submitted during the second quarter of 2012 and was approved in the first quarter of 2014.

Baseline work and studies were also performed by AECOM for the Anderson Tailings Impoundment Area (TIA) expansion. The notice of alteration was submitted in the third quarter of 2016 and approved in the second quarter of 2018. Baseline work from other Lalor related projects was used for the Lalor paste plant Notice of Alteration (NoA) which was submitted in the fourth quarter of 2016 and approved in the first quarter of 2017. At this time, there are no known environmental issues which could materially impact the ability to extract mineral resources or mineral reserves.

New Britannia Mill

Baseline work and environmental studies were conducted by AECOM and included in the (NoA) that was submitted in March 2019. The NoA was approved by Manitoba regulators in February of 2020. This approval allows Hudbay to restart the New Britannia concentrator and associated pipeline development linking the New Britannia and the Stall concentrators. The water withdrawal permit was updated to accommodate the refurbished concentrator in June 2020.

Snow Lake Satellite deposits

Due to the extensive work completed by AECOM and other existing studies completed as part of environmental effects monitoring programs at the various operations in the Snow Lake area, only limited additional baseline studies may be required for potential future modification of Hudbay's current properties. There is no present indication that future approvals will not be obtained to meet potential future construction schedules.

It is envisaged that AECOM will be utilized to complete the necessary baseline work at Watts. Based on our previous experience in the region, it is anticipated that regular processes familiar to our previous long term permitting efforts that all necessary permits will be obtained to meet the project schedule.

20.2. WASTE AND TAILINGS DISPOSAL AND WATER MANAGEMENT

Lalor Mine, Stall Mill and Paste Plant

There are no known environmental concerns which could adversely affect Hudbay's ability to operate Lalor mine and the 1901, WIM and 3 zone deposit as currently envisaged. Since the mine site and its associated satellite deposits are near existing facilities in the Snow Lake area, Hudbay will be able to utilize infrastructure, services, and previously disturbed land associated with permitted, pre-existing and current mining operations in the Snow Lake area.

Water management initiatives for the Lalor mine and associated projects are designed to minimize the potential impact on the surrounding environment by keeping the footprint of the operations as small as possible and by using existing licensed facilities for the withdrawal of water and disposal of waste. For instance, the NoA for the expansion of the Anderson TIA was prepared by AECOM and utilized the geotechnical tailings dam designs from Hudbay's ongoing Engineer of Record (performed by BGC Engineering Inc.). The permitted design allowed for the entire volume of tailings from Lalor LOM to be stored in Anderson TIA. This design did not discount the volume of tails that are and will be used in the future for paste backfill at the Lalor mine.

The initial stage of expansion is complete and subsequent raises are being contemplated based on actual production rates and forecasted life of mine plans. The actual tailings production will be measured against remaining volumes based on ongoing bathymetry surveys, as per the existing licence requirements.

The initial stage of expansion was completed in 2020. The next raise is will be initiated in 2021 and is anticipated to be complete in 2022. The planned tailings facility raises are based upon Lalor's long range plan. All base metal and gold ore tailings are incorporated into the long term storage capacity of Anderson TIA.

New Britannia Mill

The re-commissioning of the New Britannia mill will involve the placement of the Lalor gold ore tailings in the fully licensed Anderson TIA via a pipeline that will connect to the Stall mill. Trade-off studies and environmental assessments are ongoing to ensure that the use of Anderson TIA for all Lalor tailings is the most effective option. These studies include the study of potential water quality impacts of potential increases in New Britannia operating capacity. A NoA for the refurbishment and upgrade of the New Britannia concentrator was submitted in the first quarter of 2019 and was approved by Manitoba regulators in the first quarter of 2020.

Satellite deposits

Future production from the Pen II Zone would be processed in the Stall mill and the tailings would be stored at the Anderson TIA.

20.3. PERMITTING REQUIREMENTS

Lalor Mine, Stall Mill and Paste Plant

The existing Lalor mine EAL was obtained in the first quarter of 2014 and covers all facilities on the Lalor site. These include sewage, water treatment facilities and the pipelines which carry freshwater and wastewater. The Lalor mine was permitted such that it uses to the greatest extent possible, facilities that were already licensed. The sources of freshwater and discharge of wastewater at Hudbay's existing operating sites are licensed provincially. Discharge of effluent to the environment is regulated federally under the Metal and Diamond Mining Effluent Regulations (MDMER).

The permits required for the Lalor operation, including the Lalor mine, Stall concentrator, New Britannia concentrator, and Anderson TIA are current and valid licenses are held. The Anderson TIA expansion NoA was submitted for approval in the third quarter of 2016 and were approved in the second quarter of 2018. No additional tailings impoundment area will be required. Federal permits were not required for this expansion.

Lalor paste plant NoA was submitted in the fourth quarter of 2016 and was approved in the first quarter of 2017.

Environmental baseline studies have been completed by AECOM in the area of the 1901 deposit. This will augment existing environmental studies in the region and will be utilized in permitting the development and infrastructure upgrades needed for this project. The existing infrastructure in the Lalor project area will be utilized to the fullest extent to minimize the need for future construction and minimize disturbances.

New Britannia Mine Mill

The New Britannia site includes the Birch Lake Tailings Disposal Facility (BLTDF). The BLTDF is currently in care and maintenance since the closure of the mine in 2005. The seasonal discharge from BLTDF, which has an EAL, is regulated under the MDMER. Hudbay obtained a new water withdrawal permit for this site in June 2020 to accommodate the anticipated freshwater requirement to process Lalor ore. In order to initiate upgrades needed to process material from the Lalor mine; a NoA was submitted for approval in the first quarter of 2019 and was approved in the first quarter of 2020.

Satellite deposits

Hudbay will initiate baseline studies and applications will be sent to the appropriate regulatory body for approval sufficiently in advance of when these satellite deposits are anticipated to be constructed and put into production. At this stage, and given Hudbay's permitting experience and continued operation in the area, there is no indication that approvals for development of satellite deposits will not be obtained within the anticipated project schedules.

20.4. COMMUNITY SUPPORT

The main settlement in the region of the Lalor mine is the town of Snow Lake, which is an important mining and service centre for the surrounding area. Snow Lake had a population of 899 according to the 2016 Statistics Canada census. This number is not expected to have significantly changed. The majority of Snow Lake residents are directly employed or support the Lalor mine through contracting services.

Through the years, Hudbay and AECOM have carried out public consultation, including meetings to inform local communities on the progress and development of the Lalor mine, expansion of Anderson TIA, New Britannia concentrator, and the environmental effects of these projects. Manitoba Sustainable Development has taken these meetings into account in the environmental licensing process. Since the economy of Snow Lake is based on mining, opposition to the projects is seen as unlikely.

20.5. FIRST NATIONS

Based on Hudbay's long-term (more than 50 years) mining experience in the Snow Lake region, there is no known current First Nation or Aboriginal hunting, fishing, trapping or other traditional use in the zone of potential influence for the Lalor mine and potential future projects. There is no First Nation Registered Trapline District or Reserve in the area that will be affected by the Lalor operation. Although development on the Mine Site involved a loss of vegetation and habitat for wildlife, the vegetation and habitat type is common throughout the region.

The Mathias Colomb Cree Nation ("MCCN"), located 125 km northwest of Snow Lake at the community of Pukatawagan, has asserted a right to be consulted in connection with recent Hudbay projects in the Snow Lake area since the development of the Lalor Mine. During the permitting process of the New Britannia site, the Manitoba Métis Federation ("MMF") asserted a right for consultation regarding this and future projects in the area. Hudbay continues to keep First Nations and Aboriginal communities informed on potential future projects and will provide Manitoba regulators with all information necessary to support Crown Consultation decisions. Regarding First Nations and Crown Consultation it is anticipated that there will be no impact to current operations or delays in project schedule.

20.6. HERITAGE RESOURCES

Operation of the Lalor mine and construction of potential future upgrades will not affect any known site of potential historical, archaeological or cultural significance. The only known site, Trampling Lake, is located approximately 20 km south of the Lalor operation. This site is one of Manitoba's largest known concentrations of aboriginal pictographs. These paintings are thought to have been created 1,500 to 3,000 years ago by the Algonkian-speaking ancestors of the Cree and Ojibway First Nations. Activities associated with the Lalor operation will not have any impact on this historical site.

20.7. MINE CLOSURE

The Manitoba Mines and Minerals Act require a closure plan and financial assurance for any advanced exploration or mining project. Manitoba accepted Hudbay's Closure Plans prepared by SRK in 2005 and financial assurance to cover the cost of closure for all existing infrastructure that will continue to be used during operation of the Lalor mine. Existing facilities which support the Lalor mine include the Chisel North mine, which is connected by an underground ramp to Lalor, Stall Concentrator and Anderson TIA piping systems associated with milling and tailings deposition, the Chisel Open Pit and the Chisel North water treatment plant.

Prior to commercial production at the Lalor mine, Manitoba approved the Closure Plan for the Lalor AEP. The Lalor AEP Closure Plan was prepared in 2010 and approved as part of the AEP application process. As a requirement of the Lalor mine EAL, an updated closure plan was prepared by SRK and was submitted in the third quarter of 2014. It is anticipated that the site of the Lalor mine will be substantially returned to its natural state in about five to ten years post closure, after which no monitoring or other measures will be required.

The expansion of Anderson TIA, and upgrades to the New Britannia site also will require the submission of updated closure plans and financial assurance. Allowances for these applications have been included in the cash flow model supporting reserve declaration.

Post closure, all water quality and earthen structures will be monitored and inspected in order to ensure the sites' conditions meet the applicable regulatory requirements.

Expansions and upgrades of the existing operating facilities and contemplated future developments will further extend the useful life of Hudbay's assets in Northern Manitoba. Additional development of ore bodies will extend operations and deferred the closure and reclamation while these facilities are in continued use.

Satellite Deposits

The Wim and 3 Zone closure costs were estimated by Hudbay and are based on costs from Reed Mine, as the mining method and surface infrastructure requirements are similar. The major tasks for closure are assumed to be: revegetation of disturbed areas, removal of contaminated soils (in such areas where fuel tanks, maintenance shops, etc. were constructed), removal of concrete slabs used for building foundations, removal of electrical infrastructure, and capping of mine accesses (decline and raise openings). Hudbay is familiar with closure plan development for AEPs and active mine sites given its activities in Northern Manitoba in recent years. A standard practice is to leverage existing facilities already permitted and covered under existing closure plans. This reduces the capital requirements for new developments, minimizes potential impacts to new land & waterways, and reduces closure costs. It is estimated that the cost of reclamation of the satellite deposits will be in the order of \$0.9M.

Timing of Closure

The site closure of the Flin Flon Metallurgical Complex (FFMC) is tied to Hudbay's LOM of Snow Lake area operations. The FFMC will be kept on care & maintenance until all satellite ore bodies are depleted in the region, potentially allowing for the concentrator and tailings facility to be utilized. Although this is dependent on the production schedule of potential new deposits, it allows for further use of the FFMC asset and reduces the risk of permitting delays in the future. An added benefit of extending the care & maintenance period of the FFMC, is the closure costs are then deferred, allowing for additional time to improve the closure plan activities and reduce financial risk associated with uncertainties tied to future closure activities.

21. CAPITAL AND OPERATING COSTS

Unless otherwise stated, capital and operating costs are estimated in real 2021 Canadian dollars.

21.1. CAPITAL COSTS

The capital expenditures required to execute the life of mine plan at Lalor, 1901, WIM and 3 Zone can be separated between growth projects represented by the New Britannia mill refurbishment, the construction of a pipeline corridor to support the planned increase in gold production and Stall mill enhancement initiatives and sustaining capital which includes capitalized mine development, the replacement/acquisition of mining equipment and other capitalised expenditures related to mining, milling and environmental activities.

New Britannia Mill Refurbishment and Pipeline Corridor

For the New Britannia Refurbishment Project, the plan includes refurbishment costs for the existing mill building, inclusion of an outdoor crushed ore loading system, pre-aeration and leach tanks, and leach thickener. In addition, a new flotation building is being constructed to house the flotation, thickening, dewatering, and concentrate shed components required to produce copper concentrate. New major equipment purchases include outdoor feeder/hopper and conveyor, carbon screens, cyclones, kiln, acid vessel, strip vessel, EW cell, refinery furnace, DFR flotation circuit, Jameson flotation circuit, thickener, filter press, reagent packages and lime silo, and electrical equipment such as transformers MCCs and an emergency generator.

The pipeline includes three individual pipes to facilitate movement of material between the New Britannia mill and the Stall mill (with adjacent Anderson tailings facility). An 8" pipe will carry tailings from New Britannia to Stall and a 6" pipe will carry copper concentrate from Stall to New Britannia. The copper concentrate produced at Stall will be piped to New Britannia for dewatering purposes. A reclaim water line 4" will transport water from Stall to New Britannia. This arrangement facilitates environmental monitoring, maintenance, operation and efficiencies improving paste plant quality, and provides for lower concentrate moistures for the copper concentrate.

This project includes industry standard equipment and proven processing technology in a brownfield environment. Construction activities started in Q1 2020 and are 76% completed as of February 2021. Plant commissioning for the early gold mill projected for Q1 2021 and Q3 2021 for the flotation plant, ramp up in Q2 2021 for the early mill and Q4 2021 for the flotation plant. The capital expenditure summarised on Table 21-1

TABLE 21-1: NEW BRITANNIA MILL REFURBISHMENT CAPITAL COSTS

MILL REFURBISHMENT	\$M
Direct Costs	
Procurement	30.2
Site General	9.4
New Flotation Facility	13.5
New Flotation Building	10.8
Mill and Crush Facility	35.0
Pipeline Corridor	15.9
Town of Snow Lake, Raw Water, Office Upgrades	1.2
Stall Tailings Upgrades	6.5
Subtotal	122.5
Commissioning/OR	5.1
Engineering and Indirects	7.6
Owner's Costs, Indirects	27.7
Contingency	3.0
Total Mill Refurbishment	165.9

Since the initial project approval in 2019, the project budget has been increased to accommodate additional scope associated with Stall pipeline and tie-ins, fire and security systems upgrades to meet 2021 regulations and guidance (\$8M), the acceleration of the mill completion to produce early gold (\$4M) and disruption to construction associated with the impact the global pandemic (\$8M).

Sustaining Capital and Capitalized Development

The total sustaining capital cost (Table 21-2) at the Lalor mine, 1901 deposit and satellite deposits is \$858M, comprised primarily of capitalized development and mine equipment costs, totaling \$517M and \$283M, respectively. In addition, milling and other sustaining capital costs total \$58M.

Sustaining capital costs for the new mine plan are formulated assuming a focus on base metal ore extraction to preserve gold ore with higher precious metals content for extraction following commissioning of the New Britannia mill. Overall, this approach results in a superior economic value for the Snow Lake Operations due to the significantly higher precious metals recoveries at the New Britannia mill.

The largest component of capitalized development is mechanical drift costs arising from approximately 50 kilometres of development required to access stopes over the life of mines (26 km at Lalor, 6 km at 1901, 9 km at WIM, and 10 km at 3 Zone). These costs are incurred primarily in the earlier years of the mine life. Similarly, various types of raises are required for venting, material handling (ore passes), and egresses with approximately 5,400 metres required over the life of mines (3,600 m at Lalor, 600 m at 1901, 700 m at WIM, and 500 at 3 Zone).

TABLE 21-2: SNOW LAKE CAPITAL EXPENDITURES

SNOW LAKE - CAPITAL EXPENDITURES		2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Lalor Sustaining Capital	C\$M	106	123	86	59	48	58	56	47	4	4
1901 Sustaining Capital	C\$M	-	-	-	22	32	28	7	2	6	2
WIM Sustaining Capital	C\$M	-	-	-	-	-	-	-	-	57	20
3 Zone Sustaining Capital	C\$M	-	-	-	-	-	-	-	-	-	-
Total Sustaining Capital	C\$M	106	123	86	81	80	86	63	49	67	26
Total Sustaining Capital	US\$M	83	96	67	62	62	66	48	38	52	20
New Britannia Project	C\$M	90	-	-	-	-	-	-	-	-	-
Stall Recovery Improvement Project	C\$M	-	24	-	-	-	-	-	-	-	-
Snow Lake Camp	C\$M	8	-	-	-	-	-	-	-	-	-
Total Growth Capital	C\$M	98	24	-	-	-	-	-	-	-	-
Total Growth Capital	US\$M	77	19	-	-	-	-	-	-	-	-
		2031	2032	2033	2034	2035	2036	2037	LOM		
Lalor Sustaining Capital	C\$M	1	-	-	-	-	-	-	593		
1901 Sustaining Capital	C\$M	-	-	-	-	-	-	-	98		
WIM Sustaining Capital	C\$M	21	15	1	1	1	1	1	118		
3 Zone Sustaining Capital	C\$M	-	-	-	14	22	12	-	49		
Total Sustaining Capital	C\$M	22	15	1	15	23	13	1	858		
Total Sustaining Capital	US\$M	17	11	1	12	18	10	1	664		
New Britannia Project	C\$M	-	-	-	-	-	-	-	90		
Stall Recovery Improvement Project	C\$M	-	-	-	-	-	-	-	24		
Snow Lake Camp	C\$M	-	-	-	-	-	-	-	8		
Total Growth Capital	C\$M	-	-	-	-	-	-	-	123		
Total Growth Capital	US\$M	-	-	-	-	-	-	-	96		

Note: Canadian dollar expenditures converted to U.S. dollar capital expenditures at a C\$/US\$ exchange rate of 1.27 in 2021, 1.28 in 2022, 1.29 in 2023 and 1.30 long-term.

LALOR - CAPITAL EXPENDITURES		2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	LOM
Sustaining Capital												
Capitalized development	C\$M	63	65	53	50	39	35	31	32	2	2	374
Mine equipment & buildings	C\$M	25	46	20	8	8	21	23	14	1	1	167
Stall equipment & buildings	C\$M	3	0	1	1	0	0	0	0	0	0	6
New Britannia equipment & buildings	C\$M	-	1	1	1	1	1	1	1	1	1	10
Shared general plant	C\$M	2	2	-	-	-	-	-	-	-	-	4
Environmental	C\$M	14	8	11	-	-	-	-	-	-	-	33
Total Sustaining Capital	C\$M	106	123	86	59	48	58	56	47	4	4	593
Growth Capital												
New Britannia Project	C\$M	90	-	-	-	-	-	-	-	-	-	90
Stall Recovery Improvement Project	C\$M	-	24	-	-	-	-	-	-	-	-	24
Snow Lake Camp	C\$M	8	-	-	-	-	-	-	-	-	-	8
Total Growth Capital	C\$M	98	24	-	-	-	-	-	-	-	-	123

1901 - CAPITAL EXPENDITURES		2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	LOM
Sustaining Capital												
Capitalized development	C\$M	-	-	-	13	16	22	2	-	1	0	54
Mine equipment & buildings	C\$M	-	-	-	9	16	6	5	2	5	1	44
Total Sustaining Capital	C\$M	-	-	-	22	32	28	7	2	6	2	98

Note: Pre-production mine development capital have been categorized as sustaining.

WIM and 3 ZONE - CAPITAL EXPENDITURES		2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	LOM
Sustaining Capital												
Capitalized development	C\$M	-	9	15	14	6	-	11	22	12	-	88
Mine equipment & buildings	C\$M	-	48	5	8	7	0	3	1	-	-	73
New Britannia equipment & buildings	C\$M	-	-	-	-	1	1	1	1	1	1	6
Total Sustaining Capital	C\$M	-	57	20	21	15	1	15	23	13	1	167

Note: Pre-production mine development capital have been categorized as sustaining.

Equipment (new purchase, replacement, and/or rebuild) as well as infrastructure (surface and underground) costs also account for a large component of sustaining capital requirements with life of mine costs of \$283M. Replacement and rebuild costs are based on the scheduled useful life of individual equipment as well as to replace contractor provided equipment with Hudbay owned equipment as a measure to reduce operating costs.

Stall mill sustaining capital costs of \$6M over the life of mine includes scheduled replacement of key equipment, and building upkeep required to maintain production at up to 3,800 tpd. Similarly, New Britannia mill sustaining capital costs of \$16M over the life of mine is scheduled to maintain desired throughput levels. Additionally, environmental capital costs of \$33M relate to the scheduled Anderson tailings facility raises and stabilization work expected to be completed over the life of mine, to accommodate tailings from both the Stall and New Britannia mills as a result of regular operations.

21.2. OPERATING COSTS

Operating cost estimates at Lalor are based on the 2021 budget figures and were developed by Hudbay with a bottom-up approach using quotes from local suppliers, Manitoba operations experience, labour costs within the region, and actual 2020 costs. The operating cost estimates are presented below, separately for mining, milling, refining and general and administrative activities.

The operational costs at 1901 are expected to be lower than the costs experienced at Lalor. The costs associated with mining ore (direct mining, ore removal, and backfill) are like Lalor's PPCF method, with the exception that hoisting capacity (and associated costs) seen at Lalor will not be incurred. The overall PPCF mining unit operating costs are assumed to be approximately \$12 per tonne lower at 1901, and this is largely due to the difference in general mine expense (GME) costs. Personnel requirements are less, and the indirect costs tied to Lalor infrastructure that is not applicable to 1901 has been excluded in cost estimates.

The operating costs for WIM and 3 Zone reflect an owner-operator scenario for production, and contractor pricing for development similar to the way Hudbay developed the Reed mine until 2018. The longhole mining unit-costs (direct mining, ore removal, backfill) are expected to be higher than experienced at Lalor, due to lower production rates, added haulage distance, and use of smaller equipment to mine the narrow orebodies. Due to proximity to Snow Lake, the mines are operated on a "local mine" basis in which most of the employees work standard business hours and shift work. As such, GME costs are based on Reed's historical costs, with adjustment and escalation (to account for inflation) factors applied.

Mining Operating Costs

Table 21-3 presents the details of the Lalor mine and satellite deposits' operating costs. Labour costs constitute the largest component of the mining operating costs for the life of the operations, and typically include both employee and contractor costs arising from the various mining and maintenance activities. At Lalor, the staffing plan includes approximately 600 positions, comprised primarily of hourly paid employees.

TABLE 21-3: MINE OPERATING COSTS

MINE OPEX	LALOR		1901		WIM		3 ZONE	
	C\$M	C\$/t	C\$M	C\$/t	C\$M	C\$/t	C\$M	C\$/t
2021	177	112.0	-	-	-	-	-	-
2022	165	101.5	-	-	-	-	-	-
2023	189	102.0	-	-	-	-	-	-
2024	182	98.0	-	-	-	-	-	-
2025	189	101.8	-	-	-	-	-	-
2026	180	102.3	4	46.4	-	-	-	-
2027	166	112.3	29	87.3	-	-	-	-
2028	147	114.4	28	82.5	-	-	-	-
2029	152	119.2	30	84.3	-	-	-	-
2030	104	119.3	28	90.9	5	51.4	-	-
2031	25	137.9	18	121.6	28	67.1	-	-
2032	-	-	-	-	30	68.0	-	-
2033	-	-	-	-	35	80.5	-	-
2034	-	-	-	-	32	80.8	1	30.5
2035	-	-	-	-	26	80.8	14	63.8
2036	-	-	-	-	23	80.8	16	73.7
2037	-	-	-	-	4	80.8	15	80.1
LOM	1,676	107.3	138	87.1	183	74.9	46	69.8

GME includes a variety of costs such as power, mine administration, heating, ventilation, level maintenance, engineering, geology, and Snow Lake area costs that are allocated between capitalized development and mining operating costs based on accounting rules. For Lalor and the Snow Lake satellite deposits, this cost represents 30-40% of the total mining operating costs over the life of mines.

In terms of mining activities, ore extraction (direct mining, ore removal, and backfill) represents the largest cost driver accounting for 40-50% of mining operating costs over the life of mines, while mine development activities account for the remaining 20%.

Milling Operating Costs

Table 21-4 presents the details of the Stall and New Britannia operating costs. As with mining, labour costs are the largest component of milling operating costs between the mills used to process Lalor ore. The staffing plan includes a total of approximately 80 positions at Stall mill and 50 at New Britannia mill, comprised primarily of hourly paid employees. Concentrates produced at Stall and New Britannia mills are transported to Flin Flon by truck where they are loaded onto railcars and shipped to off-takers.

TABLE 21-4: MILLING OPERATING COSTS

MILL OPEX	STALL		NEW BRITANNIA		TOTAL SNOW LAKE OPERATIONS ¹	
	C\$M	C\$/t	C\$M	C\$/t	C\$M	C\$/t
2021	34	26.2	10	45.2	45	28.7
2022	32	29.4	22	40.7	54	33.2
2023	31	23.6	22	40.4	53	28.5
2024	31	23.4	22	40.4	53	28.4
2025	31	23.9	22	40.4	53	28.7
2026	31	23.4	22	40.4	53	28.3
2027	30	23.8	22	41.1	52	28.9
2028	29	26.8	22	40.7	51	31.4
2029	30	25.1	21	47.4	51	31.3
2030	28	35.2	22	44.1	50	38.6
2031	26	119.1	22	42.6	48	64.7
2032	-	-	22	50.5	22	50.5
2033	-	-	22	50.8	22	50.8
2034	-	-	22	50.5	22	50.5
2035	-	-	22	42.0	22	42.0
2036	-	-	22	44.0	22	44.0
2037	-	-	20	86.9	20	86.9
LOM	333	27.3	360	44.7	695	34.2

¹Includes \$2M allocation of Flin Flon mill costs applicable to processing of Lalor ore in 2021

Refining Costs

Until the closure of the zinc plant in Q2 2022, the Lalor zinc concentrate will be refined into finished zinc to maximize the realizable value from zinc produced. The operating costs presented in Table 21-5 summarize the costs allocated to Snow Lake Operations for the refining of zinc concentrates produced from Lalor mine's production (either through the Stall or Flin Flon mills).

TABLE 21-5: REFINING OPERATING COSTS

OPEX - REFINING		2021	2022	LOM
Snow Lake Operations				
Refining	C\$M	59	28	87
Per pound finished zinc produced	C\$/lb	0.42	0.53	0.45

Note: following the closure of the zinc plant in 2022, Lalor zinc concentrate will be sold directly to market.

General and Administration Costs

General and administration costs (Table 21-6) are comprised of an allocation of shared services, administration, and other office costs to Snow Lake Operations based on relative usage versus Flin Flon Operations. Significant costs in the shared services line item include health & safety, environmental control, logistics, and analytical services. Significant costs in the administration line item include insurance and property taxes, information technology, human resources, and finance.

TABLE 21-6: GENERAL AND ADMINISTRATION COSTS

G&A OPEX	TOTAL SNOW LAKE OPERATIONS	
	C\$M	C\$/t
2021	37	23.4
2022	42	26.0
2023	39	20.8
2024	39	20.8
2025	39	20.8
2026	38	20.3
2027	34	18.9
2028	30	18.6
2029	30	18.5
2030	24	18.2
2031	10	13.4
2032	7	14.8
2033	7	14.9
2034	7	14.9
2035	8	14.9
2036	8	14.9
2037	4	14.9
LOM	399	19.6

Note: decommissioning, pension, and severance costs are considered as corporate legacy costs and therefore not included in the economic valuation of the business plan of Snow Lake Operations.

21.3. UNIT CASH AND SUSTAINING CASH COSTS

With the benefit of the New Britannia mill, the net revenue at Lalor will shift from primarily zinc to primarily gold post 2021, positioning Lalor as a gold mine with significant zinc, copper, and silver by-products. The life of mine gross revenue from Lalor and the Snow Lake satellite deposits is approximately 57% precious metals, 24% zinc and 19% copper. Significant zinc and copper revenue provides a diversified commodity exposure with a significant upside potential from the mineral resource estimates in the Snow Lake that have not yet been converted to mineral reserve estimates both at Lalor and at other Snow Lake gold properties.

Table 21-7 illustrates that Lalor, the 1901 deposit and the Snow Lake satellite deposit's significant by-product credits will reduce its cash operating costs and sustaining cash costs both on a zinc and gold basis. During the first six years of operation with New Britannia (2022-2027), the Snow Lake operations will produce approximately 182,000 ounces of gold annually at a cash cost, net of by-product credits, of US\$411/oz. This positions Lalor to be one of the lowest cost gold mines in Canada.

TABLE 21-7: CASH AND SUSTAINING UNIT CASH COSTS

CASH COSTS	Cash Costs on a Zinc Basis			Cash Costs on a Gold Basis		
	Contained Zinc Mlbs	Cash Costs US\$/lb	Sustaining Cash Costs US\$/lb	Contained Gold '000s oz	Cash Costs US\$/oz	Sustaining Cash Costs US\$/oz
2021	134	(0.23)	0.48	115	(275)	550
2022	112	(0.77)	0.18	160	361	1,027
2023	102	(1.21)	(0.56)	190	434	784
2024	101	(1.16)	(0.54)	191	440	766
2025	78	(1.88)	(1.09)	208	393	690
2026	101	(0.88)	(0.23)	184	454	812
2027	126	(0.44)	(0.06)	162	382	680
2028	145	0.08	0.34	123	443	749
2029	158	0.26	0.58	111	488	953
2030	92	0.07	0.29	97	627	834
2031	44	0.16	0.55	42	647	1,056
2032	-	-	-	19	669	1,271
2033	-	-	-	22	518	571
2034	-	-	-	24	543	1,028
2035	-	-	-	41	642	1,074
2036	-	-	-	39	754	1,015
2037	-	-	-	25	1,140	1,170
LOM	1,193	(0.60)	(0.02)	1,753	421	812

Note:

1. Cash costs and sustaining cash costs are presented net of by-product credits. By-product credits are calculated using the following assumptions: zinc price of \$1.20/lb in 2021, \$1.15/lb in 2022, \$1.10/lb in 2023 and long-term; copper price of \$3.75/lb in 2021, \$3.30/lb in 2022, \$3.10/lb in 2023 and long-term; silver price of \$25/oz in 2021, \$23/oz in 2022, \$20/oz in 2023, \$19/oz in 2024 and \$18/oz for the long-term; C\$/US\$ exchange rate of 1.27 in 2021, 1.28 in 2022, 1.29 in 2023 and 1.30 for long-term.

22. ECONOMIC ANALYSIS

Hudbay is a producing issuer and has excluded the information required by Item 22 of Form 43-101F1 as the updated mine plan does not represent a material increase in current production.

23.ADJACENT PROPERTIES

RockCliff owns the claims located immediately to the North of the PEN II deposit. RockCliff has not reported any mineral resource estimates and no significant exploration results on these claims that would have an impact on the mineral resource estimates reported by Hudbay in this Technical Report for the PEN II property.

There are no other relevant adjacent properties in relation to the other properties discussed in this Technical Report.

24. OTHER RELEVANT DATA AND INFORMATION

There is no other relevant data or information that we are aware of that has not been presented in the other sections of this Technical Report which is necessary to make this Technical Report understandable and not misleading.

25. INTERPRETATIONS AND CONCLUSIONS

Lalor was discovered in 2007 on 100% owned land using Hudbay's innovative exploration techniques and it is the largest VMS deposit found in the Snow Lake region to-date. In 2009, the company commenced construction of Lalor with initial ramp access from the Chisel North mine. Construction of the main production shaft was approved in 2010 and was completed on time and on budget, achieving commercial production in 2014, only seven years from initial discovery.

Hudbay acquired the New Britannia gold mill in 2015 for approximately \$10 million as a potential long-term processing option for the Lalor gold and copper-gold zones. New Britannia was placed on care and maintenance in 2005 by its previous owner after producing 1.6 million ounces of gold. Since acquiring New Britannia, Hudbay has conducted significant technical work to assess the grade, tonnage, mineability and metallurgy of the gold and copper-gold zones at Lalor to support and de-risk the investment required to refurbish New Britannia and maximize the net present value of Lalor.

During the last two years, Hudbay has also dedicated considerable efforts to drilling and test mining in the gold-rich lenses at Lalor as well as advancing engineering studies on the Snow Lake satellite deposits. In addition, the recent discovery of the 1901 deposit in February 2019 has been quickly followed by exploration and infill drilling to define the mineral resource and mineral reserve estimates presented in this Technical Report. The updated estimate of mineral reserves of the integrated mining and milling complex of Snow Lake has increased in-situ contained gold by 12%, copper by 14%, zinc by 34% and silver by 30%, relative to the previous estimate of mineral reserves in Hudbay's 2020 AIF, after adjustment for 2020 mining production.

The New Britannia refurbishment project includes industry standard equipment and proven processing technology in a brownfield environment. The project is well advance and at overall completion of 76% as of February 2021 with construction completed at 70%, engineering and procurement at 99%, commissioning at 12% and with first gold planned for Q3 2021. The completion of the flotation circuit to process the higher copper grade ore is planned for Q4 2021.

In 2019 and 2020, Hudbay has conducted metallurgical testing work to demonstrate the technical and economic viability of several modifications to the Stall mill flowsheet including an increase in copper rougher and cleaner capacity with the addition of Jameson cells, the addition of talc preflotation circuit with the re-configuration of existing flotation cells and addition of a Jameson cell cleaner stage, an increase the zinc circuit cleaning capacity and froth washing through the addition of Jameson cells, and an increase in recovery of free gold through the addition of a Knelson gravity concentrator on the copper regrind cyclone underflow. Implementation of these modifications will be completed by end 2022 for a total investment estimated at \$25m with the resulting benefits planned to start in 2023.

The revised mine plan for the Lalor mineral reserves optimizes net present value by maximizing gold-rich ore for processing at the New Britannia mill and zinc-rich ore for the Stall mill, which is expected to result in significantly higher gold and copper recoveries. This revised mine plan for the Snow Lake operations supports a 17 year mine life, based solely on proven and probable reserves, and utilizes the existing mining capacity of 5,300 tonnes per day at Lalor and 1901 until 2027 and 4,650 tonnes per day for 2028 and 2029 before ramping down towards an average of 1,350 tonnes per day for the rest of the mine life. The technical work completed supports 5,300 tonnes per day as the optimal mining rate to maximize net present value, although the Lalor production shaft has the potential to hoist at higher throughput rates.

With the inclusion of the New Britannia mill, net revenue in Snow Lake will shift from primarily zinc to primarily gold positioning the integrated operations as a primary gold production center with significant zinc, copper and silver by-products. Lalor and 1901 significant by-product credits reduce its cash operating costs and sustaining cash costs on both a zinc and gold basis. During the first five years of operation with New Britannia (2022 to 2026), Recovered annual gold production will average approximately 180,000 ounces of gold at a cash cost, net of by-product credits, of \$386/oz, positioning Lalor as one of the lowest cost gold mines in Canada. In addition, significant zinc and copper revenue provides diversified commodity exposure.

Refurbishing New Britannia is expected to also enable new gold and copper-gold exploration opportunities in the Snow Lake region by having an operating processing facility with substantially higher gold and copper recoveries. It also demonstrates the opportunity to create additional value through owning multiple processing facilities in the Flin Flon and Snow Lake regions.

Since 2019, Hudbay has also implemented a more stringent approach to resource reporting for underground deposits. With this approach, the potential for economic extraction of the mineral resource estimates at Lalor and all the satellite deposits with the exception of the former New Britannia mine, are reported within the constraint of a 'stope optimization envelope' process similar in concept to a Lerchs-Grossman pit shell for an open pit deposit. This excludes from the resource estimate small individual resource blocks that may meet economic cut-off criteria on an individual basis but could not be aggregated into mineable shapes. It is anticipated that this approach will result in higher resource to reserve conversion ratios than those experienced in the past.

The updated resource models at Lalor and 1901 includes 6.5 million tonnes of inferred mineral resources, which has the potential to extend the period of the mine life when both Stall and New Britannia mills are fed at their current combined demonstrated throughput capacity of 5,300 tonnes per day. In addition, the mineral resources at Hudbay's satellite deposits in the Snow Lake region, including the 1901, Watts and Pen deposits and the former gold producing New Britannia mine could provide feed for the Stall and New Britannia processing facilities and further extend the mine life and/or maximise the utilization of the 5,300 tonnes per day processing capacity over a longer period.

In 2020, Hudbay has continued to invest in significant exploration efforts in Manitoba including major airborne and ground geophysical surveys and surface exploration drilling in the Flin Flon and Snow Lake areas. This work has identified several base metal and gold targets to be tested in 2021. In parallel, in-mine exploration at Lalor continues with the intent to convert inferred resources to indicated mineral resources and add additional inferred mineral resources in base metal and gold-rich mineralization. Hudbay will continue to advance engineering studies on Lalor and its satellite deposits in an attempt to continue to increase the tonnage of the mineral reserve estimates and the estimated operating life of the Snow Lake processing facilities at or near full capacity.

The Hudbay validation process has confirmed that the resource models are constructed using industry accepted modelling techniques and classified in accordance with the 2014 CIM Definition Standards – For Mineral Resources and Mineral Reserves. The author considers that the mineral resources and reserves as classified and reported comply with all disclosure in accordance with requirements and CIM Definitions. The author is not aware of any mining, metallurgical, infrastructure, permitting or other relevant factors that could materially affect the mineral reserve estimate.

26. RECOMMENDATIONS

Significant value has been unlocked by leveraging Hudbay's exploration, technical, and operational expertise, its existing portfolio of processing infrastructures and several inexpensive acquisitions. These efforts have highlighted the potential to further increase the tonnage of mineral reserve estimates and the estimated operating life of the Snow Lake processing facilities at or near full capacity through continued exploration success, a disciplined and methodic approach to engineering studies while maintaining excellence in tailings management and environmental stewardship.

On the exploration front, the author recommends to pursue in-mine exploration to incrementally add both base metal and gold rich inferred resources where lenses are still open in their along strike or down dip directions. The in-mine drill program should also continue to convert enough inferred mineral resource estimates to an indicated category so that they can be converted to mineral reserve estimates and replace the planned 2021 mining depletion.

Hudbay has developed a well focused surface exploration program aimed at confirming near surface exploration targets between Lalor and the past producing Chisel mine and around the past producing Snow Lake gold properties. The author supports this program which has the potential to define additional inferred resources in the Snow Lake camp over the next one to two years.

The Watts deposit located within 200km of the Snow Lake processing facilities offer additional opportunities to maximise the use of the Stall and New Britannia mills and/or extend their operating life.

Hudbay has in-mine and regional annual exploration budget of approximately \$10M to cover the cost of these drilling programs proposed for Lalor and the satellite deposits.

As the Lalor mine matures and additional satellite ore sources are identified to supplement the feed blend to the Stall and new Britannia mills, metallurgical projects are actively advanced to match the configuration of the mill to the requirement of the ore. The following are projects that are being advanced through metallurgical testwork, early stage engineering or both.

The current life of mine plan for Lalor shows an increasing Pb:Cu ratio from 0.3 to >1.0. In addition, the 1901 deposit has a Pb:Cu ratio of >1.5 indicating the need for a lead mitigation strategy. Preliminary results indicate that a traditional sequential Cu-Pb-Zn flotation circuit at Stall will mitigate the increasing lead in feed and provide an additional lead concentrate revenue stream that captures a significant portion of the copper, gold and silver that is lost into the zinc concentrate into a payable stream.

The flotation recovery of gold at Stall is limited by the mineralogy and association of gold with iron sulfides. Due to the mineralogical differences between the gold zone and base metal zone ores, the flotation tailings from the base metal ores are not compatible with the same leaching conditions as the gold zone ore flotation tailings. Metallurgical testwork is on going for this project.

Alongside the tailings Leach project, a fine grinding project is being advanced to determine the improvements in flotation recoveries when the flotation feed size is reduced from the current p80 of 120µm to 60µm.

Historical production at the New Britannia mill saw throughput rates of up to 2,000tpd based on the availability of ore from the underground operation. With the increase in the mineral reserves of the Lalor gold zones alongside additional satellite deposits, an opportunity exists to increase the throughput rate. The study to date has identified the tailings pipeline, pre-leach thickener and various minor pumping systems require upgrade to accommodate the increased tonnage rate. All the other major systems within the New Britannia mill have sufficient capacity to accommodate the increased tonnage. In addition to the processing equipment upgrades the study includes the upgrade of the pipeline corridor road to accommodate ore haulage from Stall to New Britannia to remove traffic from the public access road.

The 1901 deposit contains two gold-copper lenses which are suitable feed for the New Britannia mill and would supplement the Lalor gold zone ore. A scoping study has been initiated to advance this project

through metallurgical testwork on composites representative of the two gold lenses for which inferred mineral resource estimates have been reported for the 1901 deposit.

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